GOOGLE WORKSPACE TRAINING

Google Sheets and Intermediate

One-day course review draft

Build a reliable fictional task register. All eight labs have staged application screenshots, with full-window sources retained. Independent challenges, unresolved author checks and classroom timings still require validation.

Delivered by Asraf · Microsoft Certified Trainer

Google Sheets Fundamental and Intermediate

One-day course review draft · Review draft, version 0.4

© 2026 Asraf Jaafar Sidik. All rights reserved.

Resources: Use the practice files supplied with this review package.

Google, Google Workspace, Gmail, Google Drive, Google Docs, Google Sheets, Google Slides, Google Sites, Gemini and NotebookLM are names associated with Google. This independent training guide is not published or endorsed by Google.

Screenshots show the product at the time of writing. Google updates its apps often, so labels, availability and layouts in your account may differ. This review draft still requires full lab validation.

About the author

Asraf is a Microsoft Certified Trainer and HRD Corp-certified IT trainer based in Malaysia. He has been teaching working professionals since 2008, helping teams at banks, pharmaceutical companies, construction firms, and retailers get real work done with Microsoft 365, Power Platform, Excel, Google Workspace, and, more recently, AI tools like Copilot.

He holds a Bachelor of Information Technology (Multimedia) from the University of Malaya, along with Microsoft certifications in Power BI (PL-300) and AI for business (AB-730). Outside the classroom he writes his own automation tools in Python and Node.js, mostly to save himself from doing the same task twice.

He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

Contents

Before you begin 6

1 Create and Navigate a Workbook 7
1.1 Create the Folder and Upload the File 8
1.2 Save It as Google Sheets 9
1.3 Find Your Way Around 10
1.4 Make a Practice Tab 12
1.5 Start the Workbook Guide 15
1.6 Check Four Addresses 16
1.7 Type Three Records 19
1.8 Practise Selecting 20
1.9 Tidy the Tabs and Reopen From Drive 23

2 Enter and Format Reliable Data 28
2.1 Inspect Values and Types 28
2.2 Improve Readability 31
2.3 Inspect Formatting Effects 33
2.4 Diagnose Display, Type and Readability Problems 35
Prompts and values 41

3 Calculate and Copy Formulas 43
3.1 Calculate Variance 43
3.2 Summarise the Columns 45
3.3 Anchor an Assumption 47
3.4 Test Calculations With Zeros, Blanks and Reference Errors 50
Prompts and values 57

4 Clean and Organise Records 60
4.1 Clean a Small Import 60
4.2 Sort Whole Records 62
4.3 Freeze and Name a Range 64
4.4 Clean an Intake and Prove That Records Stay Together 66
Prompts and values 75

5 Validate and Filter the Register 77
5.1 Add a Status Dropdown 77
5.2 Create a Filter View 79
5.3 Test a Rule Safely 81
5.4 Design and Verify Entry Rules and Focused Views 84
Prompts and values 97

6 Summarise and Visualise 99
6.1 Count Statuses 99
6.2 Build a Useful Chart 101
6.3 Explain and Check 103
6.4 Build a Second Summary and Diagnose a Misleading Chart 106
Prompts and values 116

7 Share and Prepare Output 118
7.1 Inspect Access 118
7.2 Prepare the Print Range 119
7.3 Compare Output Options 122
7.4 Prepare a Readable, Scoped and Reproducible Output 123
Prompts and values 138

8 Complete the Register Challenge 140
8.1 Add a New Record 140
8.2 Repair and Reconcile 145

8.3 Explain the Result 148
8.4 Stress-Test the Extended Register and Complete the Handover 150
Prompts and values 162

INTRODUCTION

Before you begin

This review book accompanies Google Sheets Fundamental and Intermediate. Work through its chapters in order and keep the original practice inputs unchanged. The same notes and prompts also appear in the local course website.

Current Status

All chapters in this book are written drafts. Most have not yet been executed end to end in the application. The masterclass's reference brief chapter has a tested core path with a recorded recovery prompt; that does not validate the other exercises. Treat VERIFY notes and draft notices as production work still to complete.

What You Need

Use a desktop browser and the Google account approved for the class. Confirm app and AI access beforehand. Edition, administrator controls and language can affect availability.

Download the supplied fictional practice pack. No real client or personnel record is needed. Dataset week fields are relative practice weeks, not an agreed pilot start date.

How to Use the Exercises

Follow the numbered steps, attempt the output and then open the expected-result box. AI wording can vary; source facts and approval conditions must remain correct.

Values and prompts follow the notes. Replace source placeholders before submitting. Inspect proposed edits before applying them. Drafting does not authorise sending, inviting people or publishing.

Course Timing

This course is planned for 1 day, with 390 teaching minutes per day, excluding breaks and lunch. Estimates include practice, checks and discussion and need a classroom pilot.

Each course supplies independent starting inputs. Use a stated recovery file if an earlier output is missing.

Expanded Casework

Revision 0.4 adds casework, independent challenges and evidence gates to every chapter. The original short guided path alone is insufficient. Timings are targets, not measured learner results. Read each case input and keep baseline and scenario files separate.

CHAPTER 1

Create and Navigate a Workbook

Seri Maju Projects keeps its pilot task list in an Excel file, `TaskRegister.xlsx`, and the coordinator, Aina Rahman, wants the team to work on it in Google Sheets from now on. Before anyone changes a number, she wants a working copy that everyone can find, a guide tab that says what each tab is for, and somewhere safe to practise. In this lab you set that up. You keep using the same workbook in every later lab.

Estimated time: 40 minutes, planned as 5 explanation, 25 guided build and 10 independent practice. Not yet timed with learners.

Your result: a Google Sheets file, `GW - Sheets - Task Register`, with the twelve supplied tasks untouched on **Tasks**, a **Workbook Guide** tab that explains the workbook, and two practice tabs.

> **You'll practise:** uploading an Excel file and saving it as Google Sheets, moving around with the name box, duplicating, adding, renaming and moving tabs, typing records, and selecting cells, rows, columns and ranges.

What You Will Learn

- Create, name, find and reopen a Google Sheets workbook in Drive
- Move between tabs, cells and ranges with the name box
- Keep the supplied records safe and practise on copies

Before You Begin

> **Download:** TaskRegister.xlsx and save it on your computer. It has two tabs: **Tasks**, with twelve fictional tasks in rows 2 to 13, and **Summary**, which you fill in later labs.

How the workbook comes together:

Practice folder > **TaskRegister.xlsx** (the Excel original) > **GW - Sheets - Task Register** (your Google Sheets copy): **Tasks** > **Workbook Guide** > **Navigation Practice** > **Entry Practice** > **Summary**

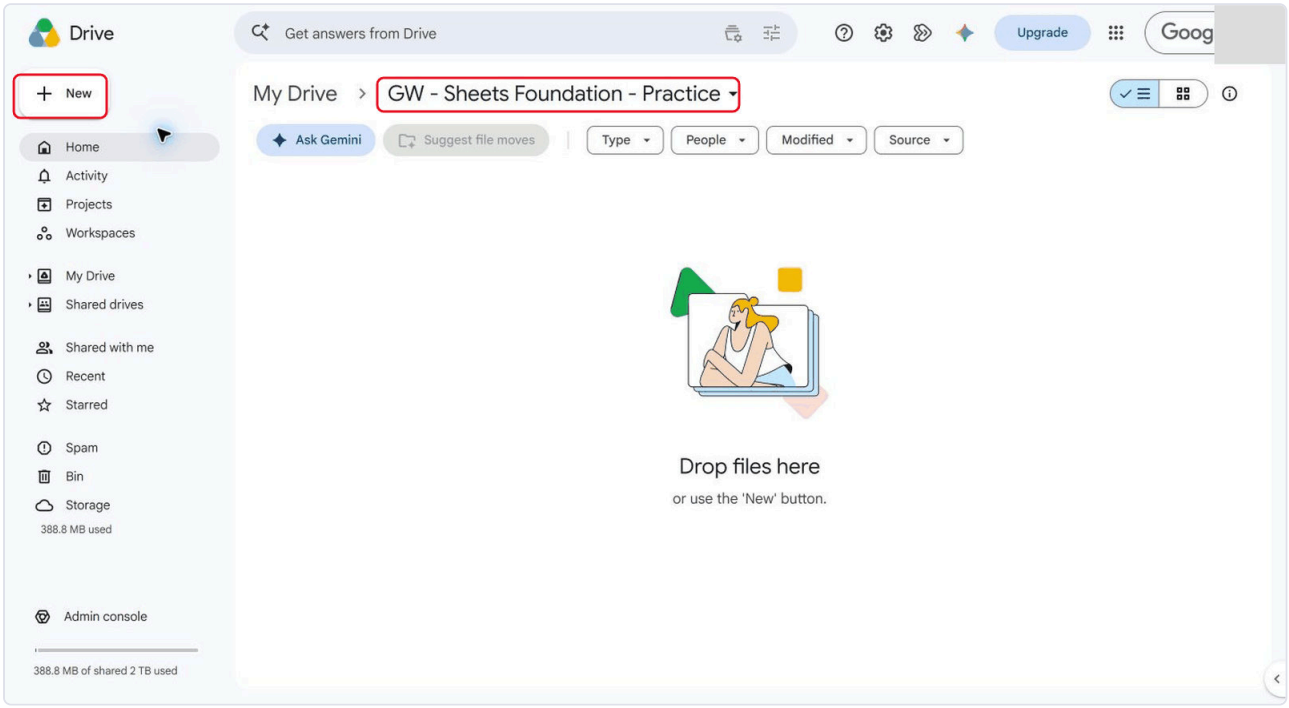
Topics

The Excel file and the Google Sheets file. Uploading `TaskRegister.xlsx` puts the Excel file in Drive. You can open it in Sheets, but it stays an Excel file (you'll see a **.XLSX** badge next to its name). **Save as Google Sheets** makes a second, separate file in Google's own format. That second file is the one you work in. The Excel original stays in the folder, unchanged, so you can always start again.

Addresses and TaskIDs. D5 is a cell address: column D, row 5. It tells you where a value is *now*. If someone sorts the list, a different task ends up in row 5. The TaskID in column A (such as **T004**) stays with its task wherever the row moves, so that's what you use to name a record.

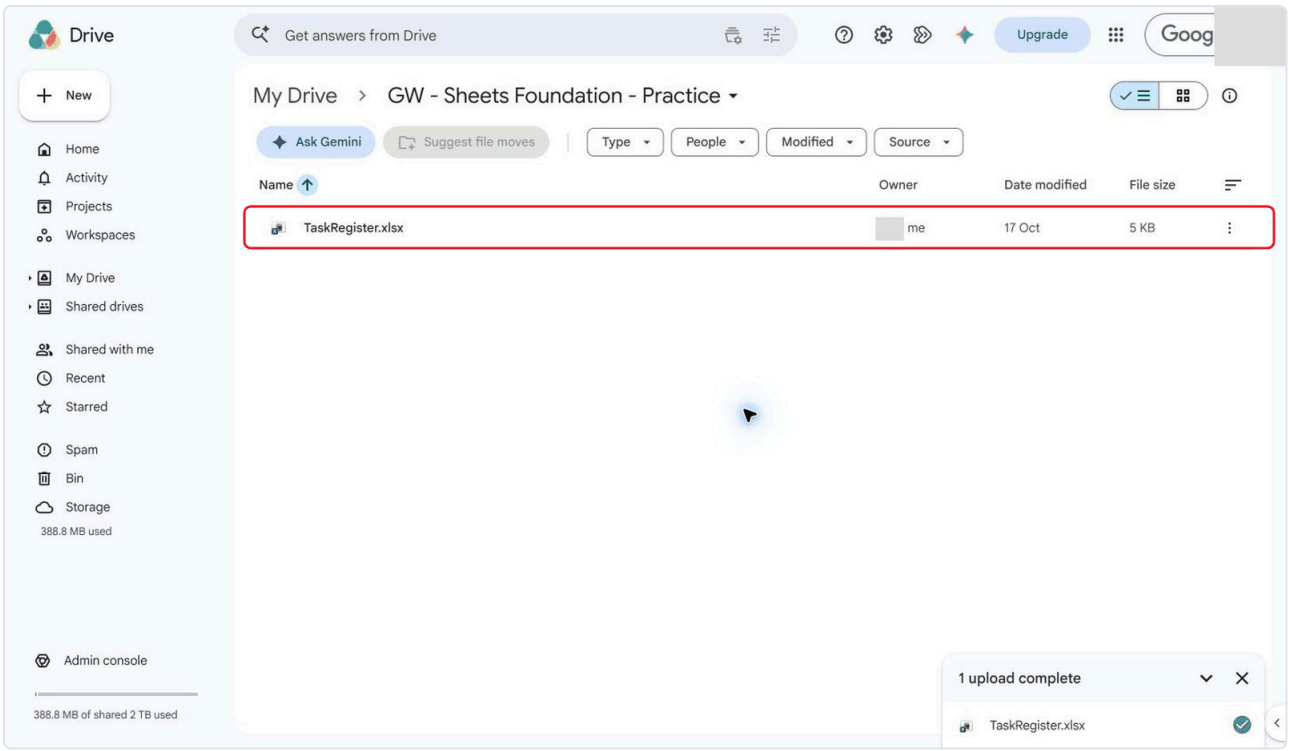
1.1 Create the Folder and Upload the File

1. Open Google Drive.
2. Click **New**, then **New folder**. Type `GW - Sheets Foundation - Practice` and click **Create**.
3. Double-click the folder to open it.



The folder name shows in the path at the top, and the folder is empty.

4. Click **New**, then **File upload**. Choose `TaskRegister.xlsx` from where you saved it and click **Open**.
5. Wait for **1 upload complete** at the bottom right.

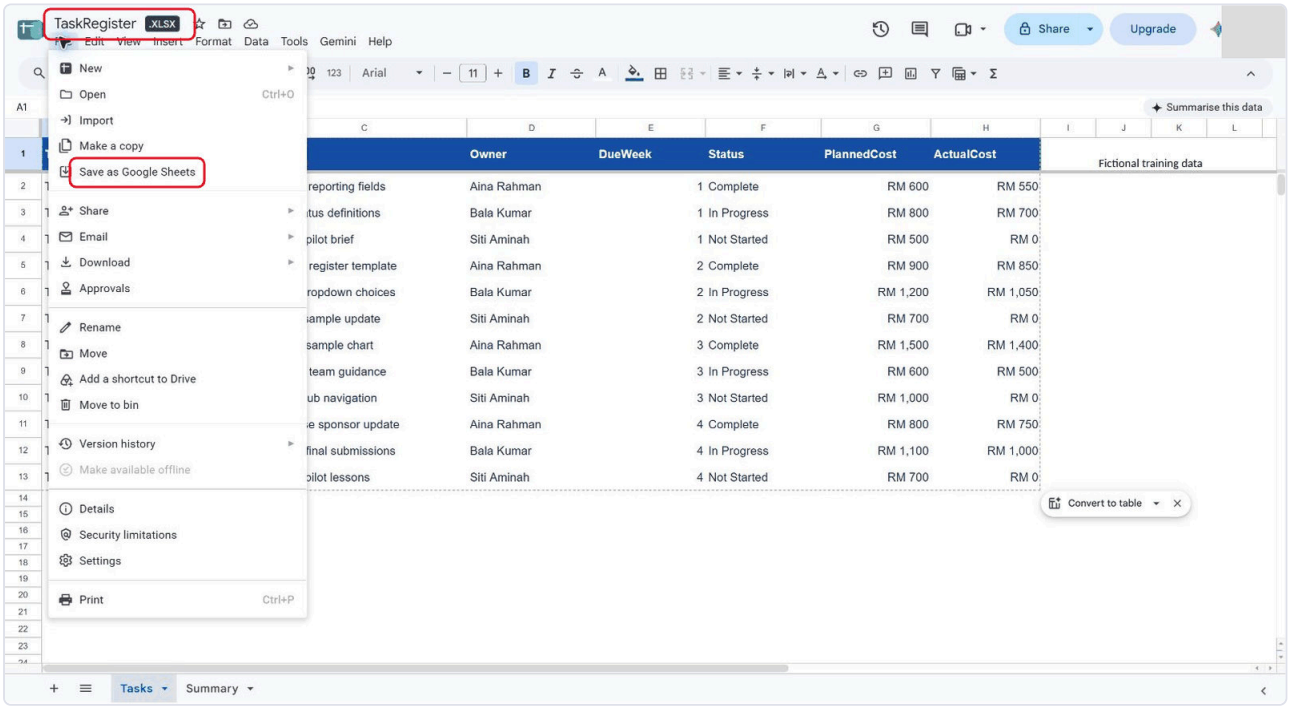


The Excel file is in the folder, and the upload box at the bottom right shows a green tick.

Check: you're inside **GW - Sheets Foundation - Practice**, and it holds one file, **TaskRegister.xlsx**.

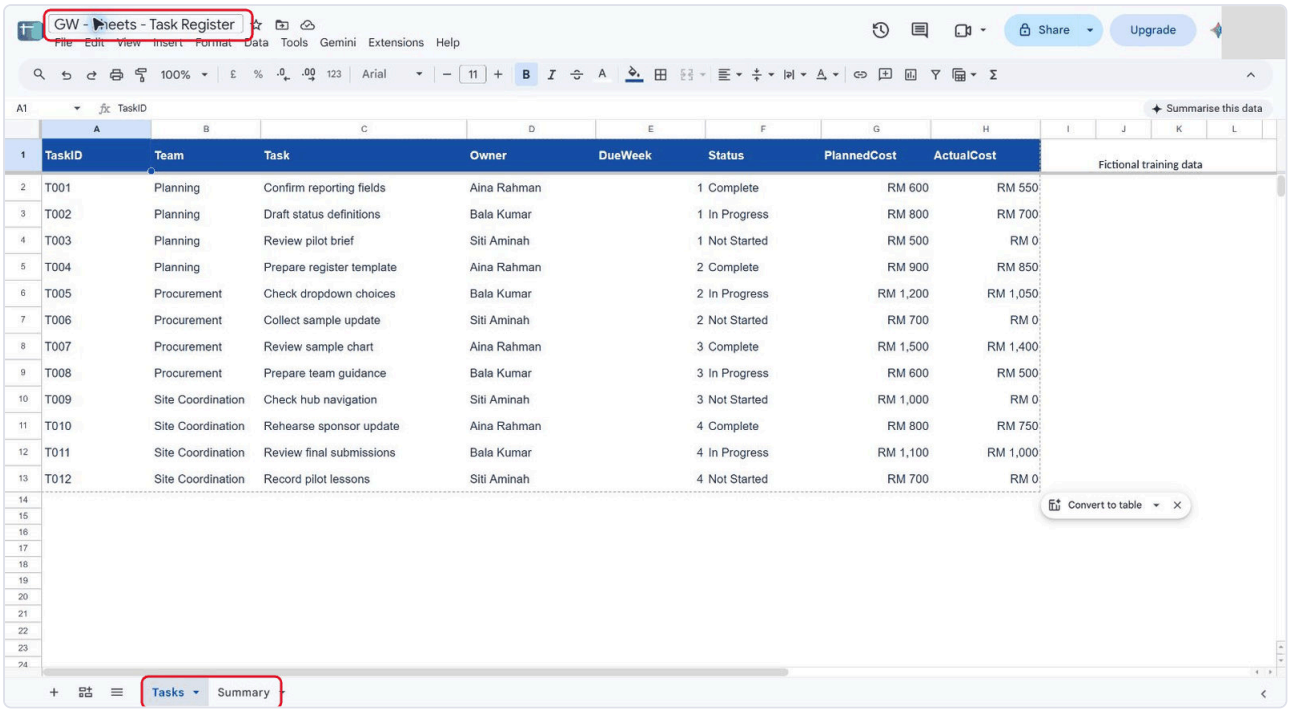
1.2 Save It as Google Sheets

1. Double-click **TaskRegister.xlsx**. It opens in Sheets with a **.XLSX** badge next to its name.
2. Click **File**, then **Save as Google Sheets**.



The .XLSX badge beside the name tells you this is still the Excel file.

3. A new Google Sheets file opens, with the same name and no **.XLSX** badge. Click its name at the top left, type `GW - Sheets - Task Register` and press **Enter**.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

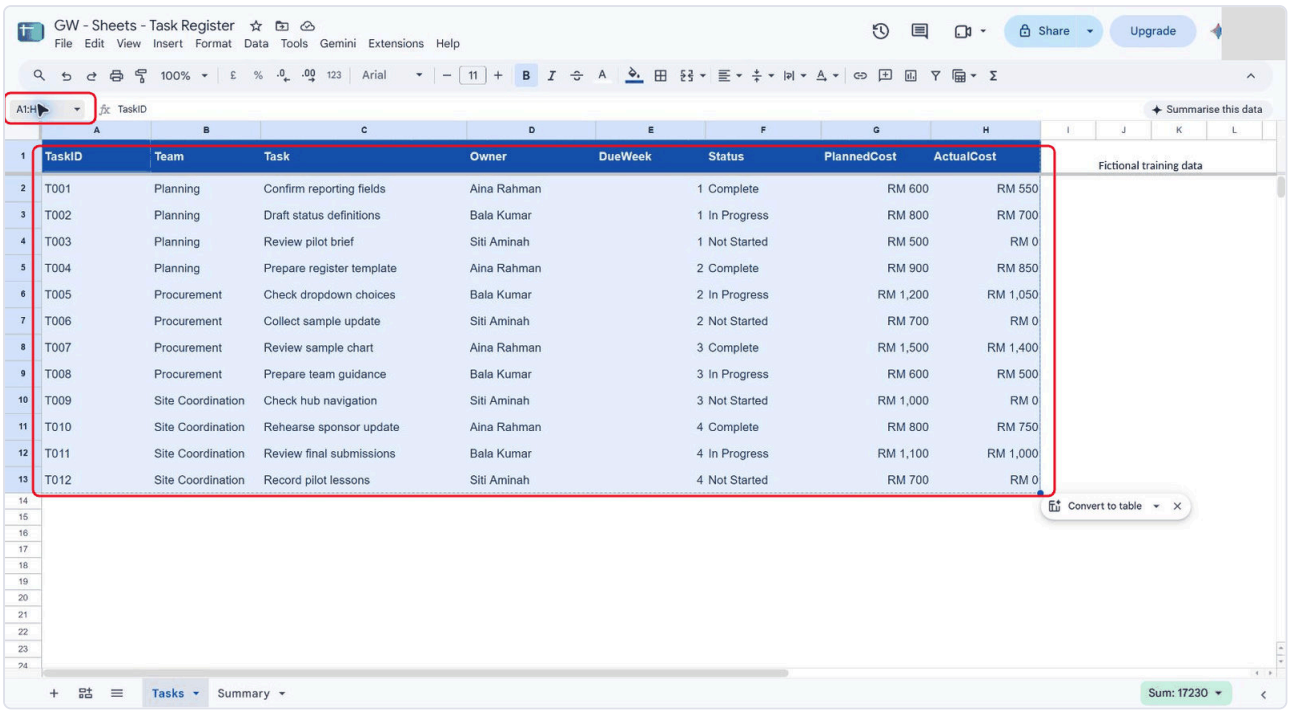
No badge beside the new name, and two tabs at the bottom: Tasks and Summary.

Check: the file is called **GW - Sheets - Task Register**, there's no **.XLSX** badge, and **Tasks** shows twelve tasks, **T001** to **T012**. Only rename this new file. Leave the Excel original as it is.

1.3 Find Your Way Around

The **name box** is the small box at the far left of the formula bar. It shows where you are, such as `A1`, and you can type an address into it to jump there.

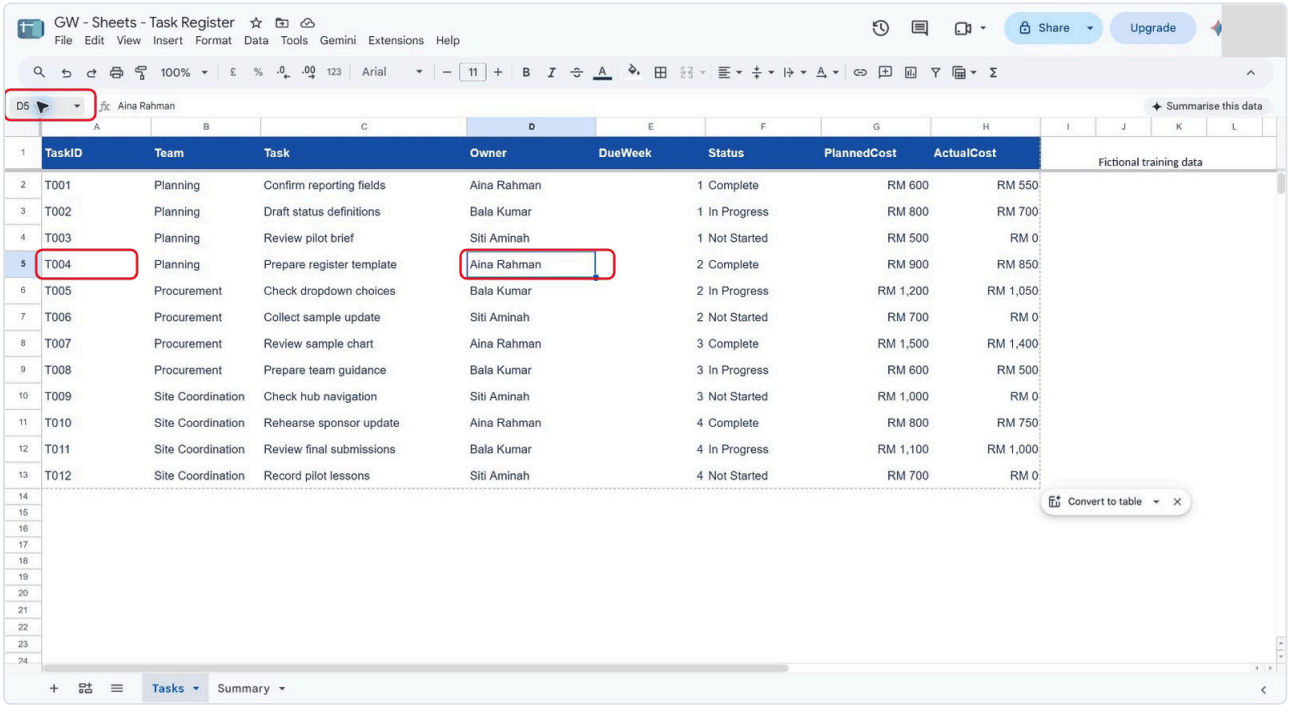
1. On **Tasks**, click the name box, type `A1:H13` and press **Enter**. The headings and all twelve tasks are selected.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

A1:H13 is a range: everything from A1 at the top left to H13 at the bottom right.

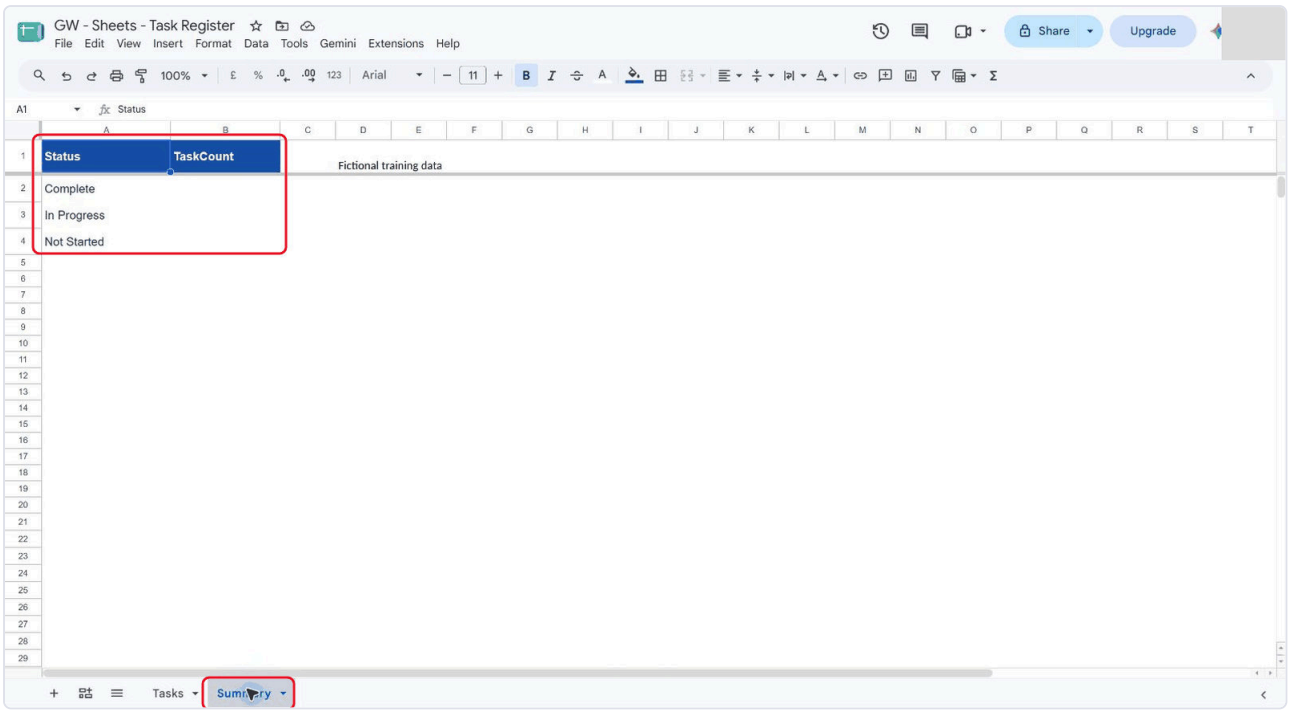
2. Click the name box, type `D5` and press **Enter**. The formula bar shows **Aina Rahman**. Look left to column A on the same row: it's task **T004**.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

D5 holds the owner of task T004.

3. Click the **Summary** tab at the bottom. It has three status names and an empty **TaskCount** column, which you fill in a later lab. Click **Tasks** to go back.



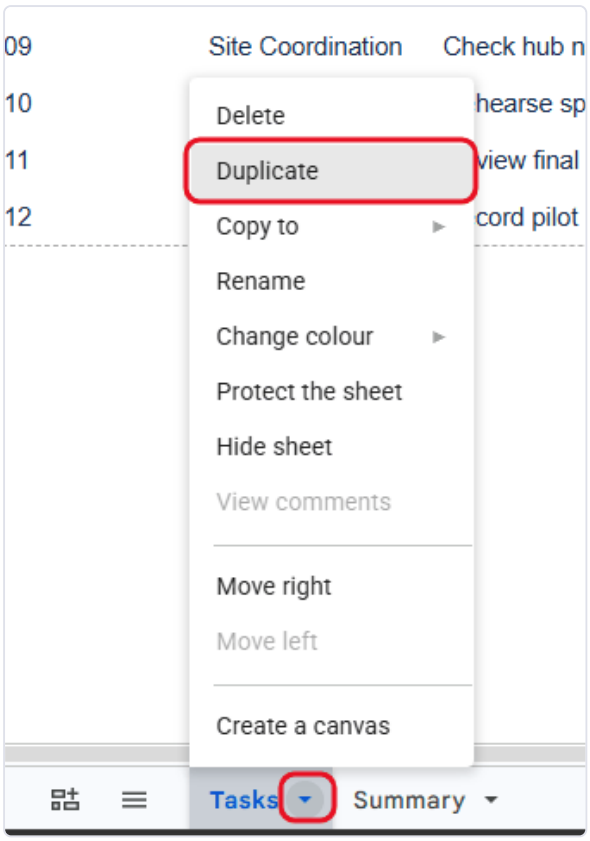
Leave Summary as it is for now.

Check: you can jump to any cell or range with the name box, and you're back on **Tasks**.

1.4 Make a Practice Tab

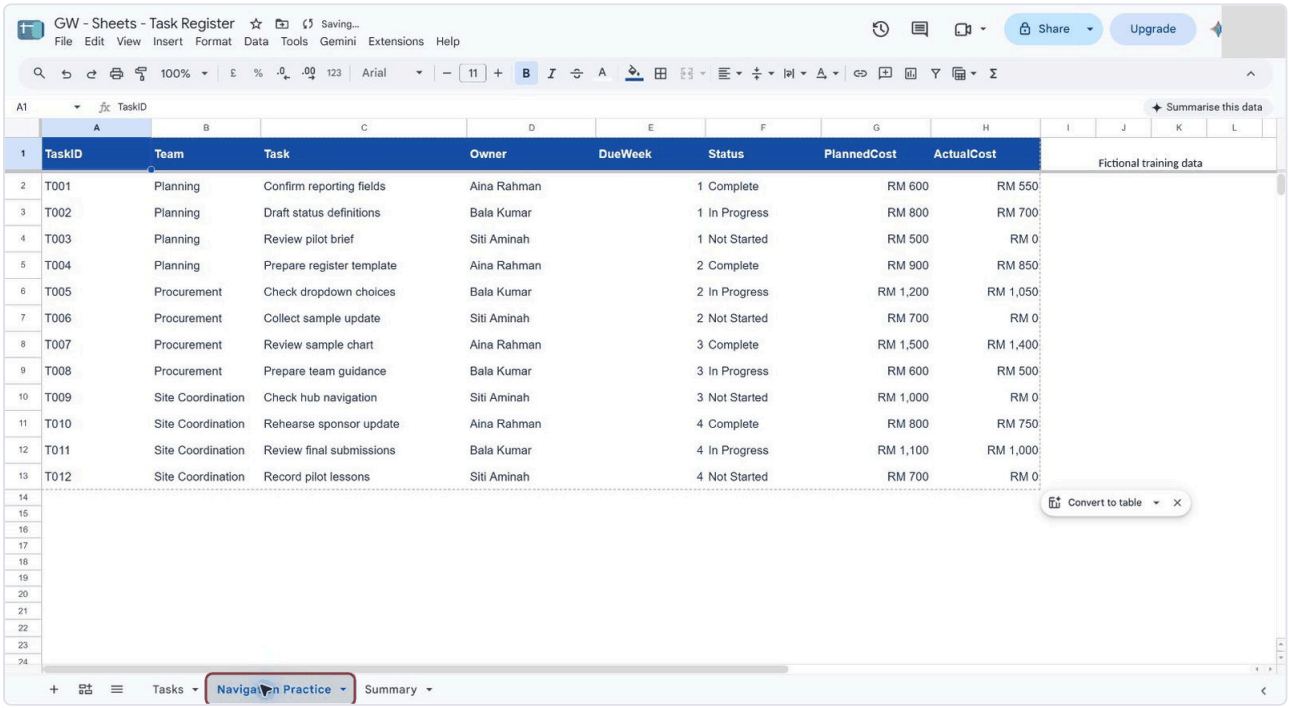
You'll try things out on a copy of **Tasks**, so the twelve supplied tasks stay exactly as they arrived.

1. Click the small arrow on the **Tasks** tab and choose **Duplicate**. A tab called **Copy of Tasks** appears.



The small arrow is on the Tasks tab itself, at the bottom of the screen.

2. Click the arrow on **Copy of Tasks**, choose **Rename**, type `Navigation Practice` and press **Enter**.

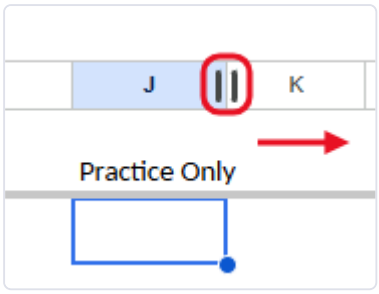


TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

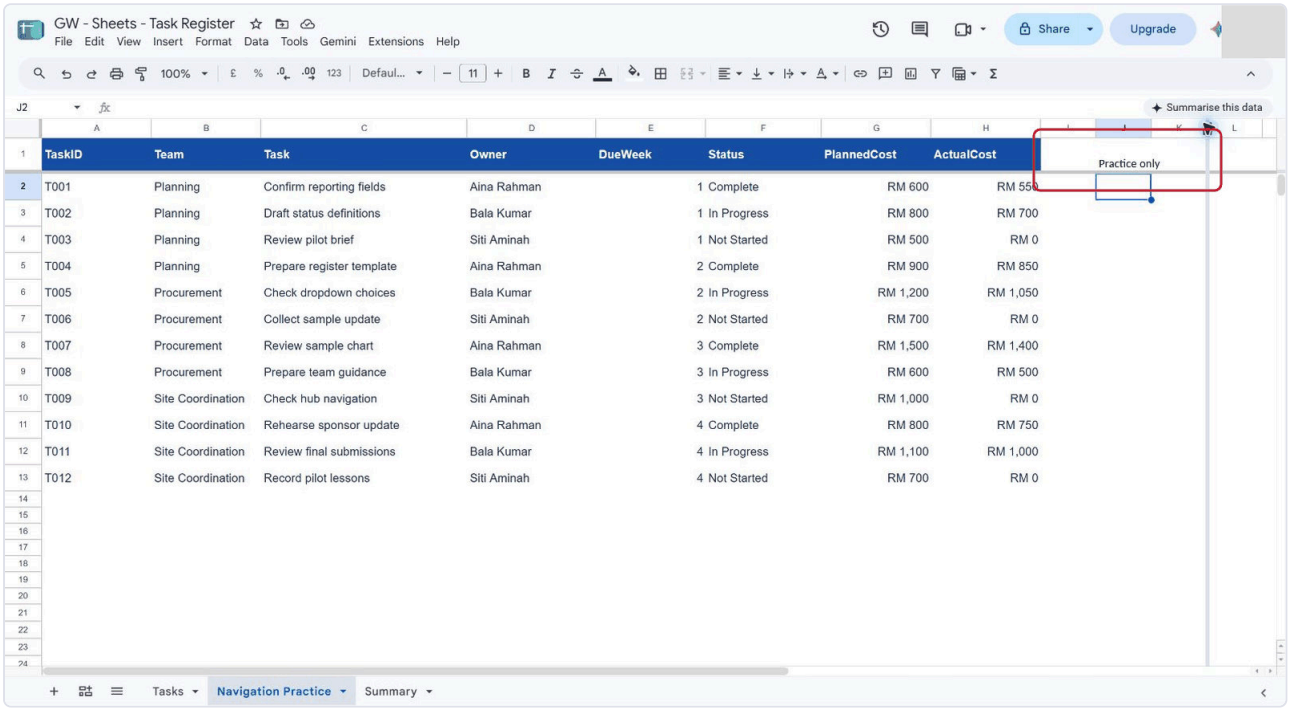
The copy has every task and the same formatting as Tasks.

3. On **Navigation Practice**, click **J1**, type `Practice only` and press **Enter**.

4. Point at the right edge of the column **J** heading, where it meets **K**. When the double bar appears, drag it to the right to make the column wider.

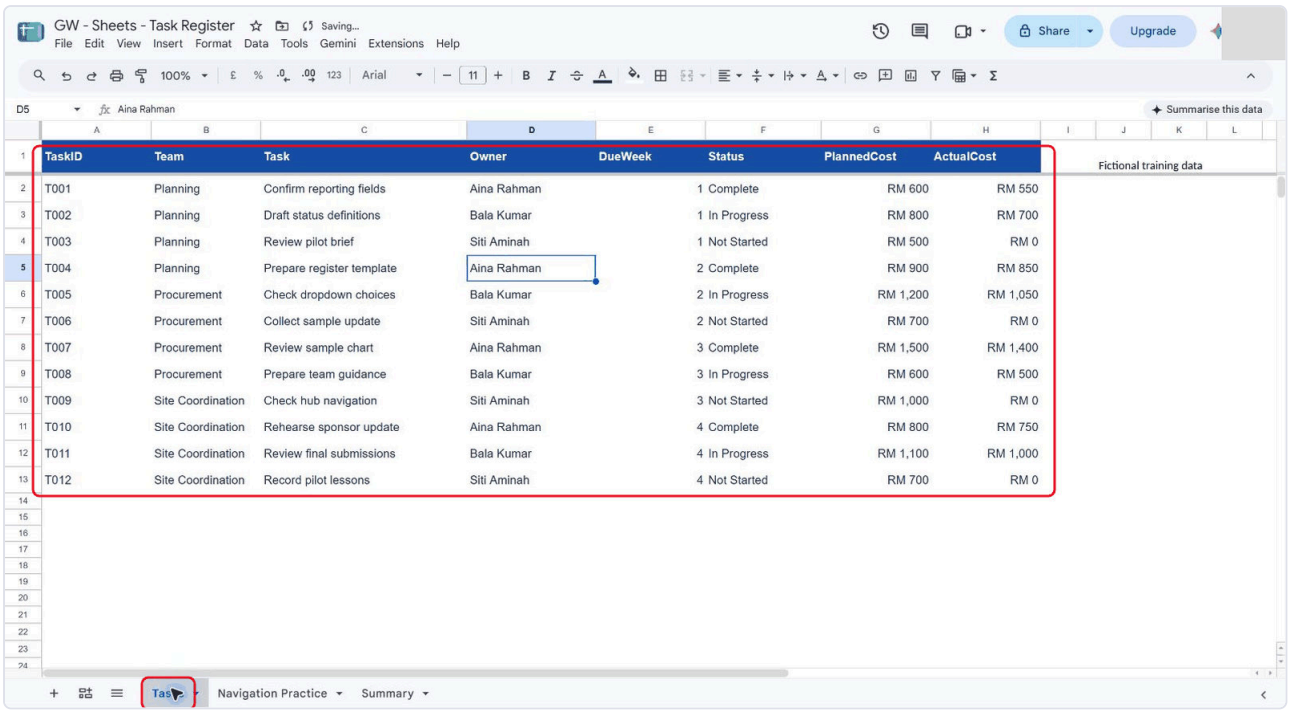


Drag the double bar to the right. Yours says Practice only, with a small o.



The change is only on Navigation Practice.

5. Click the **Tasks** tab.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

Tasks still has its twelve tasks and no Practice only label.

Check: Practice only is on **Navigation Practice**, and **Tasks** is unchanged.

1.5 Start the Workbook Guide

The Workbook Guide tells the next person what each tab is for, so they don't have to guess.

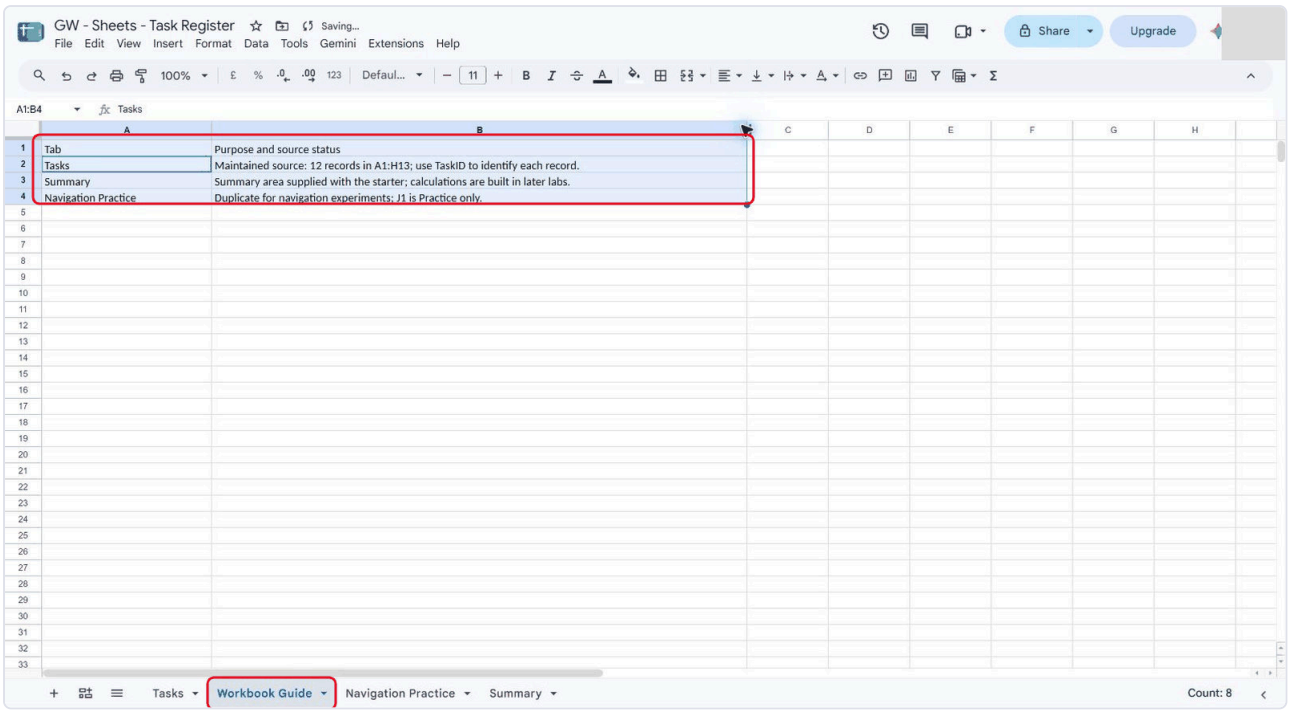
1. With **Tasks** open, click **+** (Add sheet) at the bottom left. A new tab appears next to **Tasks**.

The + is at the far left of the tab bar.

2. Double-click the new tab's name, type `Workbook Guide` and press **Enter**.
3. Copy the block below, click **A1** on **Workbook Guide** and press **Ctrl+V**. The text fills two columns, **A** and **B**.

```
Tab     Purpose and source status
Tasks   Maintained source: 12 records in A1:H13; use TaskID to identify each record.
Summary Summary area supplied with the starter; calculations are built in later labs.
Navigation Practice     Duplicate for navigation experiments; J1 is Practice only.
```

4. Drag the right edge of the column **B** heading until each line fits.



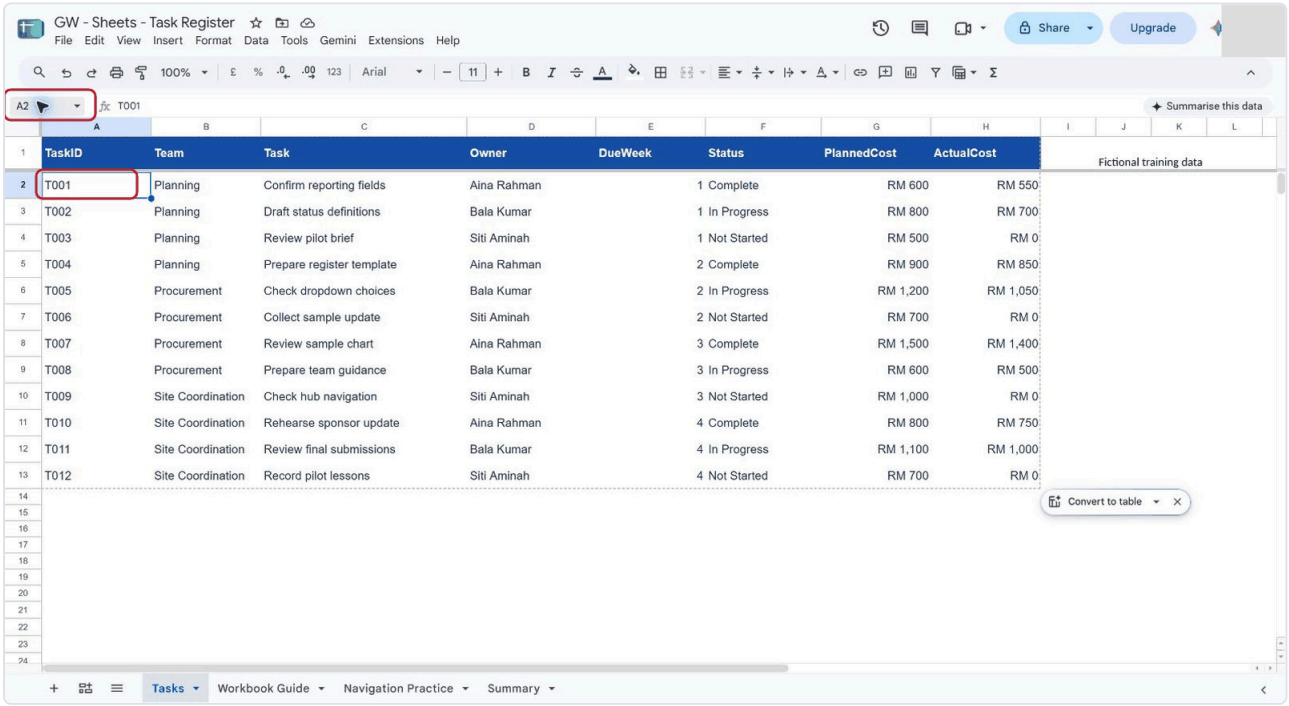
Four rows, one for each tab so far. Workbook Guide sits next to Tasks.

Check: Workbook Guide has the four rows in columns **A** and **B**, and every line is readable.

1.6 Check Four Addresses

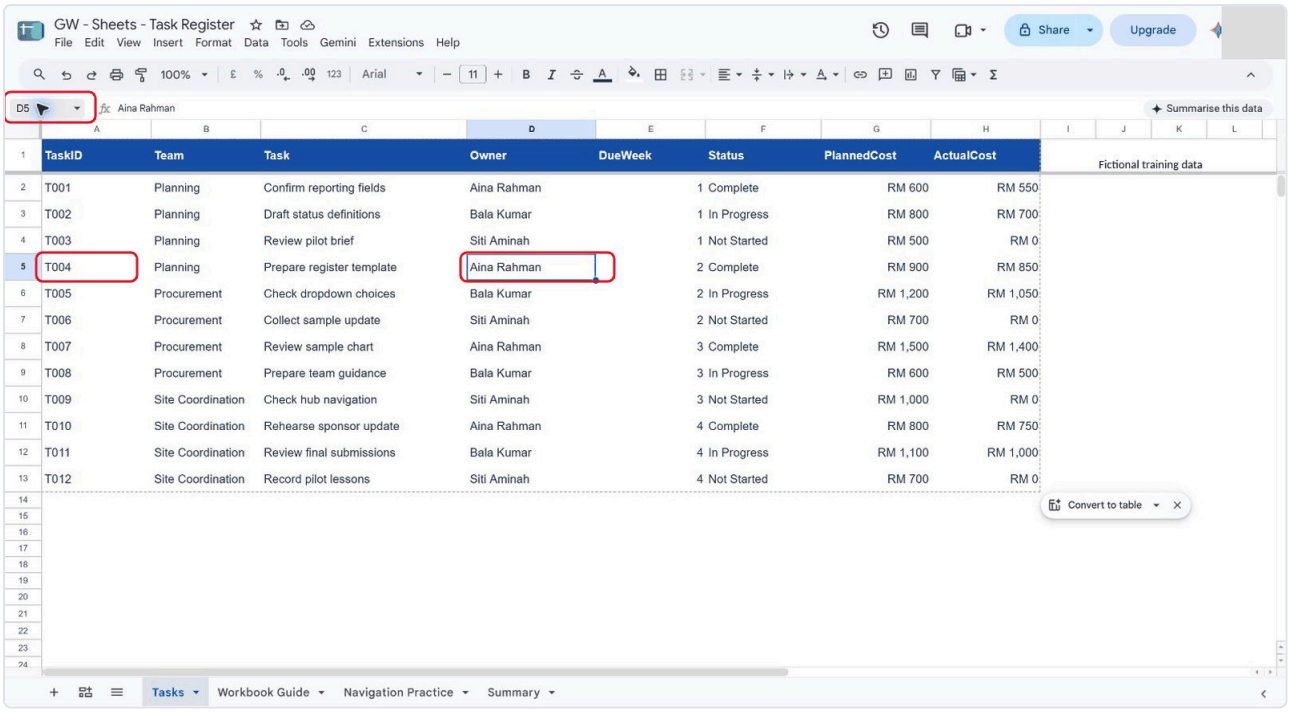
Now you look up four cells and write down which task each one belongs to. This is the habit that stops mistakes later: you name a record by its TaskID, not by its row.

1. Click the **Tasks** tab. Click the name box (top left, above column **A**), type `A2` and press **Enter**. Cell A2 is selected. Read its value in the formula bar: **T001**. In column A it's also the TaskID.



The name box shows A2, and the formula bar shows T001.

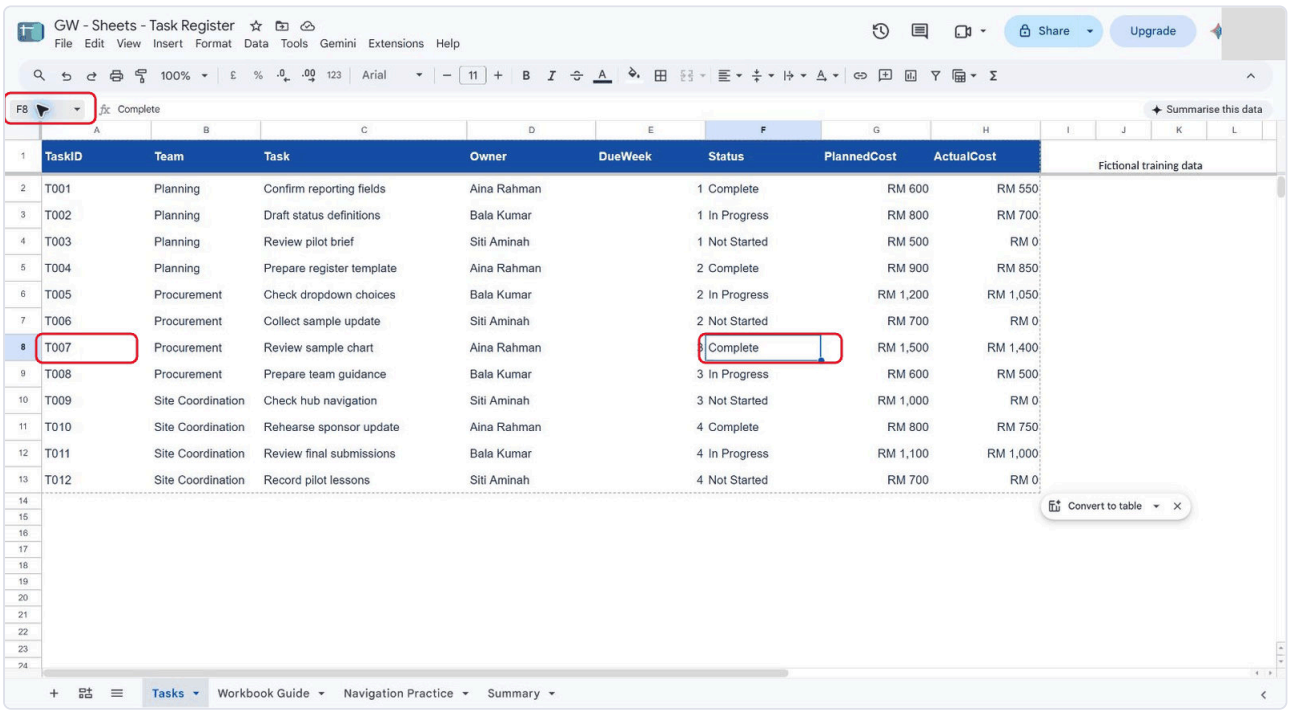
2. Click the name box again, type `D5` and press **Enter**. The formula bar shows **Aina Rahman**. Look left to column A on the same row for the TaskID: **T004**.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

D5 is the owner of T004.

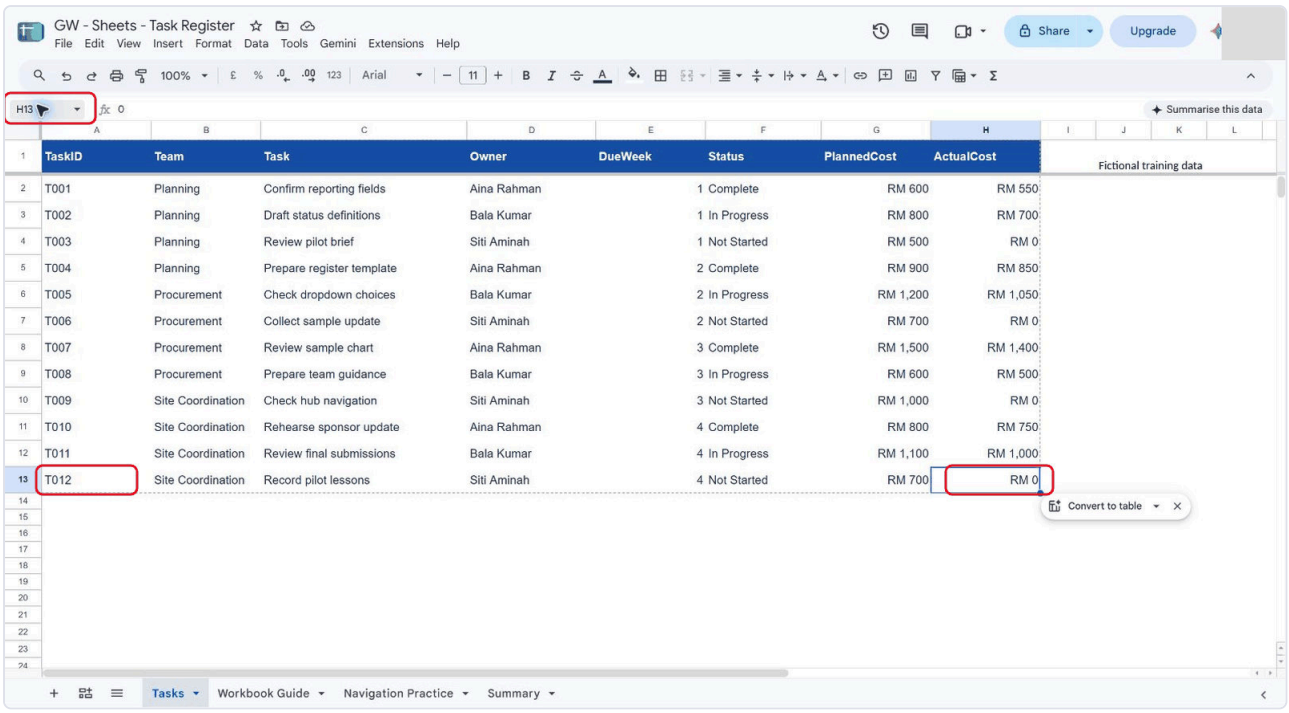
3. Do the same for `F8`. Note the value and the TaskID on that row.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

F8 is the status of T007.

4. Do the same for `H13`.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

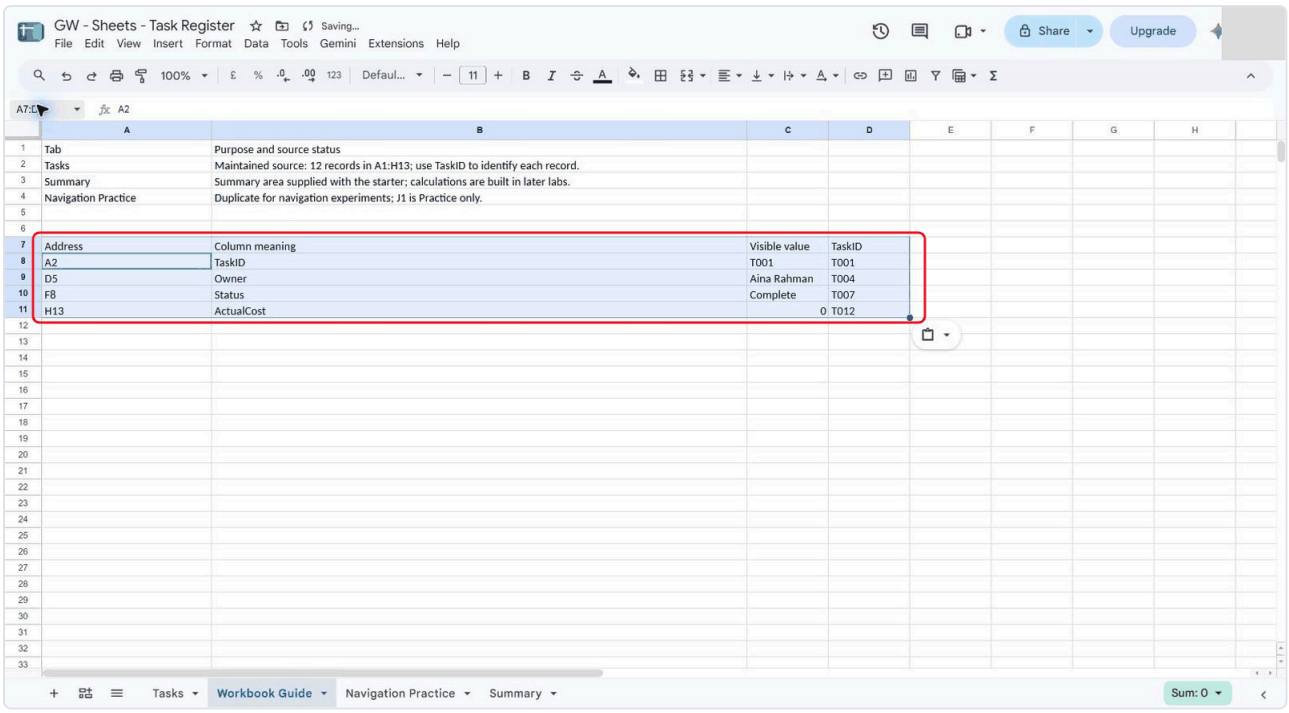
H13 shows RM 0 for T012. A zero cost is still a value, not an empty cell.

> **Tip:** press **Ctrl+Shift+Page Down** to move to the next tab and **Ctrl+Shift+Page Up** to go back. **Alt+Down arrow** and **Alt+Up arrow** do the same. On a Mac, use **Option+Down arrow** and **Option+Up arrow**.

5. Go to **Workbook Guide**. Copy the block below, click **A7** and press **Ctrl+V**. It adds the headings and the four addresses.

```
Address Column meaning  Visible value   TaskID
A2      TaskID
D5      Owner
F8      Status
H13     ActualCost
```

6. In **C8:D11**, type the value and the TaskID you found for each address.



Row 7 has the headings and rows 8 to 11 the four addresses.

What should you see?

Address	Column meaning	Visible value	TaskID
A2	TaskID	T001	T001
D5	Owner	Aina Rahman	T004
F8	Status	Complete	T007
H13	ActualCost	0	T012

Tasks shows H13 as **RM 0** because of its currency format. Typed as `0` here, it shows as plain 0.

1.7 Type Three Records

Typing a few records yourself shows you how a row holds one task, with each field in its own column.

1. Click **+** (Add sheet). Double-click the new tab's name, type `Entry Practice` and press **Enter**.
2. Copy the heading row below, click **A1** on **Entry Practice** and press **Ctrl+V**.

```
TaskID  Team    Task    Owner   DueWeek Status  PlannedCost     ActualCost
```

These are the three records you type. They're the first three rows of **Tasks**, repeated here so you can type straight from these notes:

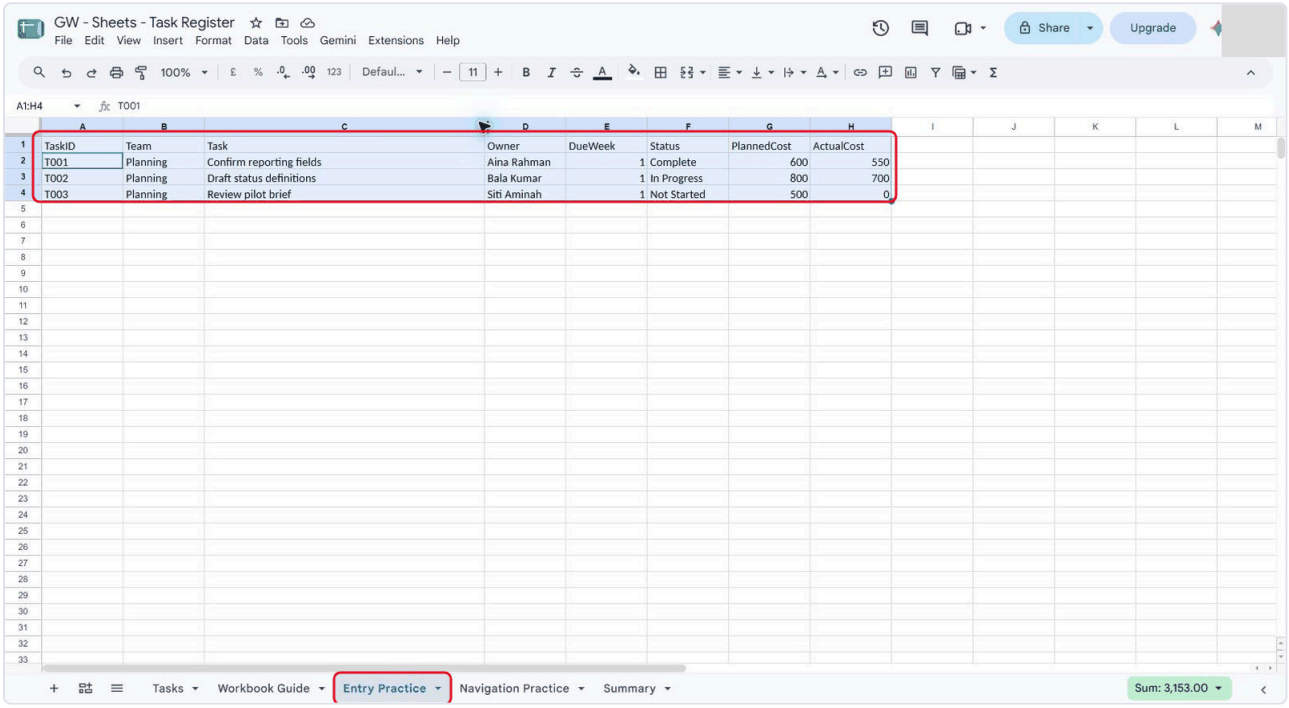
TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost

T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	600	550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	800	700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	500	0

3. Click **A2** and type the three records, all eight fields of each. Press **Tab** to move right and **Enter** at the end of a row: Sheets jumps back to column **A** on the next row. Type costs as plain numbers, such as `600`. When you start typing a word that's already higher up in the same column, such as `Planning`, Sheets suggests it: press **Enter** to accept.

> **Tip:** to see **Tasks** and **Entry Practice** at the same time, open the workbook twice. Copy its address from the browser's address bar, press **Ctrl+N** for a new window and paste the address. Show a different tab in each window, then press **Windows key+Left arrow** in one and **Windows key+Right arrow** in the other. On a Mac, hover over the green button at the top left and choose **Tile Window to Left of Screen**. Both windows are the same file, so a change in one shows in the other.

4. Drag the right edge of the column **C** heading until each task name fits.



Your three rows should match the first three rows of Tasks, field for field.

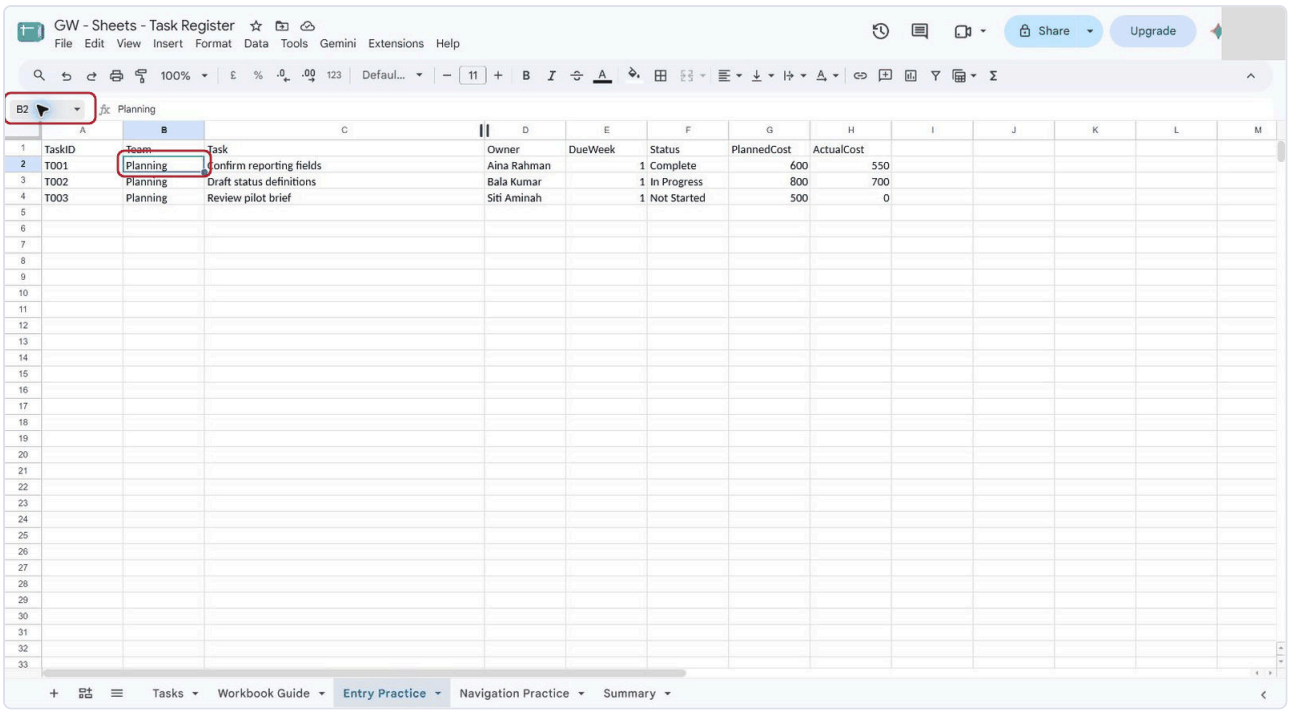
5. Compare each row with **Tasks** by its TaskID, one field at a time. If anything is different, fix it on **Entry Practice**.

Check: Entry Practice has the headings and three tasks in **A1:H4**, and every field matches **Tasks**.

1.8 Practise Selecting

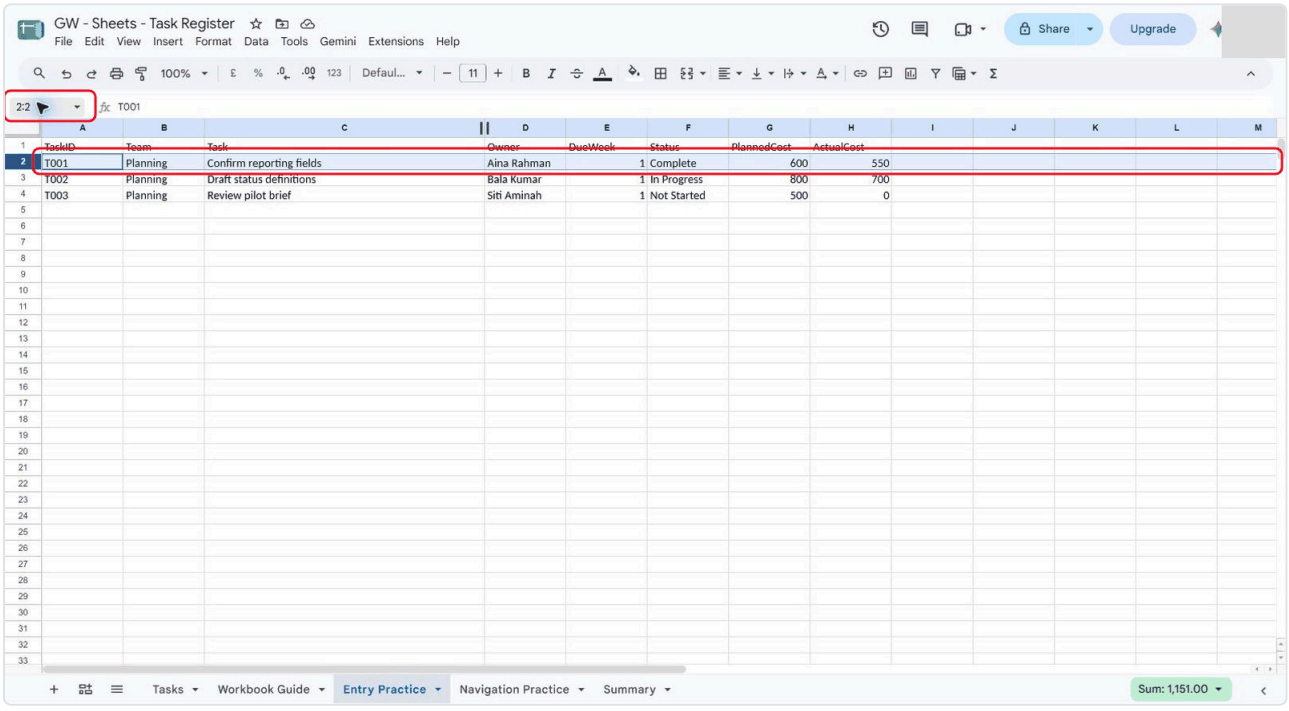
Before you format, sort or copy anything in later labs, you need to select the right thing. Use the name box on **Entry Practice** for each one.

1. Type `B2` and press **Enter**. One cell is selected.



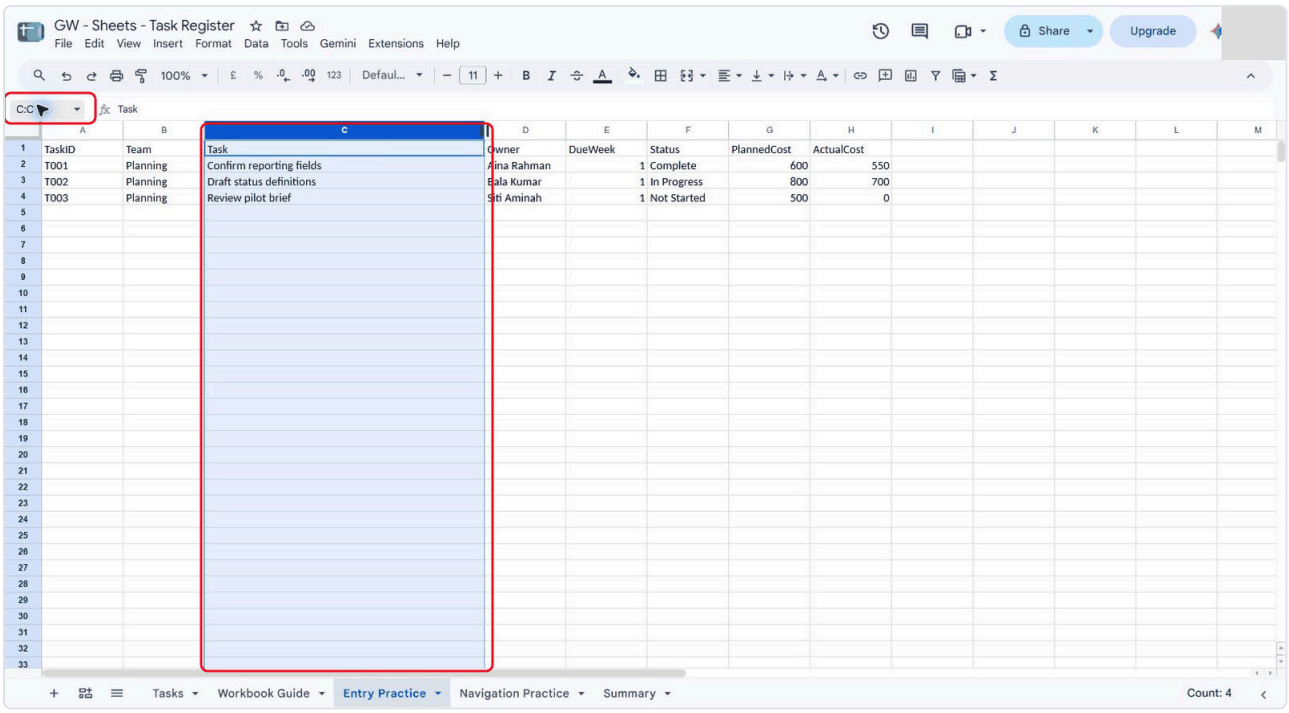
One cell: the Team for T001.

2. Type `2:2` and press **Enter**. The whole of row 2 is selected: every field of T001.



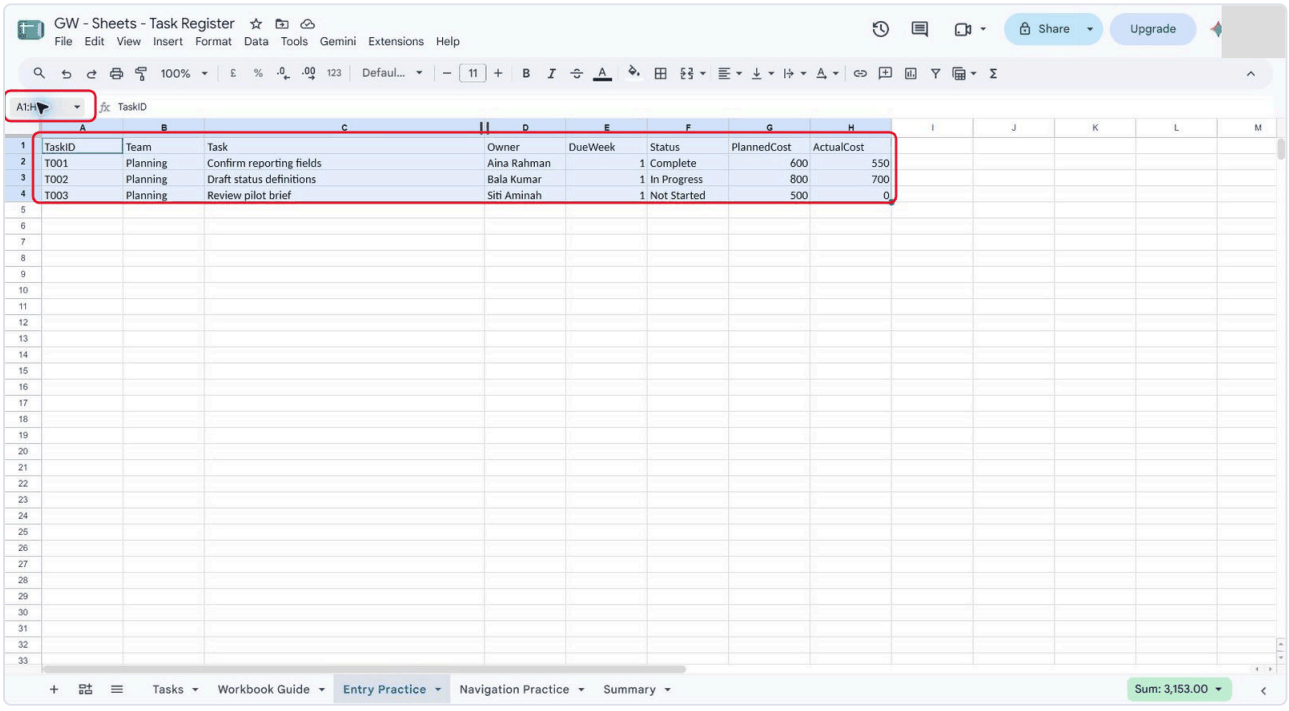
A whole row keeps a record's fields together.

3. Type `C:C` and press **Enter**. The whole Task column is selected.



A whole column is one field for every record.

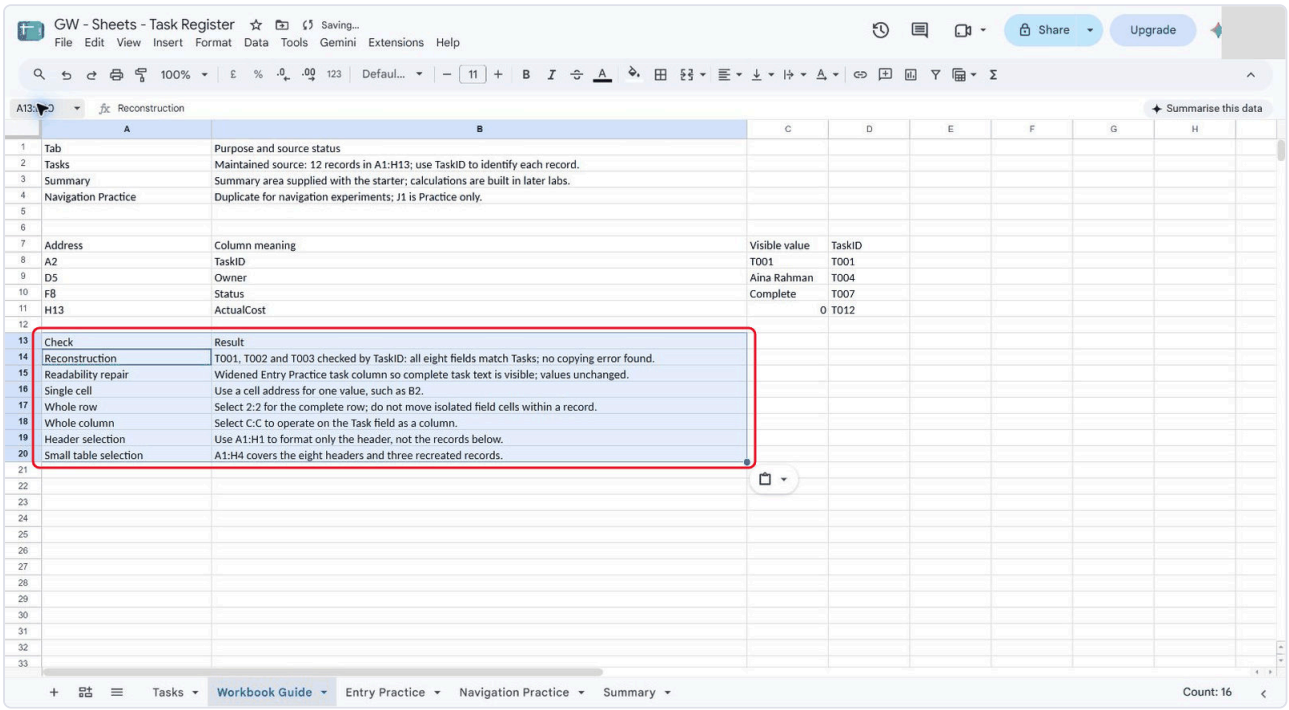
4. Type `A1:H4` and press **Enter**. The headings and your three records are selected.



A small table: the headings and three records.

5. Go to **Workbook Guide**. Copy the block below, click **A13** and press **Ctrl+V**. If you fixed a typing mistake in 1.7, change the **Reconstruction** line to say what you fixed.

```
Check   Result
Reconstruction  T001, T002 and T003 checked by TaskID: all eight fields match Tasks; no
copying error found.
Readability repair      Widened Entry Practice task column so complete task text is visible;
values unchanged.
Single cell     Use a cell address for one value, such as B2.
Whole row       Select 2:2 for the complete row; do not move isolated field cells within a
record.
Whole column    Select C:C to operate on the Task field as a column.
Header selection        Use A1:H1 to format only the header, not the records below.
Small table selection   A1:H4 covers the eight headers and three recreated records.
```

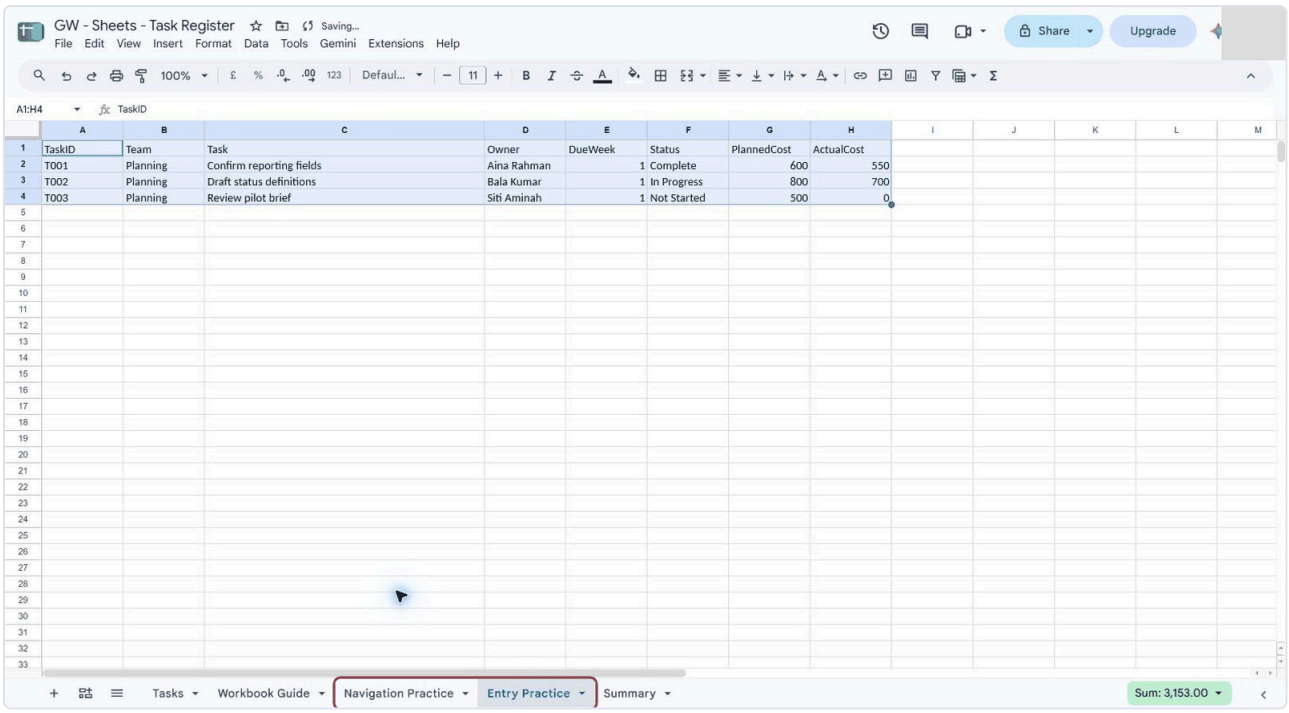


Rows 13 to 20 record what you checked and what each selection is for.

Check: you've made all four selections, and **Workbook Guide** has the check rows in **A13:B20**.

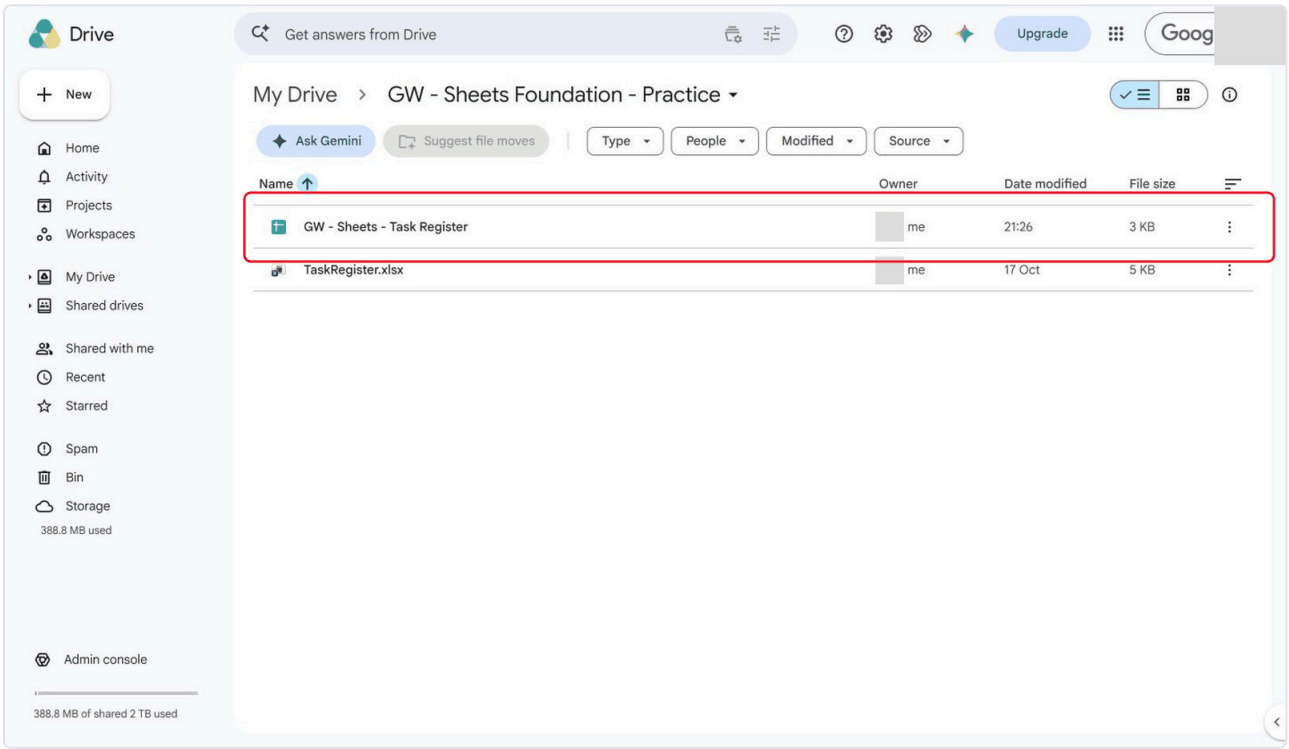
1.9 Tidy the Tabs and Reopen From Drive

1. Click the arrow on the **Entry Practice** tab and choose **Move right**. It moves one place, to after **Navigation Practice**.



The tab order now matches the map at the top of this lab.

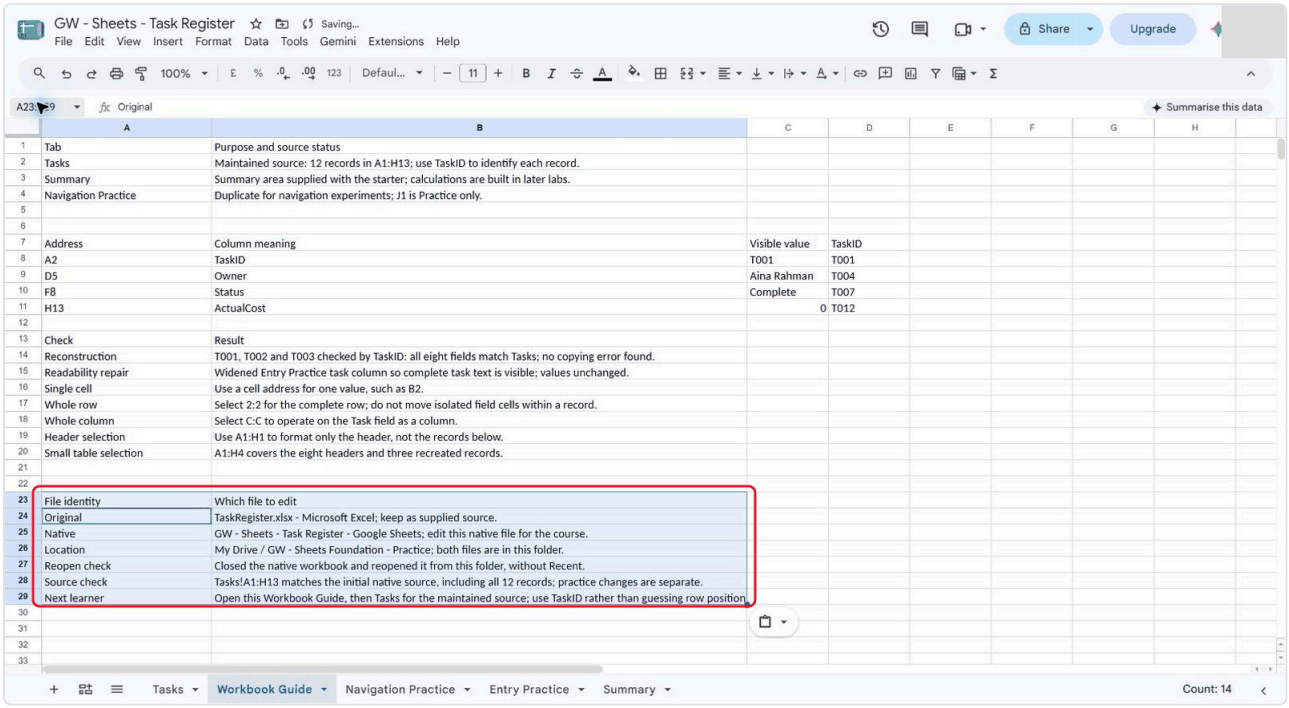
2. Close the browser tab with the workbook.
3. In Drive, open **GW - Sheets Foundation - Practice**. Don't use **Recent**: finding the file from its folder is what the next person will do.



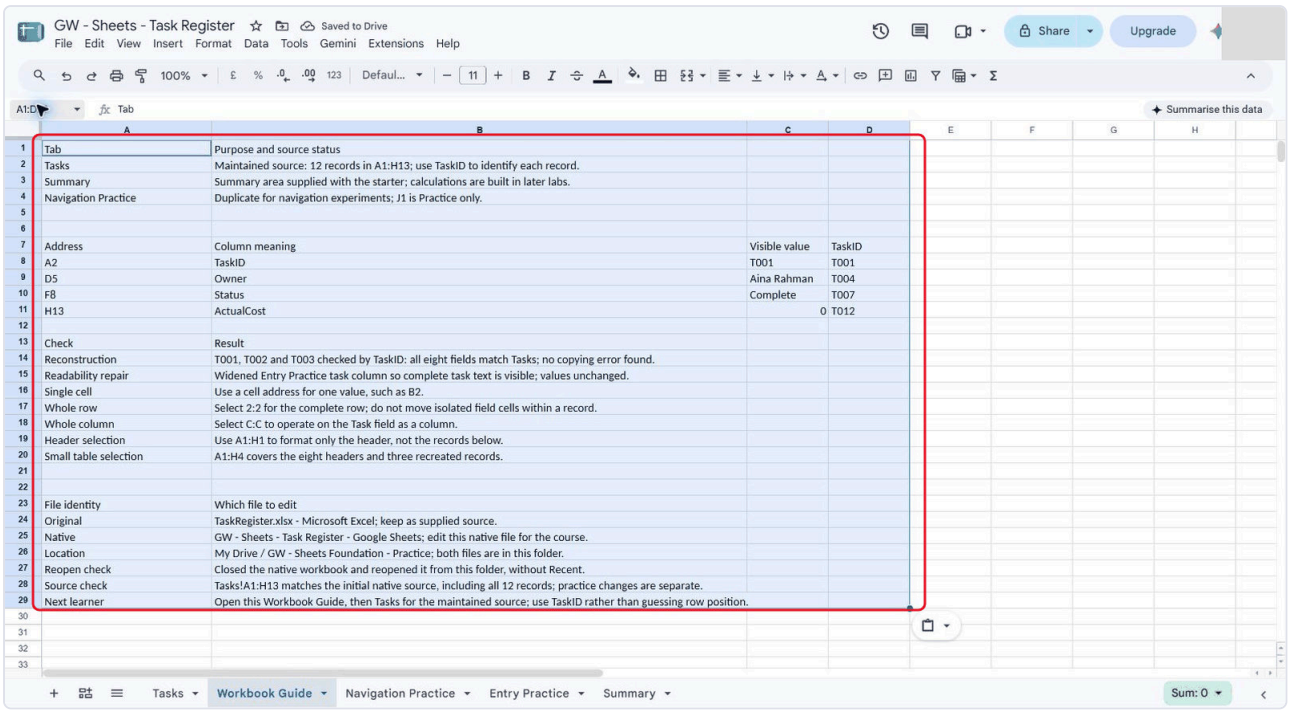
Two files: the green Sheets icon is your working file, and TaskRegister.xlsx is the Excel original.

4. Double-click **GW - Sheets - Task Register** and go to **Workbook Guide**.
5. Copy the block below, click **A23** and press **Ctrl+V**. Rows 21 and 22 stay empty as a gap.

```
File identity   Which file to edit
Original        TaskRegister.xlsx - Microsoft Excel; keep as supplied source.
Native  GW - Sheets - Task Register - Google Sheets; edit this native file for the course.
Location        My Drive / GW - Sheets Foundation - Practice; both files are in this folder.
Reopen check    Closed the native workbook and reopened it from this folder, without Recent.
Source check    Tasks!A1:H13 matches the initial native source, including all 12 records;
practice changes are separate.
Next learner    Open this Workbook Guide, then Tasks for the maintained source; use TaskID
rather than guessing row position.
```



The last block says which file to edit and where to find it.



The whole guide: tab purposes, address checks, selection checks and file identity.

What should you see?

GW - Sheets - Task Register opens from the practice folder with five tabs in this order: **Tasks**, **Workbook Guide**, **Navigation Practice**, **Entry Practice**, **Summary**. **Workbook Guide** has four blocks, starting at rows 1, 7, 13 and 23. **Tasks** still has the twelve supplied tasks in **A1:H13**.

Independent Practice

Try these without looking back at the steps.

1. Close the workbook and reopen it from the practice folder, without using **Recent**. On **Tasks**, find the owner and the actual cost of **T008**. Name the record by its TaskID, not its row.
2. Add a tab called `New Entry Form`. In row 1, type the eight headings from **Tasks**. Leave rows 2 and 3 empty, ready for two new tasks. Don't add unfinished rows to **Tasks**.
3. With a partner, take turns to point at a cell address, a range, a tab and the file. Your partner names each one and says which file to edit for the course.

What should you see?

T008 is owned by **Bala Kumar**, with an actual cost of **RM 500**. It's in row 9 today, but after a sort it could be anywhere: T008 is how you find it again.

Evidence to Hand Over

- **Workbook Guide** with the tab purposes, the four address checks, the selection checks and the file identity rows
- **Entry Practice** with three typed records that match **Tasks**
- **New Entry Form** with eight headings and two empty rows

- The owner and actual cost of T008

Troubleshooting

Problem	What to try
Save as Google Sheets isn't in the **File** menu	You're already in the Google Sheets copy. Look for the **.XLSX** badge: if it isn't there, you're in the right file.
There are two files called **TaskRegister**	One is the Excel original and one is the new copy. Rename the one without the **.xlsx** ending (green Sheets icon).
A pasted block landed in one column	Click a single cell (don't double-click into it) before you press **Ctrl+V**.
The name box didn't move you	Click the name box, type the address with no spaces and press **Enter**.
You typed on **Tasks** by mistake	Press **Ctrl+Z** until the change is gone, then check **Tasks** against the supplied file.
The upload button or **Save as Google Sheets** is missing in your account	Tell your trainer. They'll give you a Google Sheets copy to use instead.

Lesson Summary

You uploaded an Excel file, saved a Google Sheets copy to work in, and kept the original safe. You moved around with the name box, practised on copies, typed three records and wrote a Workbook Guide so the next person knows which file and tab to use. In Lab 02 you make the register easier to read without changing what it says.

Check yourself: what's the difference between a spreadsheet file and a worksheet tab?

CHAPTER 2

Enter and Format Reliable Data

In this chapter you will use dates, numbers, currency, wrapping and sensible column widths.

> **Prompts:** the prompts page has every value and prompt for this chapter, each with a Copy button.

Estimated time: 45 minutes, planned as 10 explanation and demonstration, 15 guided build and casework, 15 independent practice and 5 verification and repair, with 35 active minutes by design. Not yet timed with learners.

Your result: A readable register with numeric currency values and sensible dimensions.

> **Status:** partial author reference with 18 staged application captures through 2.4. Currency/week formats, wrapping/row fit, numeric versus text tests, SUM/COUNT and format-only behaviour checked. Values-only target-format preservation did not pass in the controlled session and needs manual verification. Source values remain unchanged; independent challenge and learner timing are pending.

What You Will Learn

- Apply formats to typed numbers.
- Use wrapping and fitted dimensions.
- Distinguish appearance from the stored value.

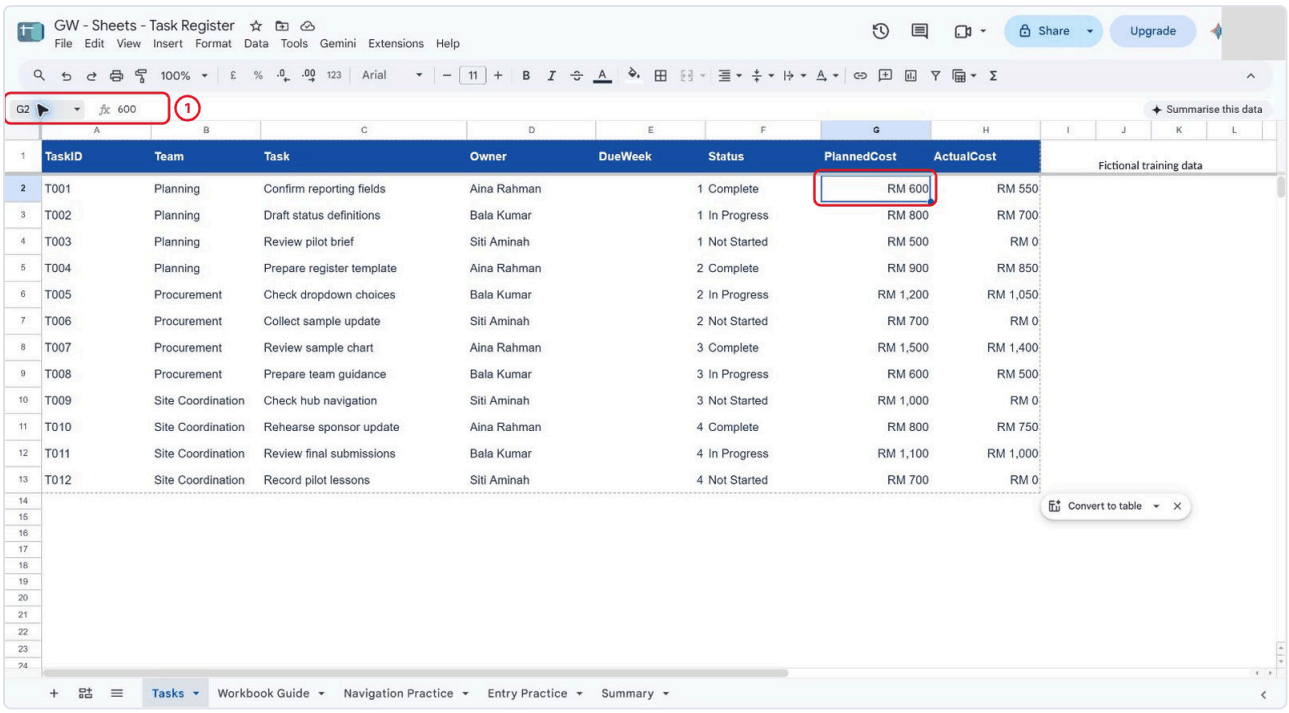
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use your original Tasks tab. If needed, reopen a fresh native copy of TaskRegister.xlsx.

2.1 Inspect Values and Types

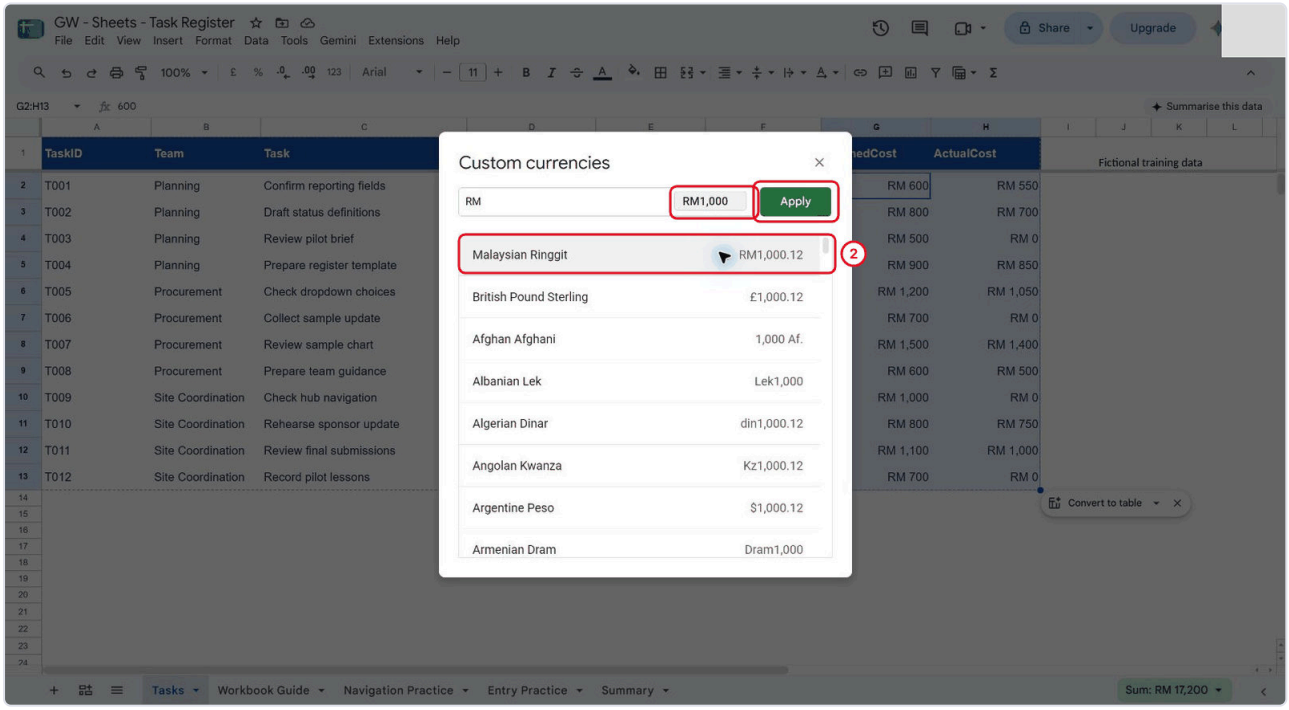
1. On **Tasks**, select **G2** through the name box. The formula bar shows numeric **600**, while the cell displays its currency. Do not type the RM label into the source number.



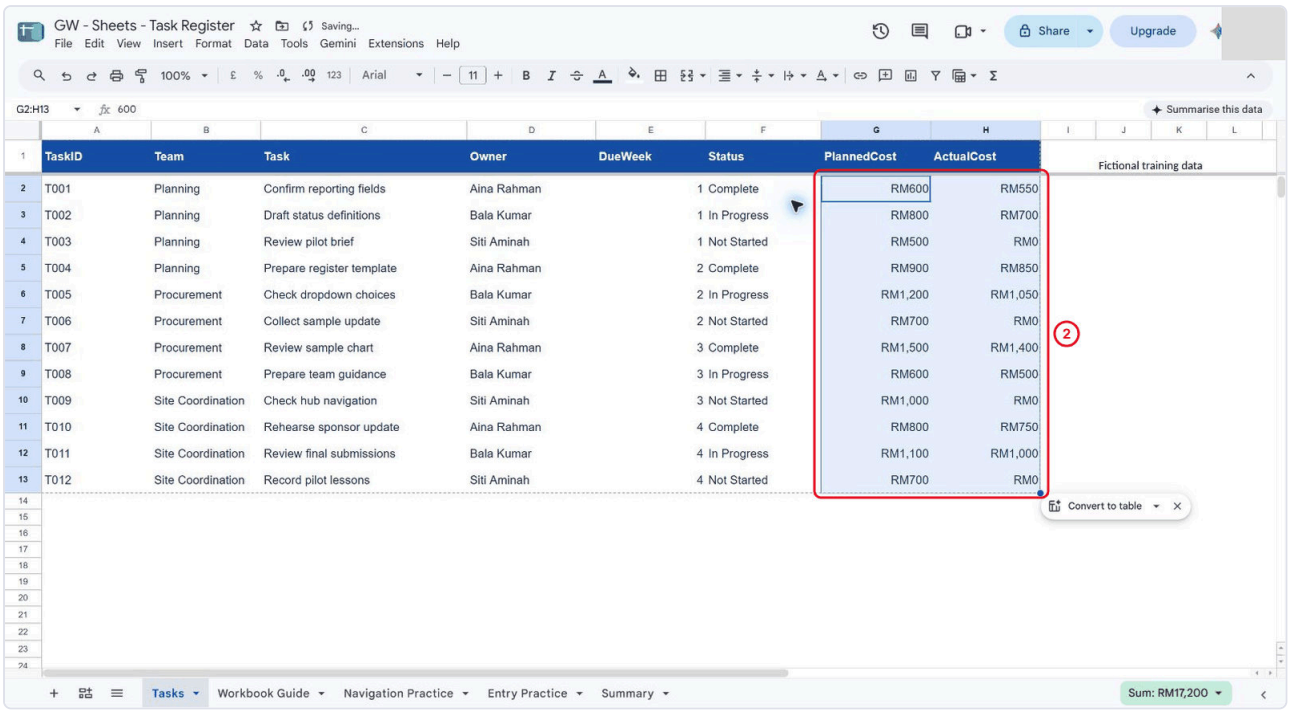
TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

Inspect numeric 600 in the formula bar.

2. Select **G2:H13**. Open **More formats > Custom currency**, choose **Malaysian Ringgit**, select **RM1,000** without decimals, and **Apply**. The toolbar currency shortcut follows the workbook locale, so it may offer pounds; select RM explicitly.



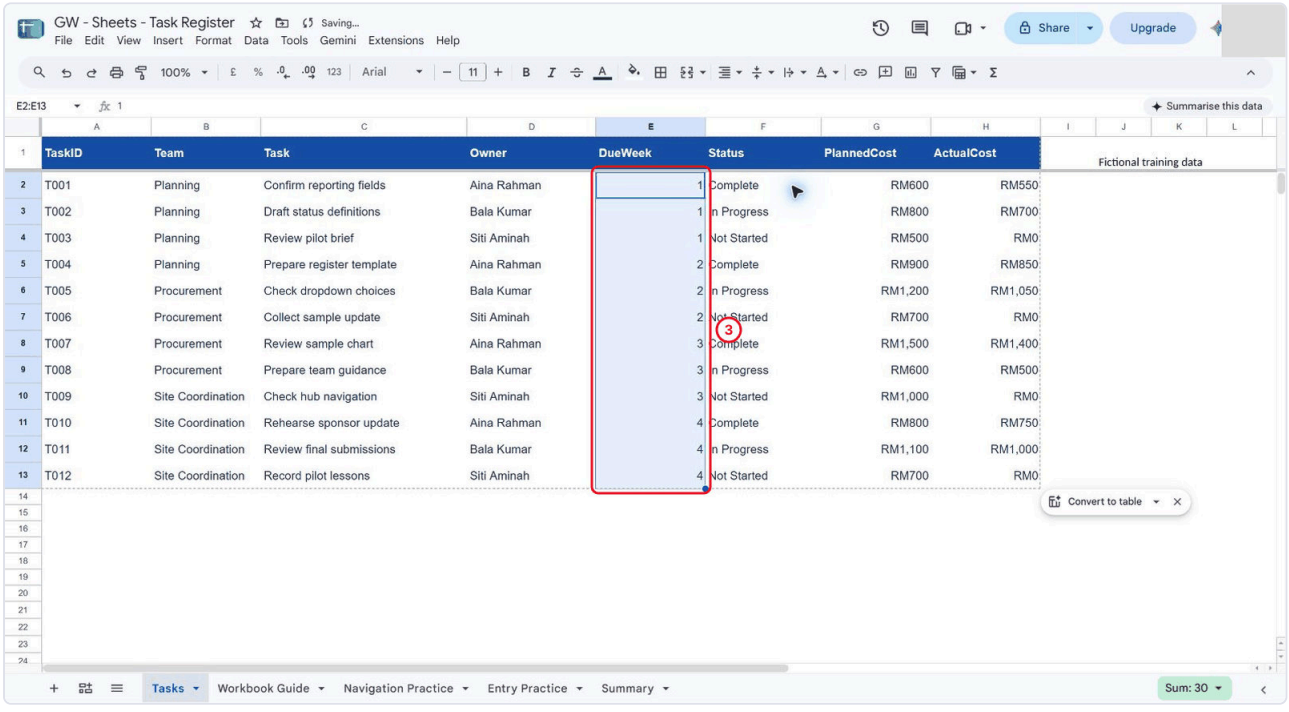
Select Malaysian ringgit without decimal places.



	TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
2	T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
3	T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
4	T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
5	T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
6	T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
7	T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
8	T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
9	T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
10	T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
11	T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
12	T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
13	T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

Apply the RM display to the source cost range.

3. Select **E2:E13**, use **More formats > Custom number format**, enter **0** and **Apply**. DueWeek remains an index from 1 to 4, not a calendar date.



	TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
2	T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
3	T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
4	T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
5	T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
6	T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
7	T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
8	T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
9	T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
10	T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
11	T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
12	T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
13	T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

Keep DueWeek as a whole-number index.

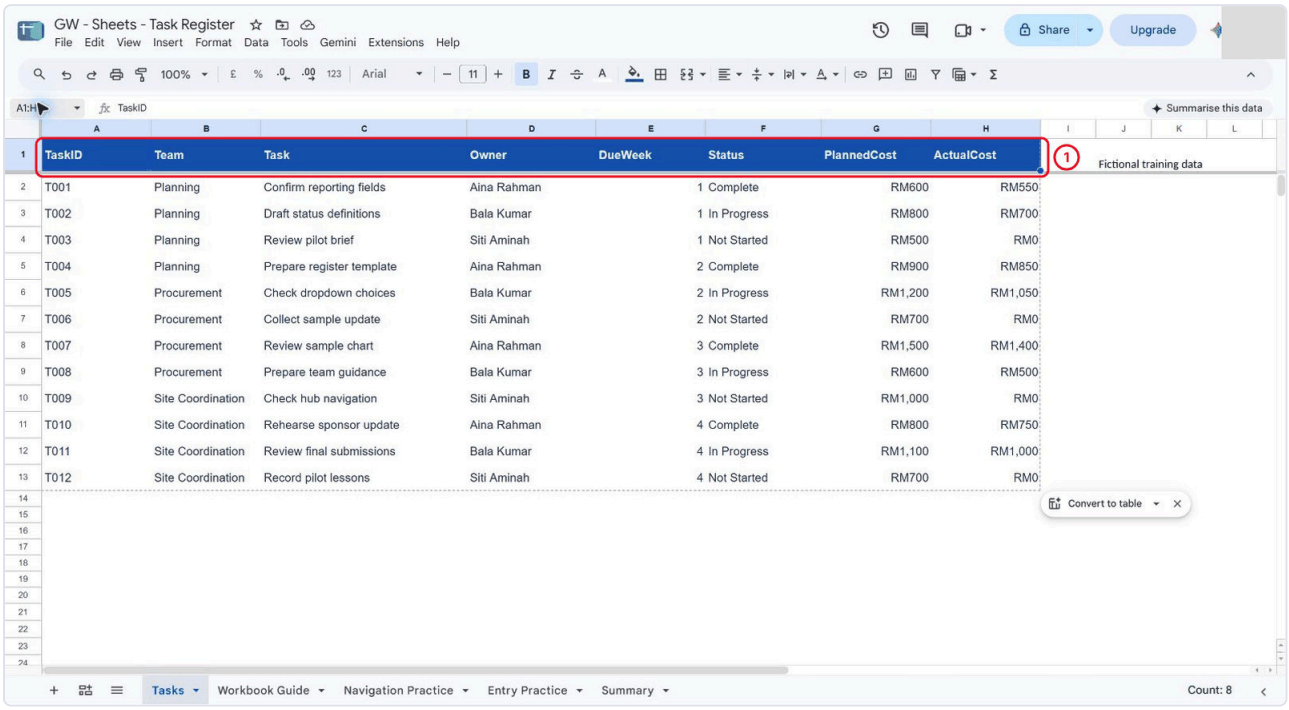
What should you see?

Costs show their currency while remaining numeric. Week values still range from 1 to 4.

2.2 Improve Readability

1. Select **A1:H1** and inspect its contrast. A reasoned no-change decision is preferable to changing a readable header just to demonstrate a button.

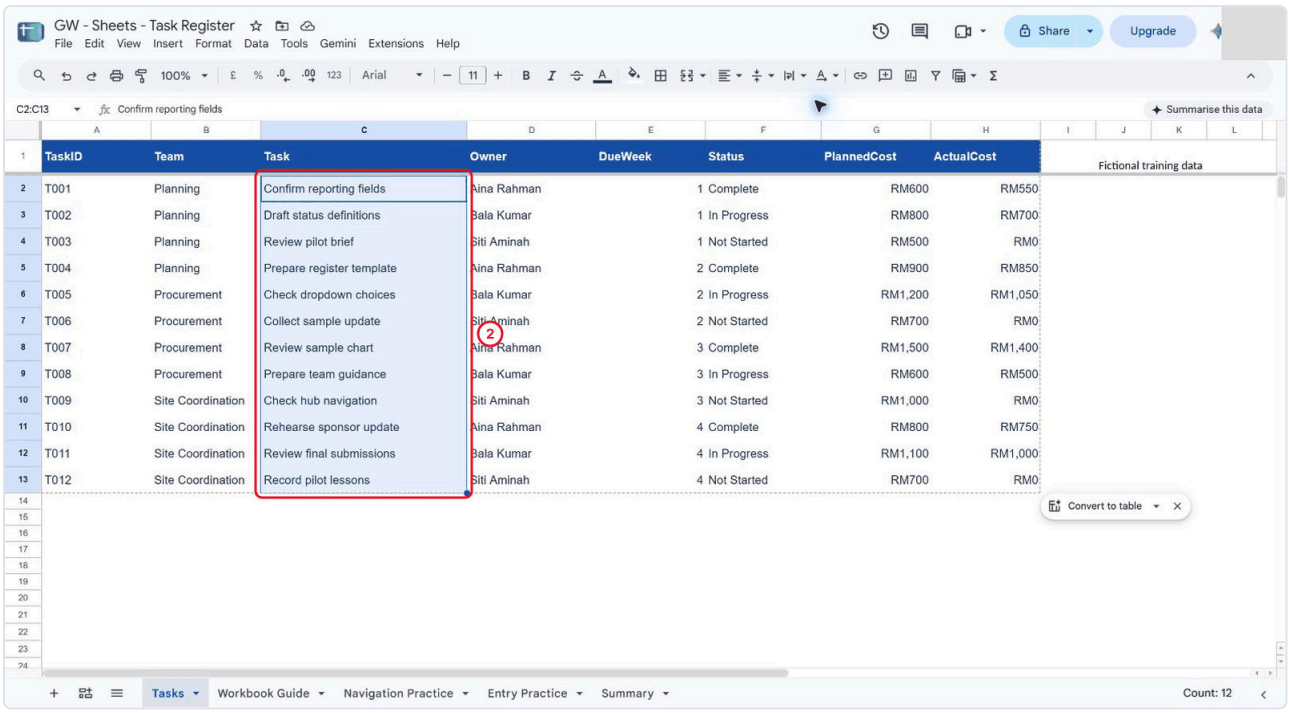
> **Reference run:** The starter already has bold white labels on a dark blue fill, so the reference retains that suitable style.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

Retain the existing readable source header contrast.

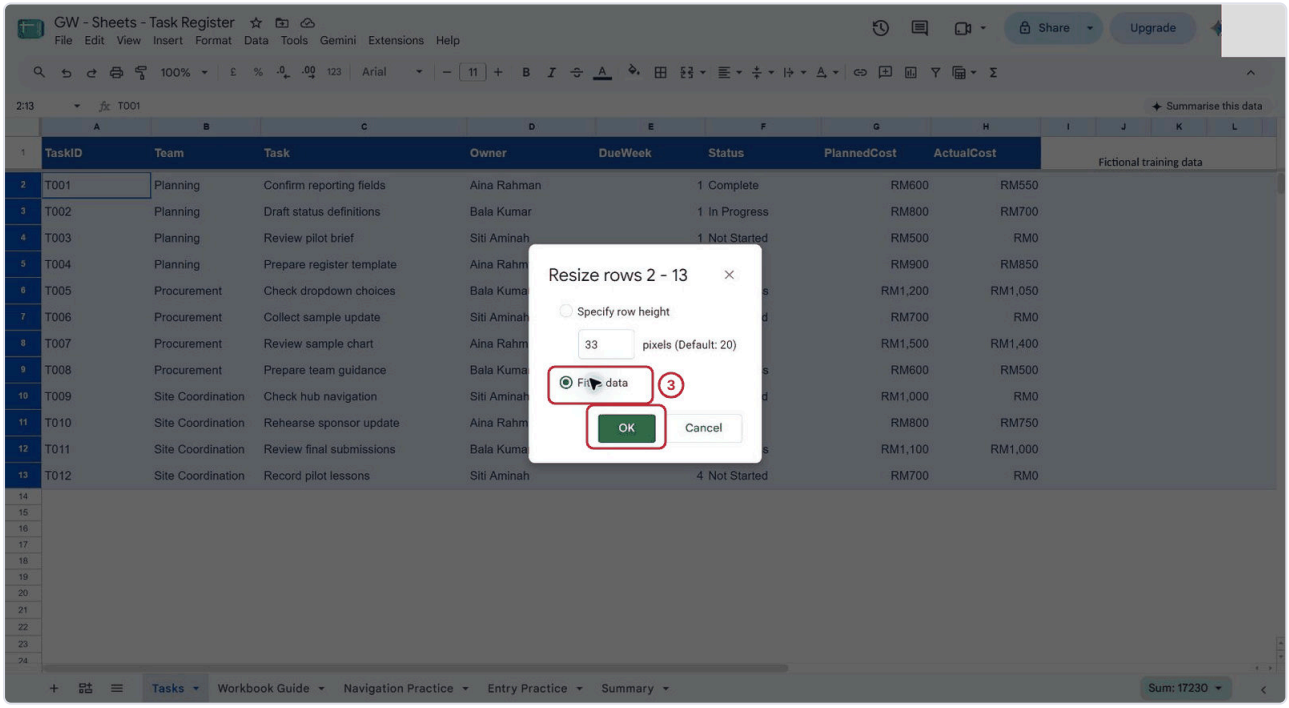
2. Inspect Task and Owner widths and widen only if text is cut off. The supplied source widths already show each task and owner. Select **C2:C13**, choose **Text wrapping > Wrap**, and inspect complete descriptions.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

Wrap source task descriptions without merging fields.

3. Select rows **2:13**, open the row-header menu, choose **Resize rows 2–13 > Fit to data > OK**. Keep each data column separate; do not merge cells in the register.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

Fit source row heights to data.

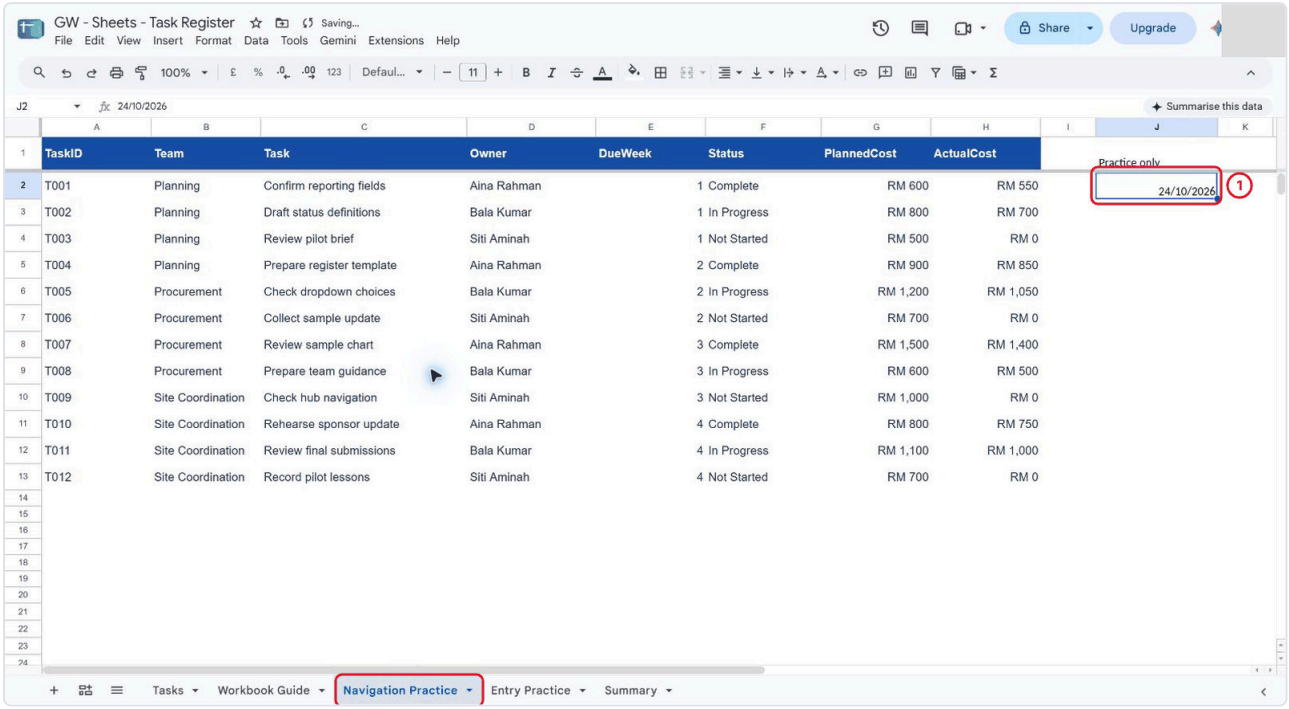
What should you see?

Headers, names and cost figures are readable without cutting off text.

2.3 Inspect Formatting Effects

1. On **Navigation Practice**, enter **2026-10-24** in **J2**, commit it, then use **More formats > Date**. This is a practice example only, not an agreed pilot start.

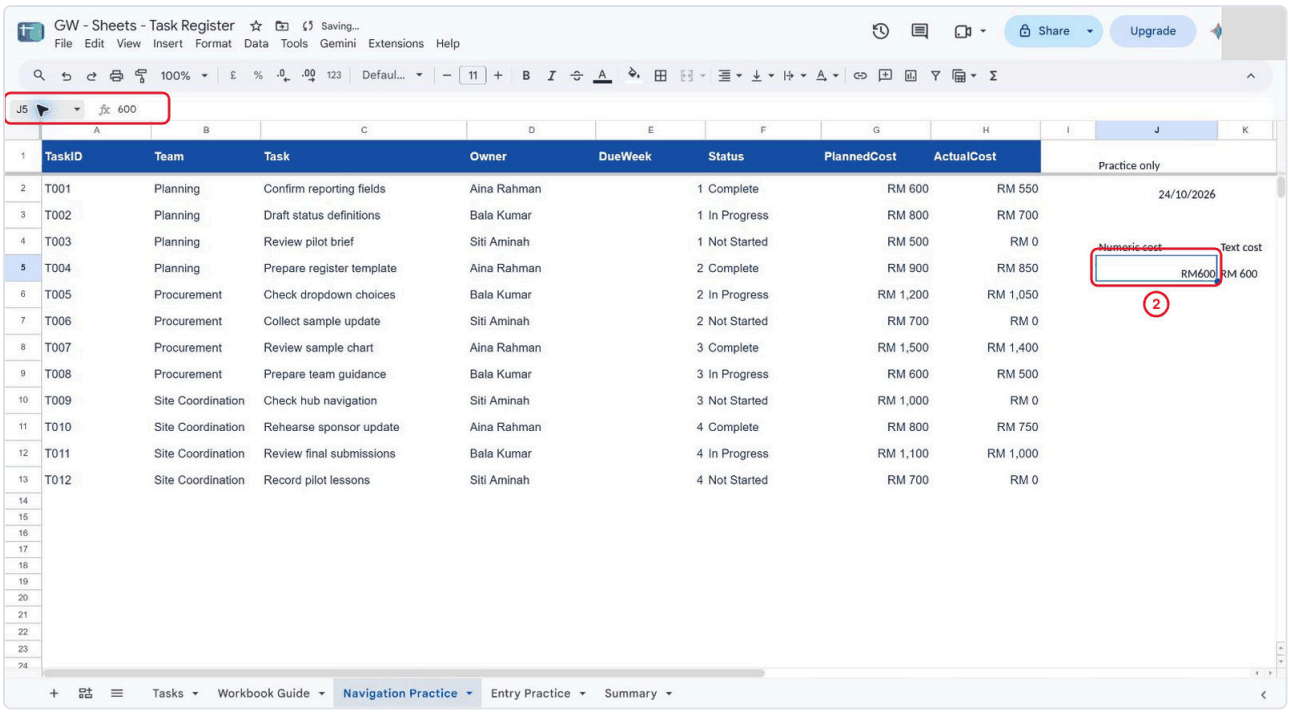
> **Reference run:** In the reference locale it displays **24/10/2026**.



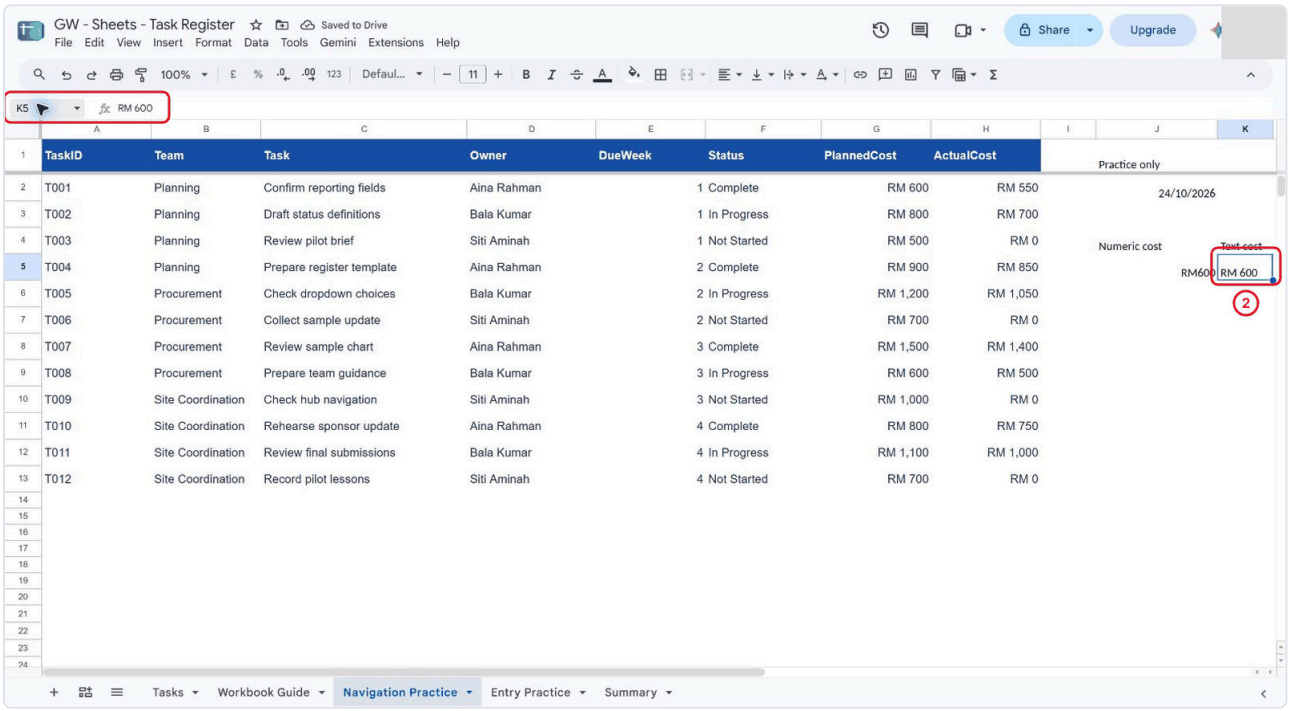
TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

Keep the example date outside the source register.

2. Enter **Numeric cost / Text cost** in **J4:K4**, then numeric **600** in **J5** and text **RM 600** in **K5**. Apply RM currency to J5. Select each cell and compare its formula-bar value.



Inspect the numeric practice cost.



Inspect the typed text cost.

3. Keep these clearly labelled practice entries as evidence, or undo them when no longer useful. Compare the original source values before and after formatting; display changes must not alter the twelve records.

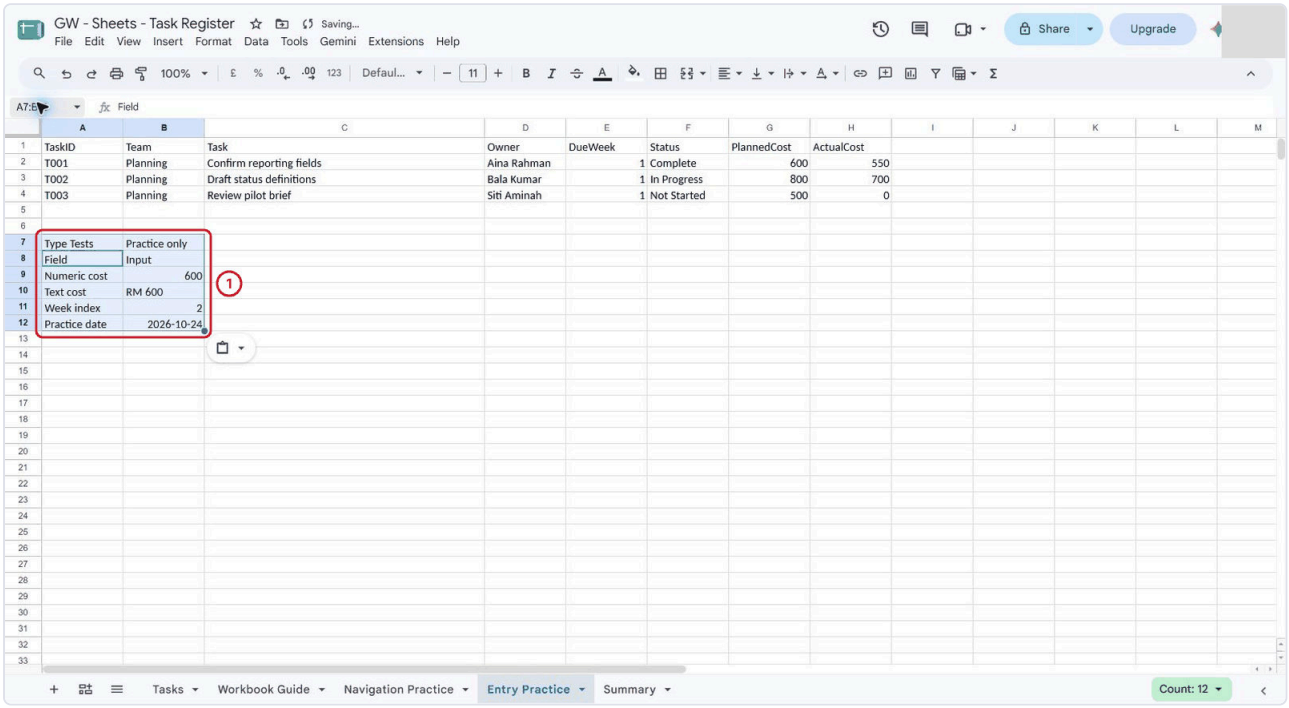
> **Reference run:** The reference retains them outside the baseline.

What should you see?

You can explain why number formatting does not convert arbitrary text into a number.

2.4 Diagnose Display, Type and Readability Problems

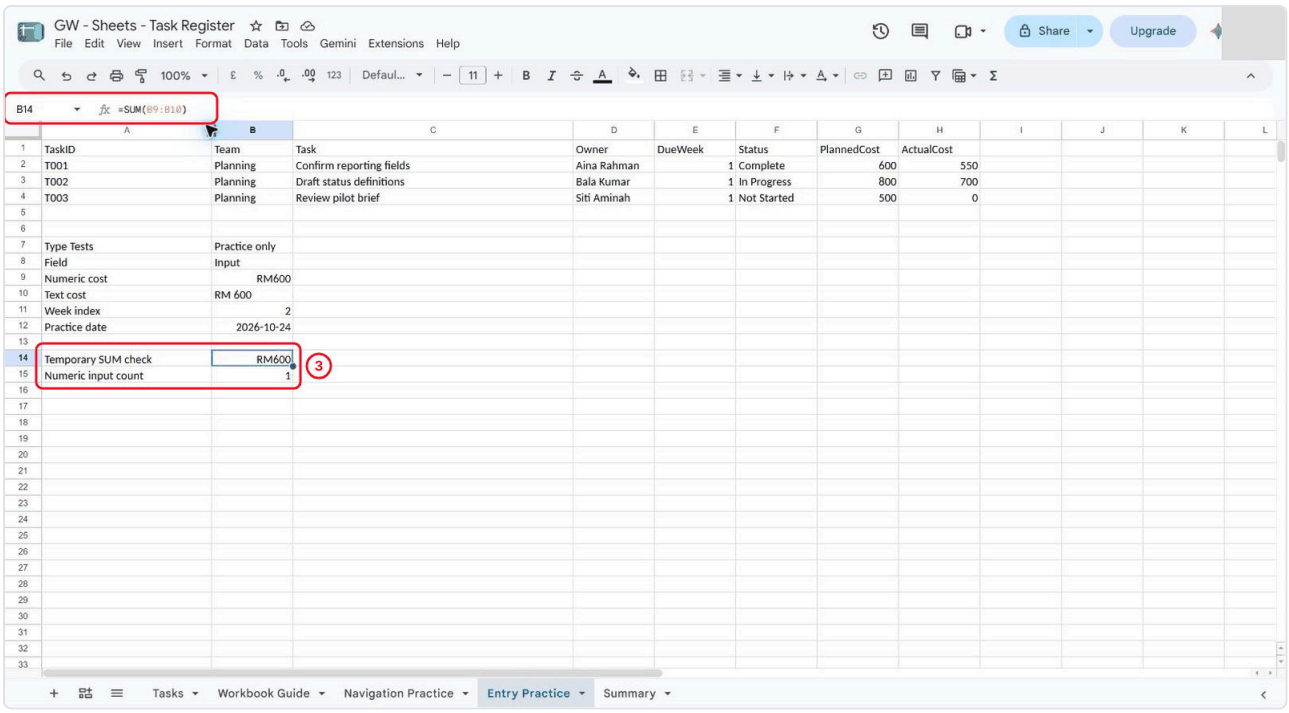
1. On **Entry Practice**, keep **A1:H4** as the copied three-record table. Create **Type Tests** in **A7:B12** with the labels and values in Part 3 of the prompts page: numeric **600** in B9, text **RM 600** in B10, week index **2** in B11 and practice date **2026-10-24** in B12.



Create labelled type tests outside the copied records.

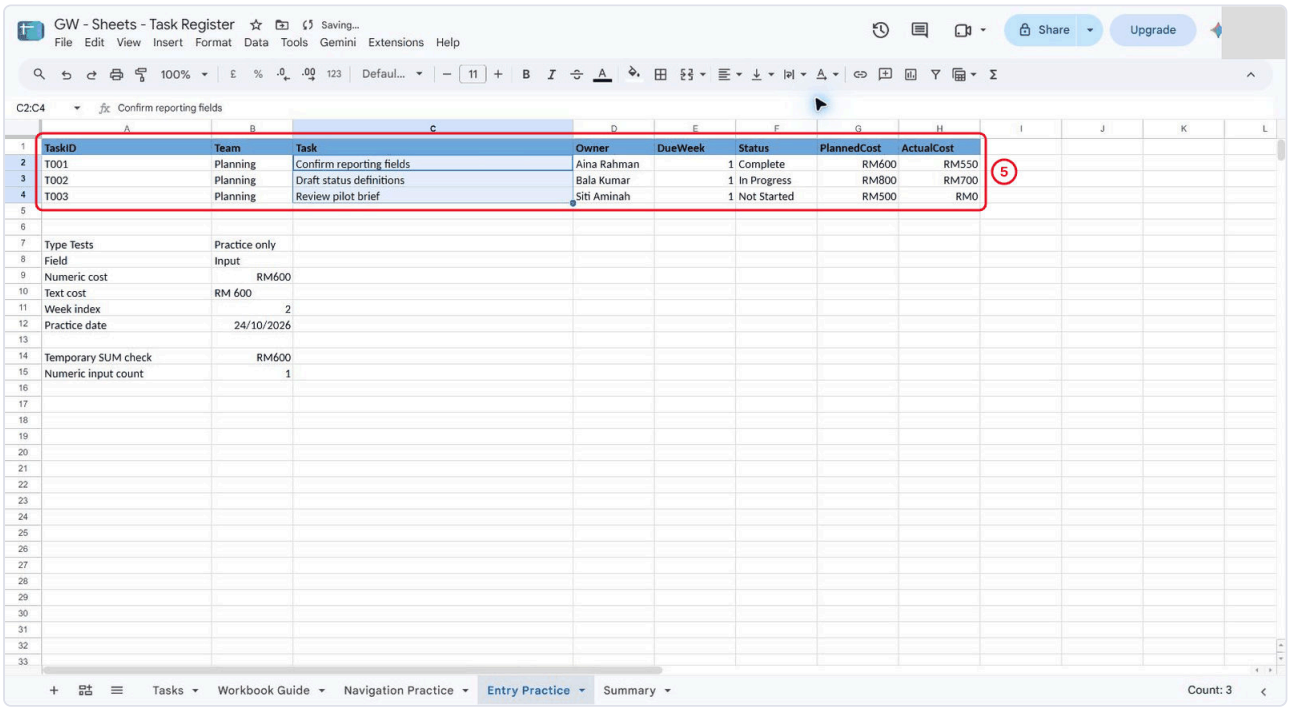
2. Apply RM currency to **B9:B10**, then inspect both values. B9 remains a number and B10 remains text. Give B12 a date display; keep the practice date outside the source week column.
3. Type `Temporary SUM check` in A14 and `Numeric input count` in A15. Add **=SUM(B9:B10)** in B14 and **=COUNT(B9:B10)** in B15. Widen the label column if needed. These checks remain on the practice tab.

> **Reference run:** The reference results are **600** and **1**: numeric 600 contributes, while text RM 600 is ignored by this range SUM.



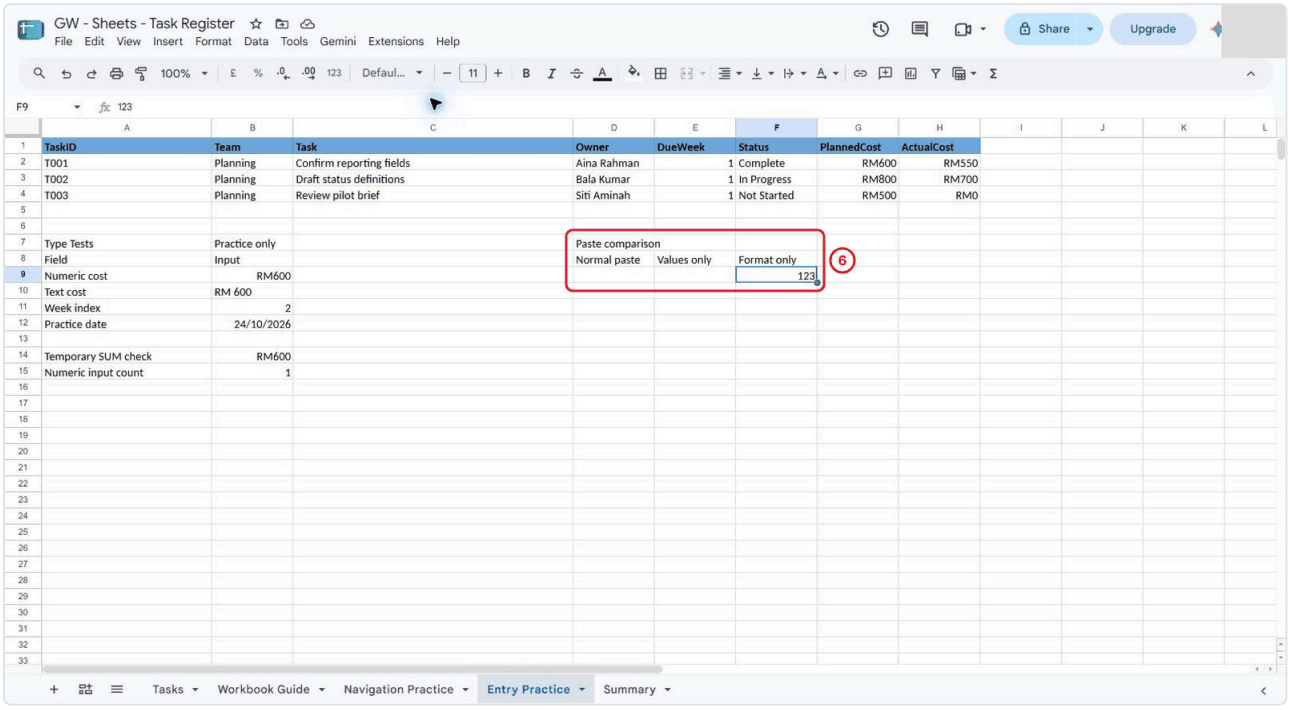
Observe SUM 600 and COUNT 1 for numeric and text costs.

4. Give **G2:H4** the same custom RM currency display and **E2:E4** custom number format **0**. B11 remains a whole-number practice index. Dates stay in B12 and Navigation Practice J2; they do not supply missing source schedule evidence.
5. Apply a bold, light-blue header with readable dark text to **A1:H1**, wrap **C2:C4**, and retain the widened task column. Make each complete description readable. No source cells are merged.



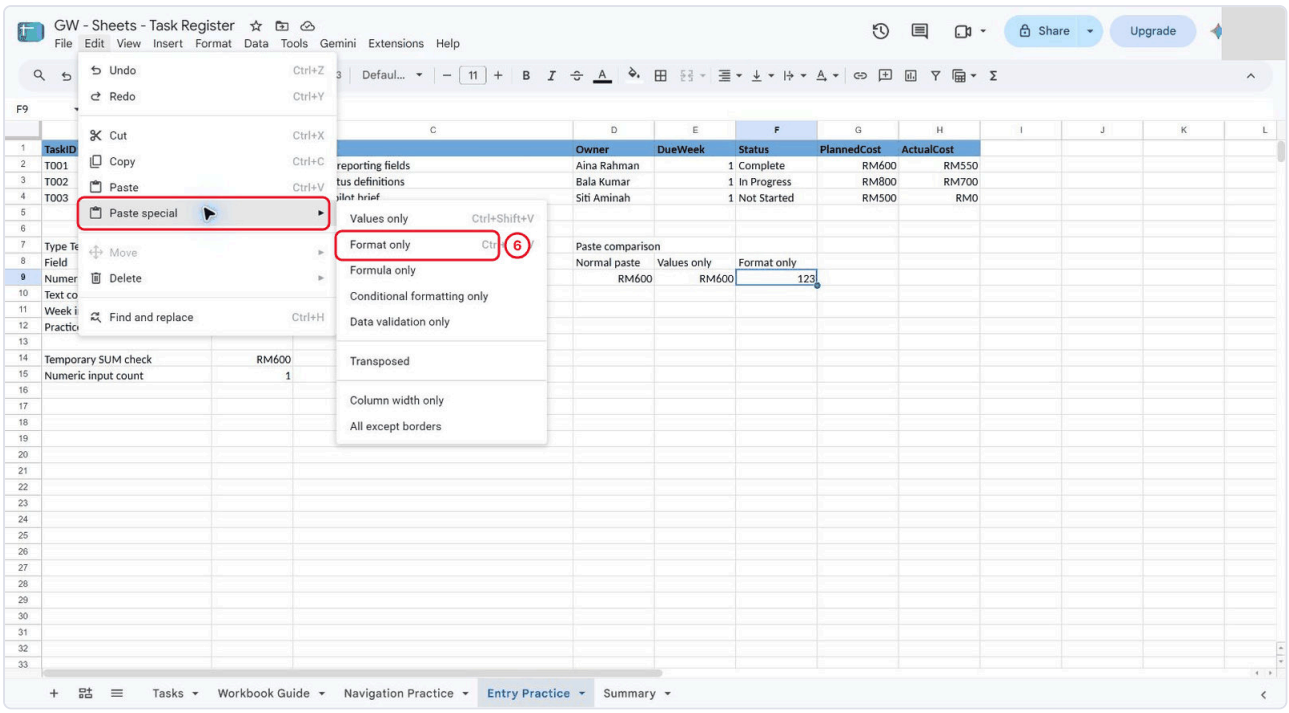
Format the practice table with readable headers and wrapped task text.

6. Prepare a separate paste comparison: type `Paste comparison` in D7 and the headers **Normal paste / Values only / Format only** in **D8:F8**. Type **123** in F9. Select E9 and set **More formats > Custom number format** to **0**, so you can see whether a values-only paste keeps that format. Then select B9 (numeric 600 with RM display) and press **Ctrl+C**:
   - select D9 and press **Ctrl+V** (normal paste)
   - select E9 and choose **Edit > Paste special > Values only**
   - select F9 and choose **Edit > Paste special > Format only**

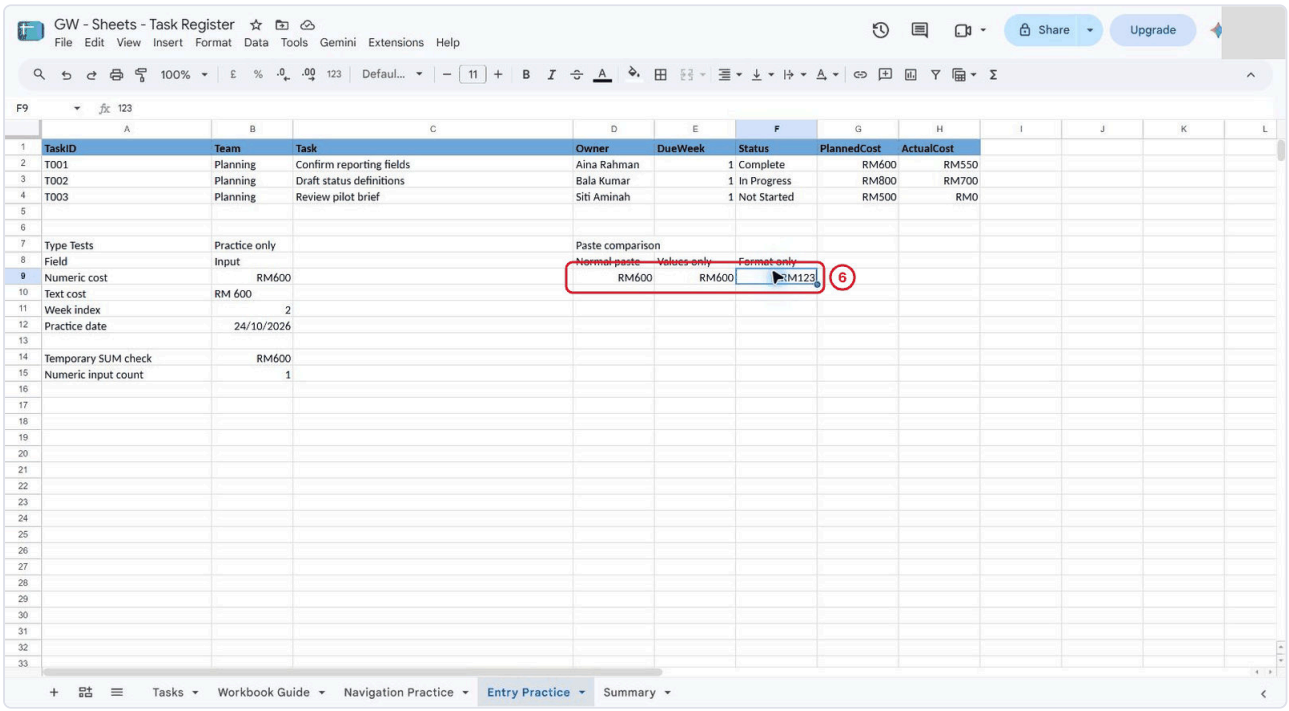


Prepare separate paste targets including the existing 123.

Compare the three cells. Normal paste D9 should be numeric 600 with RM display. Format-only F9 keeps numeric **123** and displays **RM123**. A values-only paste should bring 600 into E9 without its RM format.



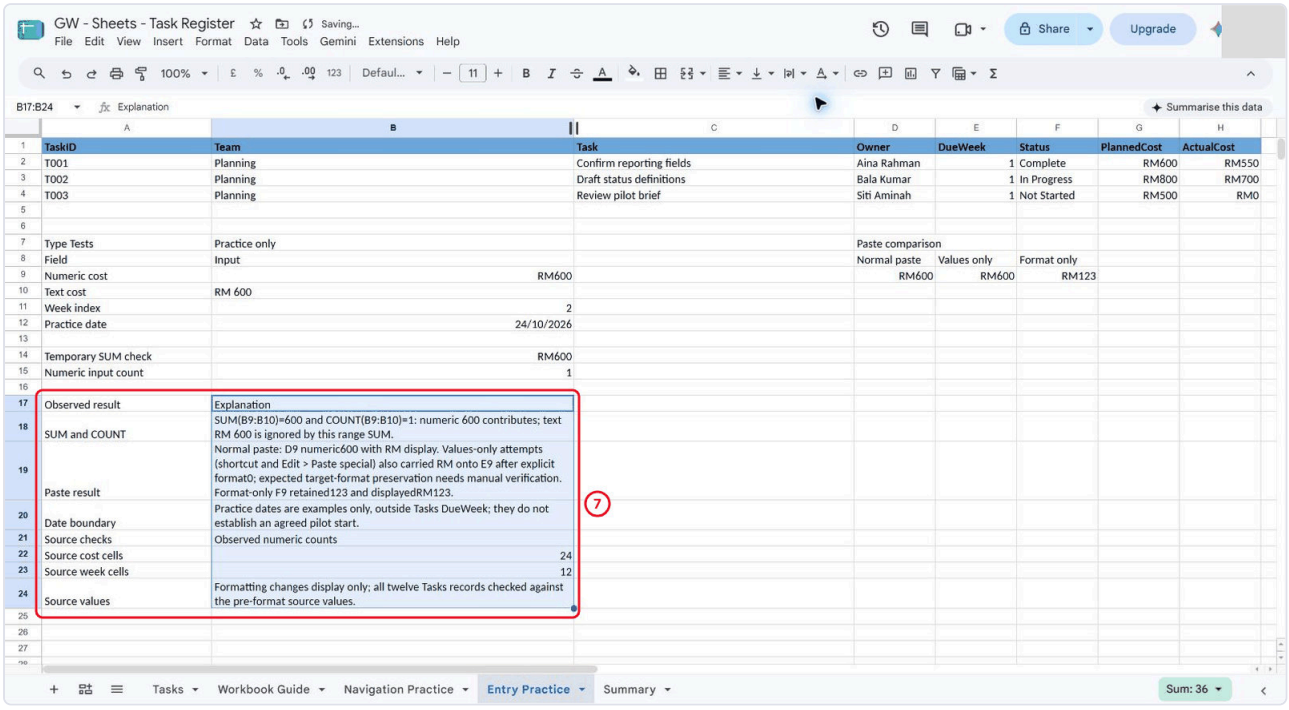
Inspect the actual Paste special menu.



Verify format-only retains 123 while displaying RM123.

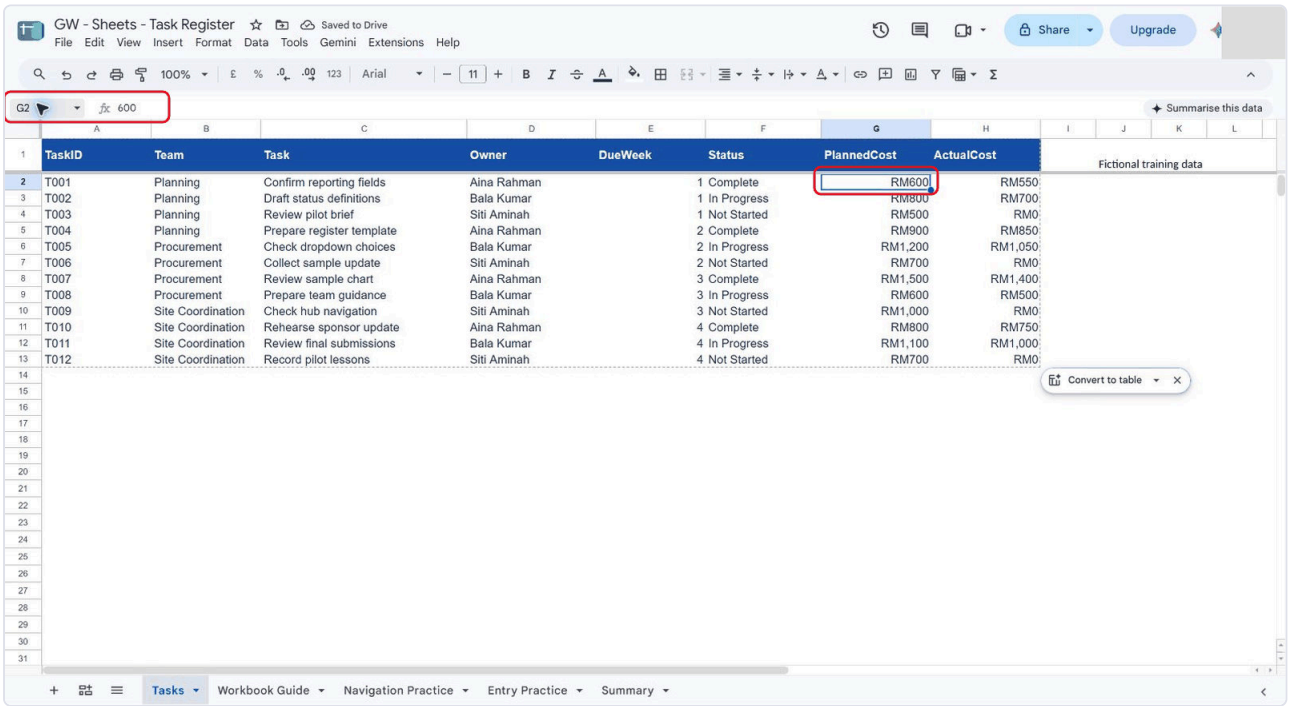
> **Verification limitation:** the controlled session's values-only shortcut and direct menu both carried the RM format onto E9 even after its target was explicitly set to number format 0. The number remained 600, but expected preservation of the target format was **not verified**. Repeat this step manually, compare E9's number format before and after, and record the outcome. Do not treat visually similar cells or an attempted command as proof that the property transfer passed.

7. Record the results and the unresolved paste check in **A17:B24** of the practice tab, using the labels in Part 4 of the prompts page. **=COUNT(Tasks!G2:H13)** returns **24** numeric cost cells; **=COUNT(Tasks!E2:E13)** returns **12** numeric week cells. A bounded comparison confirmed that every original **Tasks!A1:H13** underlying value remained unchanged. Wrap the observation text so it is readable.



Record source counts and the unresolved values-only format behaviour.

Return to **Tasks > G2** and inspect **600** again. Formatting did not create a confirmed calendar date or change the twelve records.



Inspect the unchanged source value after formatting.

What should you see?

Type-test block with observed calculation behaviour. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Practice

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Repair the three proposed entries in the case input (Part 1 of the prompts page) in a separate intake table. Preserve raw input and record the clean value or clarification needed.
2. Prepare a readable three-record handout view with consistent currency and whole-week display. Prove the underlying numeric values remain numeric.
3. Give a partner two visually similar cost cells and ask them which one can be calculated reliably. Use formula-bar and SUM evidence in your explanation.

Evidence to Hand Over

- Type-test block with observed calculation behaviour.
- Raw-to-clean intake table and readable entry-practice layout.
- Independent numeric evidence and unchanged 12-row source.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
Currency looks correct but totals are wrong	Inspect numeric type and formula-bar value before changing more display formats.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Typed values and readable formatting prepare the register for calculation.

Check yourself: Why should you avoid merging cells inside a dataset?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Use the supplied source and the explicit practice values below. Keep dates and type tests outside the original register.

Part 1: Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Formatting intake: PlannedCost “RM 600” supplied as text; DueWeek “next Monday” with no agreed calendar start; a long Task description clipped in a narrow column. Treat type, missing date evidence and presentation as three different issues.
```

Part 2: Exact Practice Values

- Source costs: Tasks!G2:H13, custom Malaysian Ringgit **RM1,000** without decimals.
- Source week indices: Tasks!E2:E13, custom number format **0**.
- Practice date: **2026-10-24**, displayed **24/10/2026** in the reference locale; not a confirmed project start.
- Navigation Practice: J2 date; J4:K4 Numeric cost / Text cost; J5 numeric 600, K5 text RM 600.

Entry Practice type block: A7:B12, labelled numeric 600 / text RM 600 / week 2 / practice date. B14 **=SUM(B9:B10)** gives 600; B15 **=COUNT(B9:B10)** gives 1. Both formulas reference the cells as a range; the observed result does not establish behaviour for every possible way of passing text to a function.

Paste comparison: D9 normal paste from B9 gives numeric 600 with currency format; F9 format-only retains 123 and displays RM123. **Values-only target-format preservation remains unverified**: both shortcut and menu carried the RM format in this controlled session after setting E9 to number format 0. Repeat manually and compare formats as well as values. Do not claim this property-transfer check passed.

Source controls: **=COUNT(Tasks!G2:H13)** gives 24; **=COUNT(Tasks!E2:E13)** gives 12. All twelve source records retain their original underlying values. The source header already had readable contrast and its widths were retained; wrapping and Fit to data were applied. Entry Practice header/widths/wrapping were improved. Independent intake repair, partner test and classroom timing remain pending.

Part 3: Type Tests Block

Section 2.4, step 1. Type these into **A7:B12** of Entry Practice (column A, then column B):

```
Type Tests      Practice only
Field   Input
Numeric cost    600
Text cost       RM 600
Week index      2
Practice date   2026-10-24
```

Part 4: Results Block Labels

Section 2.4, step 7. Type these labels in **A17:A24** and `Explanation` in B17, then write your own observation for each row in column B. Use **=COUNT(Tasks!G2:H13)** in B22 and **=COUNT(Tasks!E2:E13)** in B23.

```
Observed result
SUM and COUNT
Paste result
Date boundary
Source checks
Source cost cells
Source week cells
Source values
```

CHAPTER 3

Calculate and Copy Formulas

In this chapter you will use arithmetic, SUM, AVERAGE and COUNT; explain relative and absolute references.

> **Prompts:** the prompts page has every value and prompt for this chapter, each with a Copy button.

Estimated time: 65 minutes, planned as 15 explanation and demonstration, 22 guided build and casework, 20 independent practice and 8 verification and repair, with 50 active minutes by design. Not yet timed with learners.

Your result: Copied variance formulas, reconciled totals and a tested fixed reference.

> **Status:** author reference with 23 staged application captures through guided sections 3.1–3.3 and casework 3.4. Row formulas, summaries, fixed-reference change/restoration, zero/blank/text checks, deliberate reference repair and reconciliation verified. Independent estimate/omitted-range challenge and learner timing remain pending.

What You Will Learn

- Write arithmetic and aggregation formulas.
- Copy relative references correctly.
- Anchor a shared assumption.

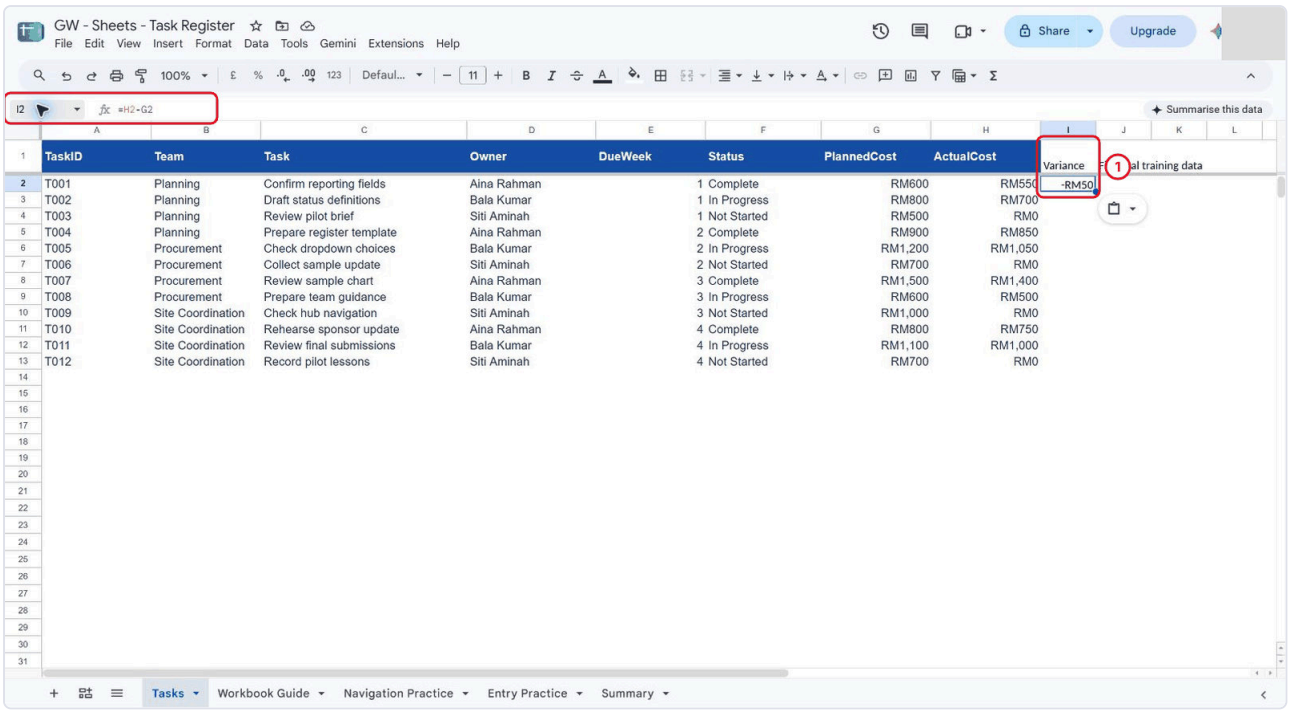
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks A1:H13. The workbook has no prefilled answer formulas.

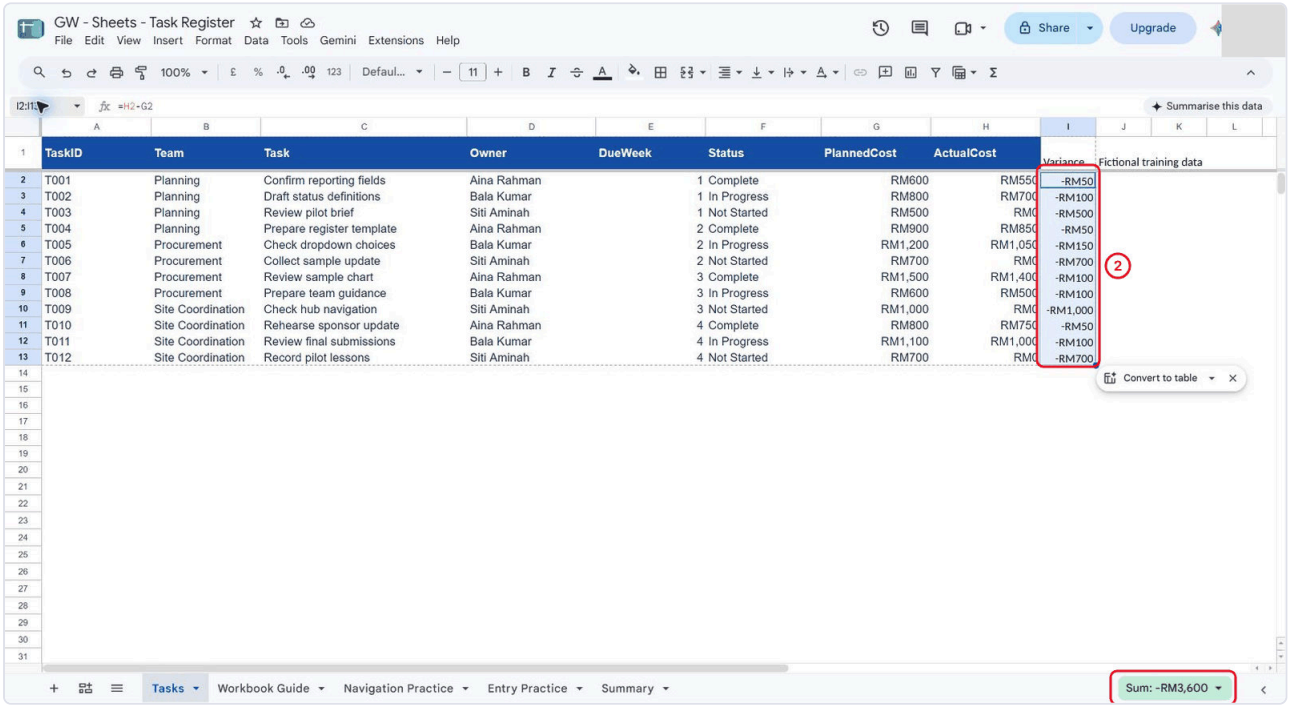
3.1 Calculate Variance

1. On **Tasks**, use the name box to select **I1** and enter **Variance**. In **I2** enter **=H2-G2** and commit it. T001 displays **-RM50**; the formula bar shows the underlying calculation.



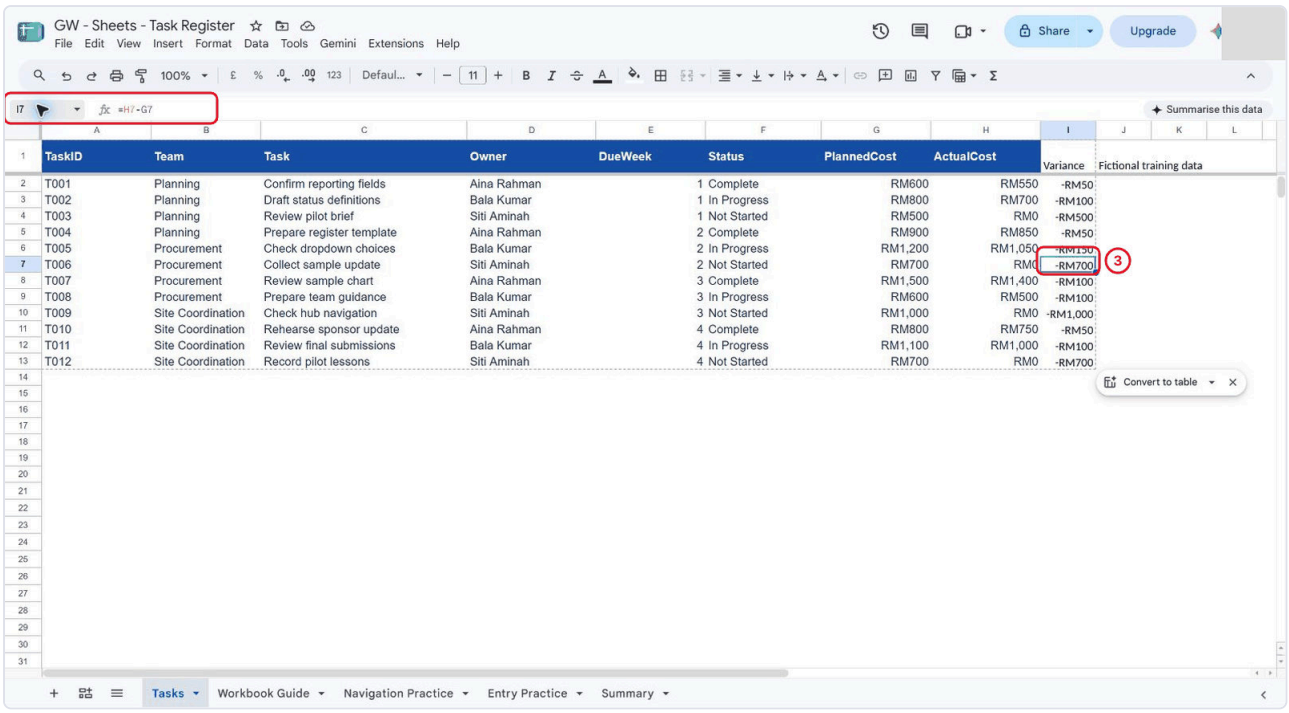
First variance result before filling.

2. Select **I2:I13** with the name box and press **Ctrl+D** to fill down from the first cell. The fill handle or normal copy/paste is an alternative. The range sum is **-RM3,600**.

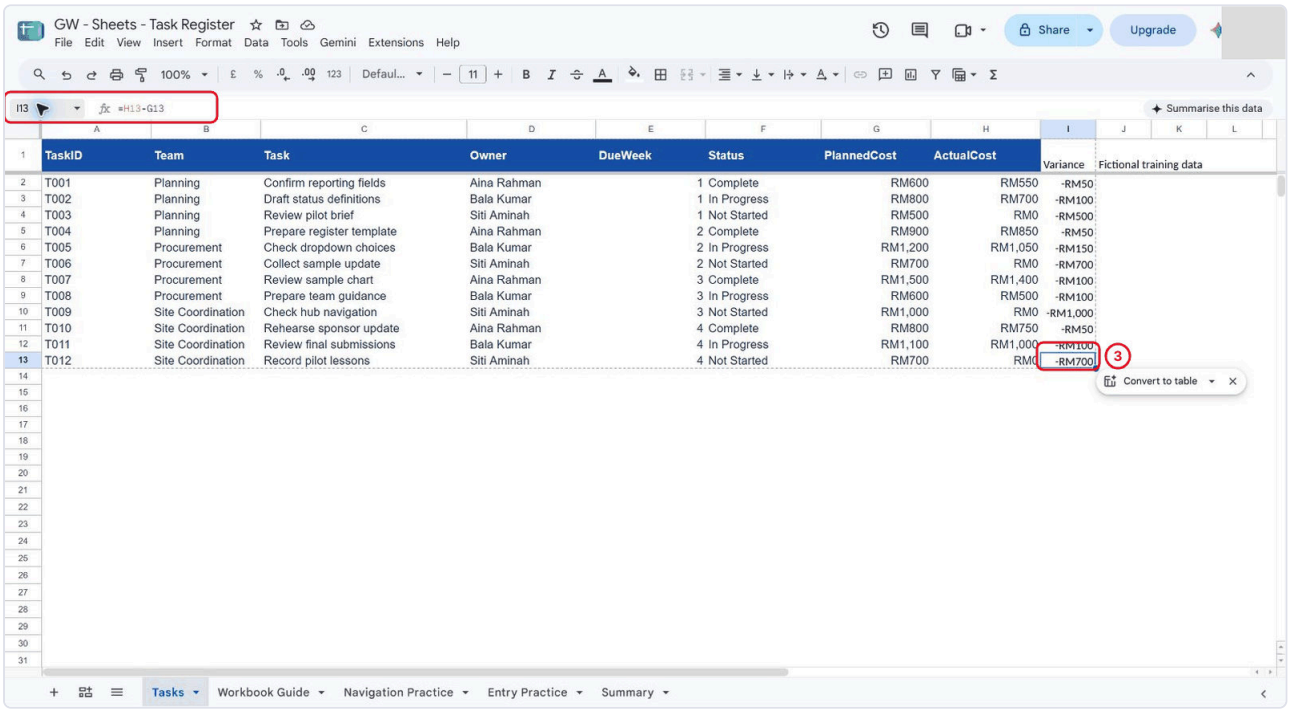


Fill all twelve row variances.

3. Inspect **I2**, **I7** and **I13** in the formula bar. Each refers to its own row: **=H2-G2**, **=H7-G7** and **=H13-G13**. The final two checks display **-RM700**. A negative variance means actual cost is below planned; it does not by itself establish a final saving or project completion.



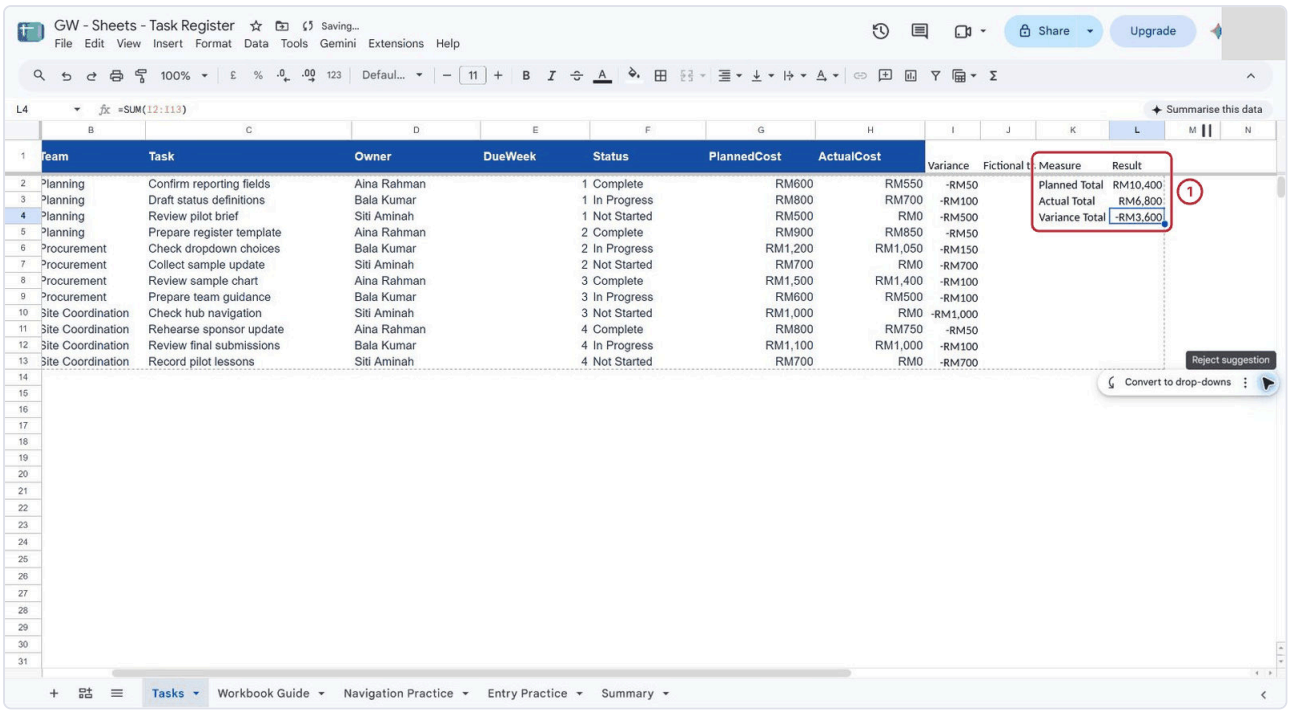
Inspect the middle row reference.



Inspect the final row reference.

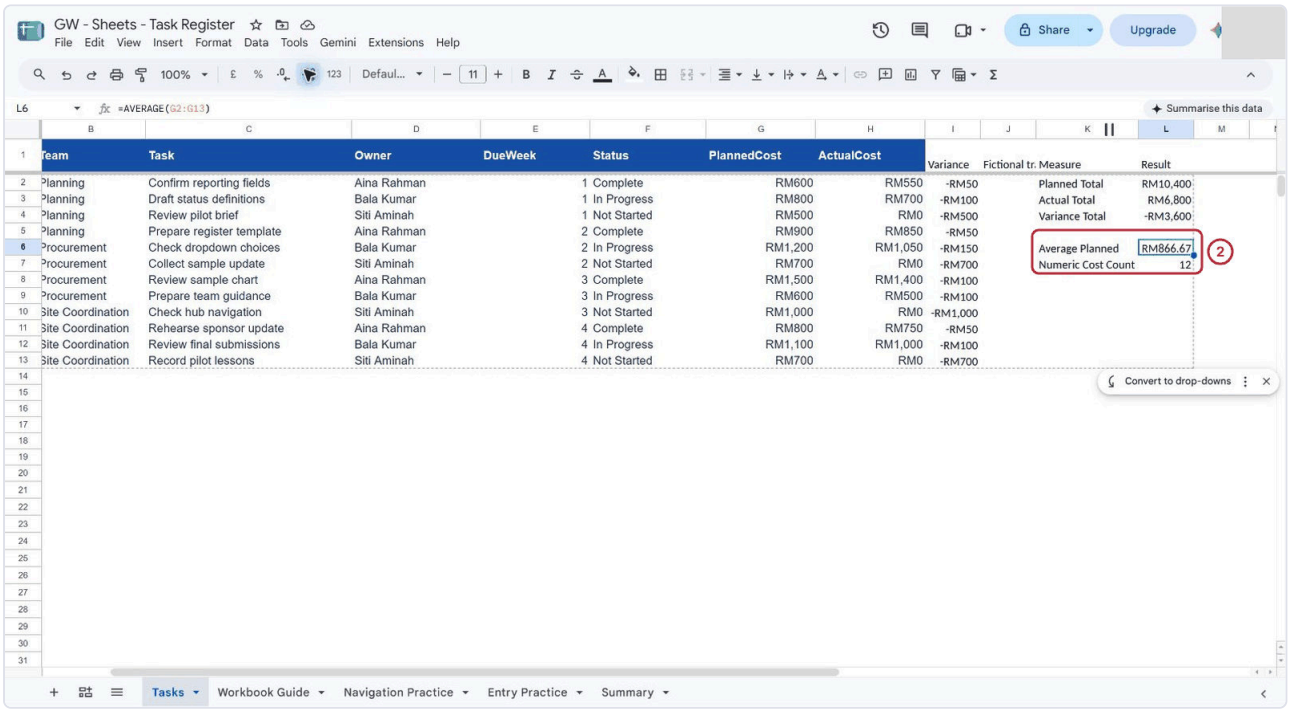
3.2 Summarise the Columns

1. Still on **Tasks**, in **K1:L1** enter **Measure / Result**. Label **K2:K4 Planned Total / Actual Total / Variance Total**. In **L2:L4** enter **=SUM(G2:G13)**, **=SUM(H2:H13)** and **=SUM(I2:I13)**. Double-click the right edge of the K column header to fit its labels. Scroll horizontally when needed to show the complete summary.



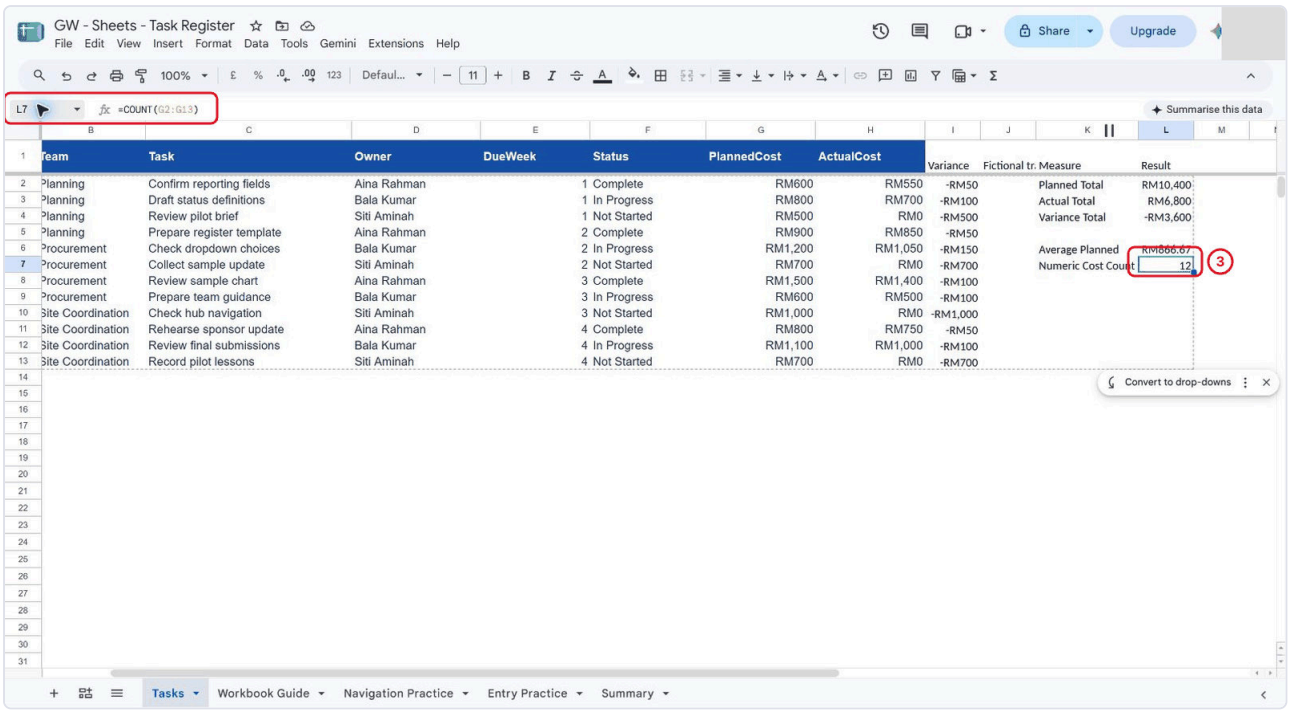
Reconcile planned, actual and variance totals.

2. Add **Average Planned** in K6 with **=AVERAGE(G2:G13)** in L6, and **Numeric Cost Count** in K7 with **=COUNT(G2:G13)** in L7. Fit K again after adding the longer labels. Select L6 and increase decimal places twice if it inherited the source's whole-ringgit display. The displayed average is **RM866.67**; its underlying calculation remains 10,400 divided by 12.



Display the average with two decimals.

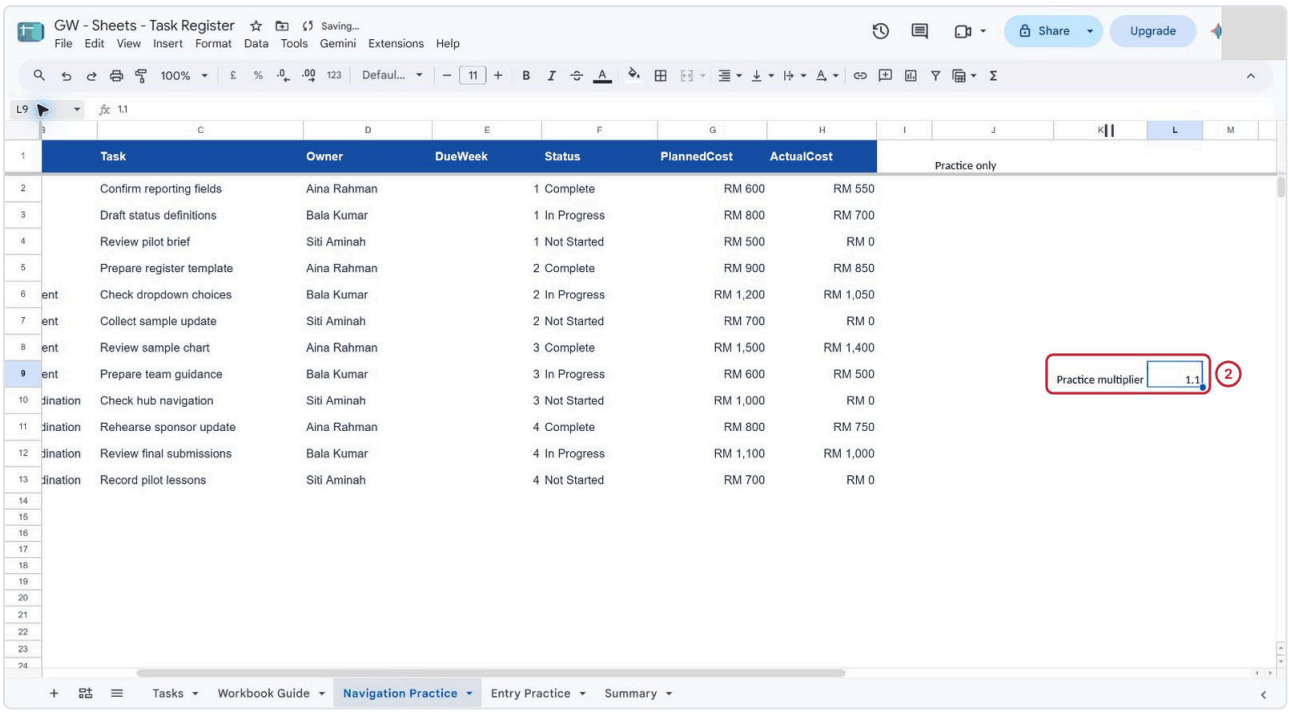
3. Inspect L7's formula and **12** result. The totals are **RM10,400 planned**, **RM6,800 actual** and **-RM3,600 variance**. Do not replace the formulas with those expected numbers.



Count twelve numeric planned costs.

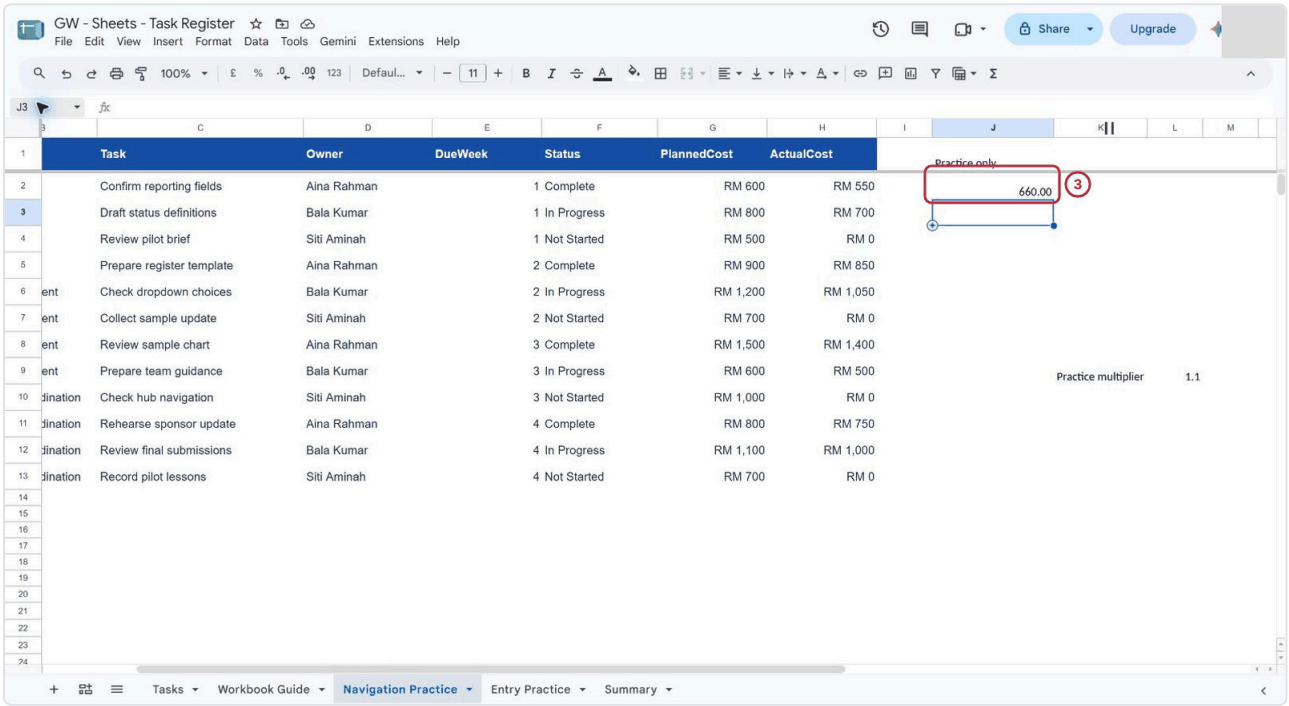
3.3 Anchor an Assumption

1. On **Navigation Practice**, reuse the clearly labelled practice area: clear the earlier Lab 2 date/cost examples in **J2:K6**. Their earlier screenshots remain in Lab 2. Set **J2:J13** to **More formats > Number**, so the old J2 date format cannot disguise a numeric calculation as a date. Keep the original Tasks source and the copied A:H records unchanged.
2. Enter **Practice multiplier** in **K9** and numeric **1.1** in **L9**. Fit K to show the complete label. This assumption is hypothetical; it is not an approved rate or a tax rule.



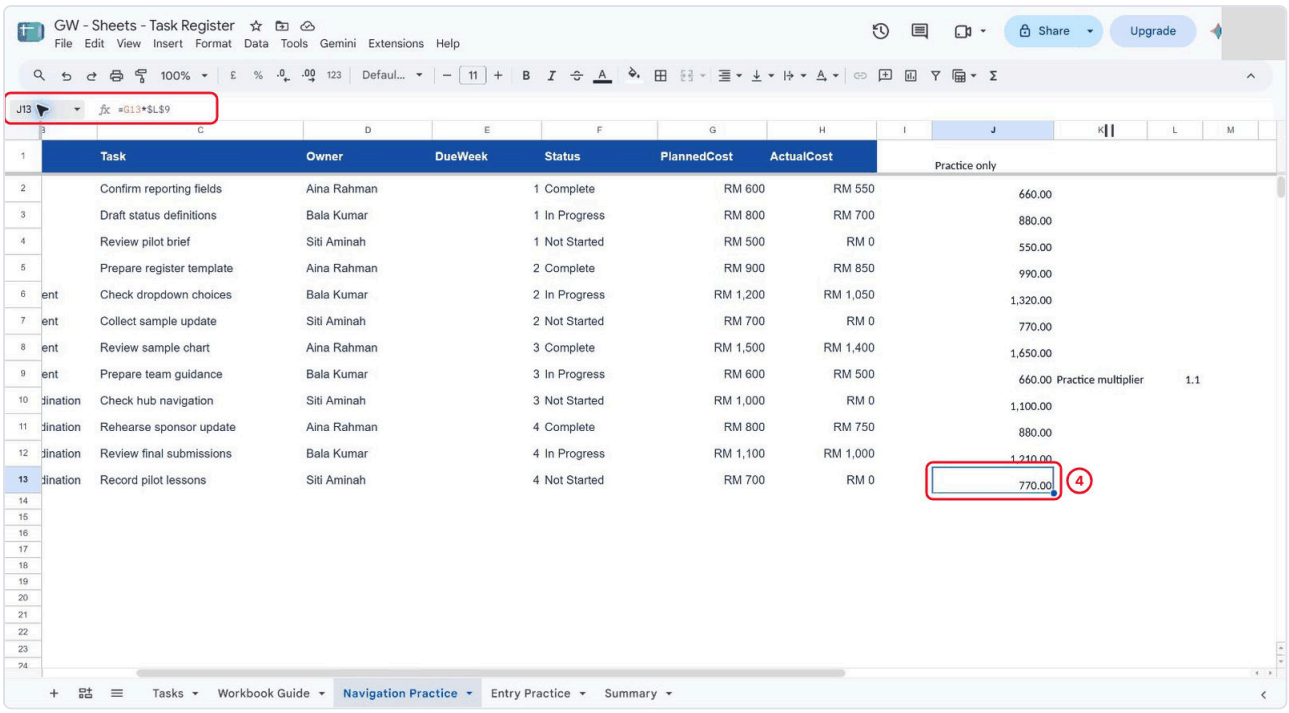
Enter the hypothetical practice multiplier.

3. In **J2**, enter `=G2*$L$9` and commit it. The first result is **660.00** before filling the remaining rows.



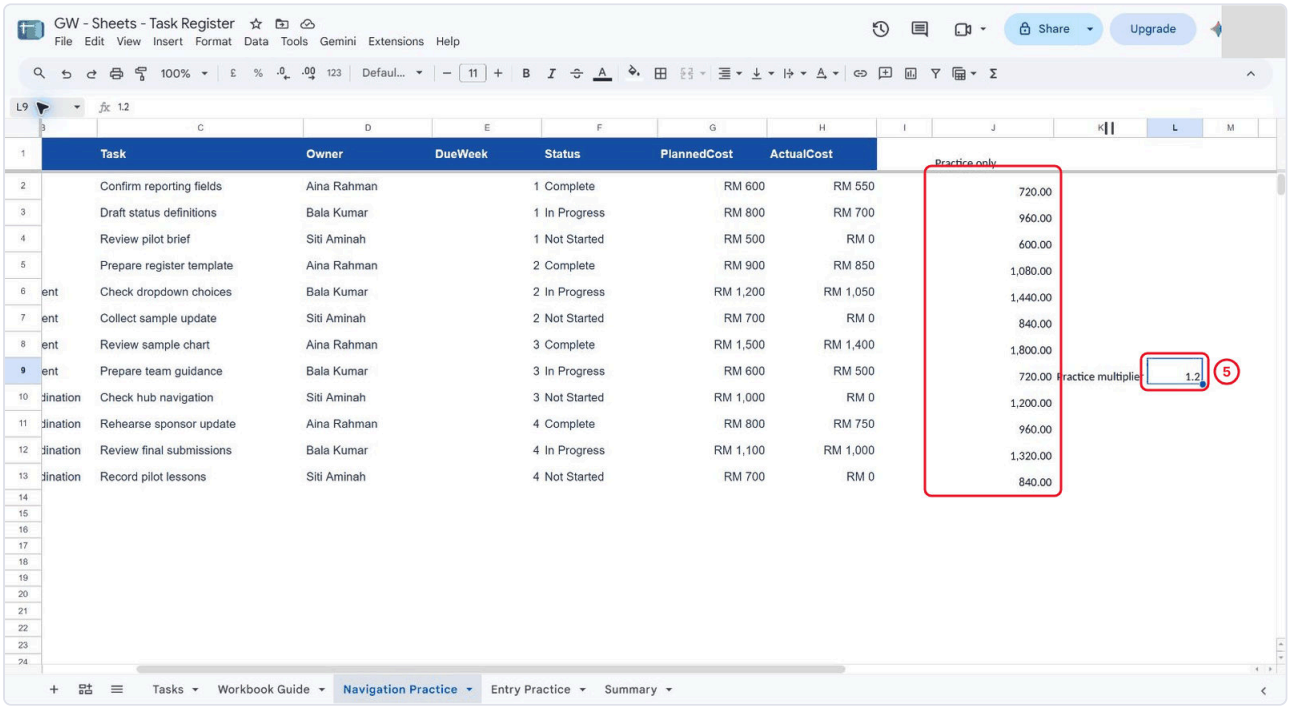
First anchored calculation result.

4. Select **J2:J13** and use **Ctrl+D**. Inspect **J13**: `=G13*$L$9` displays **770.00**. G changes with the row; $L$9 stays fixed.



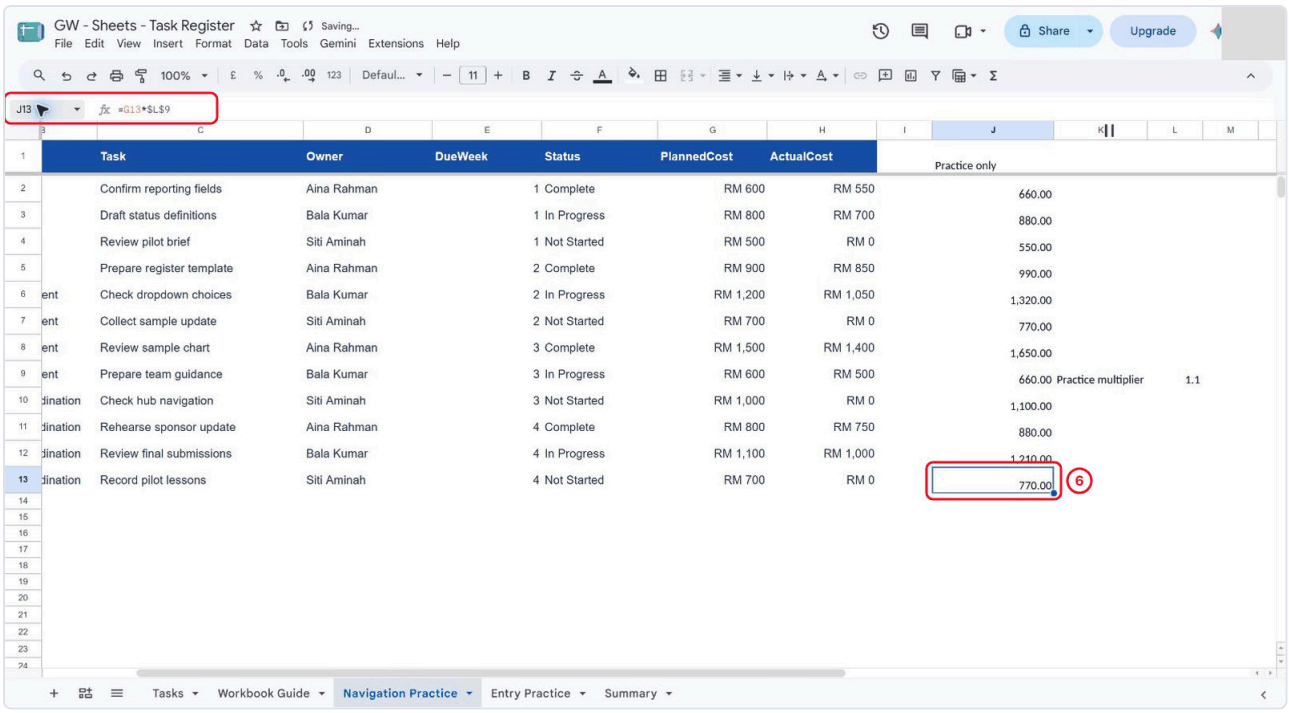
Inspect the final anchored reference.

5. Change **L9** to **1.2** and commit it. The first and final results become **720.00** and **840.00**.



Change the multiplier to 1.2.

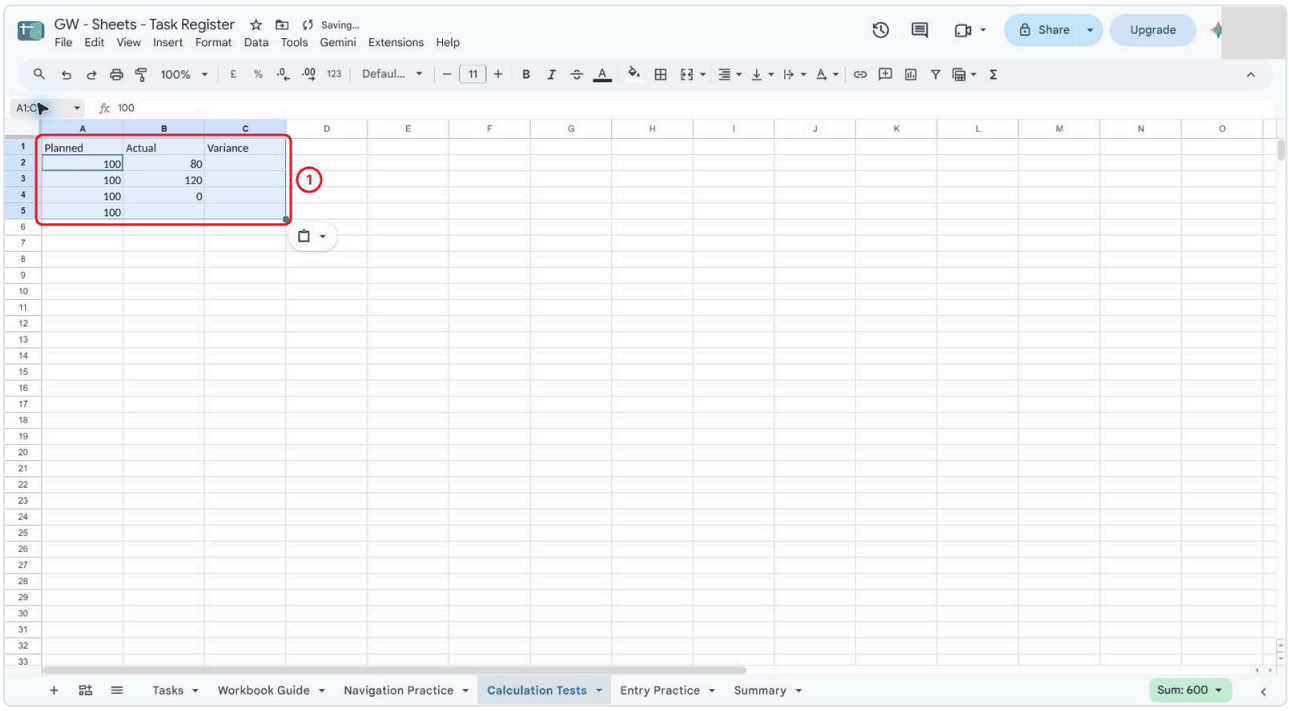
6. Restore **L9** to **1.1**. Reinspect J13's fixed reference and **770.00** result.



Restore the multiplier to 1.1.

3.4 Test Calculations With Zeros, Blanks and Reference Errors

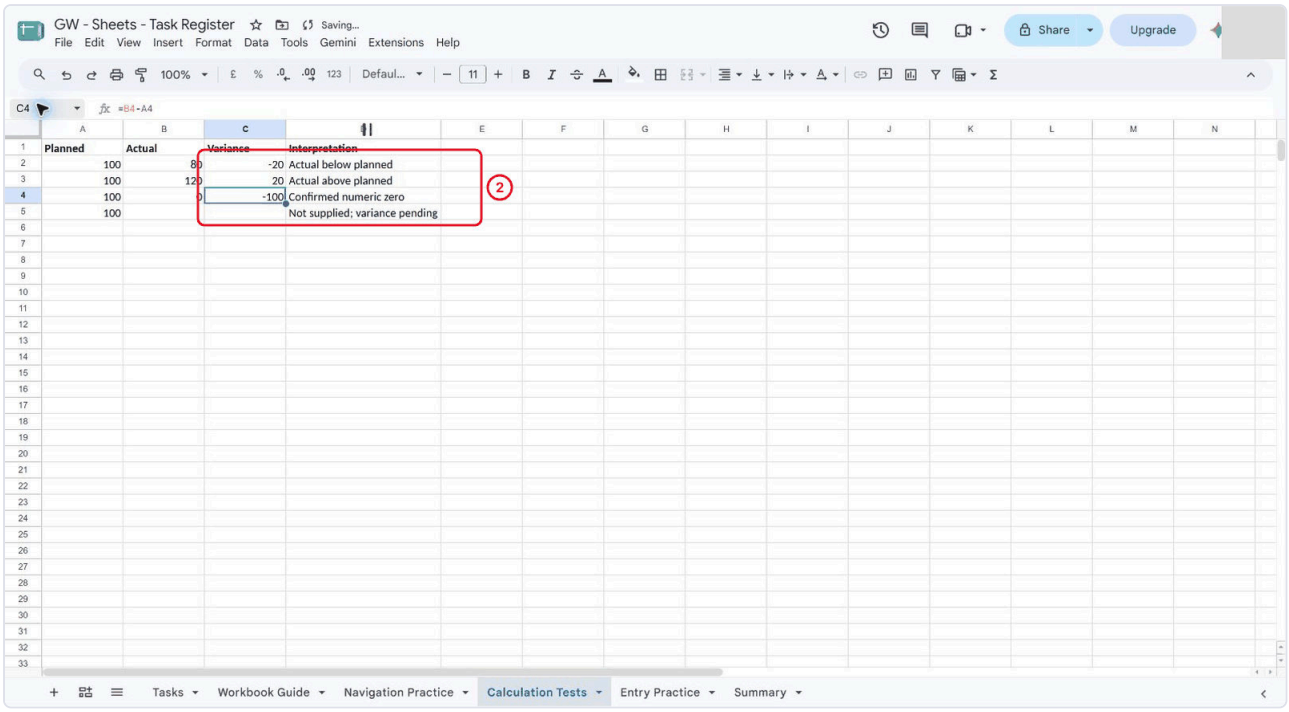
1. Use **Add Sheet**, then double-click the new tab and name it **Calculation Tests**. Enter **Planned / Actual / Variance** in **A1:C1**, and the four planned/actual pairs **100/80**, **100/120**, **100/0**, **100/blank** in rows 2–5. Leave **B5** genuinely empty.



Enter four calculation input cases.

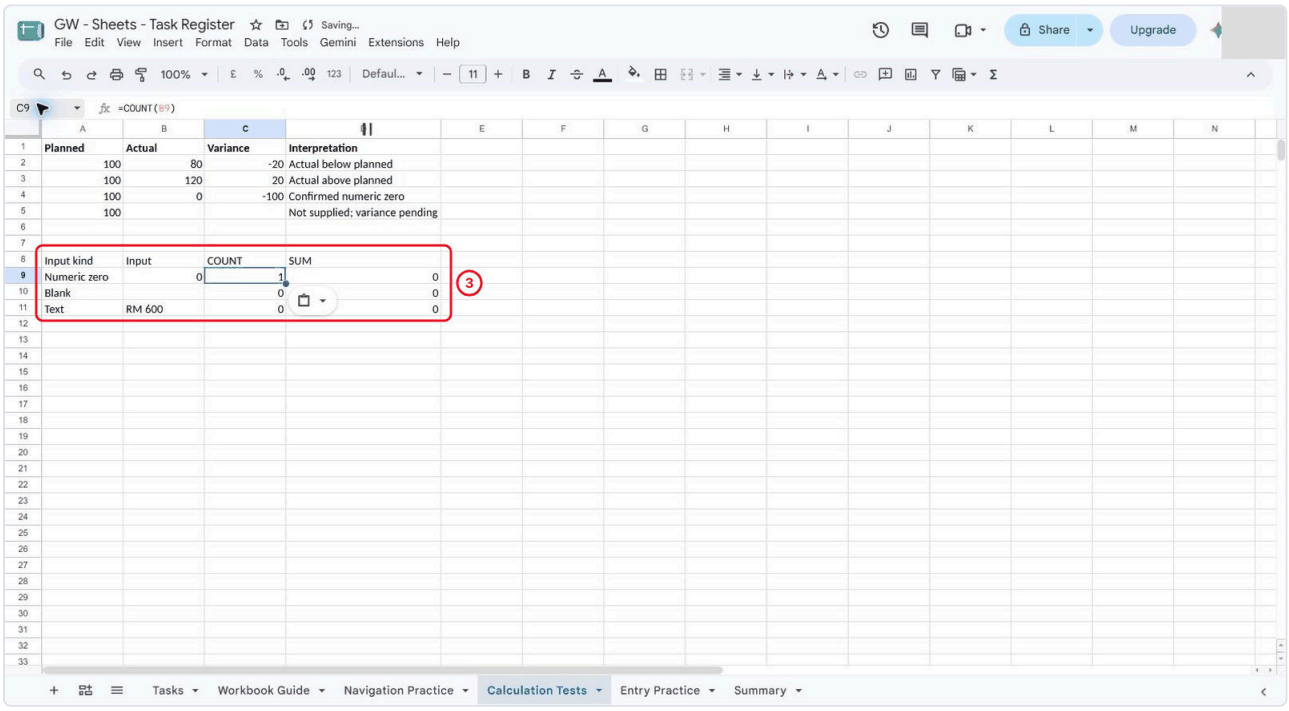
2. Enter **=B2-A2** in C2 and fill **C2:C4**. Results are **-20, 20, -100**. Leave C5 empty because its actual is not supplied. Type `Interpretation` in D1 and the four interpretations from Part 4 of the prompts page

in D2:D5, then fit the column. A formula that treats an empty reference as zero would not establish confirmed zero expenditure.



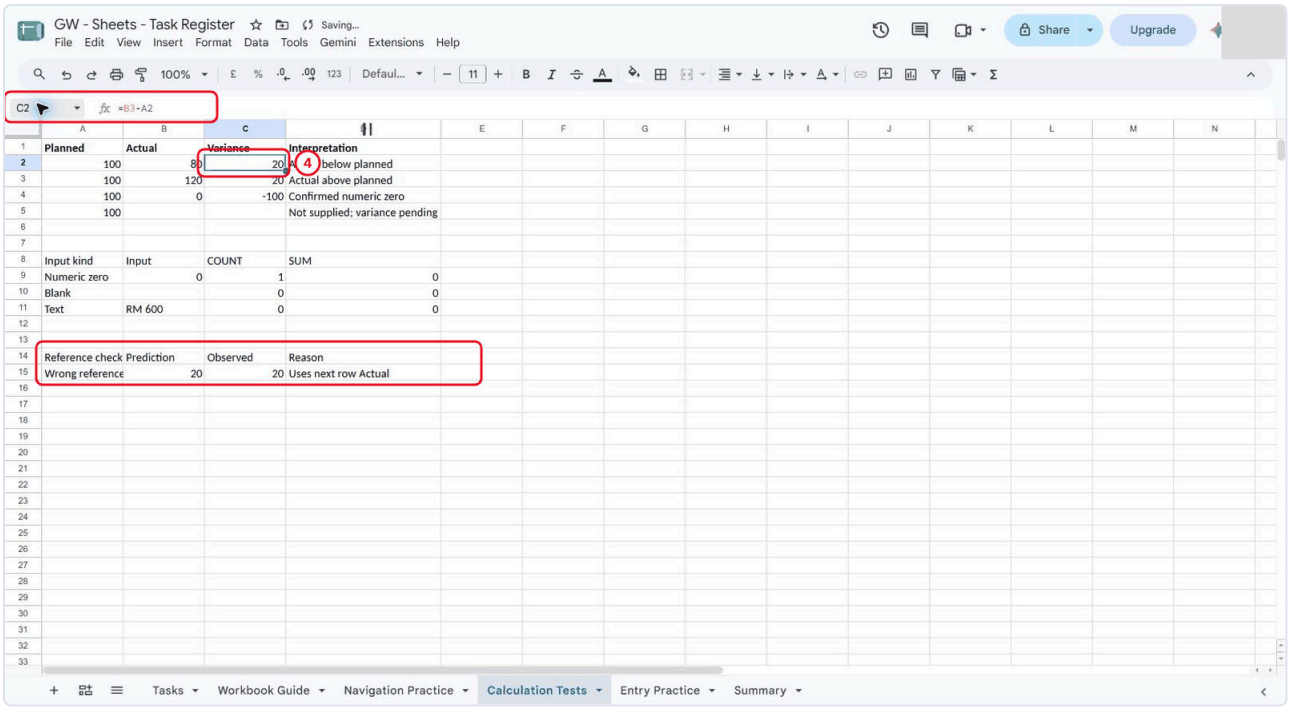
Distinguish a confirmed zero from missing information.

3. In **A8:D11**, create a separate test with headers **Input kind / Input / COUNT / SUM** and the row labels from Part 4 of the prompts page. B9 is numeric **0**, B10 stays blank, B11 is text **RM 600**. In C9 and D9 enter **=COUNT(B9)** and **=SUM(B9)**, then fill through row 11. COUNT returns **1, 0, 0**; SUM returns **0, 0, 0**. SUM alone cannot distinguish these three inputs.



Compare COUNT and SUM for zero, blank and text.

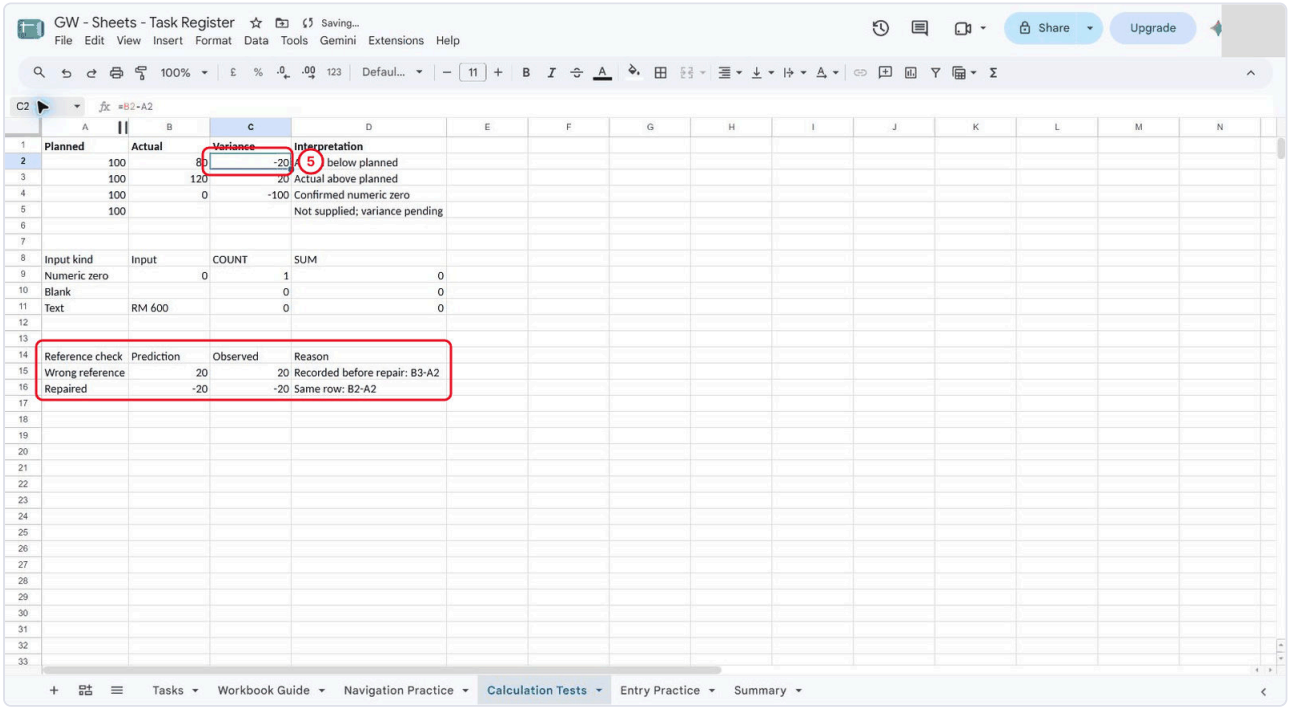
4. Predict that the intentionally wrong **=B3-A2** in C2 gives **20** by using the next row's actual. Enter it, inspect the formula bar and compare it with the first pair's intended **-20**. Record prediction and observation in **A14:D15** (headers and labels are in Part 4 of the prompts page).



Inspect a deliberately wrong row reference.

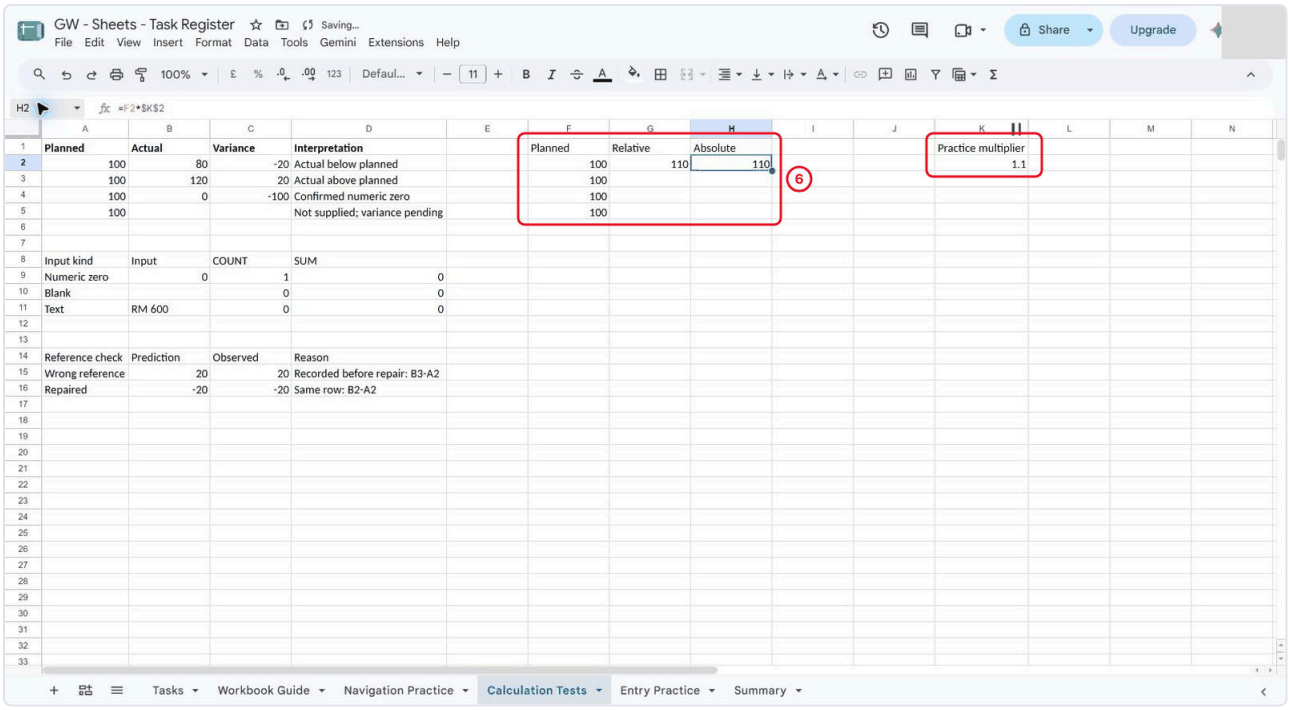
5. Repair C2 to **=B2-A2** and verify **-20**. Keep the wrong test's observed **20** as a labelled historical value, rather than a live link that changes after repair. Add the repaired observation in row 16.

> **Reference run:** The reference uses **=C2** for the current repaired result.



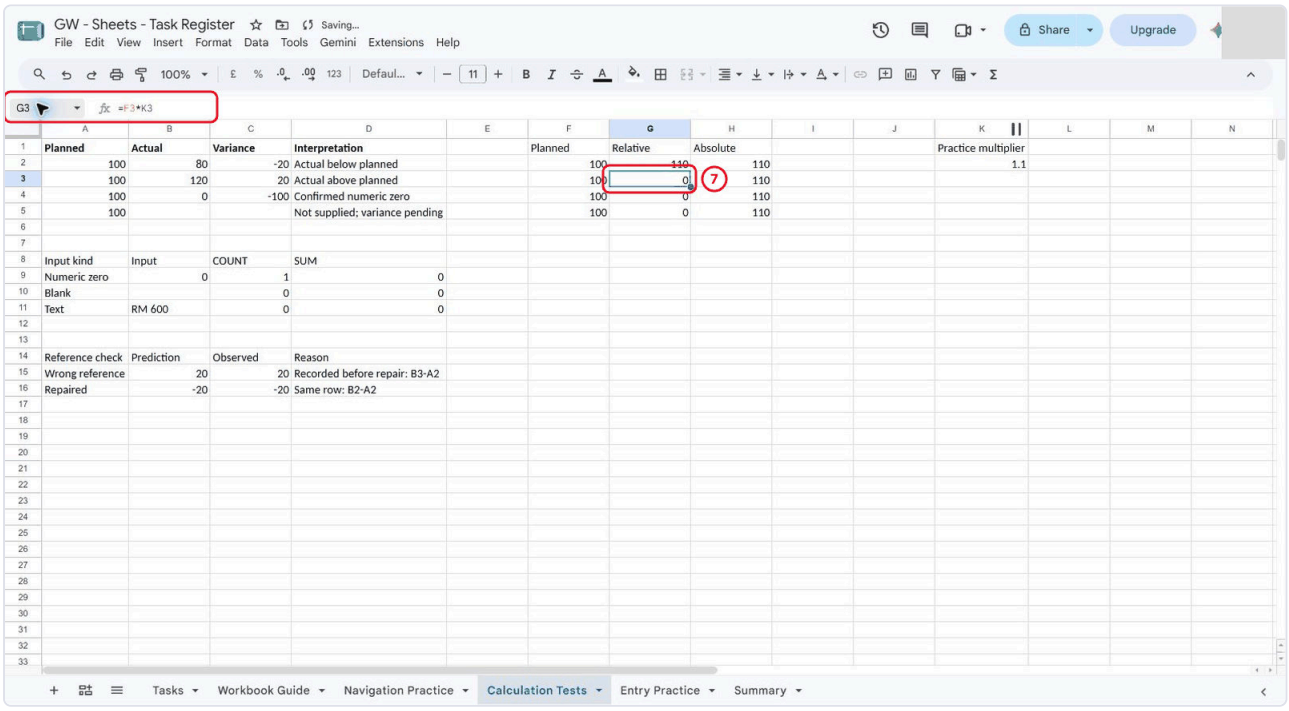
Repair the reference and retain the recorded test.

6. Build the reference comparison in **F1:H5**: headers **Planned / Relative / Absolute**, and four planned inputs of **100**. Put **Practice multiplier** in K1 and **1.1** in K2. Enter `=F2*K2` in G2 and `=F2*$K$2` in H2. Both initially return **110**.

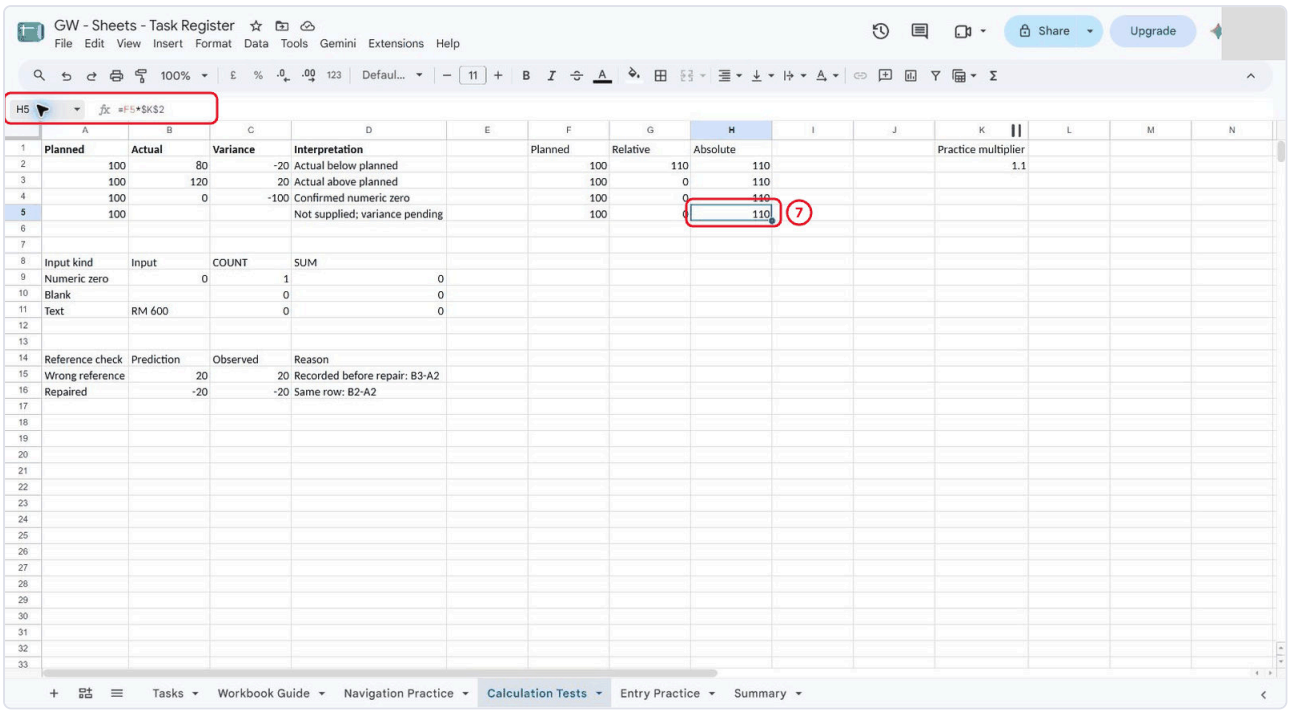


Prepare relative and absolute formulas.

7. Fill **G2:H5** using Ctrl+D. Inspect **G3** `=F3*K3` and **G5** `=F5*K5` : both refer to blank multiplier cells and give **0**. Inspect **H3** `=F3*$K$2` and **H5** `=F5*$K$2` : both keep the intended assumption and give **110**. Retain the relative column as a labelled teaching counterexample.

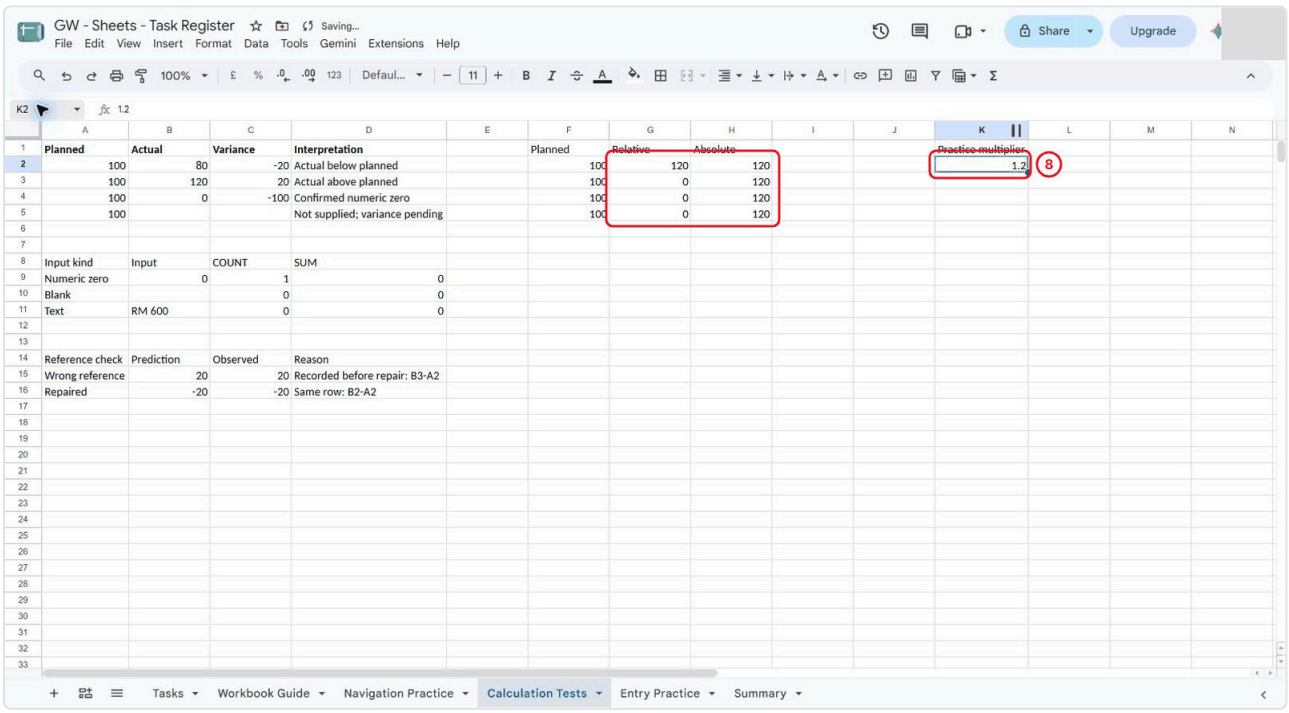


Inspect relative reference drift to a blank cell.

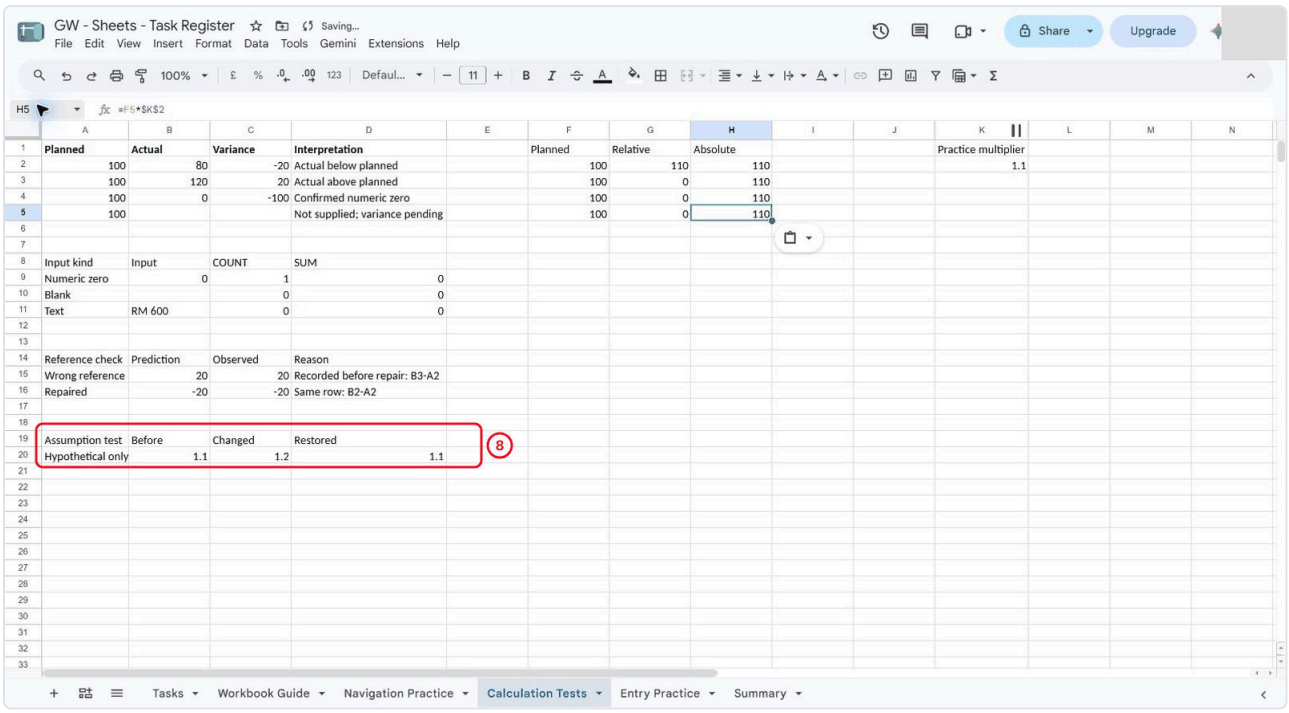


Inspect the retained absolute reference.

8. Change K2 to **1.2**. G2 and all four absolute results become **120**; the later relative results remain **0**. Record that observation, restore K2 to **1.1**, and keep the before/change/restored values in **A19:D20** (labels in Part 4 of the prompts page). This is hypothetical practice only.

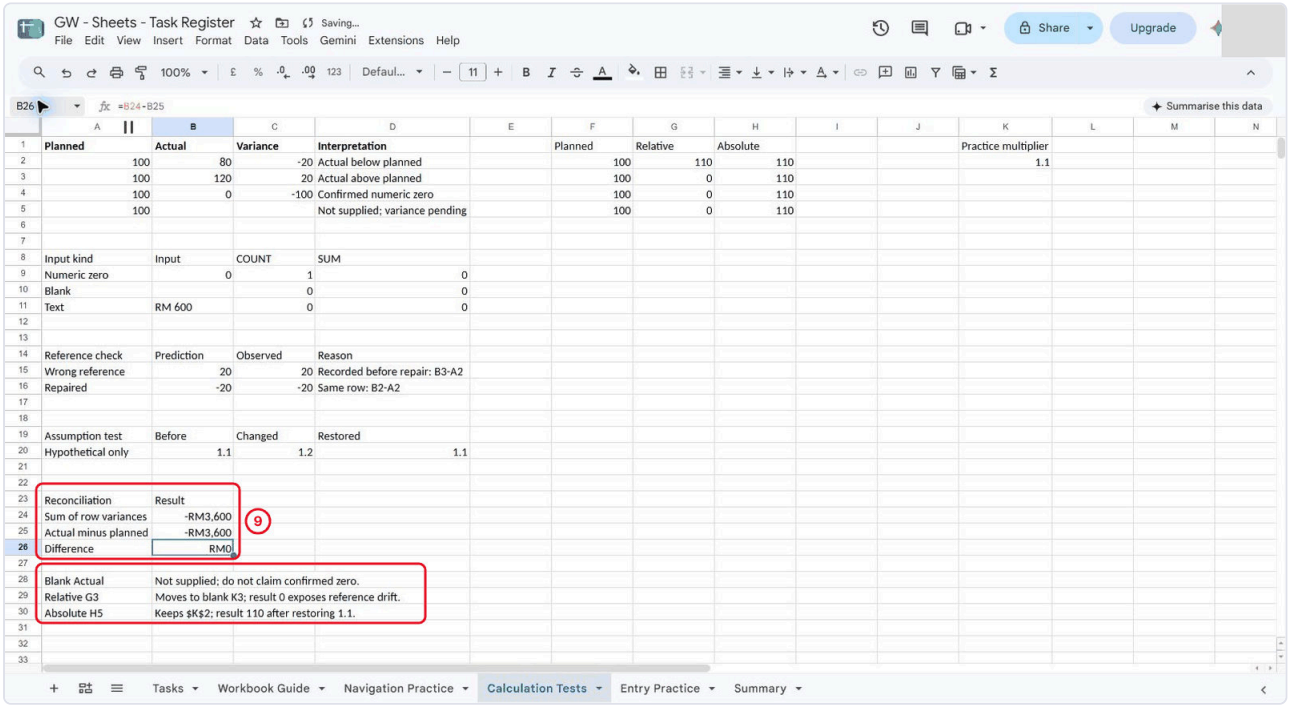


Change the casework multiplier to 1.2.



Restore 1.1 and record the test.

9. In **A23:B26**, reconcile the original source with **=SUM(Tasks!I2:I13)**, **=SUM(Tasks!H2:H13)-SUM(Tasks!G2:G13)** and the difference **=B24-B25**. Both methods give **-RM3,600** and their difference is **RM0**. Use the labels in Part 4 of the prompts page and fit A to show them. Record the blank-information and reference-drift explanations in A28:B30. All original **Tasks!A1:H13** values remain unchanged.



Reconcile both source calculations to zero difference.

> **Validation boundary:** these are author UI reference checks. The independent estimate table, deliberately omitted total range, independent diagnosis and learner timing pilot have not been completed by this reference run.

Independent Practice

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Create an independent three-row estimate table with at least one zero actual cost and one positive variance. Calculate and explain the results without copying a finished answer column.
2. Shorten one total range on the practice table, then use a second calculation to detect and repair the omission.
3. Write a short explanation of negative variance, numeric zero, blank information and a fixed reference, each supported by an actual test cell.

Evidence to Hand Over

- Four-case calculation test with a documented blank-information limitation.
- Reference-error and omitted-range diagnosis with repaired formulas.
- Reconciled baseline totals and independent estimate table.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A plausible total hides a copied-row error	Inspect row references and compare sum of variances against the difference of totals.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You calculated and reconciled the register using visible source values and references.

Check yourself: Which part of a copied formula should move, and which should stay fixed?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Variance
```

```
=H2-G2
```

```
=SUM(G2:G13)
```

```
=SUM(H2:H13)
```

```
=SUM(I2:I13)
```

```
=AVERAGE(G2:G13)
```

```
=COUNT(G2:G13)
```

```
Practice multiplier
```

```
=G2*$L$9
```

Part 2: Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Calculation pairs Planned/Actual: 100/80; 100/120; 100/0; 100/blank. The final blank means
“not supplied” for this test, not a confirmed zero. Use this to distinguish calculation
mechanics from data meaning.
```

Part 3: Captured Reference Ranges

Tasks: I1 Variance; I2 **=H2-G2**, fill I2:I13. Summary block on Tasks K1:L7: L2 **=SUM(G2:G13)** →10400; L3 **=SUM(H2:H13)** →6800; L4 **=SUM(I2:I13)** →-3600; L6 **=AVERAGE(G2:G13)** →866.666… (display RM866.67); L7 **=COUNT(G2:G13)** →12.

Navigation Practice: clear prior Lab 2 examples J2:K6 and reset J2:J13 to Number. K9 Practice multiplier; L9 1.1; J2 `=G2*$L$9`, fill to J13. Change 1.1→1.2→1.1; first/final results 660/770→720/840→660/770.

Calculation Tests: A1:D5 four pairs and interpretation, C2:C4 **=B2-A2** filled down gives -20,20,-100; B5/C5 stay empty. A8:D11 zero/blank/text tests give COUNT 1/0/0 and SUM 0/0/0. C2 deliberately wrong **=B3-A2** gives 20, repaired **=B2-A2** gives-20. A14:D16 retains the observed test history. F2:F5 are 100, G2 `=F2*K2`, H2 `=F2*$K$2`, fill G2:H5. K2 1.1→1.2→1.1; G3/G5 drift to empty K3/K5 while H3/H5 stay anchored. A19:D20 records restoration. B24 **=SUM(Tasks!I2:I13)** and B25 **=SUM(Tasks!H2:H13)-SUM(Tasks!G2:G13)** both give-3600; B26 **=B24-B25** gives 0.

The source twelve records remain unchanged. Independent estimate/omitted-range challenge and classroom timing are pending. Lab 2 values-only target-format preservation remains a separate unresolved check.

Part 4: Calculation Tests Labels

Section 3.4. Type each block where the step says. Columns are separated by tabs, so you can paste a block into its first cell.

D1:D5 (step 2):

```
Interpretation
Actual below planned
Actual above planned
Confirmed numeric zero
Not supplied; variance pending
```

A8:A11 (step 3), with Input / COUNT / SUM as the other headers in B8:D8:

```
Input kind
Numeric zero
Blank
Text
```

A14:D16 (steps 4 and 5). Type the values in columns B and C yourself:

```
Reference check Prediction      Observed        Reason
Wrong reference                 Recorded before repair: B3-A2
Repaired                        Same row: B2-A2
```

A19:D20 (step 8):

```
Assumption test Before  Changed Restored
Hypothetical only
```

A23:A26 (step 9), with `Result` in B23:

```
Reconciliation
Sum of row variances
Actual minus planned
Difference
```

A28:A30 (step 9), then your own explanation for each in column B:

```
Blank Actual
Relative G3
Absolute H5
```

CHAPTER 4

Clean and Organise Records

In this chapter you will find and replace, trim text, sort whole records and freeze the header row.

> **Prompts:** the prompts page has every value and prompt for this chapter, each with a Copy button.

Estimated time: 50 minutes, planned as 10 explanation and demonstration, 20 guided build and casework, 15 independent practice and 5 verification and repair, with 40 active minutes by design. Not yet timed with learners.

Your result: A cleaned text example, whole-record sort and named cost range.

> **Status:** partial author reference with 22 staged application captures through 4.4. Raw text preservation, TRIM, scoped case repair, whole-record sorts, frozen header, named-range reconciliation and immediate Undo of an unsafe sort verified. The reference used manual raw-value entry; CSV upload/import, independent boundary challenge and learner timing remain unverified.

What You Will Learn

- Clean text without altering source evidence.
- Sort whole records.
- Keep headers visible while scrolling.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

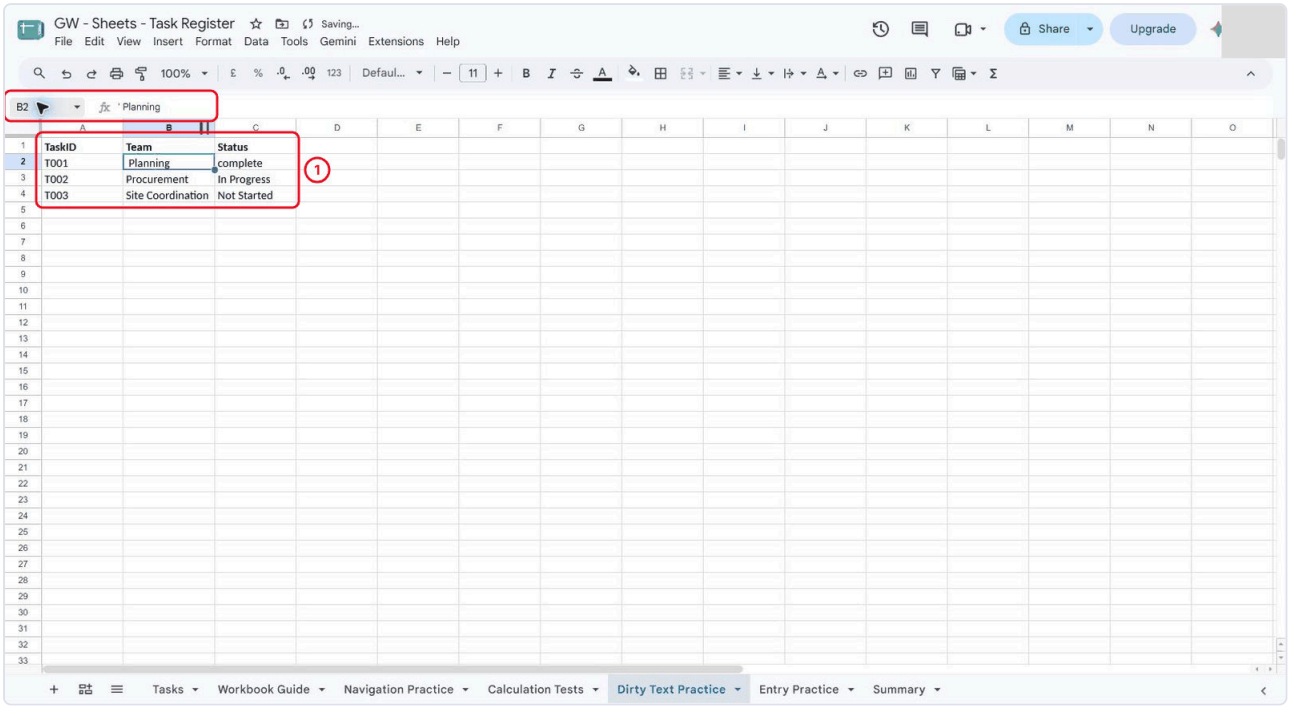
Use Tasks and download DirtyText.csv. Keep cleaning practice separate from the original register.

4.1 Clean a Small Import

1. Click **File > Import**, choose **Upload** and select your downloaded **DirtyText.csv**. Set **Import location** to **Insert new sheet(s)**, leave the separator on **Detect automatically**, untick **Convert text to numbers, dates, and formulas**, then click **Import data**. Rename the new tab **Dirty Text Practice**. It should have the headings **TaskID / Team / Status** in A1:C1. Click B2 and check in the formula bar that the spaces around Planning survived the import. If the upload isn't available, or the spaces were removed, use the manual fallback below.

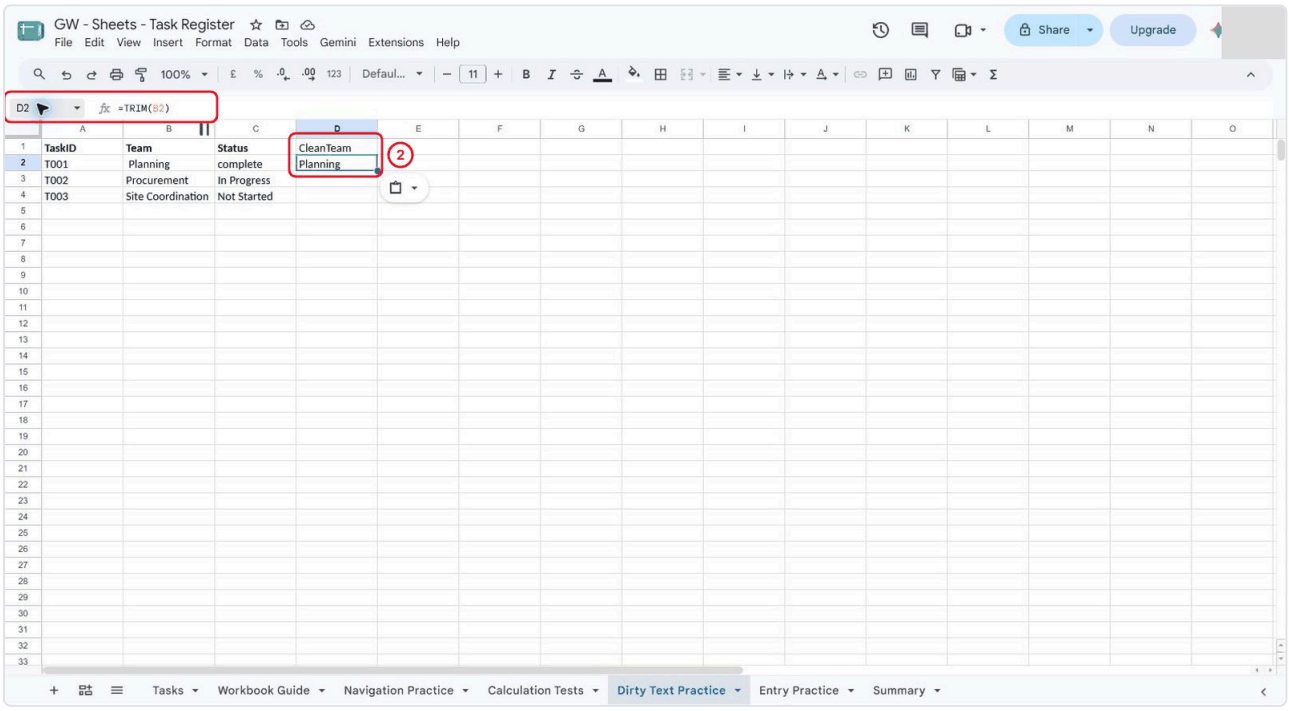
> **Reference run:** The captured reference uses a **Dirty Text Practice** tab and manual entry of the supplied values because browser file upload remained unavailable.

Manual fallback: add a tab named **Dirty Text Practice** and enter the three records from Part 3 of the prompts page into A1:C4. Check the raw spaces: B2 must contain one leading and one trailing space around Planning, and B4 one trailing space after Site Coordination. The controlled paste stripped those spaces, so they were explicitly restored through the cell editor with a leading apostrophe before the literal text. The apostrophe is an input prefix, not part of the stored team. Do not clean the raw cells themselves.



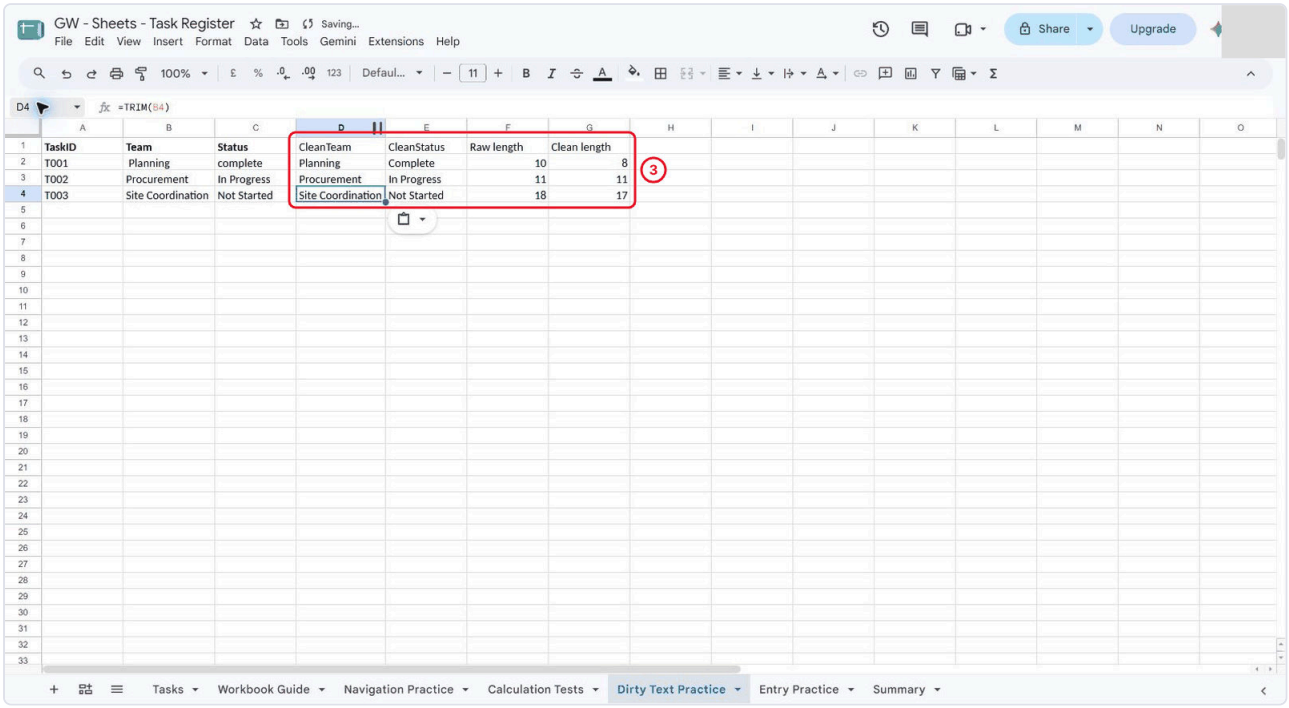
Preserve raw CSV text and its surrounding spaces.

2. Enter **CleanTeam** in D1 and **=TRIM(B2)** in D2. Inspect its formula before filling **D2:D4** with Ctrl+D.



Inspect the first TRIM formula.

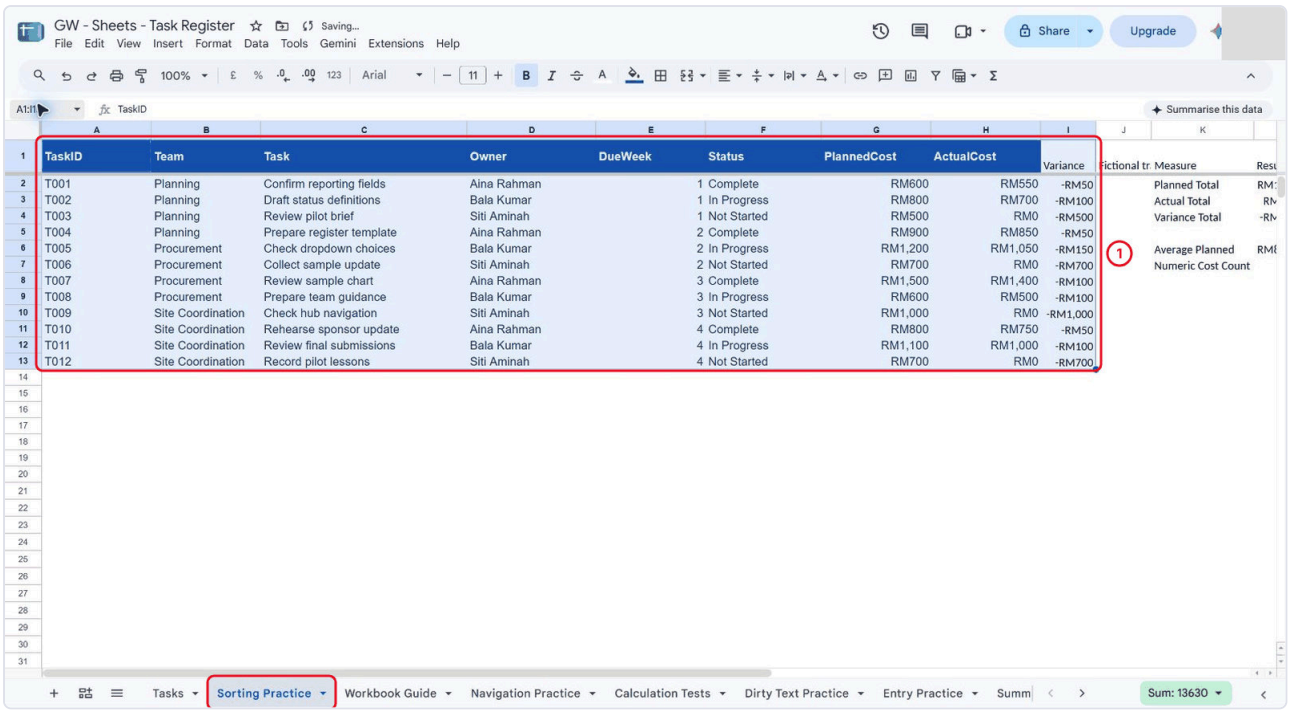
3. Put **CleanStatus** in E1 and the canonical values **Complete / In Progress / Not Started** in E2:E4, retaining the original C column. Fit B and D so complete team names are visible. Add optional **Raw length / Clean length** checks in F:G using **=LEN(B2)** and **=LEN(D2)**, filled through row 4. Planning changes from length **10 to 8** and Site Coordination from **18 to 17**; Procurement remains **11**. These checks make otherwise invisible spaces testable.



Compare cleaned text and raw versus clean lengths.

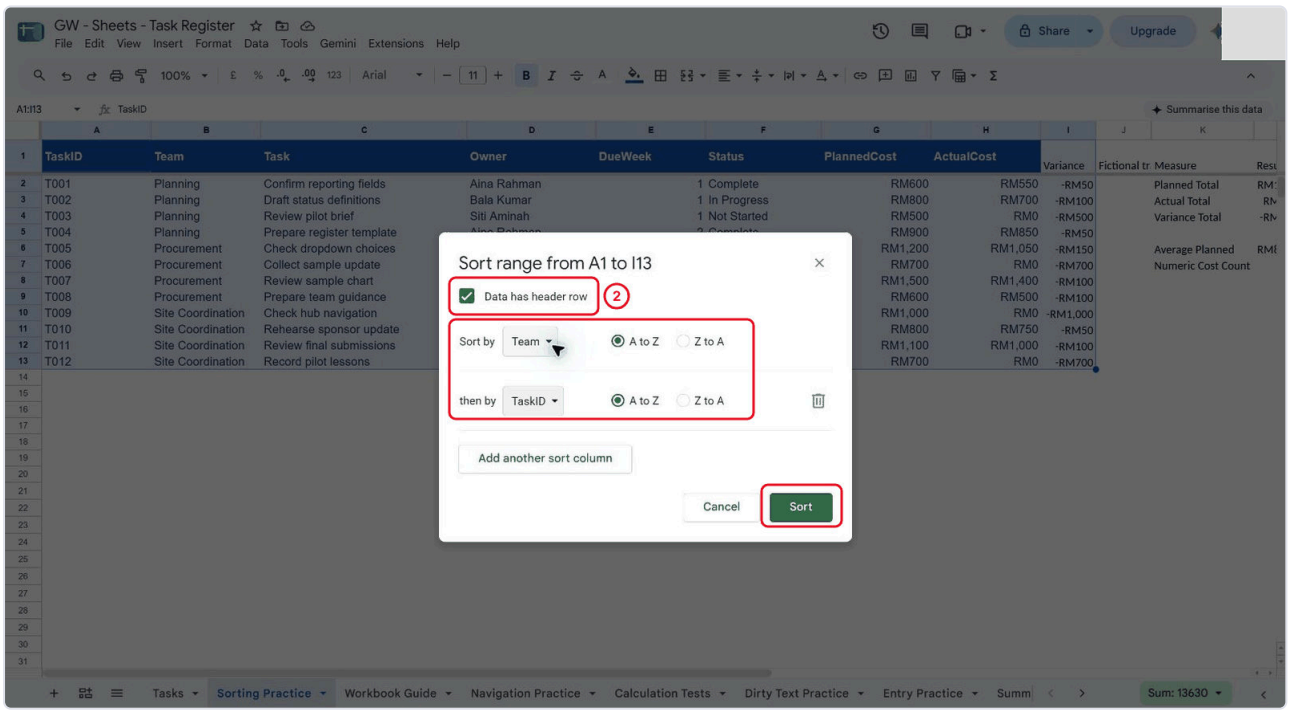
4.2 Sort Whole Records

1. Right-click **Tasks > Duplicate**, then rename the copy **Sorting Practice**. Work on the copy to preserve the original register for later chapters. Select **A1:I13**, including the variance column.



Select all record columns in an isolated sorting copy.

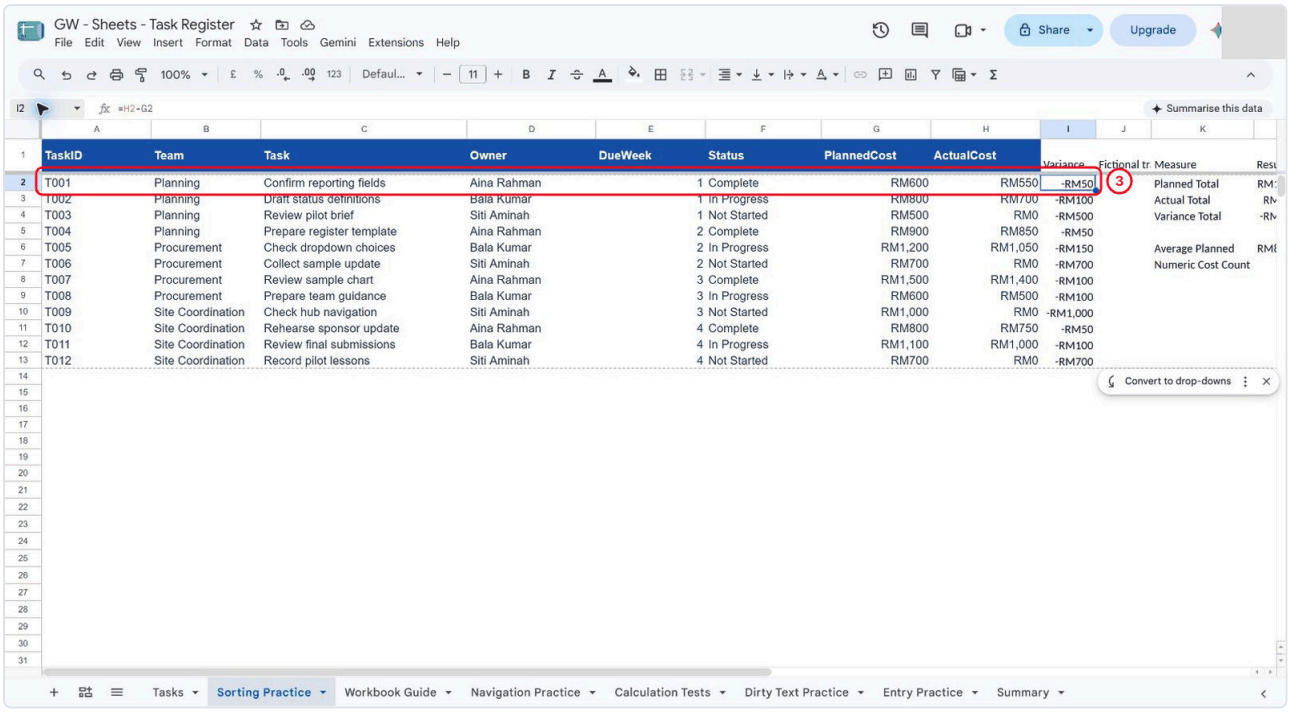
2. Open **Data > Sort range > Advanced range sorting options**. Select **Data has header row**. Choose **Team, A to Z**, add another sort column, and explicitly choose **TaskID, A to Z**; the automatic second choice may be a different column. Inspect the full range and both keys before clicking **Sort**.



Configure Team then TaskID with a header row.

3. Locate **T001** and compare **Aina Rahman**, planned **600**, actual **550**, and **-50 variance** with the source. All twelve record relationships remain intact. This supplied register already follows Team/TaskID order,

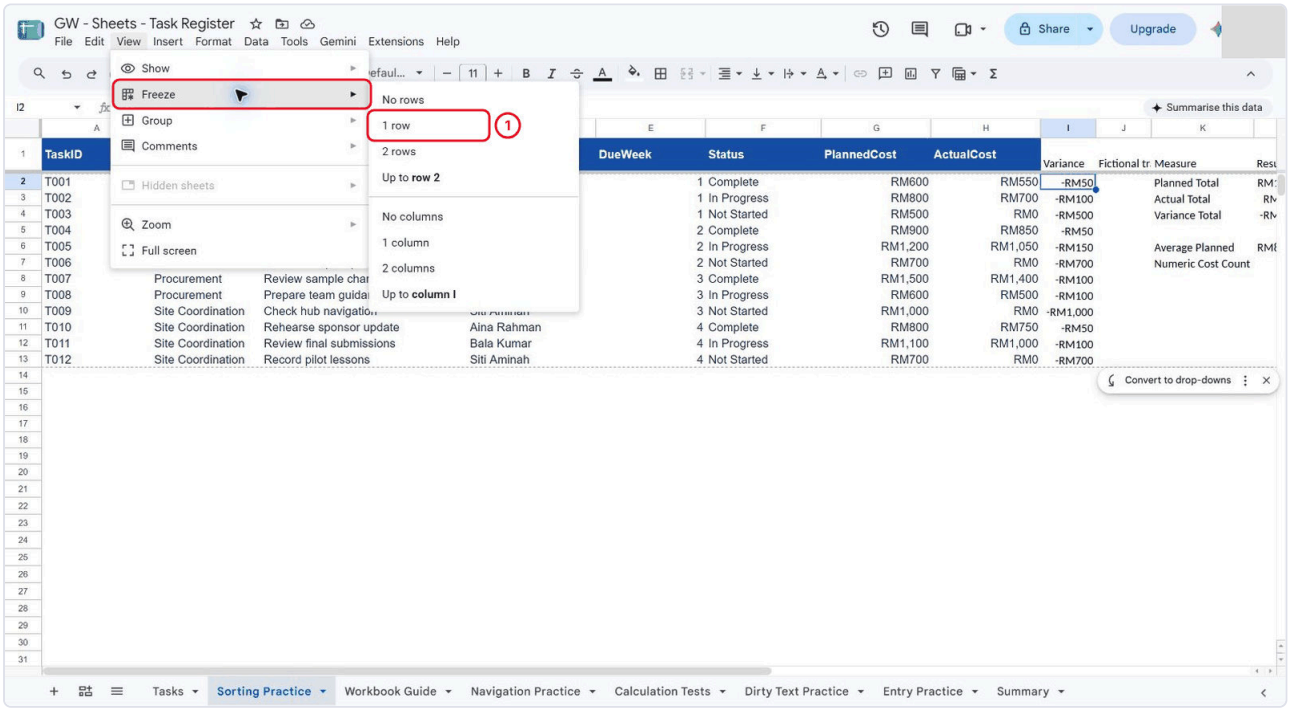
so this ascending sort legitimately leaves its order unchanged. The intake in 4.4 supplies a visible reorder test.



Inspect the intact T001 costs, owner and variance.

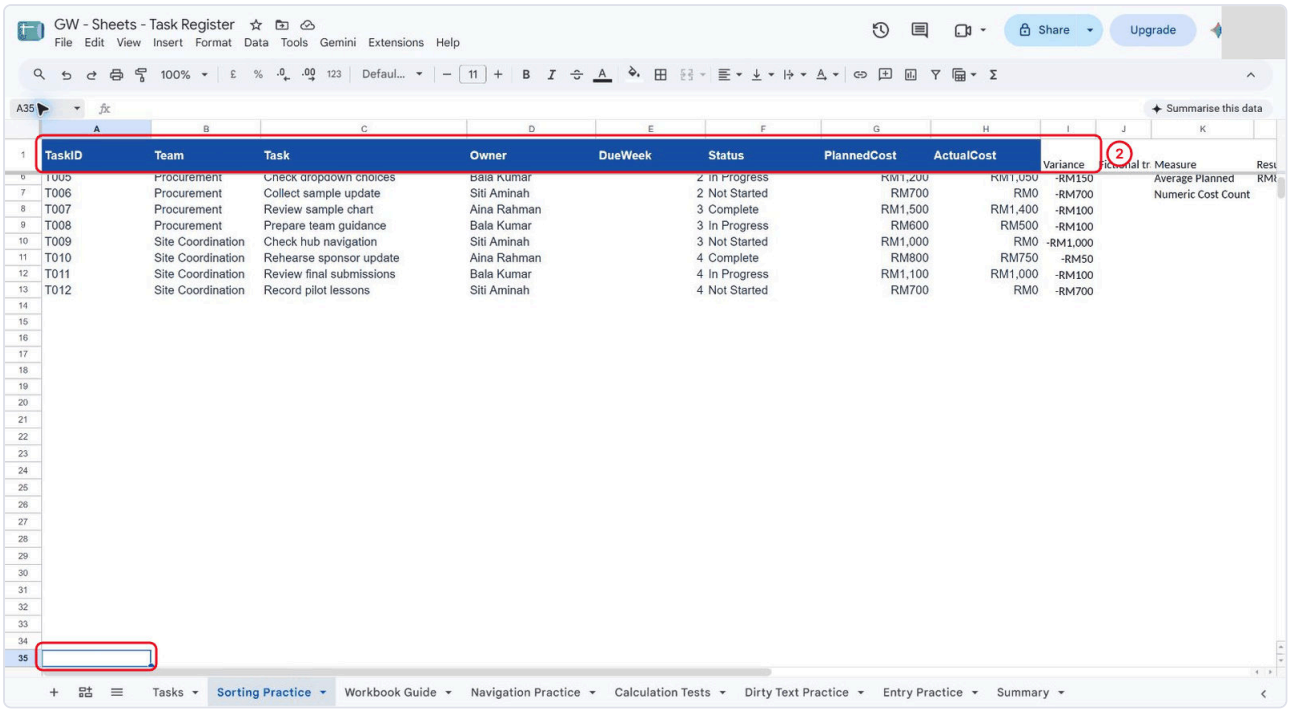
4.3 Freeze and Name a Range

1. On **Sorting Practice**, use **View > Freeze > 1 row**. The copy already inherited a frozen header; retaining or reselecting one row is appropriate. Do not freeze additional rows merely to produce a visible change.



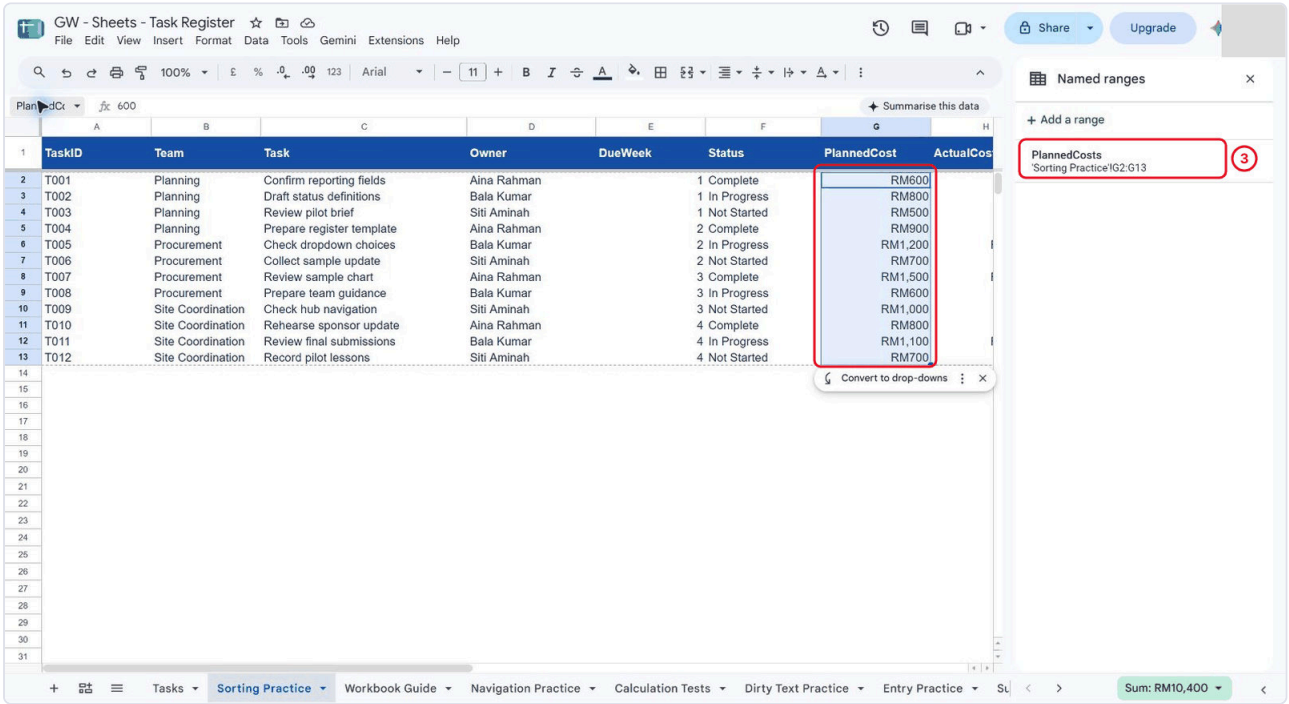
Choose a one-row freeze.

2. Scroll down, or navigate to **A35**, and verify row 1 remains visible above the scrolled rows. Return to the record range afterwards.



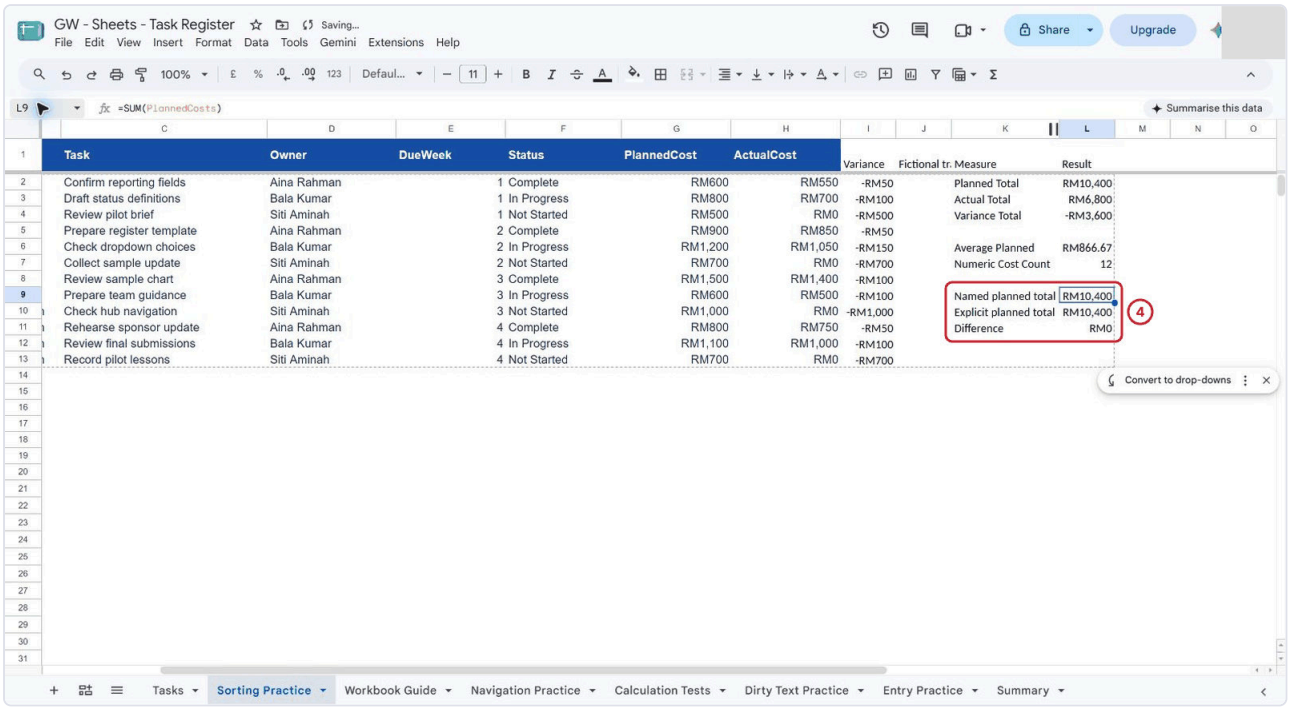
Verify the frozen header after scrolling.

3. Select **G2:G13** on **Sorting Practice**, use **Data > Named ranges**, enter **PlannedCosts** and save with **Done**. Inspect the bounds: **'Sorting Practice'!G2:G13**, twelve numeric cost cells. This name deliberately refers to the practice copy, not a changed baseline source.



Inspect the exact PlannedCosts bounds.

4. In K9:L11 add labels **Named planned total / Explicit planned total / Difference**, with **=SUM(PlannedCosts)**, **=SUM(G2:G13)** and **=L9-L10**. Fit K to show the complete labels and scroll horizontally when needed. Both totals are **RM10,400**; the difference is **RM0**.

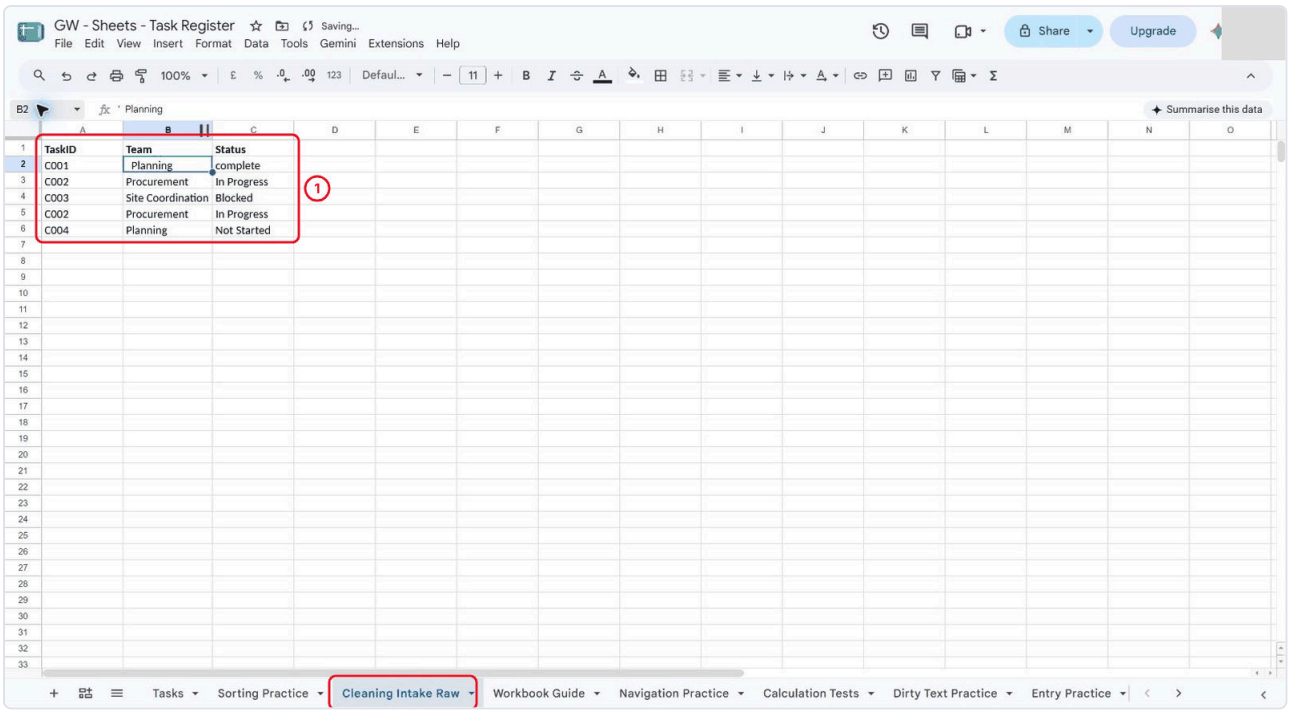


Compare named and explicit totals.

4.4 Clean an Intake and Prove That Records Stay Together

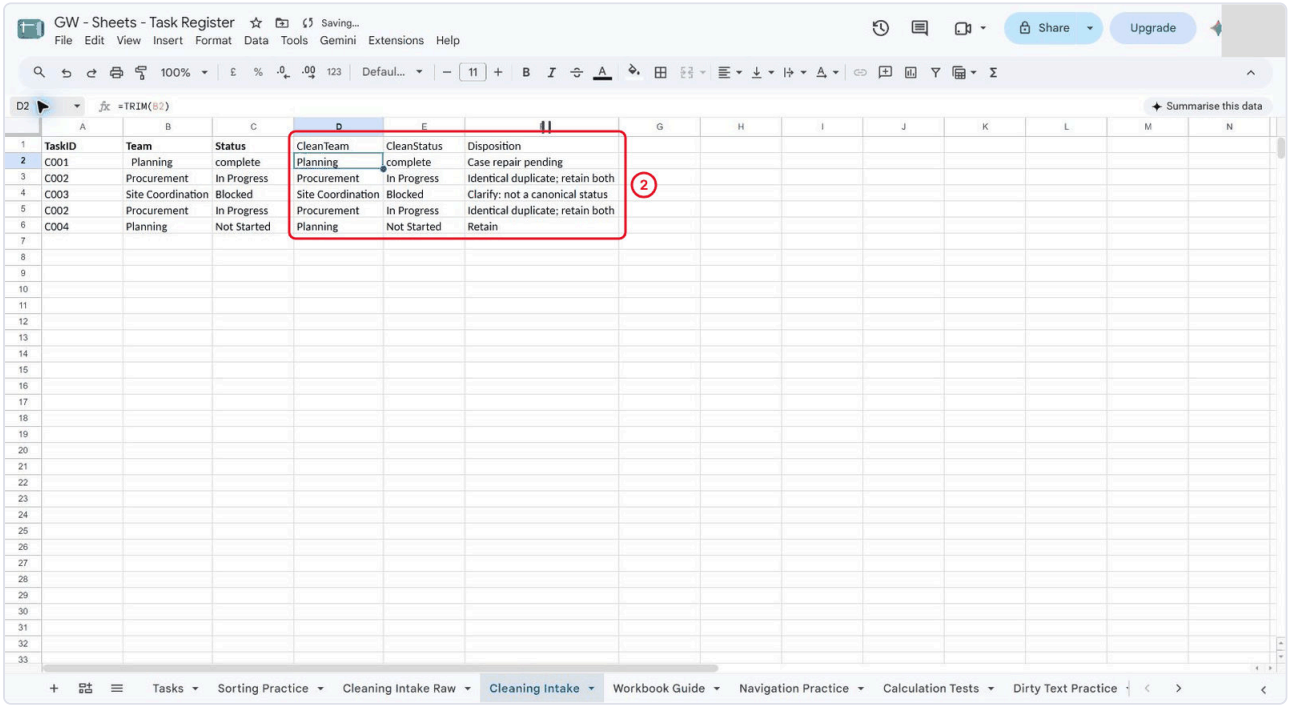
1. Create **Cleaning Intake Raw**, type the headers **TaskID / Team / Status** in A1:C1 and enter the five records from the case input (Part 2 of the prompts page) in A2:C6, and preserve the surrounding spaces in C001's team. Inspect the raw cell value, not only its appearance. Duplicate this raw tab and rename the working copy **Cleaning Intake**. Keep the raw tab unchanged.

> **Reference run:** The reference uses two leading and two trailing spaces around Planning.



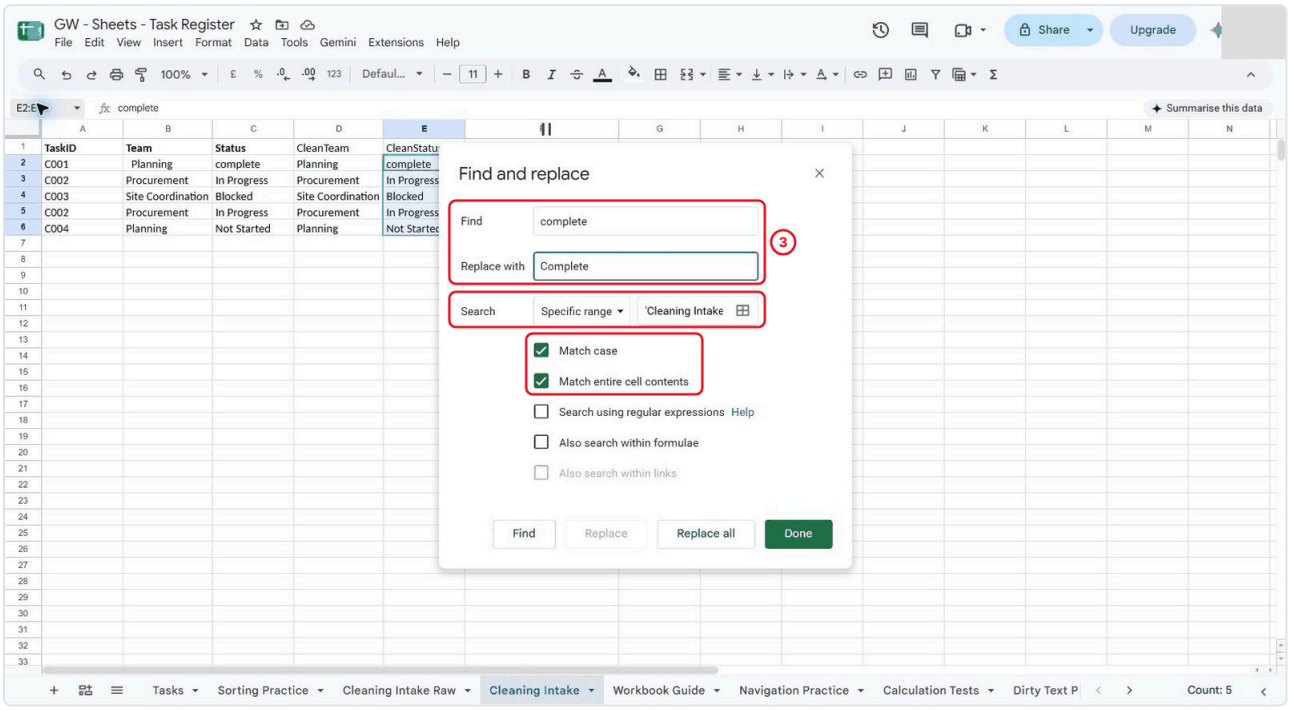
Preserve the five-row raw intake.

2. Add **CleanTeam / CleanStatus / Disposition** in D:F. Use **=TRIM(B2)** through D6. Initially copy the status values into E, so C001's cleaned status is still lowercase **complete** before the controlled repair. Compare both C002 records across all three supplied fields: they are identical. Keep both and label their disposition **Identical duplicate; retain both**. Retain **Blocked** on C003 and flag it for clarification; no canonical mapping is supplied. No owner or cost fields are invented for this intake.



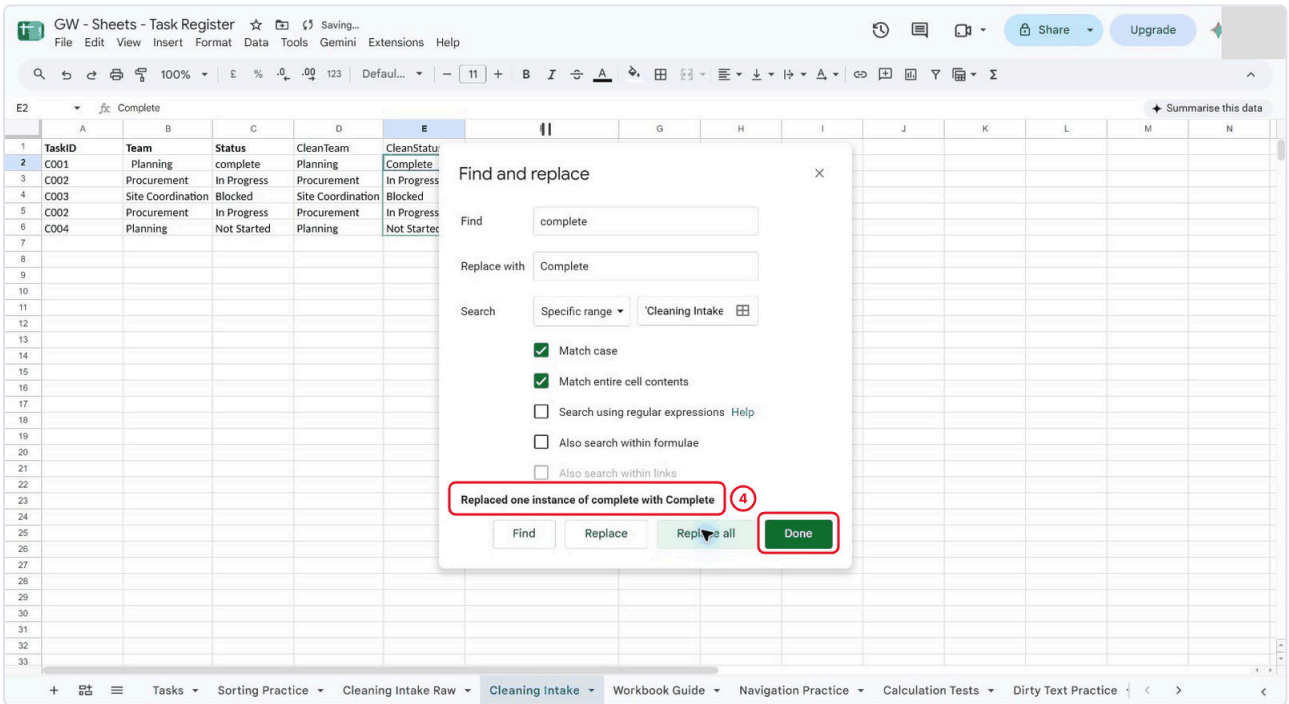
Retain duplicates and flag the ambiguous status.

3. Select **E2:E6** and open **Find and replace** with Ctrl+H. Find **complete**, replace with **Complete**, and confirm **Specific range 'Cleaning Intake'!E2:E6**. Enable **Match case** and **Match entire cell contents**; leave regular expressions and formula search off. This is the known case repair supplied by the exercise, rather than an invented spelling error.



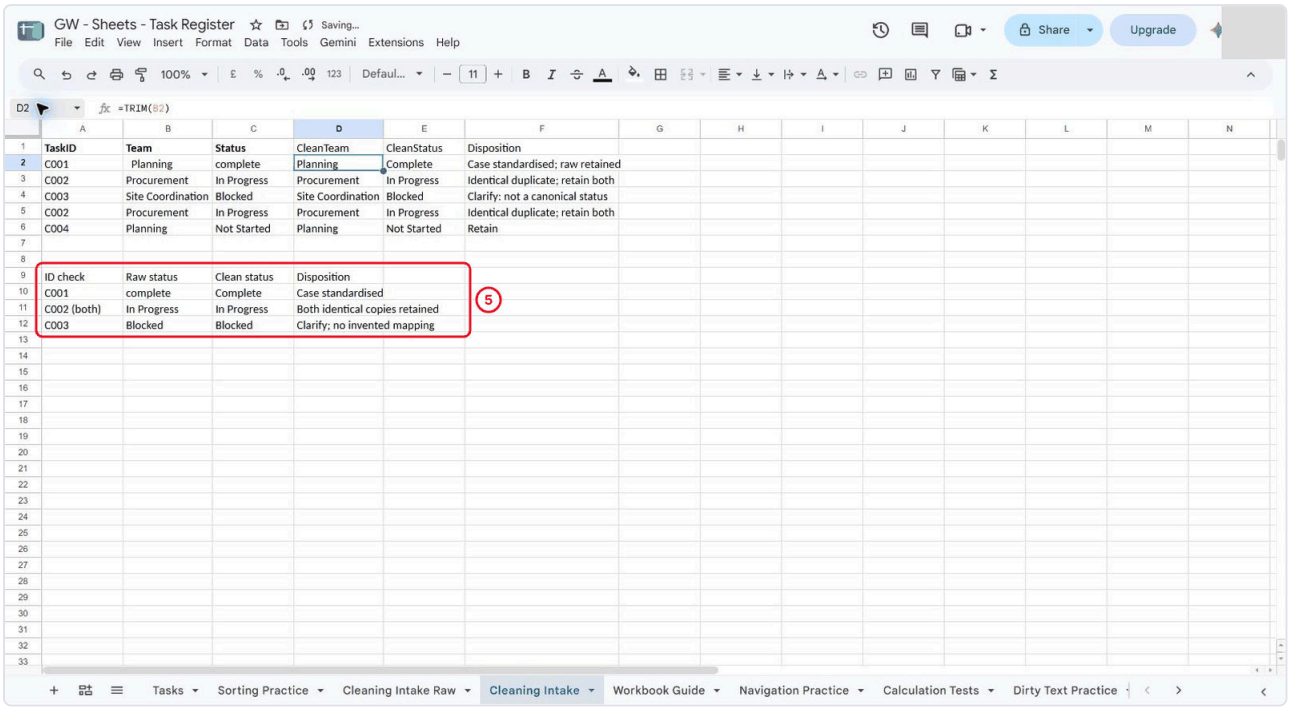
Restrict case repair to cleaned statuses.

4. Preview with **Find**, then use **Replace all**. Verify **Replaced one instance of complete with Complete**, then **Done**. The raw C2 and raw tab still contain lowercase complete. Update C001's disposition to **Case standardised; raw retained**.



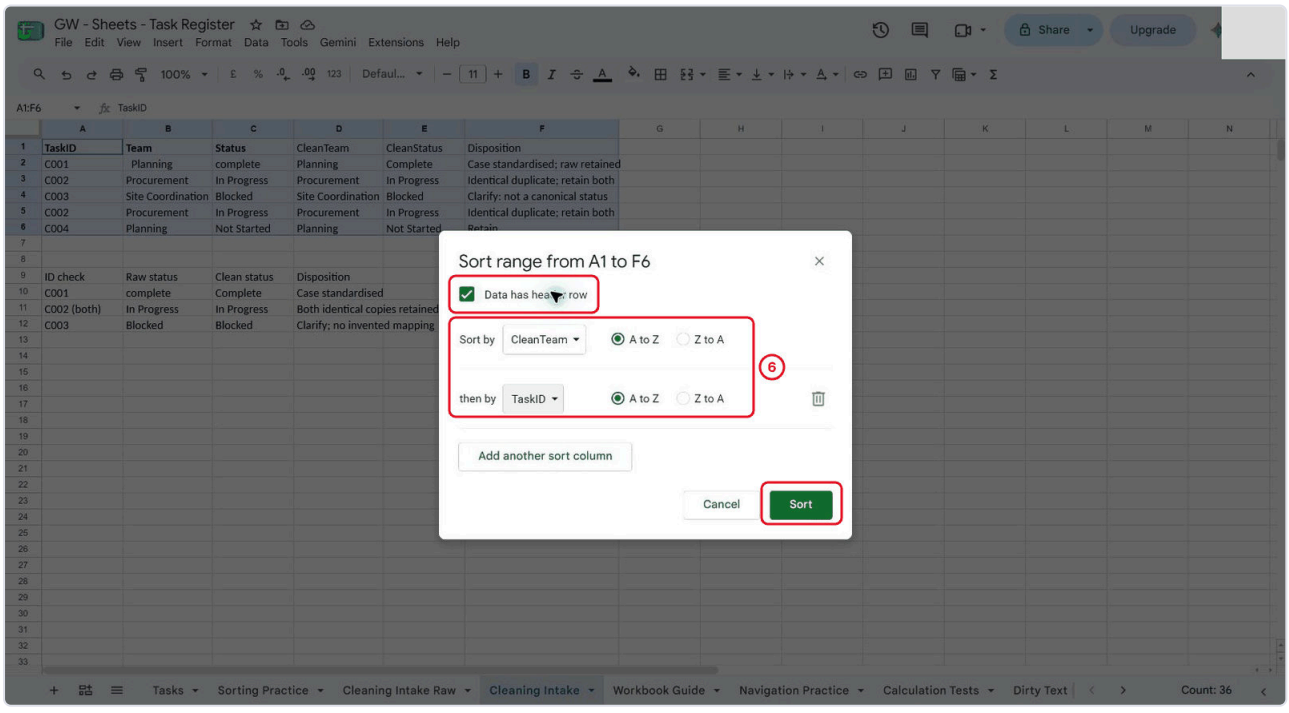
Verify exactly one replacement.

5. Record three TaskID-to-status checks before sorting in A9:D12, using the labels in Part 5 of the prompts page: C001 complete→Complete, both C002 In Progress→In Progress, and C003 Blocked→Blocked with clarification pending. The supplied intake has no owners, so status relationships provide the identity check.



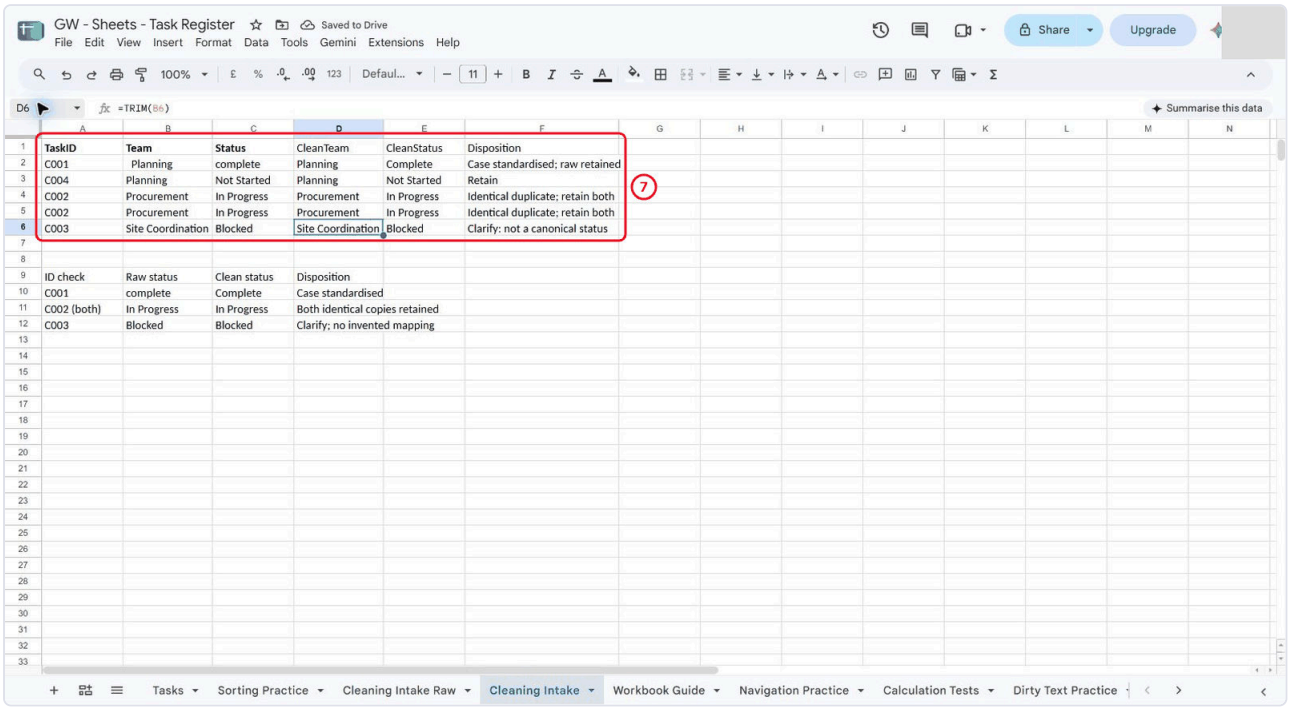
Record identity checks before sorting.

6. Select **A1:F6**, including raw fields, cleaned fields and dispositions. Open advanced sorting, mark the header row, and sort **CleanTeam A to Z**, then **TaskID A to Z**. Using the cleaned team avoids making whitespace part of the sort key. Keep the check table below row 6 outside the sort.



Sort the complete intake by CleanTeam and TaskID.

7. Verify the sorted ID order **C001, C004, C002, C002, C003**. Inspect the copied TRIM formulas: each still points to its own row's raw team. Compare the three identity checks again and record them in A14:D17 (labels in Part 5 of the prompts page). Both identical C002 rows remain; C003's unresolved status remains attached to C003.

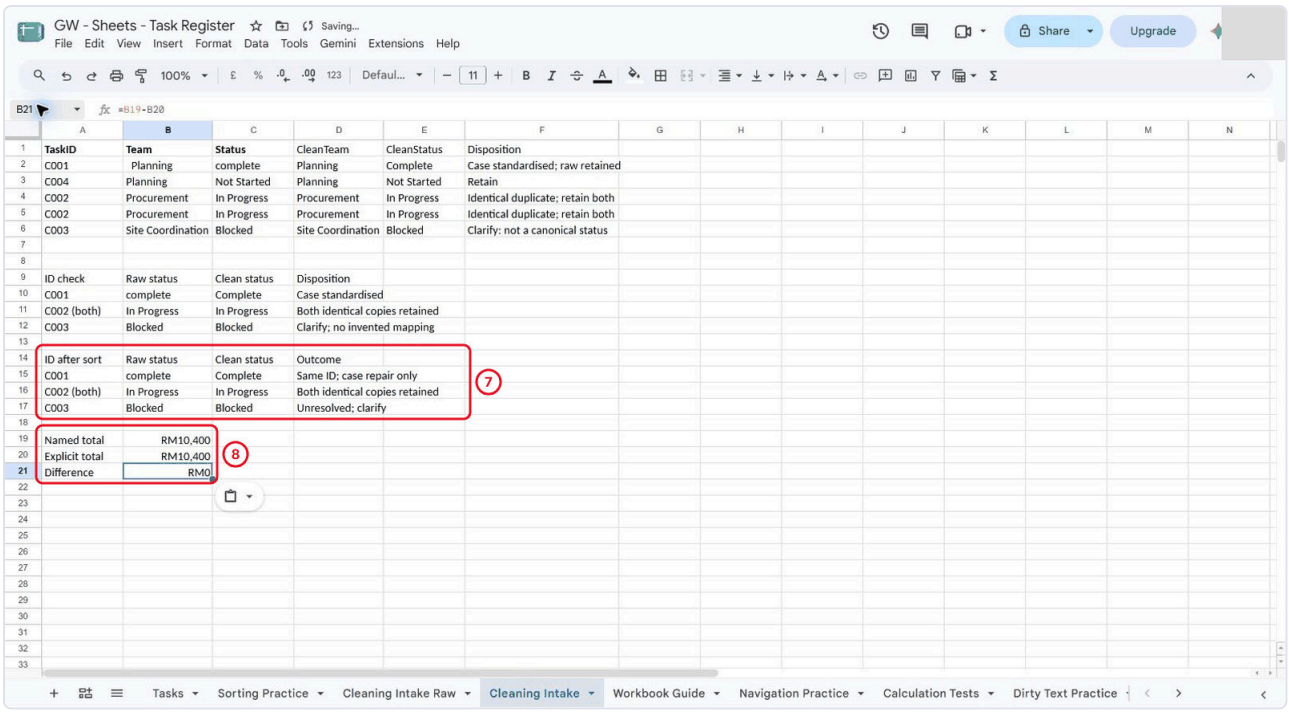


TaskID	Team	Status	CleanTeam	CleanStatus	Disposition
C001	Planning	complete	Planning	Complete	Case standardised; raw retained
C004	Planning	Not Started	Planning	Not Started	Retain
C002	Procurement	In Progress	Procurement	In Progress	Identical duplicate; retain both
C002	Procurement	In Progress	Procurement	In Progress	Identical duplicate; retain both
C003	Site Coordination	Blocked	Site Coordination	Blocked	Clarify: not a canonical status

ID check	Raw status	Clean status	Disposition
C001	complete	Complete	Case standardised
C002 (both)	In Progress	In Progress	Both identical copies retained
C003	Blocked	Blocked	Clarify; no invented mapping

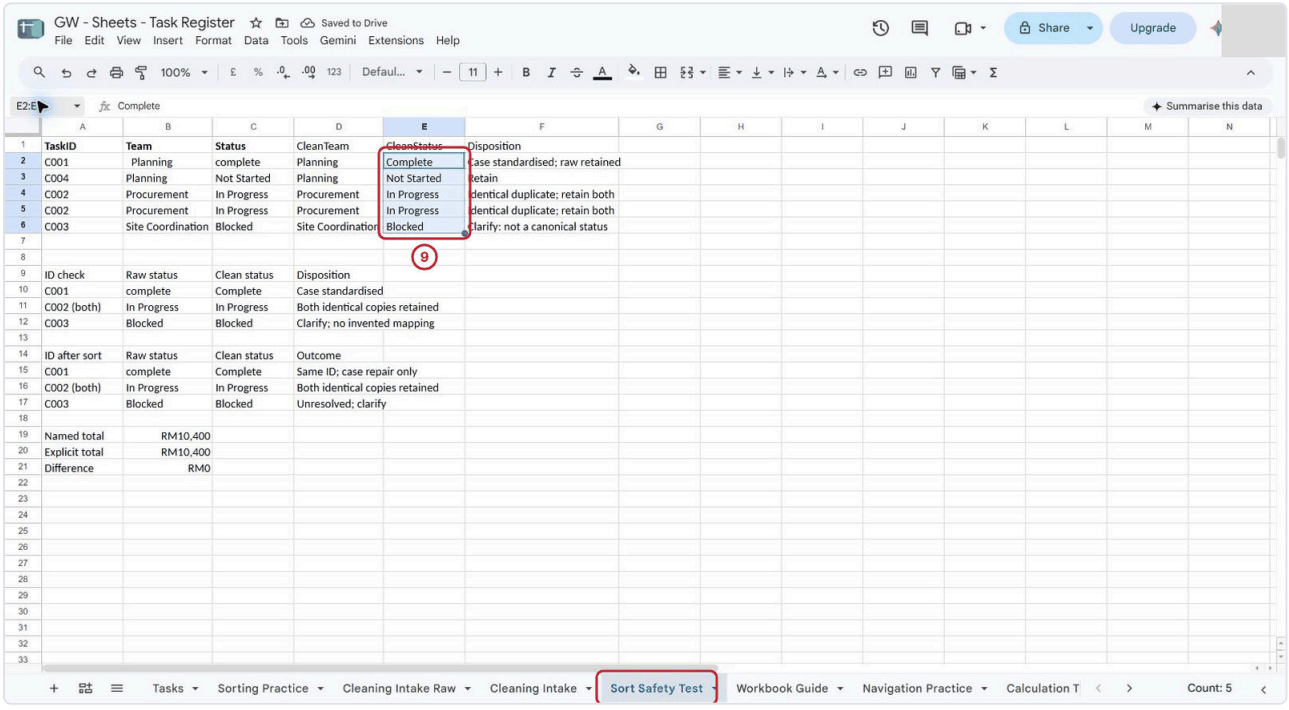
Inspect identities after the whole-record sort.

8. Reuse **PlannedCosts**, since this intake has no numeric cost column. In A19:B21 (labels in Part 5 of the prompts page) compare **=SUM(PlannedCosts)** with **=SUM('Sorting Practice'!G2:G13)**, then subtract the two results. Both give **RM10,400** and the difference is **RM0**. The exact twelve-cell bounds were inspected in 4.3. This does not test automatic expansion when new rows are added.



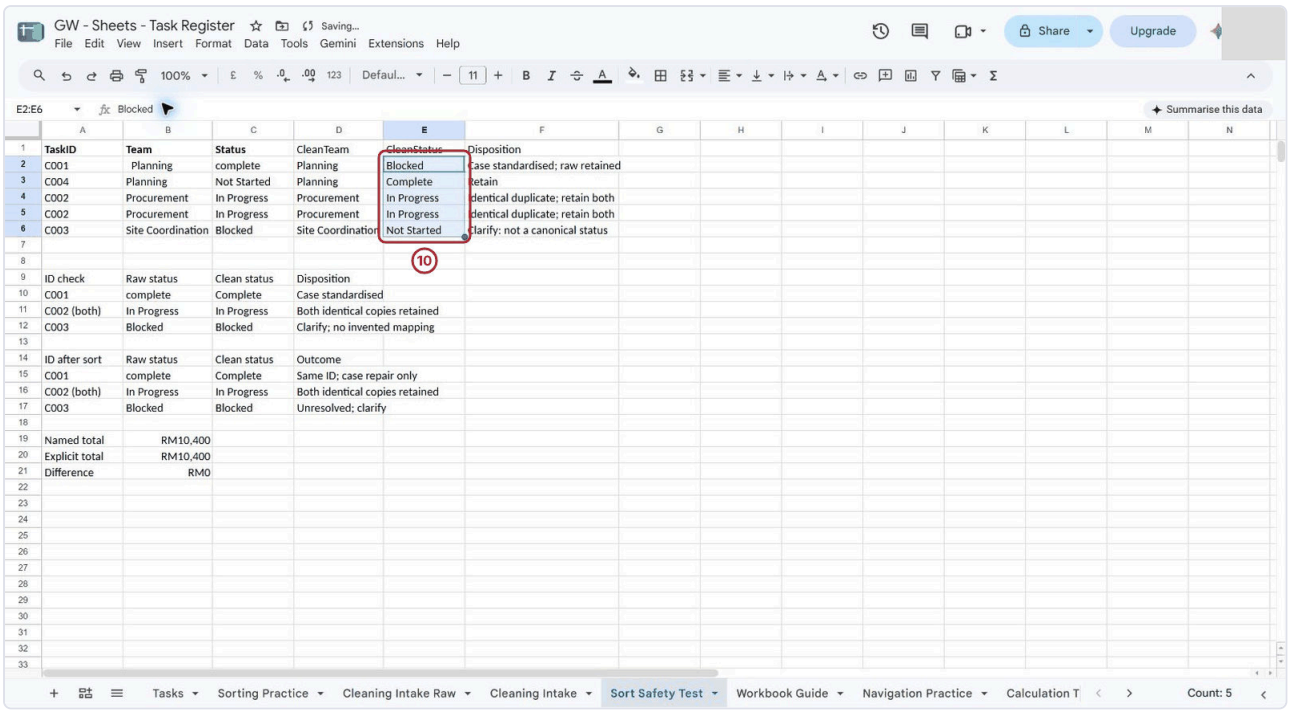
Compare identity checks and named-range reconciliation.

9. Duplicate the cleaned tab as **Sort Safety Test**. Select **only E2:E6** on this disposable copy. Keep the raw and clean intake tabs out of the test.



Select only one column in a disposable copy.

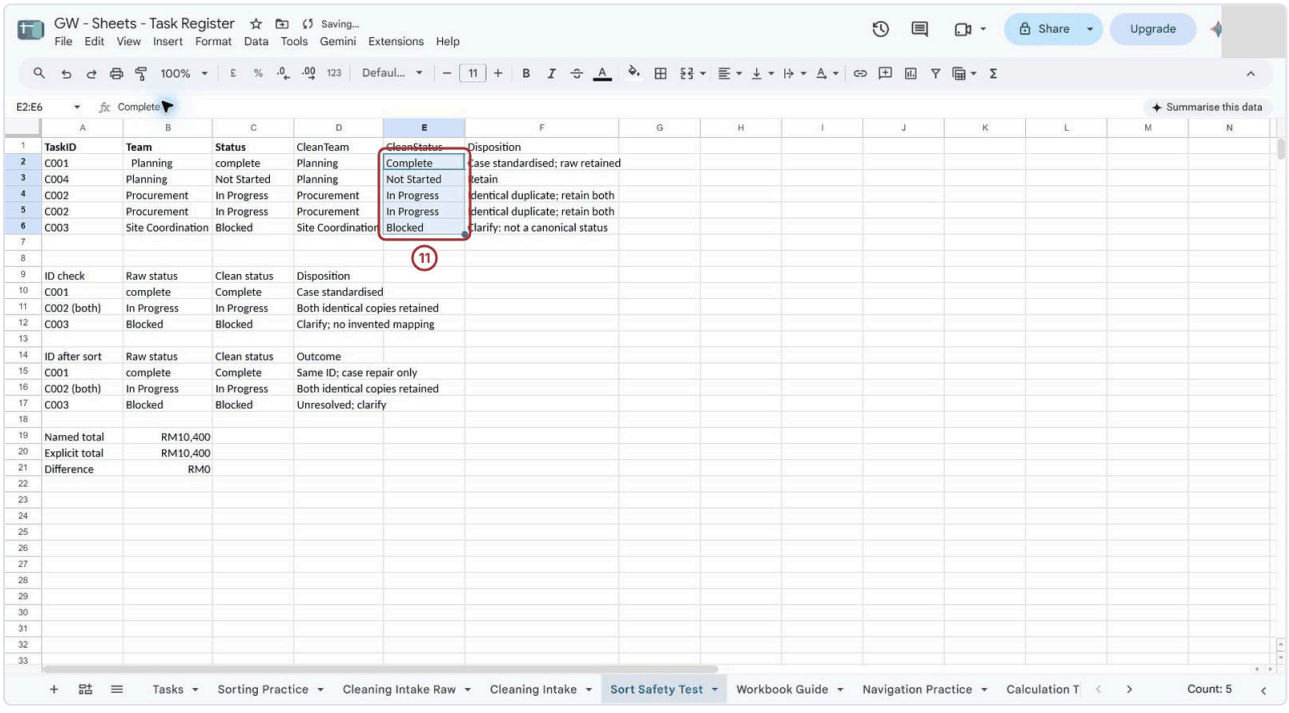
10. Use **Data > Sort range > Sort range by column E (A to Z)** on that isolated selection. Inspect the broken relationship: C001 now incorrectly has **Blocked**, C004 has **Complete**, and C003 has **Not Started**. The two C002 statuses happen to remain In Progress; checking only those would miss the defect.



Observe broken relationships from the isolated-column sort.

11. **Undo immediately** with Ctrl+Z, before making any other edit. Compare all six columns and the three ID checks with Cleaning Intake.

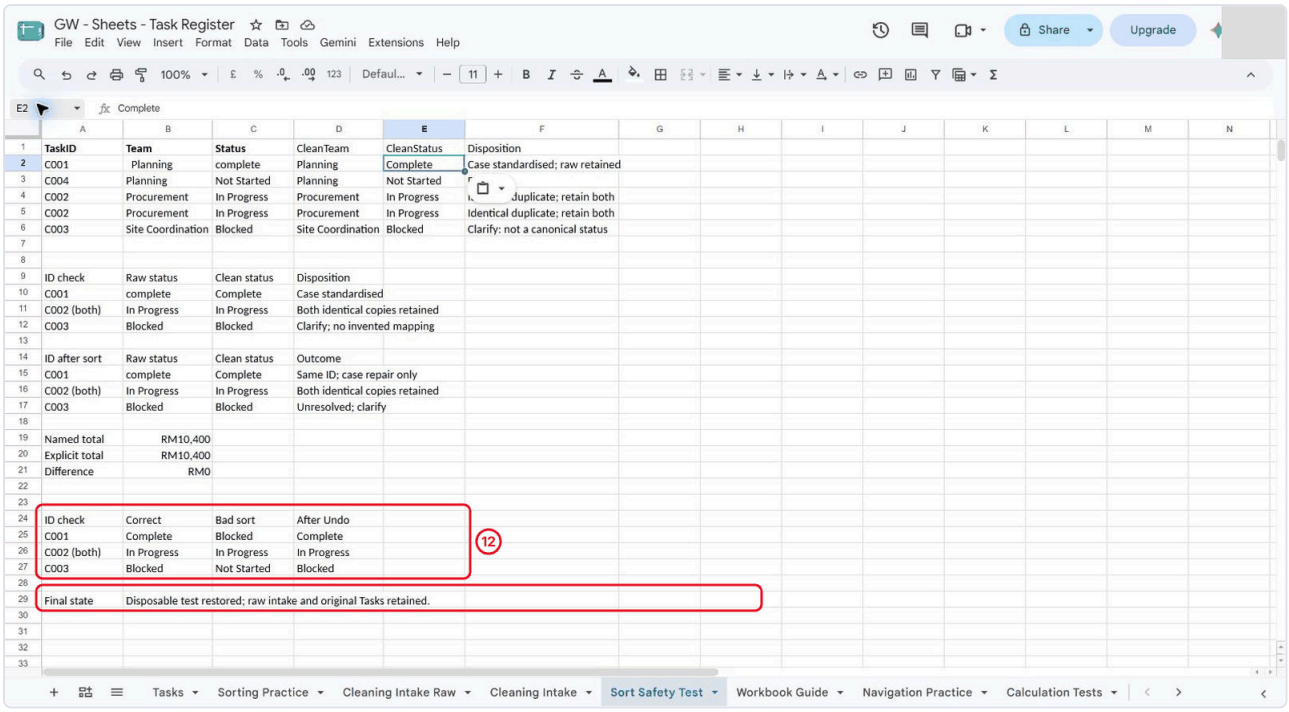
> **Reference run:** The reference verified an exact A1:F6 match after Undo.



Undo the unsafe sort immediately.

12. Record the observed correct / bad-sort / after-Undo statuses in A24:D27, and a one-line final state in A29 (labels in Part 5 of the prompts page). C001 is restored to Complete, both C002 to In Progress,

and C003 to Blocked. Keep the restored disposable tab as labelled evidence. The original Tasks values/formulas and raw intake remain unchanged.



Record restored identities and final state.

> **Validation boundary:** the reference verifies the guided and casework operations above. CSV file upload/import, the independent named-range expansion test, an independent cleaner's result and classroom timing remain pending.

Independent Practice

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Produce a clean intake and an exception list independently. Keep unresolved category and conflicting-duplicate decisions visible.
2. Add one new numeric row only to a named-range practice copy and inspect whether the named range includes it. Repair its bounds and compare the two total methods.
3. Hand over the raw copy, clean result and three row-identity checks so another person can verify the cleaning without trusting your description.

Evidence to Hand Over

- Preserved raw intake, justified clean values and explicit exceptions.
- Whole-record sorting evidence and a restored isolated-column demonstration.
- Named-range boundary test and independently reconciled total.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A clean-looking row is associated with the wrong task	Compare complete TaskID relationships before and after the sort, not only totals.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Cleaning, whole-record sorts and visible headers make the data safer to analyse.

Check yourself: What must stay together when sorting a register?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
CleanTeam
```

```
=TRIM(B2)
```

```
PlannedCosts
```

```
=SUM(PlannedCosts)
```

Part 2: Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Cleaning intake (TaskID | Team | Status): C001 |  Planning  | complete; C002 | Procurement |
In Progress; C003 | Site Coordination | Blocked; C002 | Procurement | In Progress; C004 |
Planning | Not Started. C001 has surrounding team spaces. The duplicate is identical; Blocked
remains unresolved. These are separate practice records.
```

Part 3: Exact DirtyText.csv Records

For manual entry, treat each CSV field below as a separate cell and preserve the spaces inside its quotation marks. Do not paste the entire comma-separated block into one cell.

```
"TaskID","Team","Status"
"T001"," Planning ","complete"
"T002","Procurement","In Progress"
"T003","Site Coordination ","Not Started"
```

Part 4: Manual Raw-Entry Fallback

CSV import remains unverified in the captured browser session. If entry trims the surrounding spaces, use a leading apostrophe in the cell editor to preserve the literal raw text; do not include that prefix in the intended stored value. DirtyText.csv raw team values are `Planning`, `Procurement`, and `Site Coordination`, with lengths 10,11,18. TRIM produces Planning/Procurement/Site Coordination, lengths 8,11,17. Keep raw A:C and cleaned D:E separate.

Cleaning Intake Raw A1:C6 contains the five supplied C records. The reference explicitly preserves two spaces before and after C001's Planning. Duplicate it as Cleaning Intake. D2:D6 uses TRIM of B on each row; E initially copies raw status. Restrict Find and replace complete→Complete to E2:E6, matching case

and entire cell; one replacement. Both C002 records are identical and retained. C003 Blocked is unresolved, with no invented mapping.

Sort A1:F6 by CleanTeam then TaskID, with the header option: C001,C004,C002,C002,C003. Inspect C001/C002/C003 status relationships before and after. In a disposable Sort Safety Test copy, sorting only E2:E6 A→Z makes C001 Blocked and C003 Not Started; Undo restores Complete and Blocked respectively. C002 remains In Progress through this test.

PlannedCosts bounds: **'Sorting Practice'!G2:G13**, twelve numeric cells. SUM(PlannedCosts)=10400, explicit SUM of that range=10400, difference 0. No new-row expansion test is claimed. Tasks A1:L13 values/formulas remain unchanged. Independent challenge and learner timing remain pending.

Part 5: Cleaning Intake Check Tables

Section 4.4. Columns are separated by tabs, so you can paste a block into its first cell, then fill in what you observe.

A9:D12 (step 5):

```
ID check        Raw status      Clean status    Disposition
C001
C002 (both)
C003
```

A14:D17 (step 7):

```
ID after sort   Raw status      Clean status    Outcome
C001
C002 (both)
C003
```

A19:A21 (step 8), with the two SUM formulas and their difference in B19:B21:

```
Named total
Explicit total
Difference
```

A24:D27 and A29 (step 12, on Sort Safety Test):

```
ID check        Correct Bad sort        After Undo
C001
C002 (both)
C003
```

```
Final state
```

CHAPTER 5

Validate and Filter the Register

In this chapter you will add a status dropdown, identify invalid entries and create a filter view.

> **Prompts:** the prompts page has every value and prompt for this chapter, each with a Copy button.

Estimated time: 50 minutes, planned as 10 explanation and demonstration, 20 guided build and casework, 15 independent practice and 5 verification and repair, with 40 active minutes by design. Not yet timed with learners.

Your result: A status dropdown and two named filter views.

> **Status:** author reference with 34 staged application captures through 5.4. Strict status validation, actual rejection tests, whole-week boundaries/fractions, clearing limitations, two saved views, visible TaskID checks, ordinary SUM under filtering and restoration verified. Independent coverage/range repair and learner timing remain pending.

What You Will Learn

- Restrict status values.
- Create and name a filter view.
- Check hidden rows before interpreting totals.

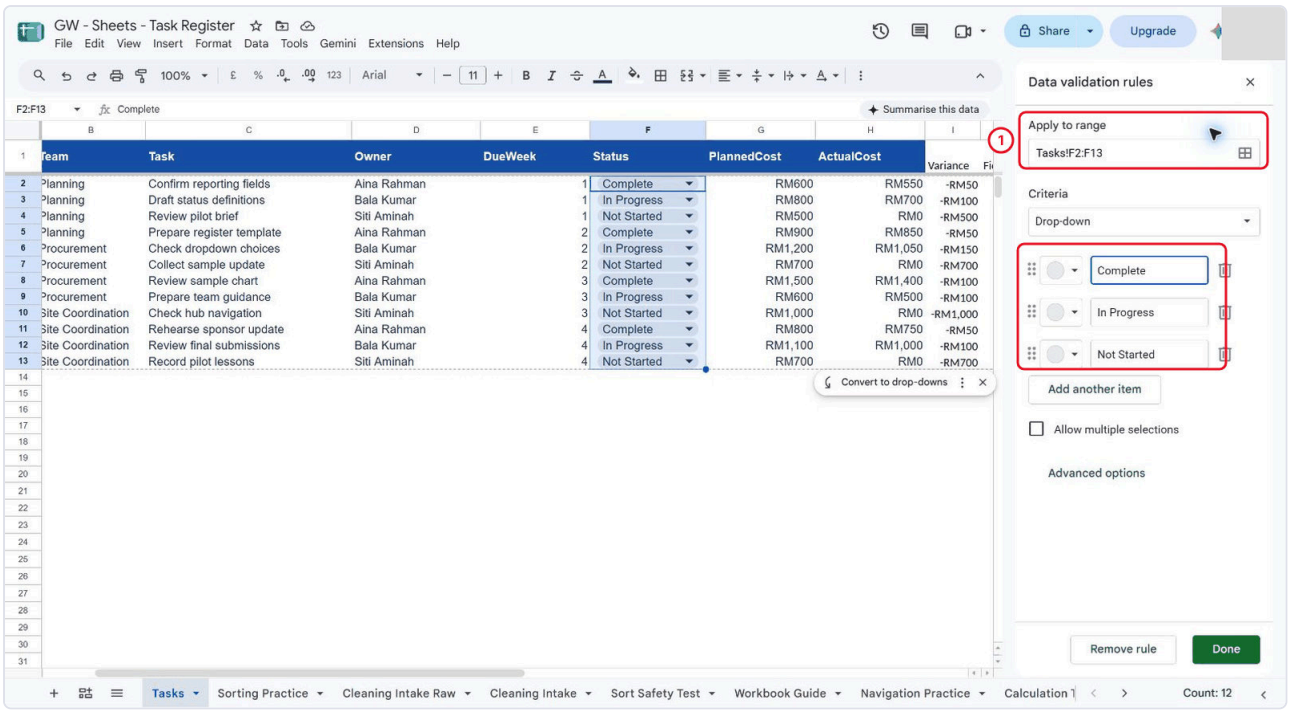
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the original Tasks table, with all 12 baseline records still present.

5.1 Add a Status Dropdown

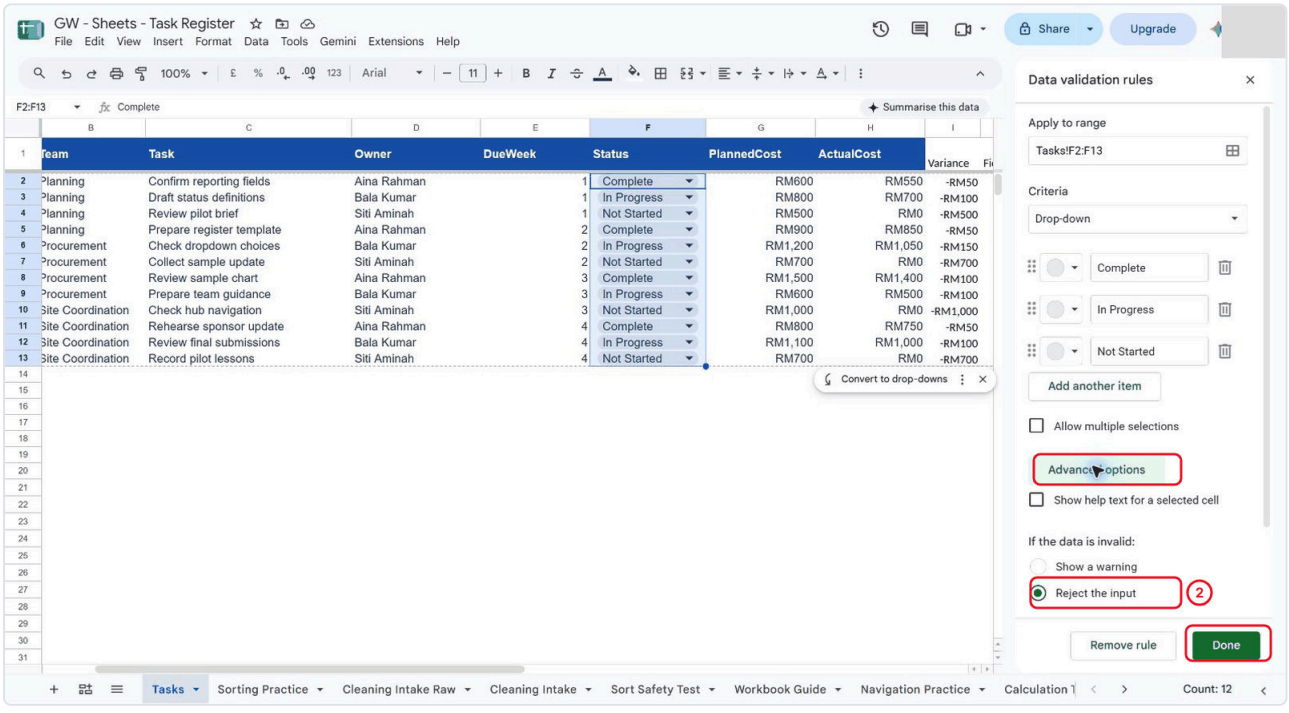
1. On **Tasks**, select **F2:F13** and open **Data > Data validation > Add rule**. Confirm **Apply to range Tasks!F2:F13**. In the current interface the default Drop-down criterion populated its choices from the selected cells; inspect rather than assuming those inferred choices are correct. Use exactly **Complete / In Progress / Not Started**.



Inspect status rule coverage and canonical options.

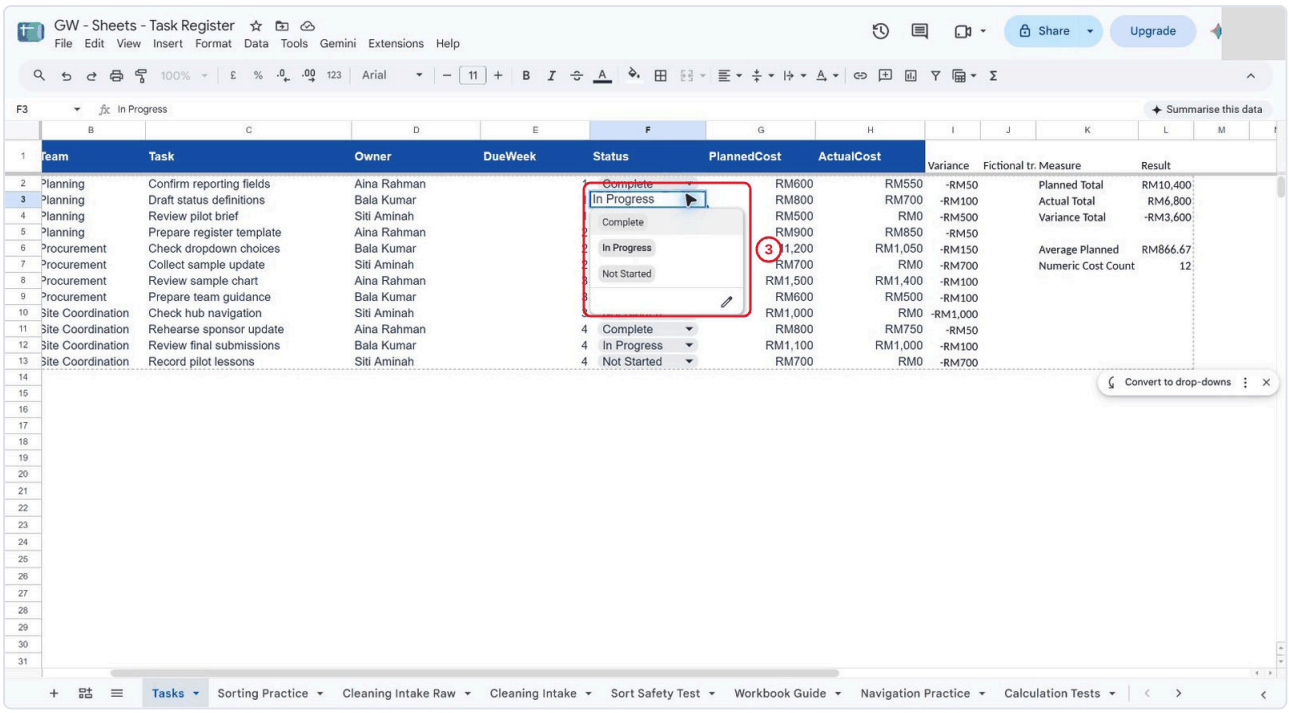
2. Open **Advanced options**, confirm **Reject the input**, and keep **Allow multiple selections** off. Click **Done**, then close the sidebar.

> **Reference run:** The reference uses the Chip display.



Verify Reject the input and single selection.

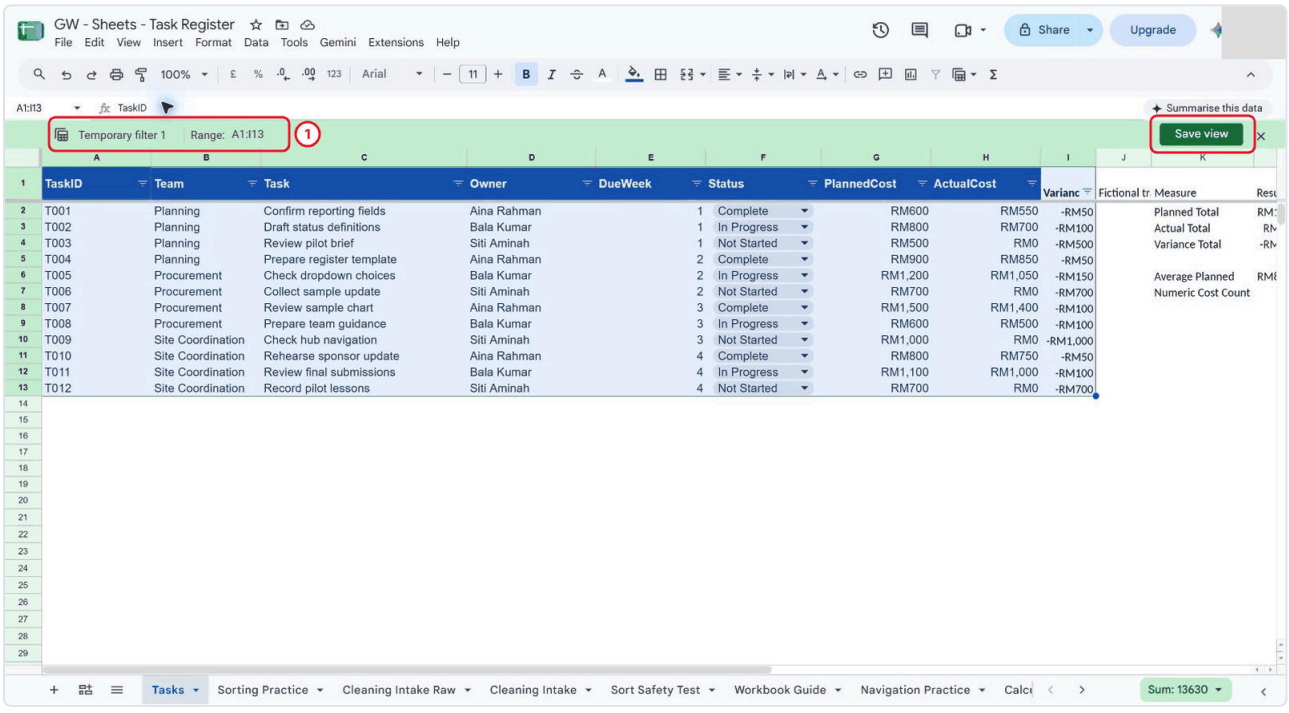
3. Inspect **F3** and open its dropdown. Its existing **In Progress** value remains unchanged, and the three canonical choices are available. Adding a rule must not rewrite the twelve source statuses.



Inspect an unchanged existing status dropdown.

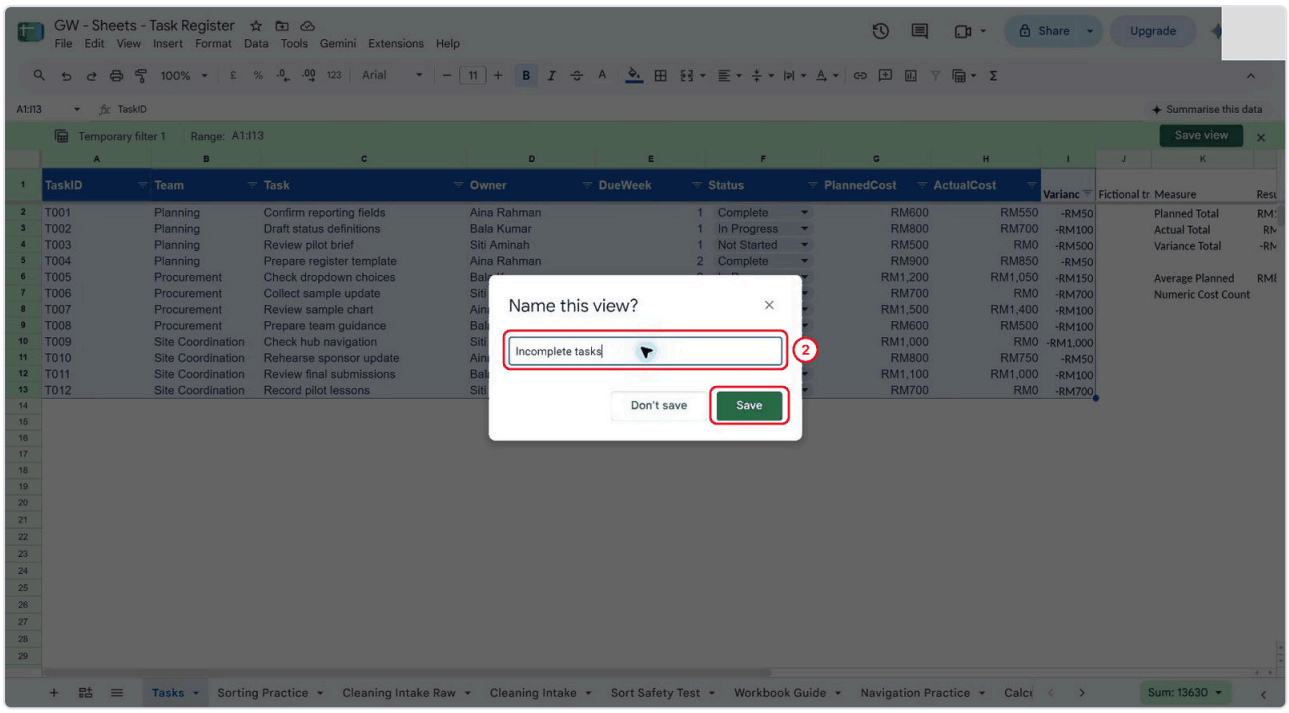
5.2 Create a Filter View

1. Select **A1:I13**, including the header and variance. Use **Data > Create filter view**, or the toolbar's **Filter views > Create filter view**. This interface initially creates **Temporary filter 1**; inspect its range before saving.



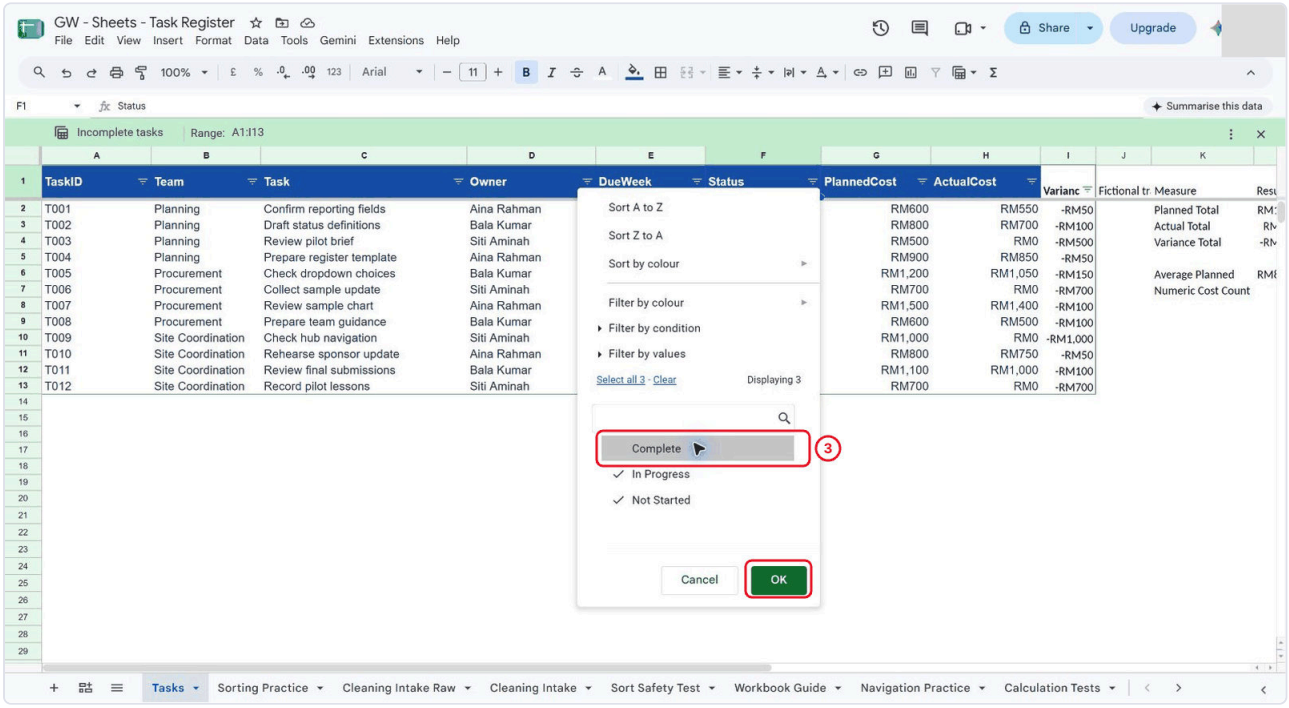
Inspect the temporary filter range.

2. Click **Save view**, enter **Incomplete tasks** in **Name this view?**, then **Save**. This save-and-name step replaces the older guide's assumption that a new view is named immediately.



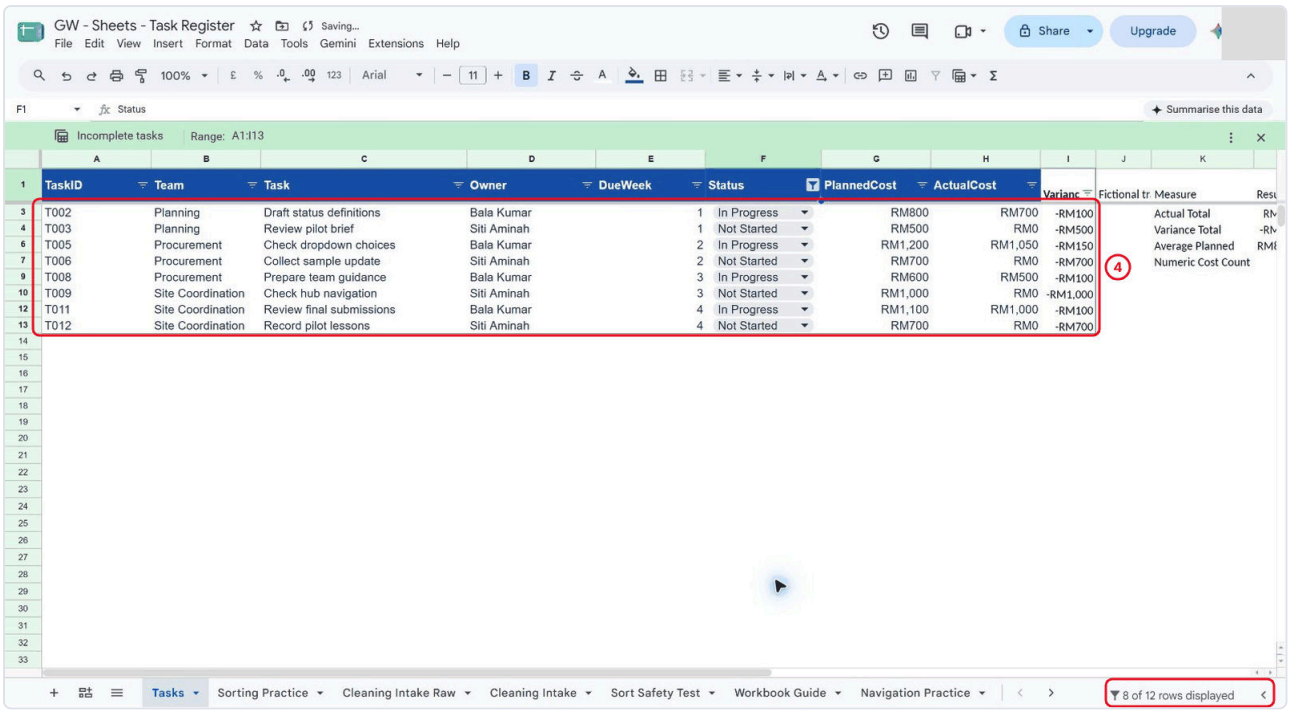
Name the saved incomplete view.

3. Open the Status header filter, uncheck **Complete**, and click **OK**. Leave **In Progress** and **Not Started** selected.



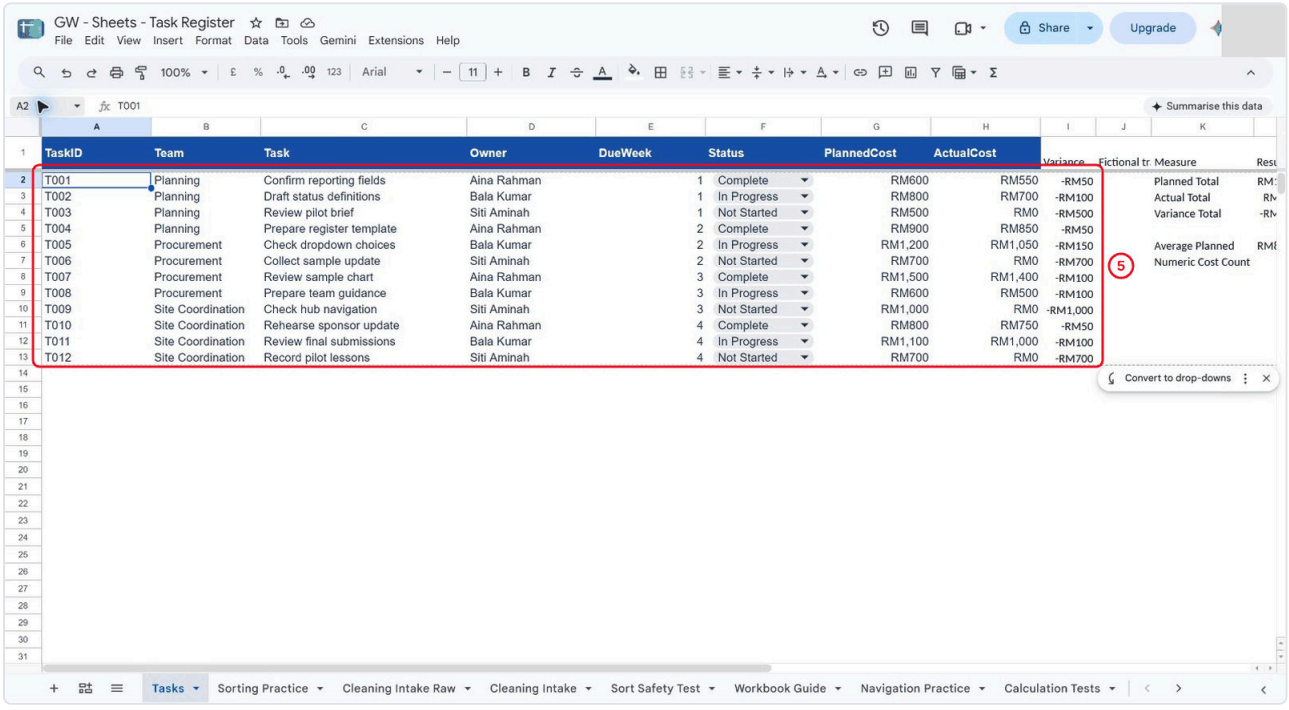
Exclude Complete from the Status filter.

4. Count the visible IDs: **T002, T003, T005, T006, T008, T009, T011, T012**. The lower-right indicator reports **8 of 12 rows displayed**. Compare these with the original source statuses.



Verify eight incomplete task IDs.

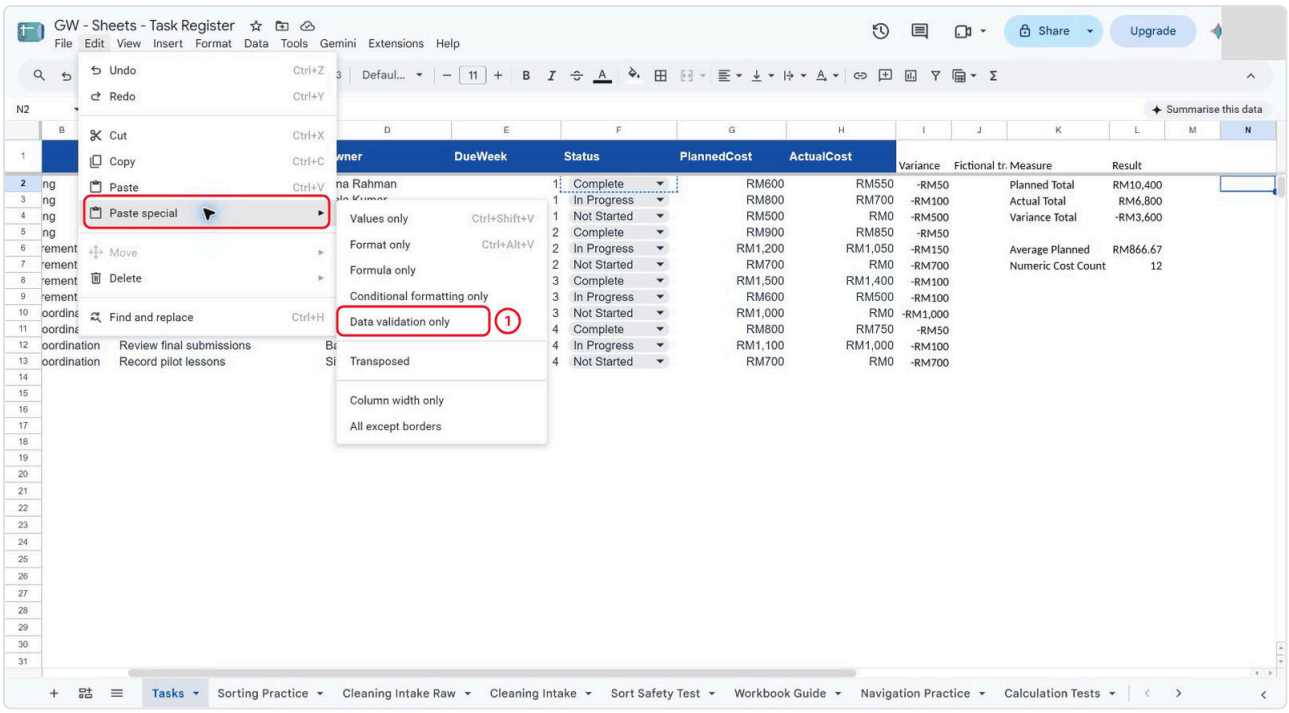
5. Use **Exit** at the right of the green filter-view bar. All twelve records return. Closing the view does not delete its saved definition or any hidden records.



Restore all twelve rows after closing the view.

5.3 Test a Rule Safely

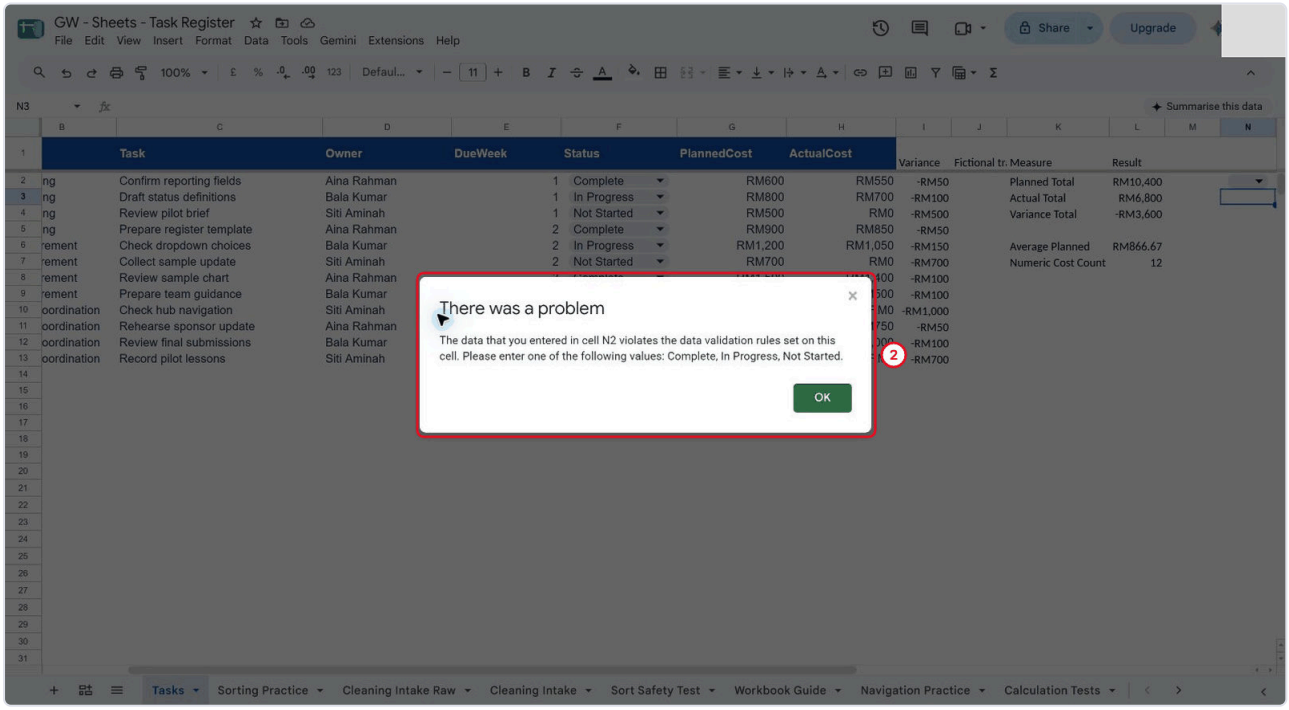
1. Select **F2**, copy it, and select the unused **N2** outside the source register. Use **Edit > Paste special > Data validation only**. This transfers the rule without copying a source status value. Keep the copied clipboard intact while selecting the destination; pasting a cell address would replace it.



Copy only data validation to the spare cell.

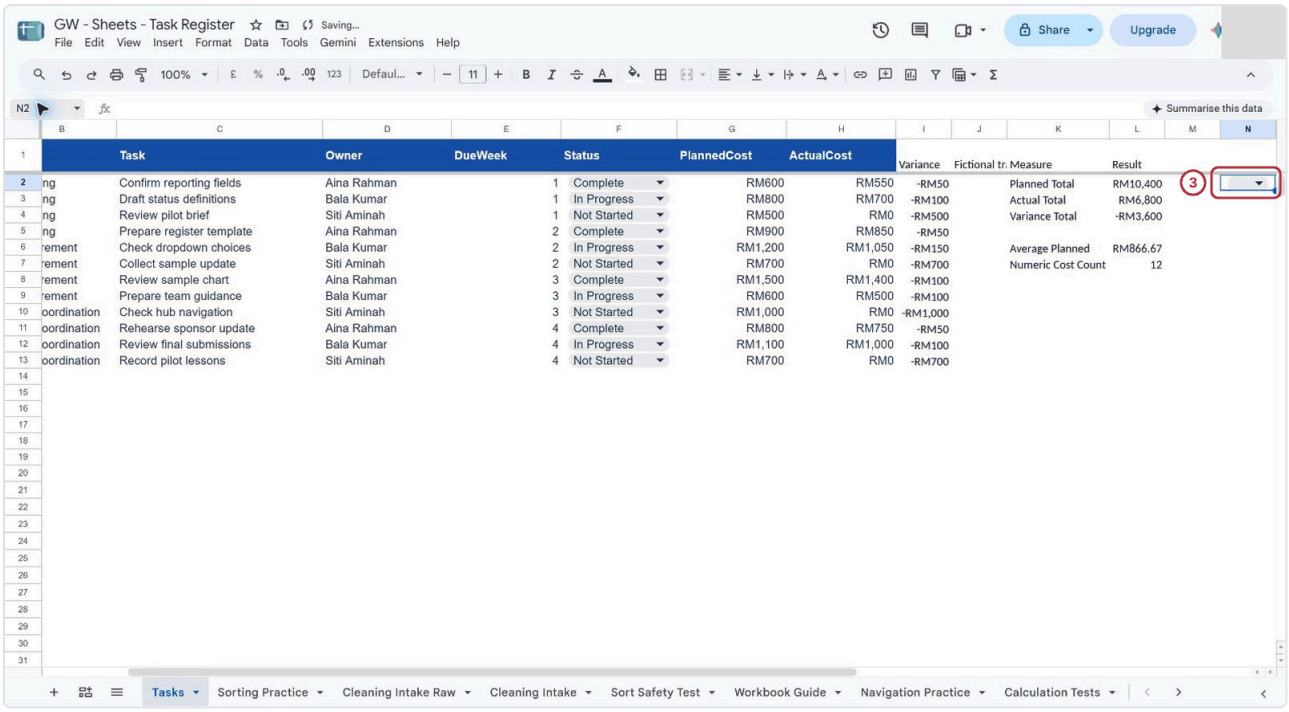
2. Edit N2 with **F2** (the edit-cell key), enter **Done**, and commit. This is rejection, not a warning that leaves an invalid value stored.

> **Reference run:** The reference shows **There was a problem** and identifies the allowed values **Complete, In Progress, Not Started**.



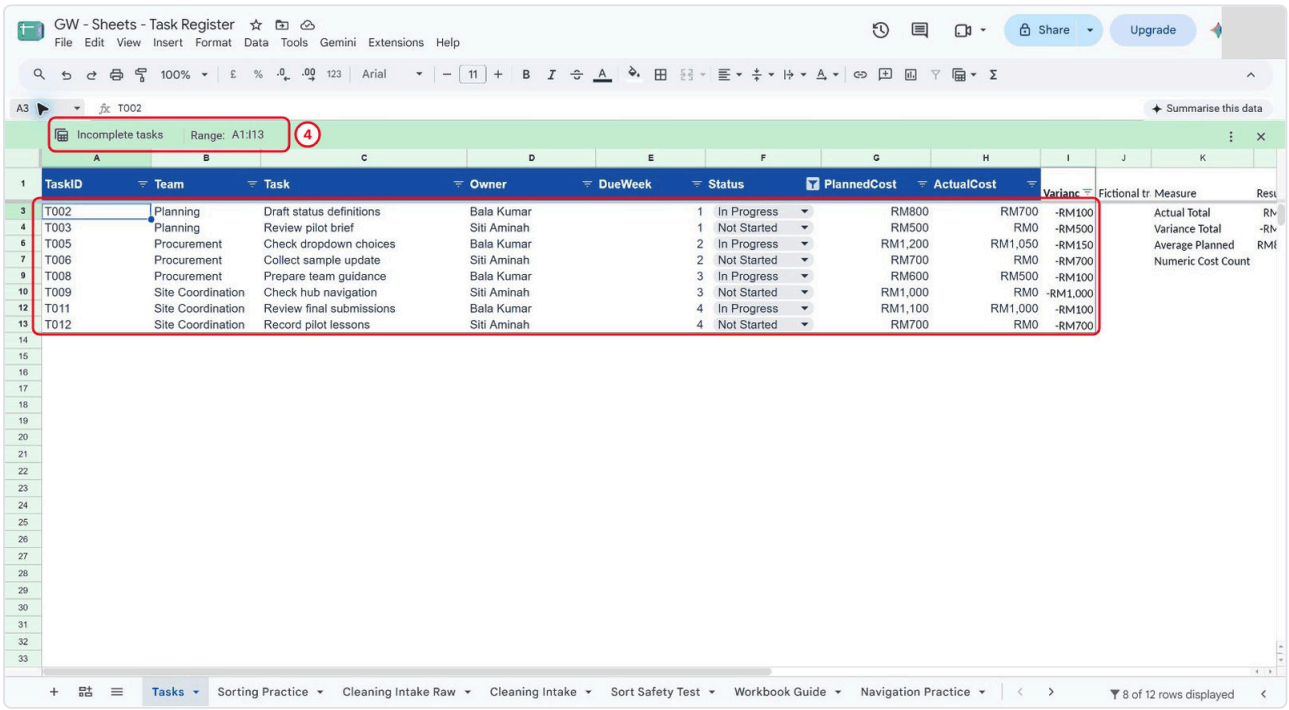
Observe Done rejected in N2.

3. Dismiss the message, exit editing if necessary, and clear N2. Its value is empty; its copied validation remains available for safe demonstrations. No baseline status was changed.



Clear the spare test cell.

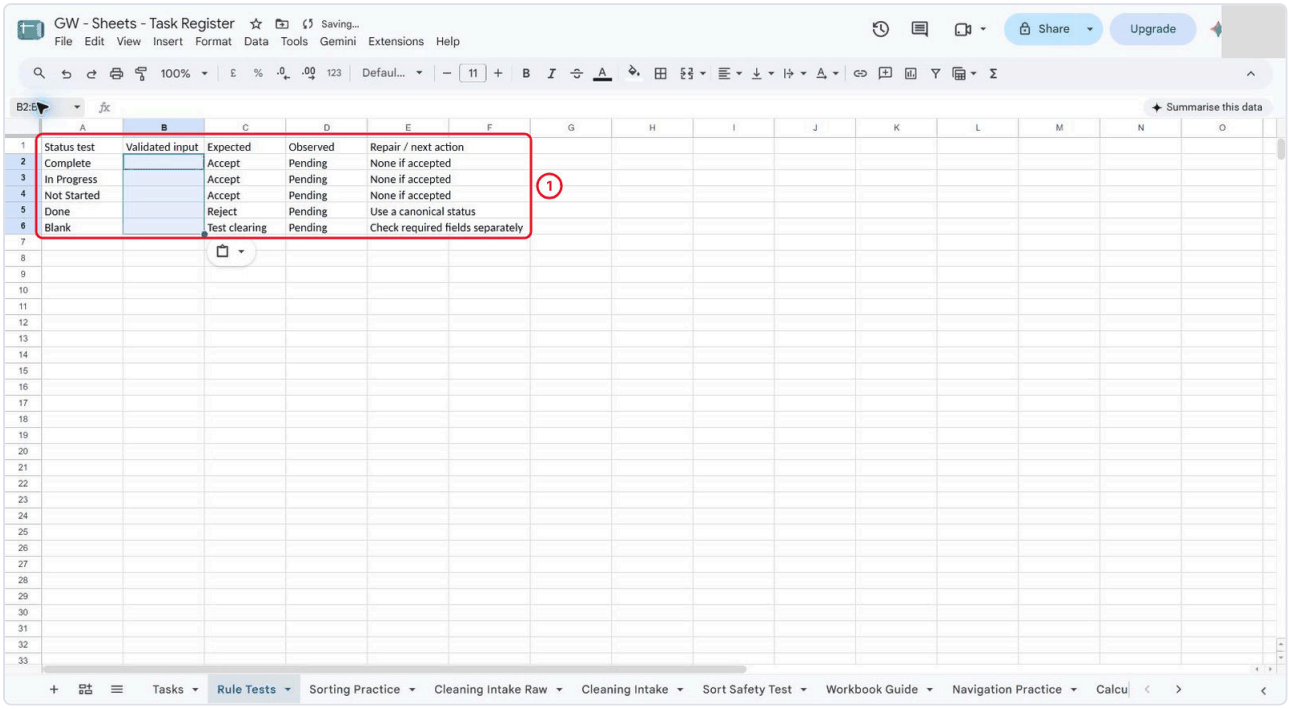
4. Open the toolbar's **Filter views** menu and choose **Incomplete tasks**. It still shows the same eight task IDs. Exit the view before continuing.



Reopen the saved incomplete view.

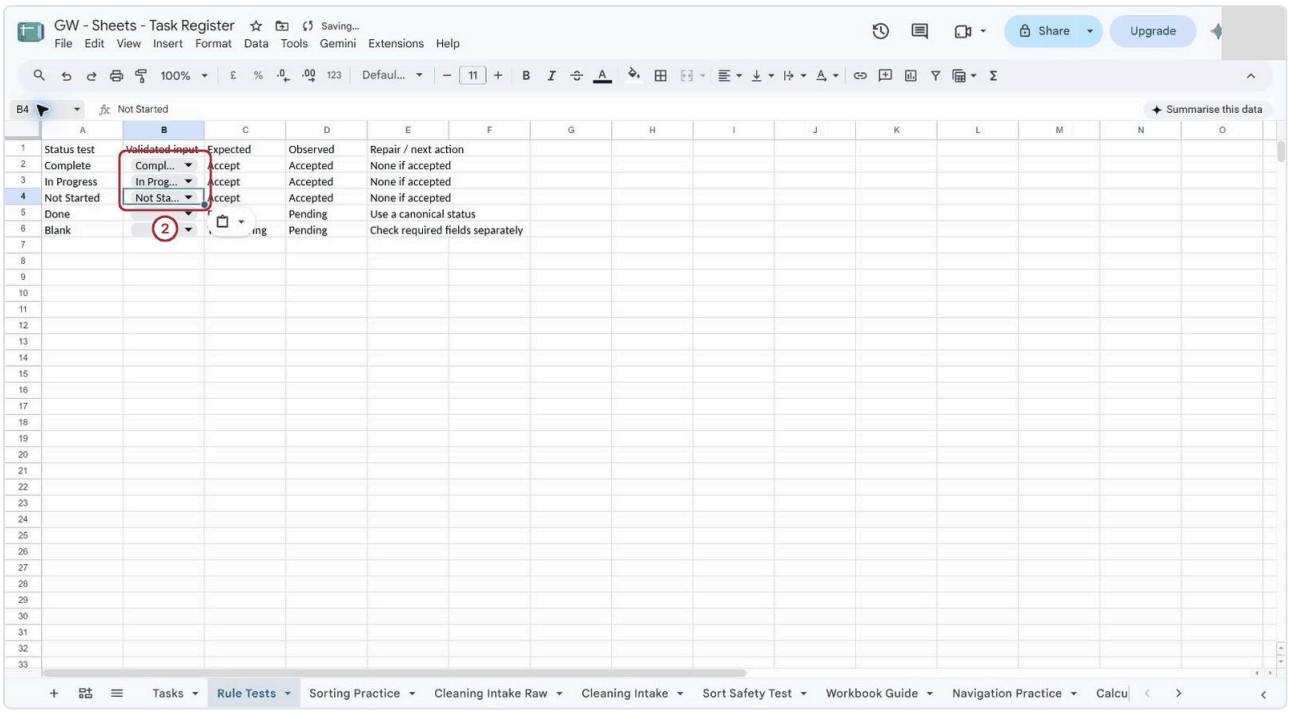
5.4 Design and Verify Entry Rules and Focused Views

1. Add a sheet named **Rule Tests** and paste the status test plan from Part 4 of the prompts page into A1: headers **Status test / Validated input / Expected / Observed / Repair / next action**, with **Complete, In Progress, Not Started, Done and Blank** in rows 2–6. Predict acceptance for the canonical statuses and rejection for Done; leave observed results pending. Treat clearing as an actual test, not an assumed required-field guarantee.



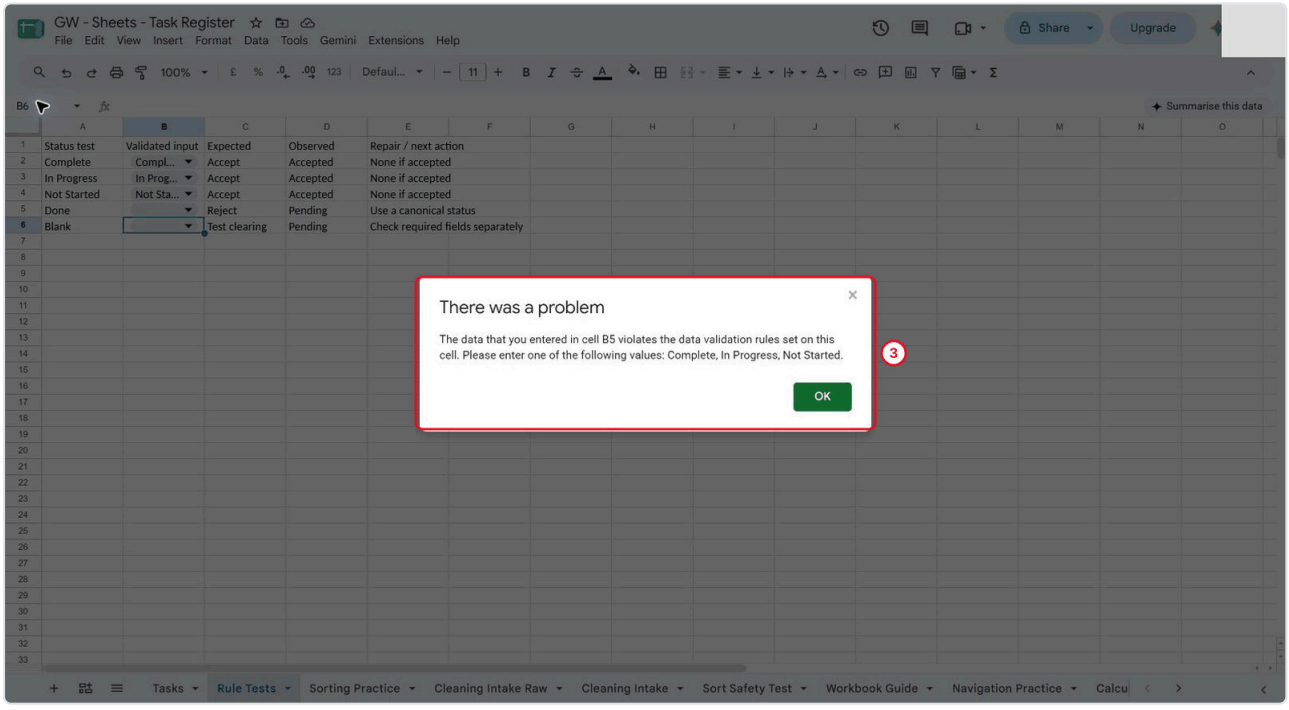
Plan status tests before running them.

2. Copy validation from Tasks F2 to **Rule Tests B2:B6** using **Paste special > Data validation only**. Enter test values through cell-edit mode, preserving the rule. The three canonical statuses are accepted in B2:B4.



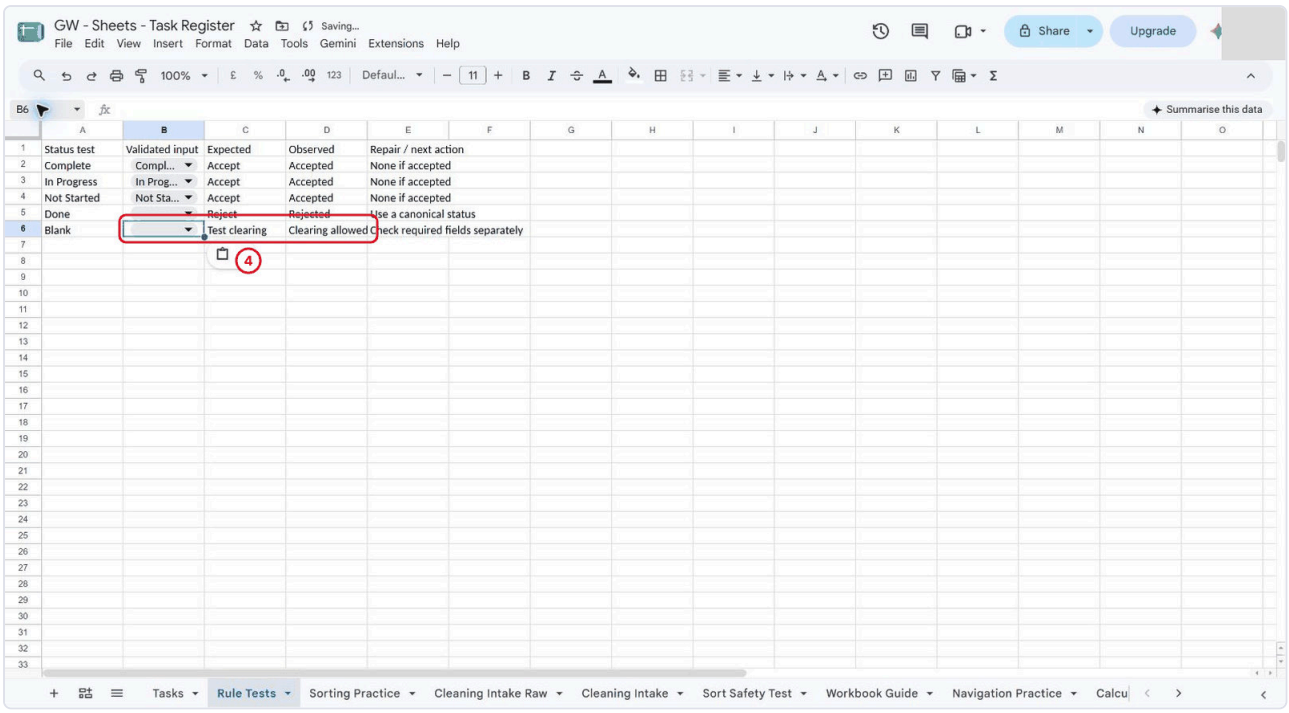
Verify all three canonical statuses accepted.

3. Attempt **Done** in B5. It is rejected with the same allowed-value message. Dismiss the message; B5 stays empty. Record **Rejected** separately from the attempted value.



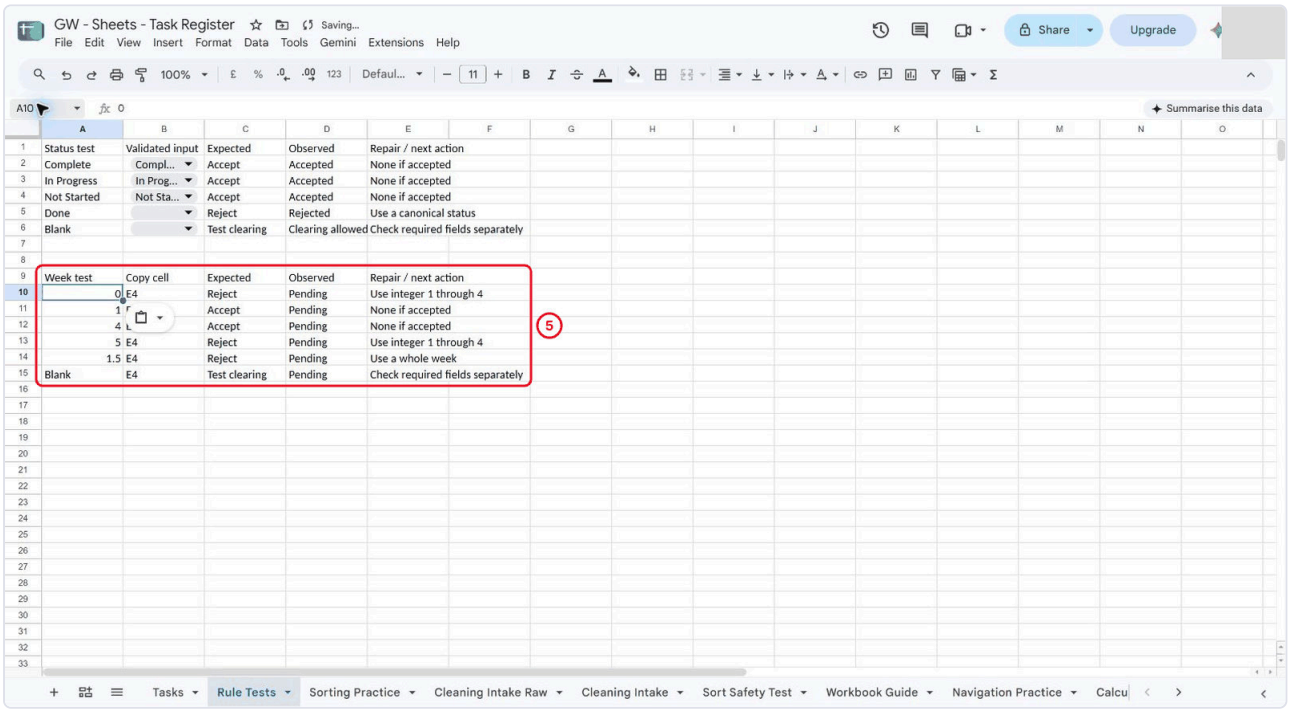
Reject Done in the planned matrix.

4. Enter a valid status in B6, then clear it with Backspace. Clearing is allowed. Record that observation and keep B6 empty. The tested rule limits nonempty status entries but does not prove that every required field is complete.



Record that status clearing is allowed.

5. On **Rule Tests**, paste the week test plan from Part 4 of the prompts page into **A9**: **0 reject / 1 accept / 4 accept / 5 reject / 1.5 reject / blank clearing to test**. Then right-click the **Entry Practice** tab, choose **Duplicate** and rename the copy **Entry Rule Practice**. The test target is **E4** on that copy; the original three copied records remain unchanged.

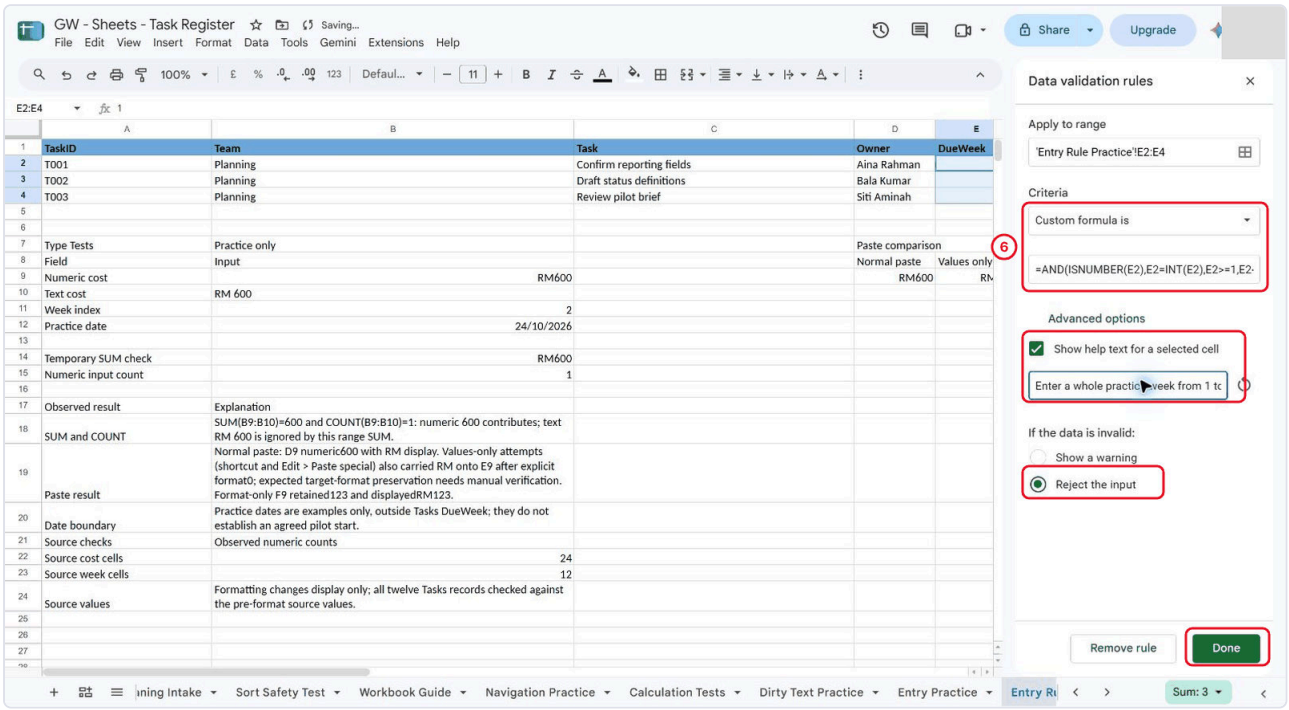


Plan week boundaries and fraction tests.

6. On **Entry Rule Practice**, select **E2:E4** only, open **Data validation > Add rule**, choose **Custom formula is**, and enter:

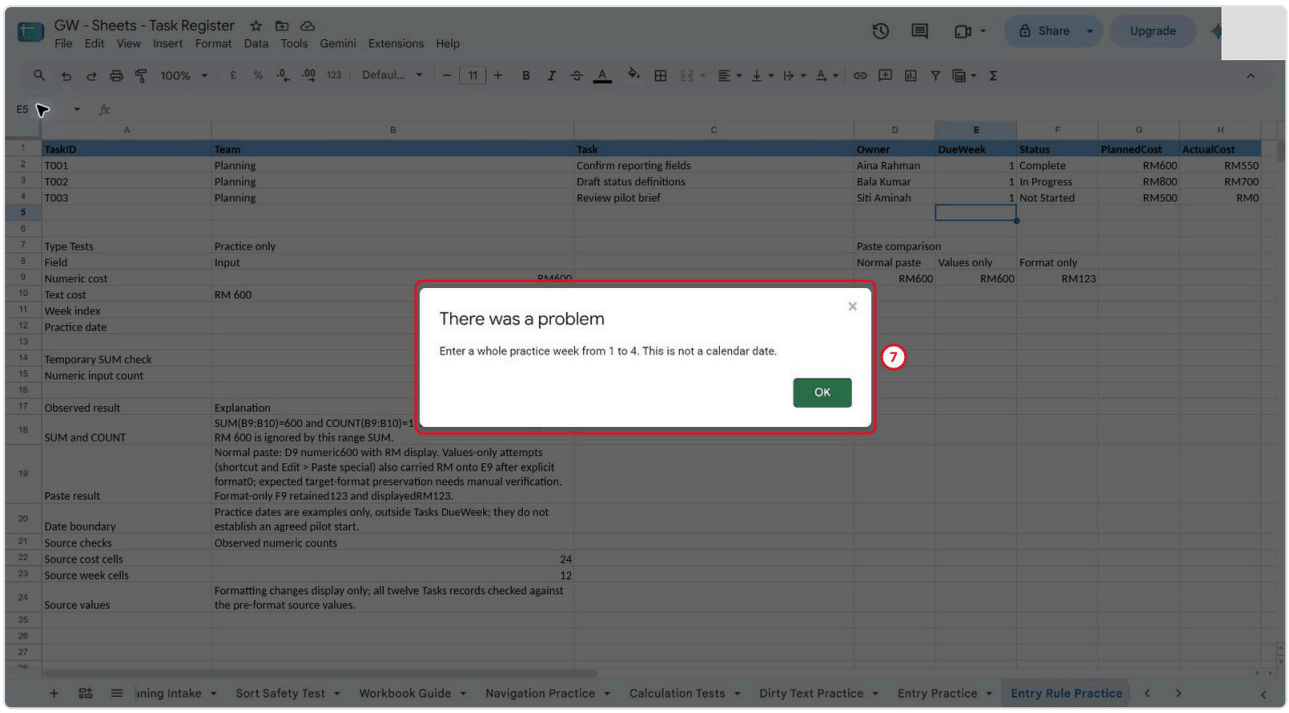
```
=AND(ISNUMBER(E2),E2=INT(E2),E2≥1,E2≤4)
```

Use the top-left cell E2 as the relative reference. This checks numeric type, a whole number and both bounds; a simple between rule alone would not express the whole-week requirement. Explicitly choose **Reject the input**: the custom criterion initially offered **Show a warning** in this session. Enable help text **Enter a whole practice week from 1 to 4. This is not a calendar date.**, then save and close the sidebar. E2:E4 covers the three-record practice table, not the notes and type tests farther down that tab.



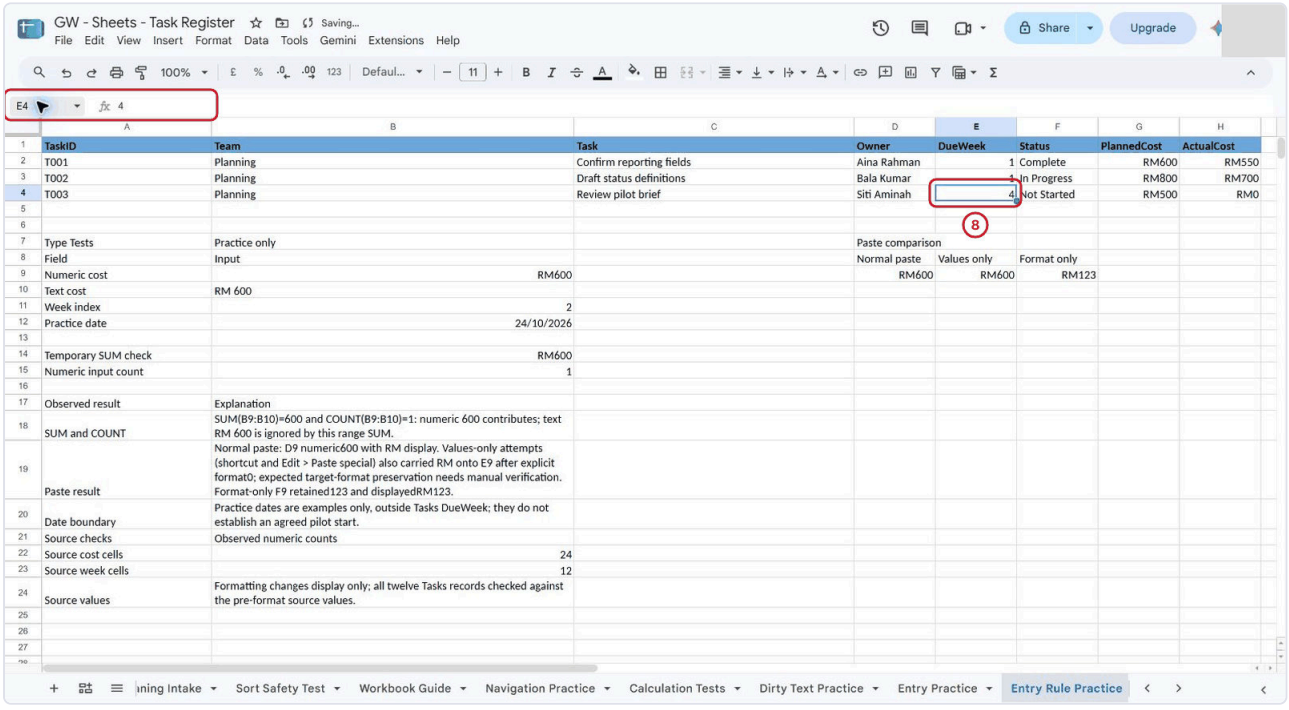
Configure the strict whole-week custom formula.

7. Reuse E4 for each planned test. Enter edit mode, select its previous content, type the candidate and commit. **0** is rejected with the configured help text. Dismiss the message before the next test.



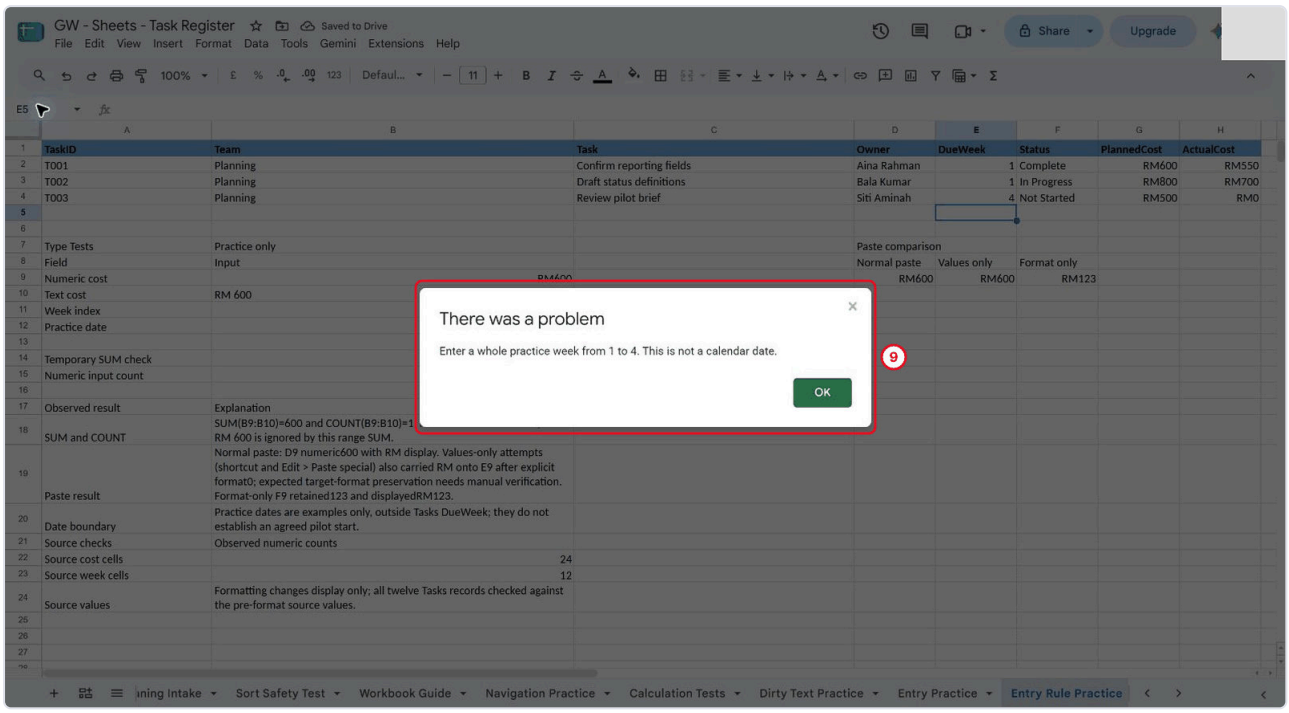
Reject week zero.

8. Test **1**, then **4**; both are accepted. Inspect the formula bar after committing each input. The captured upper-bound stage shows E4 holding numeric **4**.

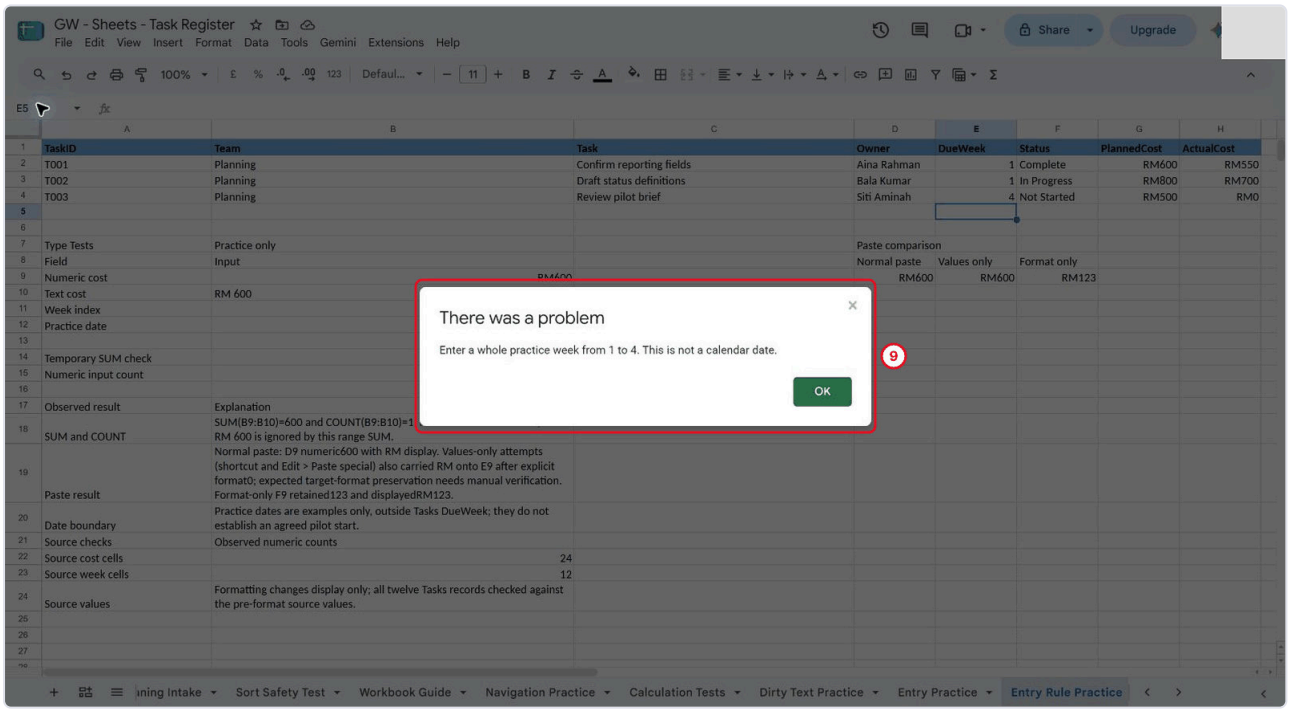


Accept the upper boundary of four.

9. Test **5** and **1.5** separately. Both are rejected. A fractional value must not be accepted merely because it lies between the bounds.

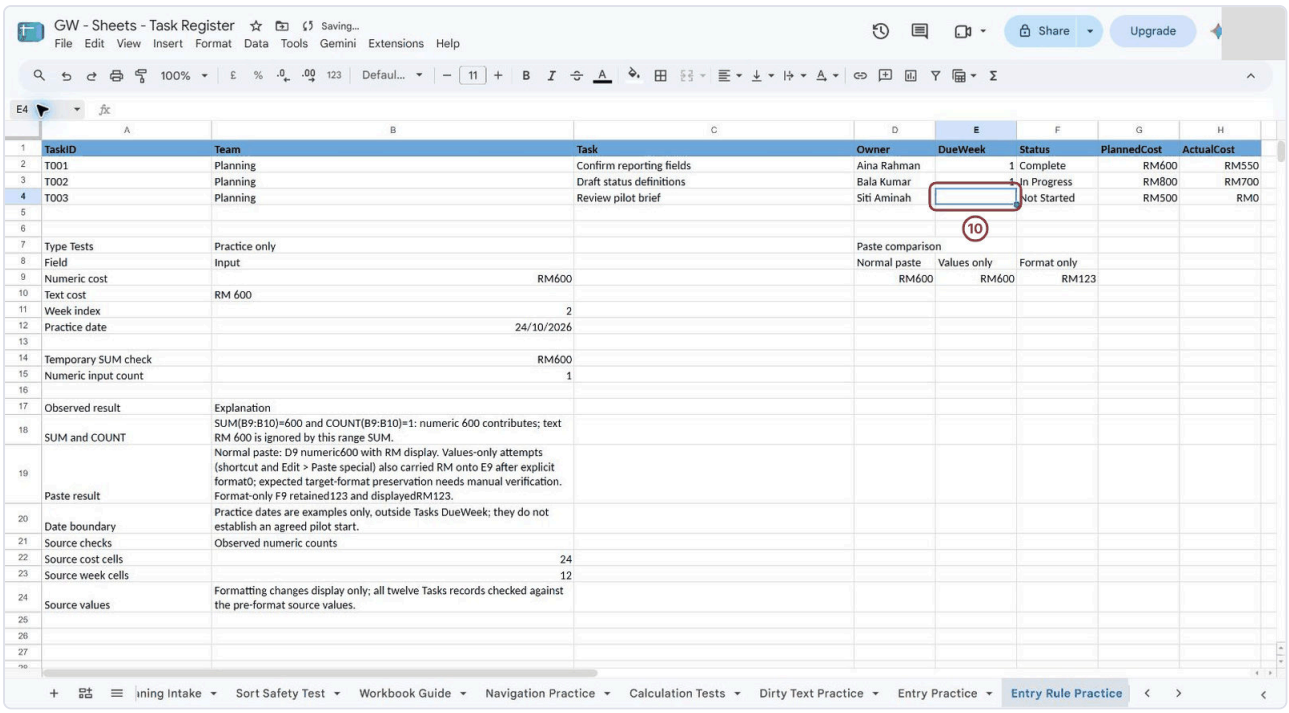


Reject week five.



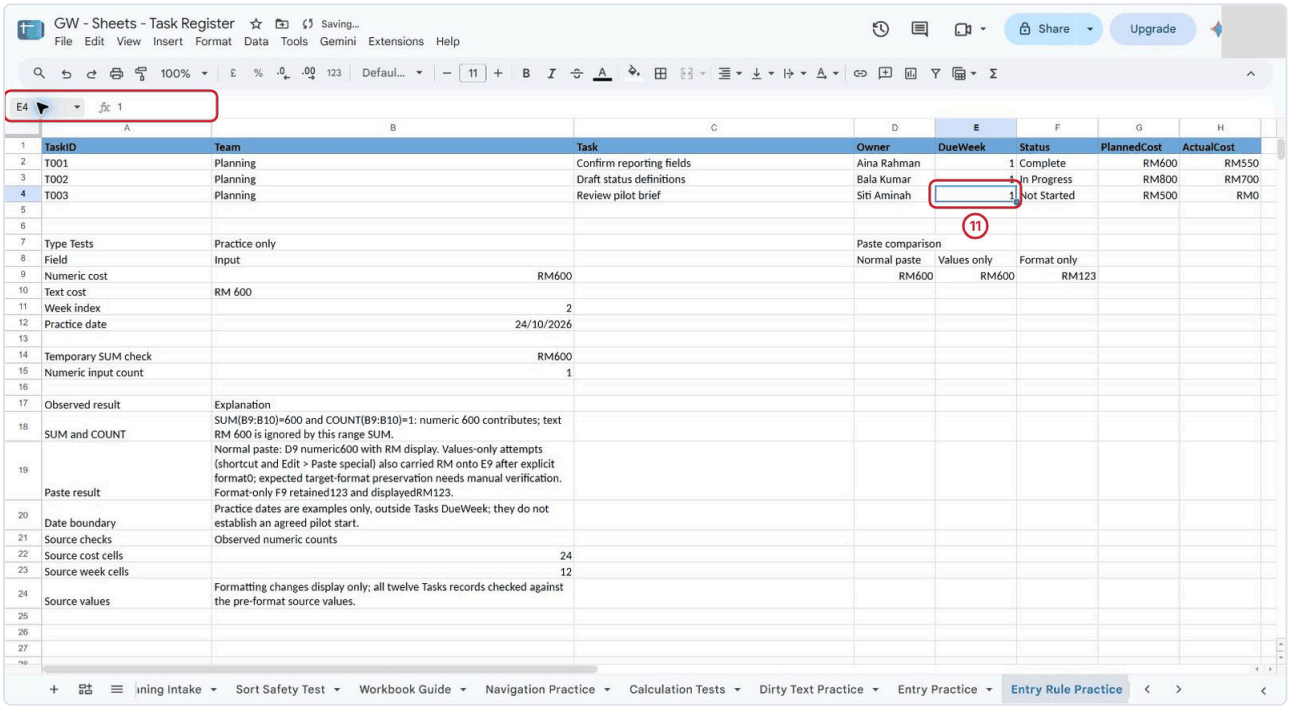
Reject a fractional week.

10. Clear the previously valid week and observe that deletion is allowed. This clearing test does not establish acceptance of every possible blank-input mechanism. Required-field completeness still needs an explicit review.



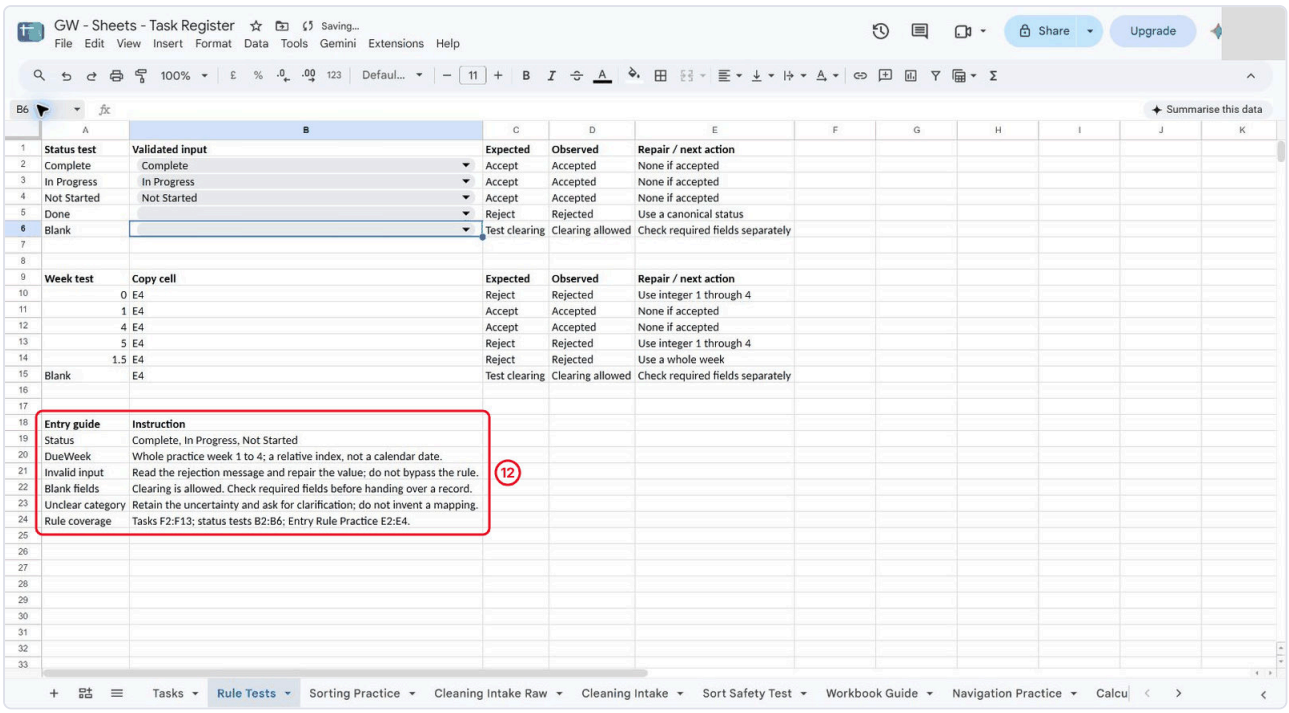
Observe that clearing a week is allowed.

11. Restore E4 to its original numeric **1**. Keep the other practice-record values unchanged.



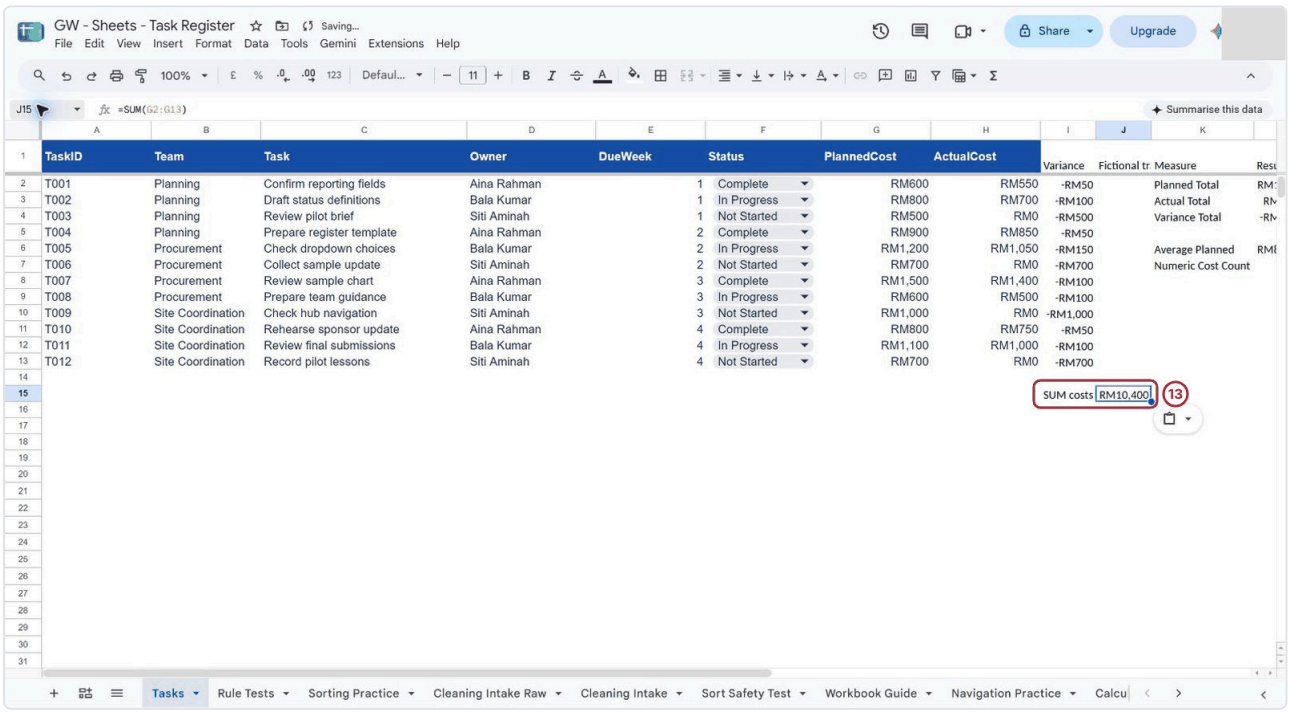
Restore the original practice week.

12. Record all observed results in Rule Tests. Add an entry guide in A18:B24 (row labels in Part 4 of the prompts page) covering the canonical statuses, whole relative weeks, actual rejection messages, clearing limitation, unclear-category escalation and precise rule ranges. Fit the columns so labels and instructions are readable. Do not invent a mapping for an unclear category.



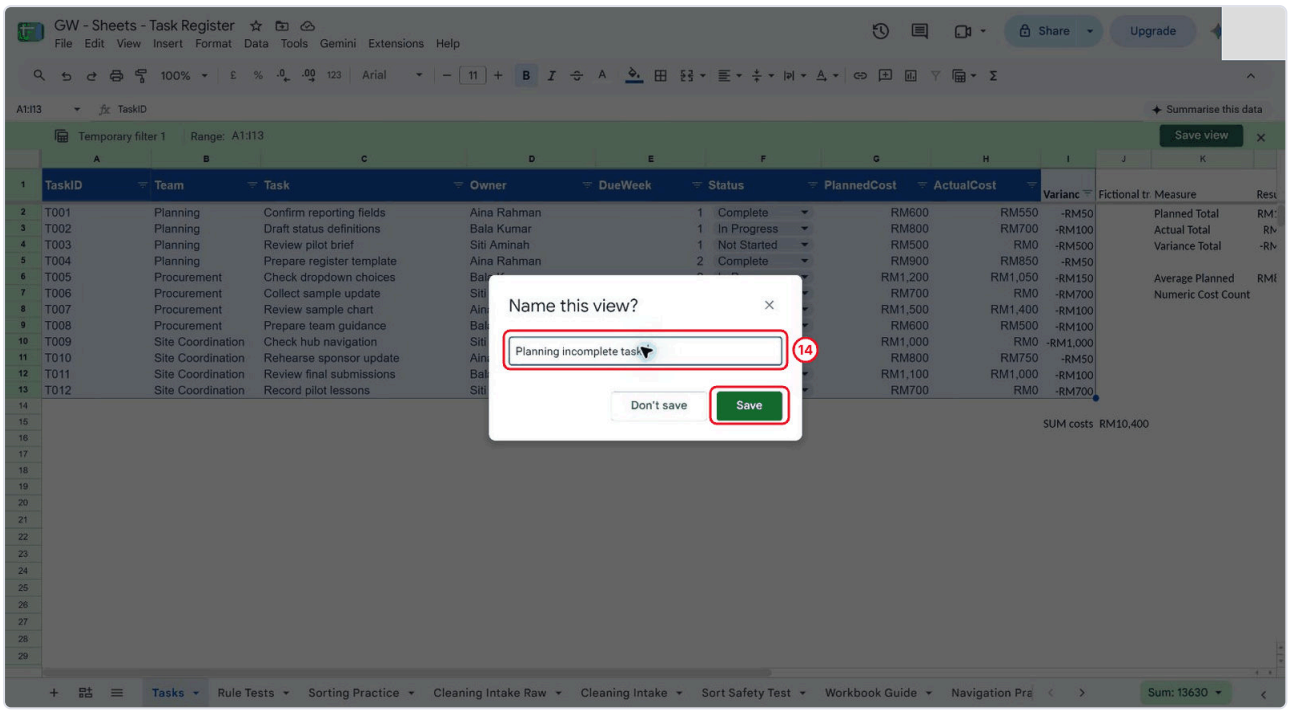
Review the completed rule matrix and entry guide.

13. On **Tasks**, enter **SUM costs** in I15 and **=SUM(G2:G13)** in J15. This labelled check sits outside the bounded A1:I13 filter range and stays visible while rows are hidden. Before filtering its result is **RM10,400**.



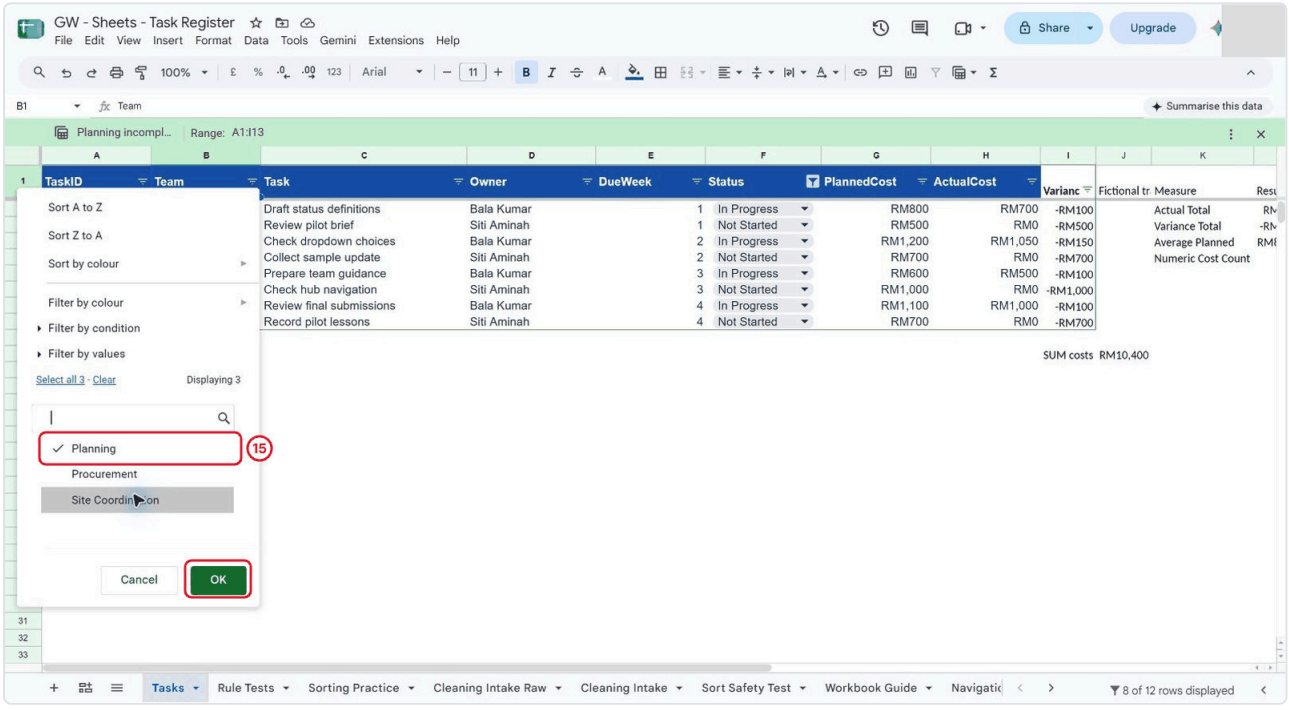
Record the ordinary SUM before filtering.

14. Create a second filter view over **A1:I13**, save it as **Planning incomplete tasks**, and exclude Complete in Status as before.



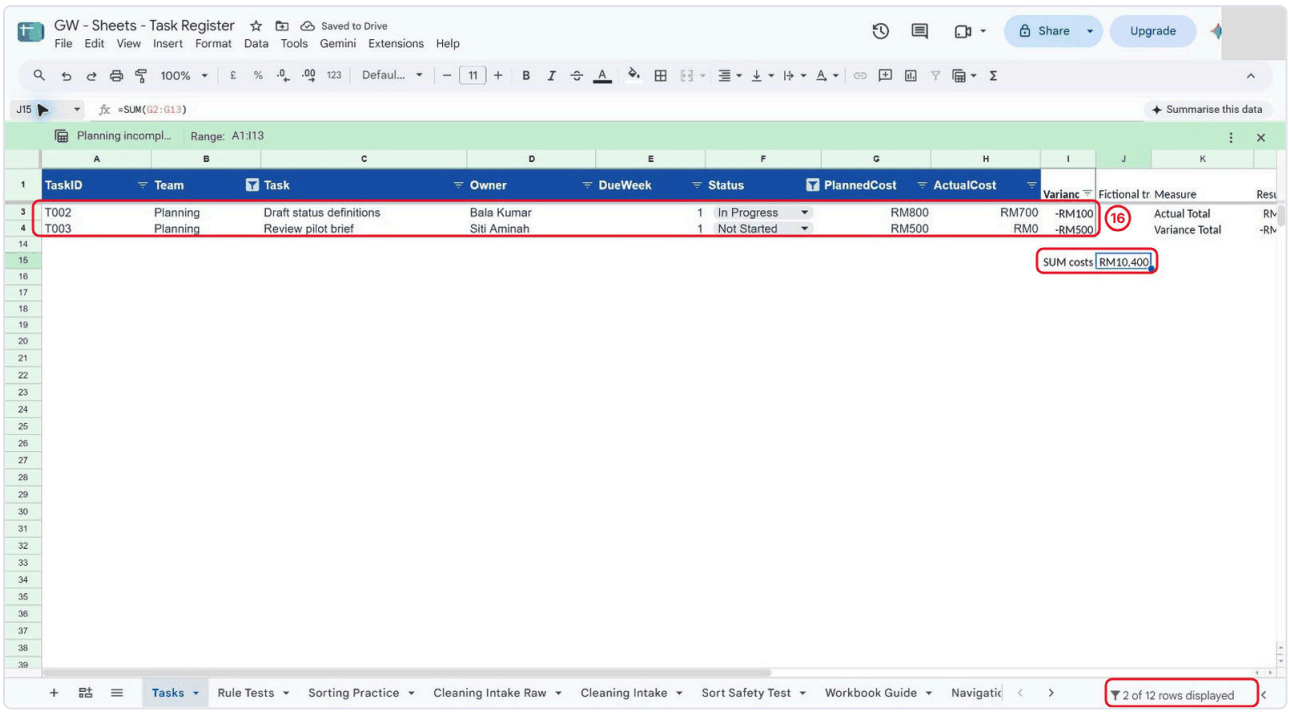
Name the Planning-focused view.

15. In the Team header filter leave only **Planning** selected. Both Team and Status criteria must apply to the same view.



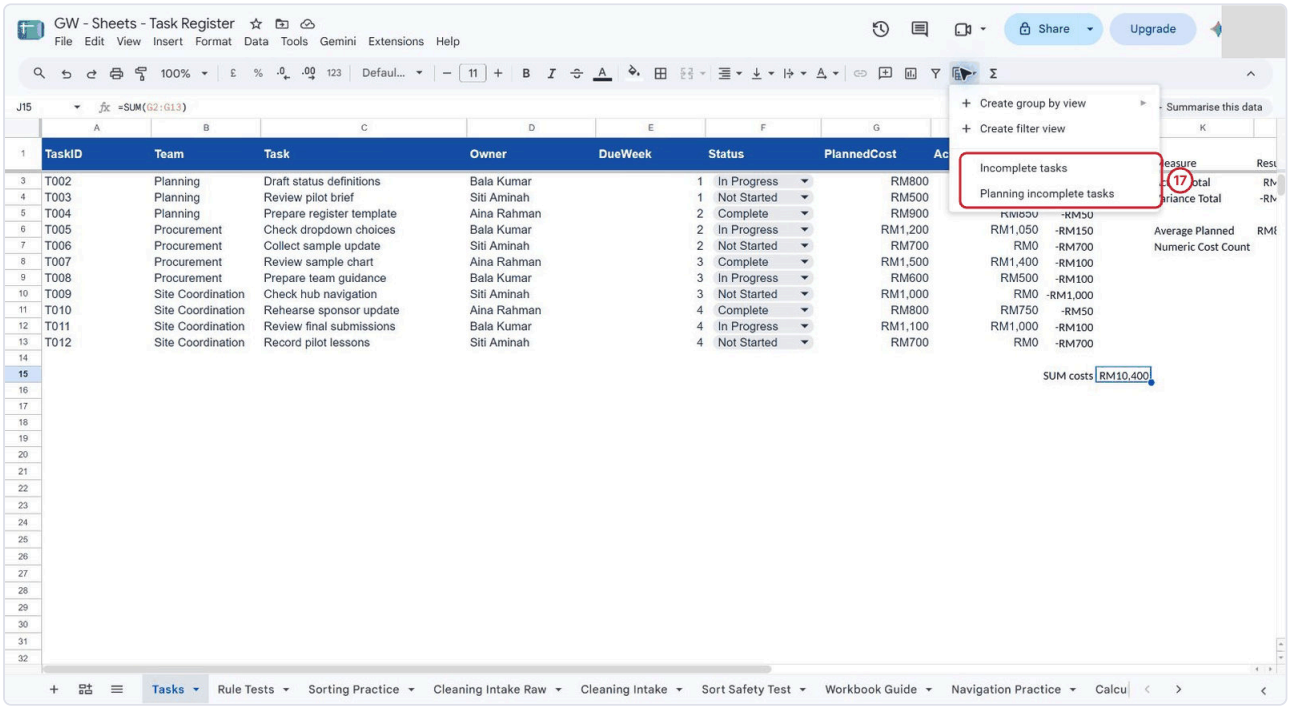
Keep only Planning in the Team filter.

16. Verify **T002 and T003**, and the lower-right **2 of 12 rows displayed** indicator. A manual source scan confirms these are the two Planning records not Complete. J15 still shows **RM10,400** because an ordinary SUM includes values in hidden rows; it is not the total of only the two visible records.



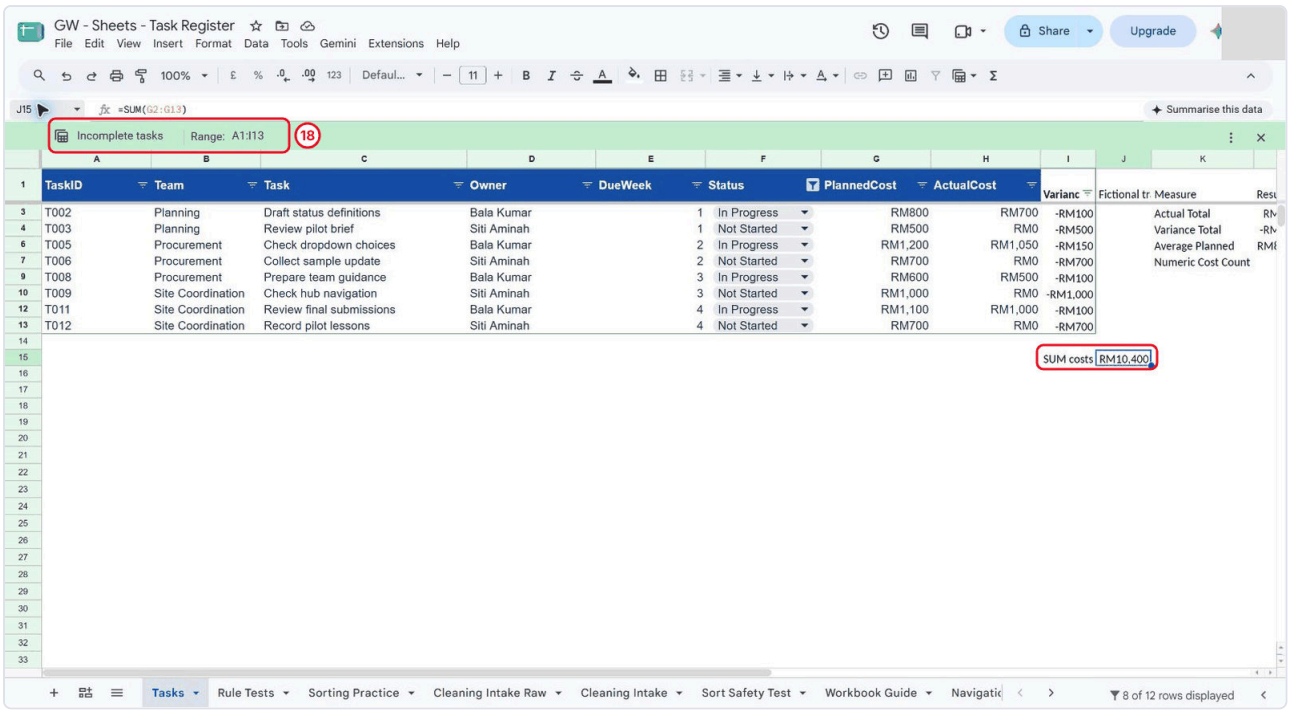
Verify two Planning tasks with an unchanged SUM.

17. Exit the view and inspect the **Filter views** menu. Both saved names remain available.

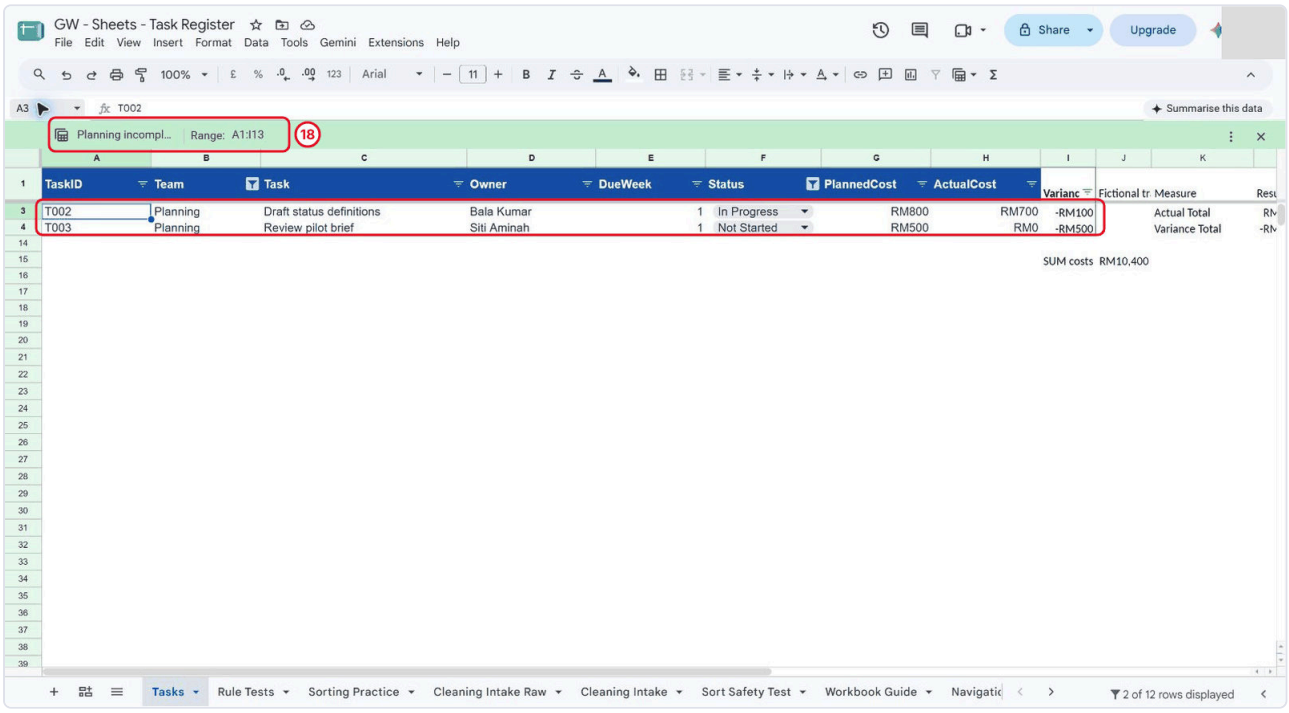


Inspect both saved view names.

18. Reopen **Incomplete tasks** and verify eight IDs with J15 still **RM10,400**. Exit it, reopen **Planning incomplete tasks**, and verify the same two IDs. Do not use the mere presence of a view name as proof that its criteria persisted.

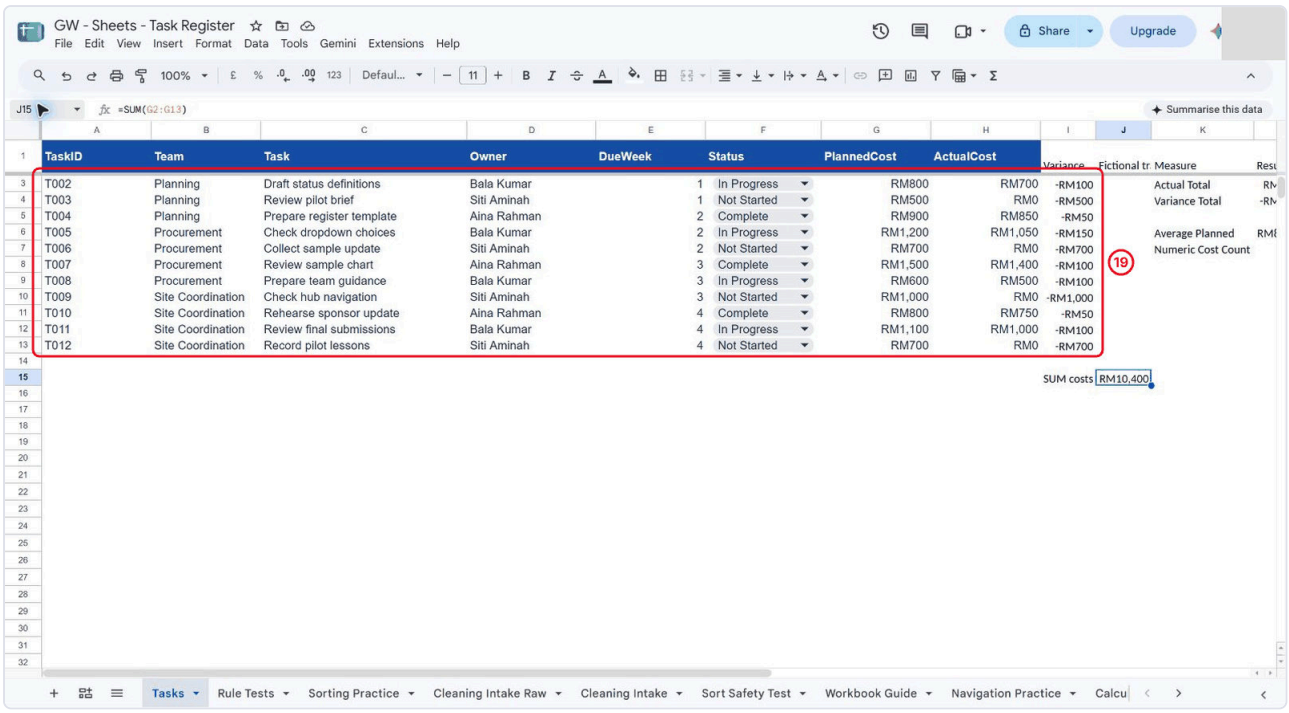


Reopen eight incomplete tasks with unchanged SUM.



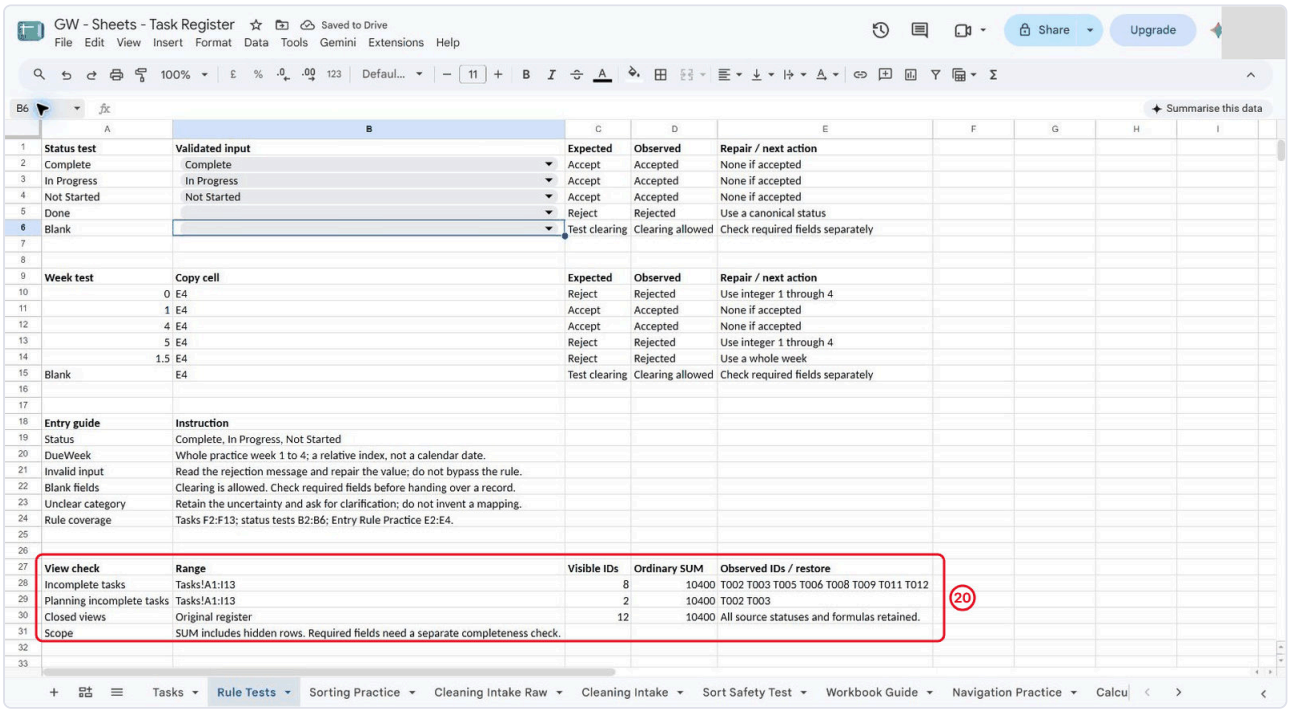
Reopen the Planning view and confirm two tasks.

19. Exit the view and leave all twelve source records visible. The source A1:L13 values and formulas remain unchanged; the validation rules and separate J15 control are additions.



Restore all source rows after closing both views.

20. Record the two views' ranges, visible counts, observed IDs and ordinary SUM in **Rule Tests A27:E31**, using the labels in Part 4 of the prompts page. All source rows return after closing the views. Inspect rule coverage separately from view coverage: an input outside a bounded rule may escape validation, and hiding a row does not remove it from an ordinary SUM.



Review the final rule and view evidence.

> **Validation boundary:** these are author UI reference checks. The independent Procurement view, new-row coverage/range repair challenge, independent entry test and learner timing pilot remain pending. Earlier CSV-import and values-only-format limitations remain separate.

Independent Practice

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently build a small entry test with one valid row, one invalid status and one out-of-range week. Demonstrate the rules rather than merely listing them.
2. Create a named view answering “Which Procurement tasks are not complete?” and list its TaskIDs as evidence.
3. Check whether each new practice row lies inside the rule and view ranges. Repair a missing bound on the practice copy and retest.

Evidence to Hand Over

- Expected-versus-observed rule matrix including week boundaries and blanks.
- Two useful named views with independently checked visible records.
- Entry guide and a repaired coverage/range test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A new row accepts invalid values	Inspect rule coverage as well as the dropdown definition; retest the actual new row.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Validation prevents inconsistent entries and filter views support focused inspection.

Check yourself: Does hiding a row delete it or remove it from an ordinary SUM?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Incomplete tasks
```

```
Done
```

Part 2: Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Rule test values: status Complete, In Progress, Not Started, Done and blank; DueWeek 0, 1, 4, 5 and 1.5. DueWeek is a relative index for this training exercise, not an approved calendar schedule.
```

Part 3: Captured Rule and View Reference

- Tasks status range **F2:F13**; single-selection dropdown Complete / In Progress / Not Started, **Reject the input**. Spare N2 has the copied validation and is empty after the rejected Done test.
- Rule Tests status inputs **B2:B6** use that same strict rule. B2:B4 retain Complete / In Progress / Not Started; attempted Done in B5 was rejected, and clearing B6 was allowed. A2:A6 retain the attempted test values; observations are separate in D.
- Entry Rule Practice **E2:E4**: custom formula **=AND(ISNUMBER(E2),E2=INT(E2),E2>=1,E2<=4)**, reject invalid input. Help text: Enter a whole practice week from 1 to 4. This is not a calendar date. Test E4 with 0/1/4/5/1.5, then clearing. Observed reject/accept/accept/reject/reject/clearing allowed. Restore E4 to 1. The bounded rule does not cover the copied notes below the three-record table.
- Tasks **I15 SUM costs / J15 =SUM(G2:G13)** displays 10400 before and during either view. Ordinary SUM includes hidden rows.
- **Incomplete tasks**, Tasks A1:I13: exclude Complete; eight IDs T002,T003,T005,T006,T008,T009,T011,T012.
- **Planning incomplete tasks**, same range: Team Planning and Status excluding Complete; two IDs T002,T003.
- Both views were reopened and checked, then closed with all 12 rows visible. Entry guide and view observations are in Rule Tests A18:B24/A27:E31.

Source Tasks A1:L13 values/formulas and the original Entry Practice records remain unchanged. Testing clearing does not guarantee all required fields are complete. Independent Procurement/new-row coverage challenges and classroom timing remain pending.

Part 4: Rule Tests Layout

Section 5.4. Columns are separated by tabs, so you can paste each block into its first cell. Leave **Validated input** (column B) empty until you test it, and change **Pending** to what you actually observe.

Status test plan, A1:E6 (step 1):

```
Status test     Validated input Expected        Observed        Repair / next action
Complete                Accept  Pending None if accepted
In Progress             Accept  Pending None if accepted
Not Started             Accept  Pending None if accepted
Done            Reject  Pending Use a canonical status
Blank           Test clearing   Pending Check required fields separately
```

Week test plan, A9:E15 (step 5). Column B names the cell you test on Entry Rule Practice:

```
Week test       Copy cell       Expected        Observed        Repair / next action
0       E4      Reject  Pending Use integer 1 through 4
1       E4      Accept  Pending None if accepted
4       E4      Accept  Pending None if accepted
5       E4      Reject  Pending Use integer 1 through 4
1.5     E4      Reject  Pending Use a whole week
Blank   E4      Test clearing   Pending Check required fields separately
```

Entry guide, A18:A24 (step 12). Write your own instruction for each row in column B:

```
Entry guide
Status
DueWeek
Invalid input
Blank fields
Unclear category
Rule coverage
```

View check, A27:E31 (step 20):

```
View check      Range   Visible IDs     Ordinary SUM    Observed IDs / restore
Incomplete tasks
Planning incomplete tasks
Closed views
Scope
```

CHAPTER 6

Summarise and Visualise

In this chapter you will build a simple conditional total and chart; ask Gemini to explain a checked formula if available.

> **Prompts:** the prompts page has every value and prompt for this chapter, each with a Copy button.

Estimated time: 55 minutes, planned as 10 explanation and demonstration, 25 guided build and casework, 15 independent practice and 5 verification and repair, with 45 active minutes by design. Not yet timed with learners.

Your result: A conditional summary and a source-linked status chart.

> **Status:** author reference with 29 staged application captures through 6.4. Conditional summaries, two labelled charts, Gemini explanation, actual source-change propagation, reconciliations and repaired chart/category faults checked. Source status and cost restored. Independent dashboard/reviewer and learner timing remain pending.

What You Will Learn

- Calculate a conditional count and total.
- Choose and label a simple chart.
- Use AI to explain a formula without surrendering the check.

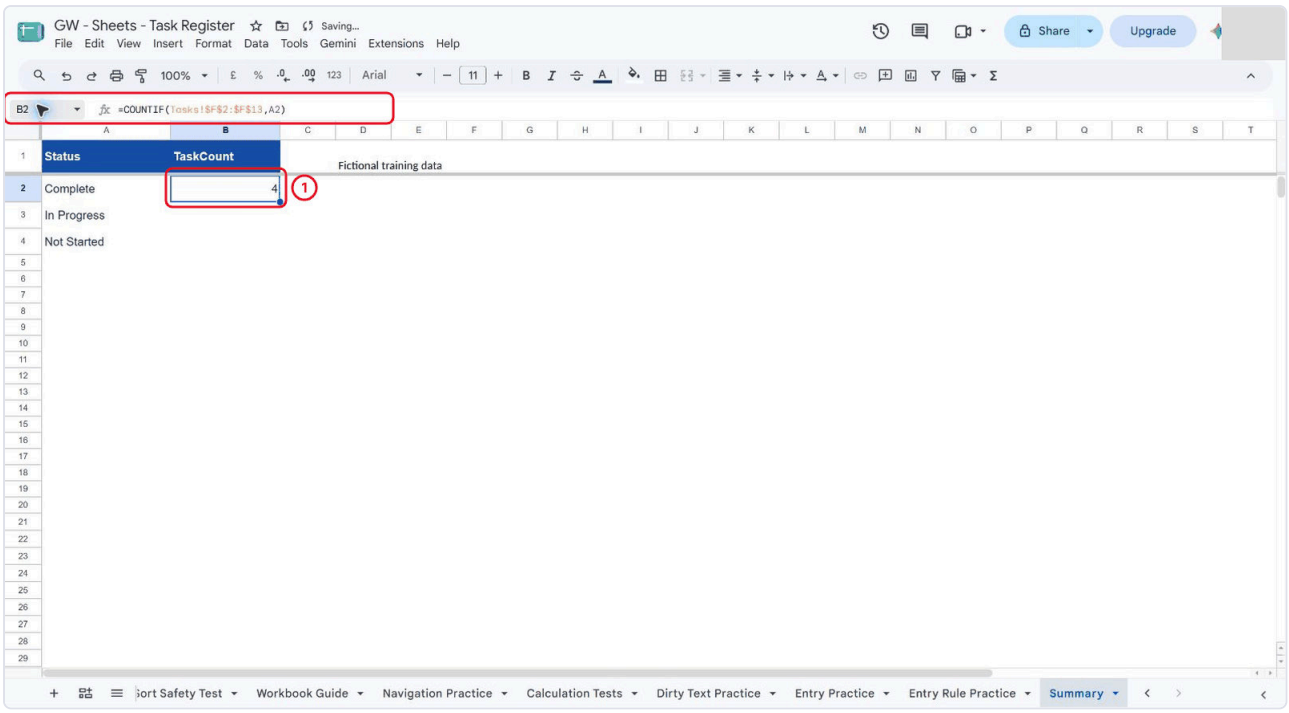
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks and the starter Summary tab. Existing baseline status counts are four per category.

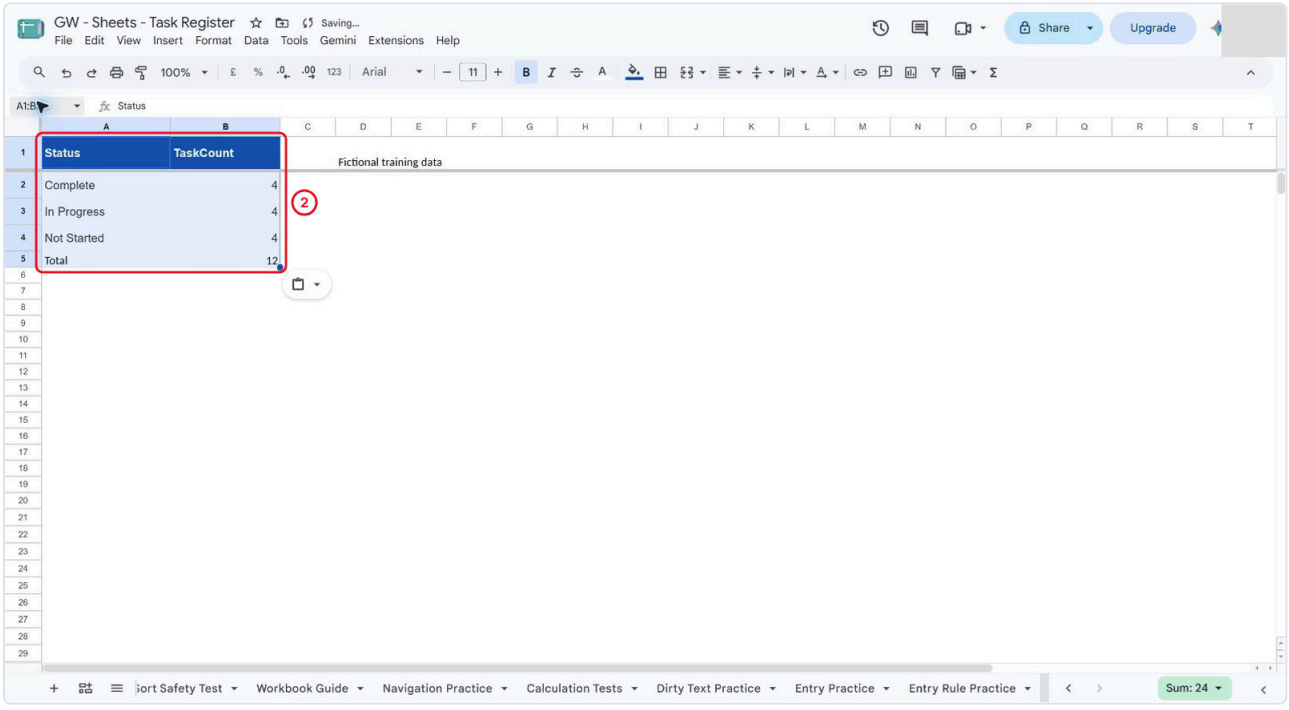
6.1 Count Statuses

1. Open the supplied **Summary** tab. A1:B4 already contains Status / TaskCount and the three canonical status labels; the counts are blank. Enter **=COUNTIF(Tasks!$F$2:$F$13,A2)** in B2. The fixed range covers the twelve task statuses; A2 supplies the criterion.



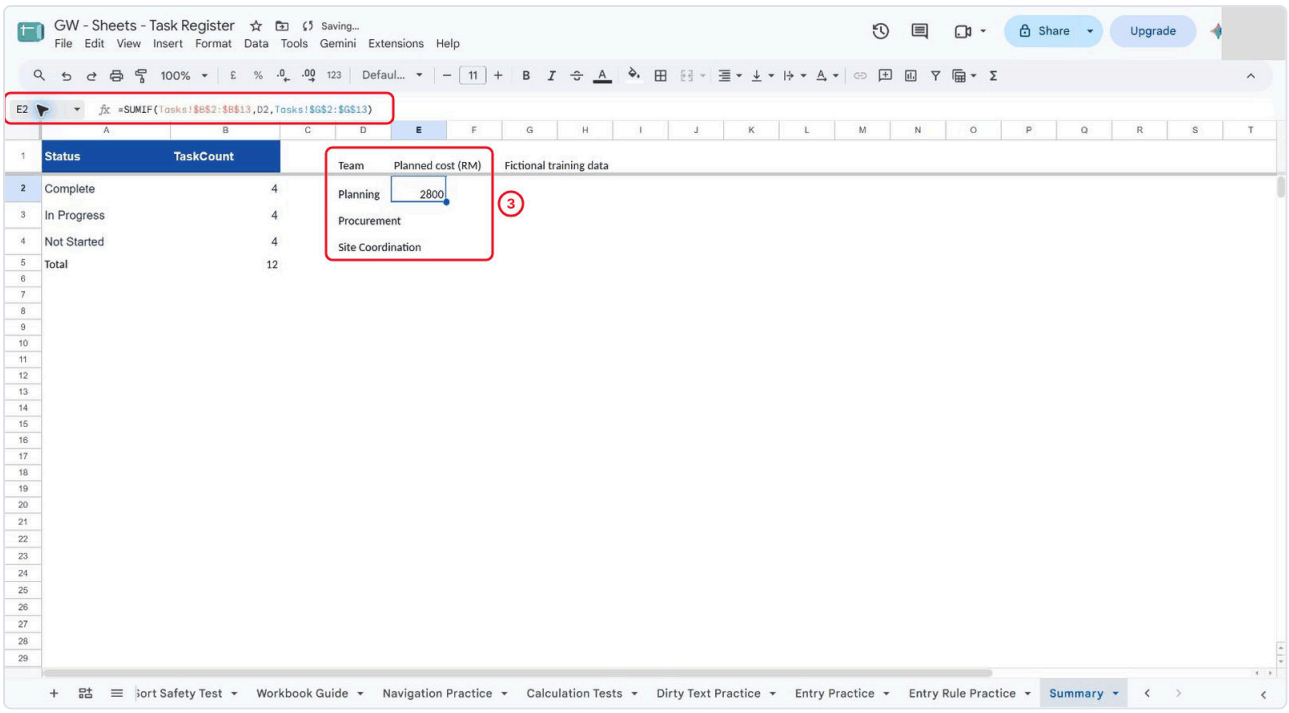
Enter the first anchored COUNTIF.

2. Select B2:B4 and use **Fill down** (Ctrl+D). Inspect B3: the range remains fixed and the criterion becomes A3. Enter **Total** in A5 and **=SUM(B2:B4)** in B5. Verify **4 / 4 / 4**, total **12**.



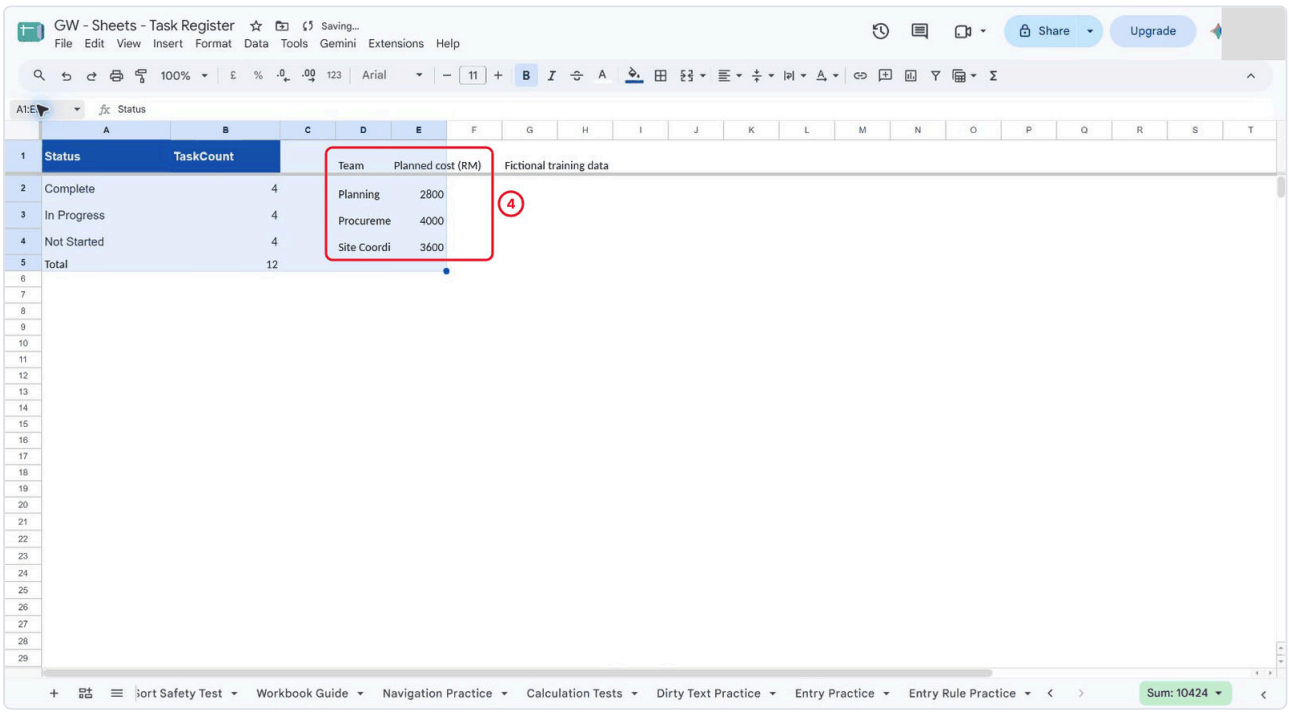
Fill three status counts and total twelve.

3. Move the fictional-data marker from D1 to G1 before using D:E. Enter **Team / Planned cost (RM)** in D1:E1 and **Planning / Procurement / Site Coordination** in D2:D4. In E2 enter **=SUMIF(Tasks!$B$2:$B$13,D2,Tasks!$G$2:$G$13)**.



Enter the first anchored SUMIF.

4. Fill E2 through E4. Verify **2800 / 4000 / 3600**, which sum to **10400**. Fit D:E to the labels so Site Coordination and the RM heading remain readable. These are planned amounts, not actual expenditure or savings.

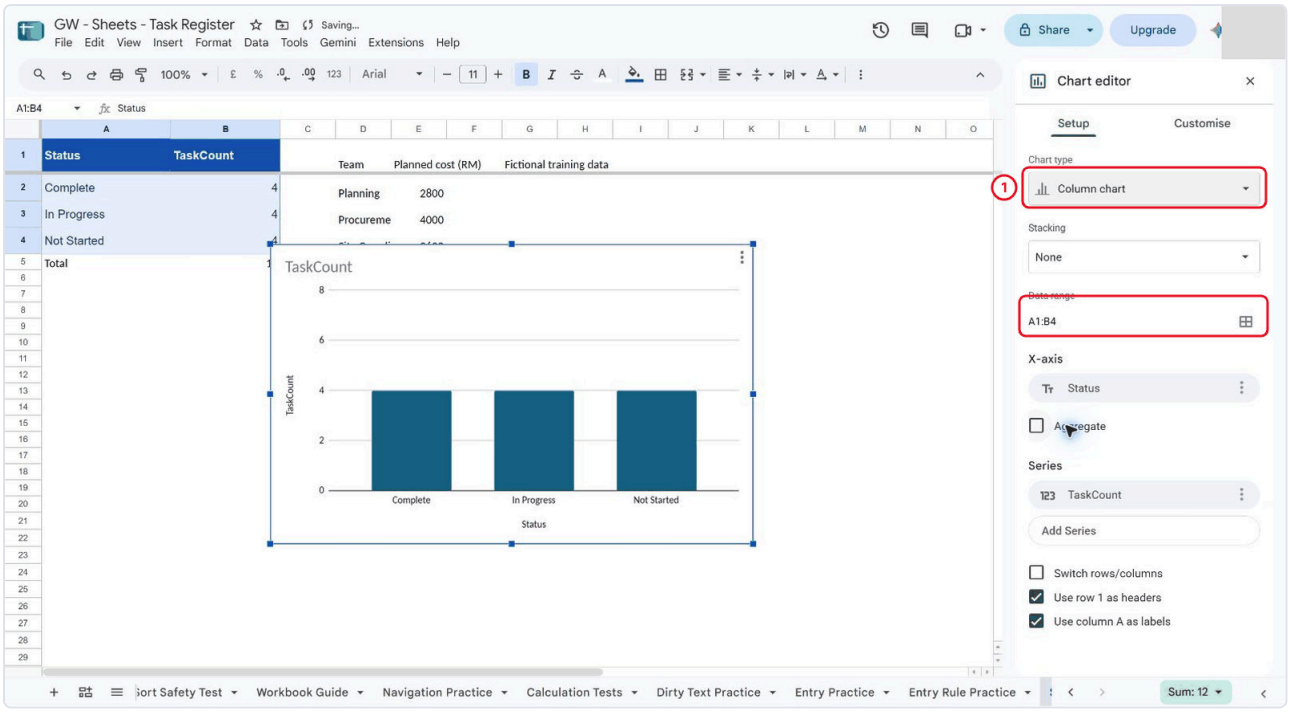


Inspect both conditional summaries.

6.2 Build a Useful Chart

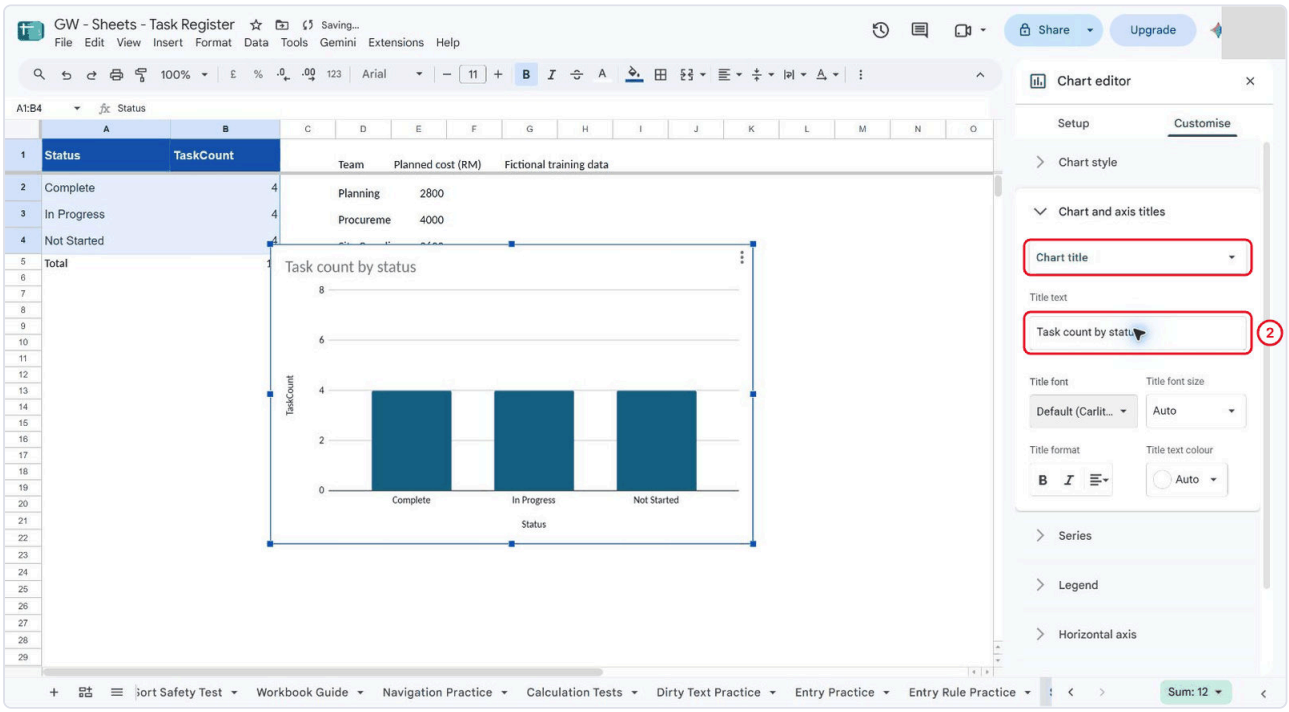
1. Select **A1:B4**, excluding Total, then **Insert > Chart**. This session initially suggested a pie chart; explicitly choose **Column chart** in **Chart editor > Setup**. Confirm **A1:B4**, row 1 headers and column A

labels.



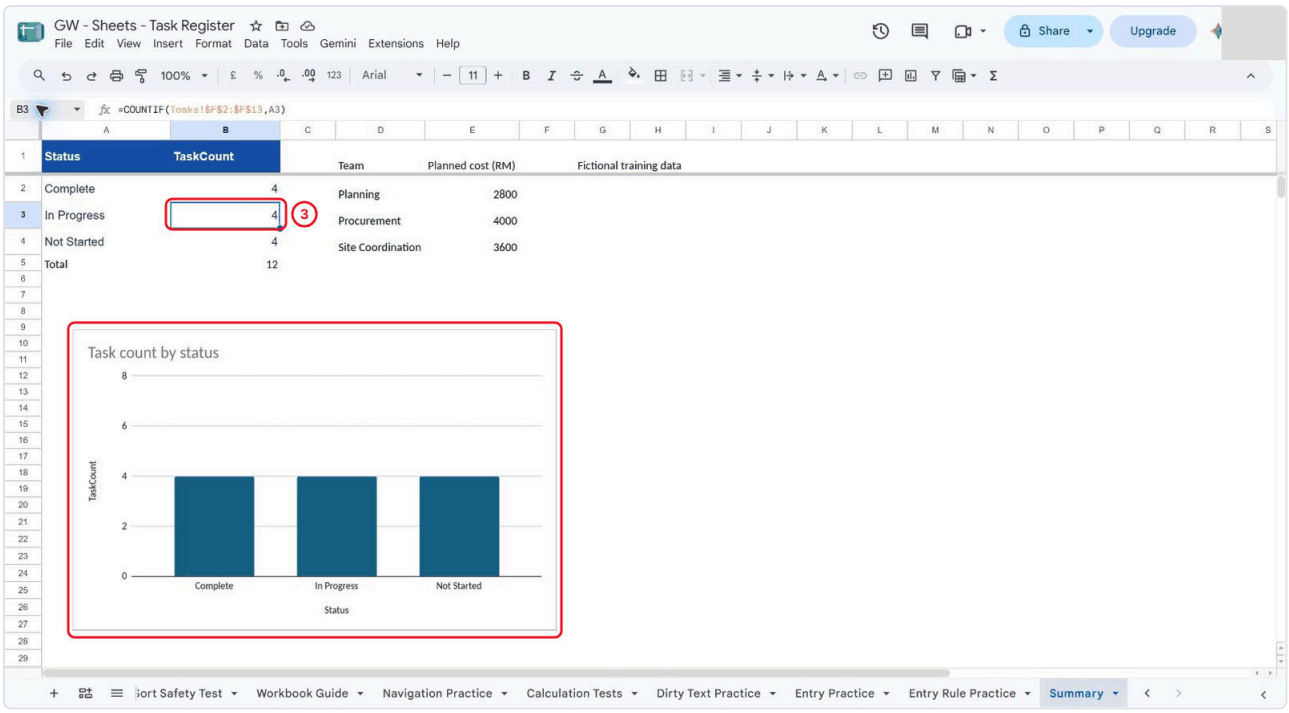
Set column chart and bounded status range.

2. In **Customise > Chart and axis titles**, set the chart title to **Task count by status**. Close the editor. Select the chart once, then drag its blank upper area to place it below the summaries without covering cells.



Set a descriptive status chart title.

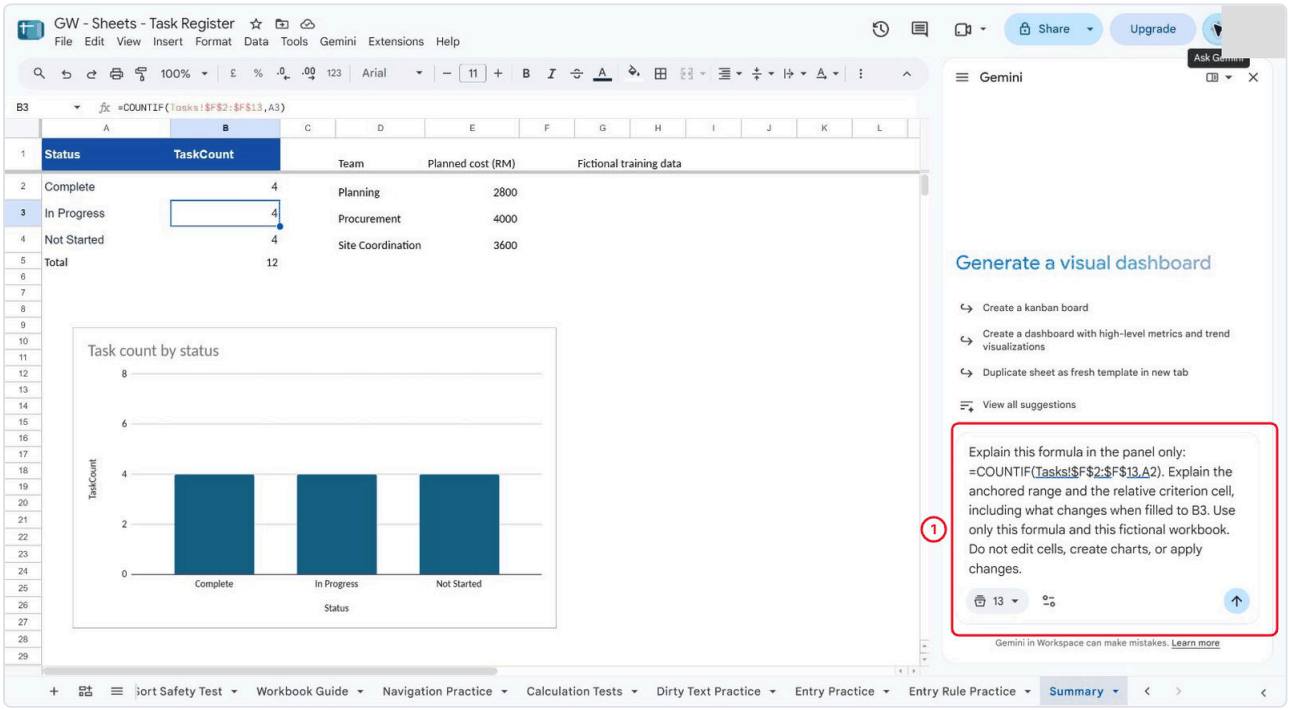
3. Verify exactly three categories, each at **4**. No blank category or Total should be plotted. Inspect B3 in the formula bar alongside the chart; its criterion is A3.



Verify three equal categories and the B3 relative criterion.

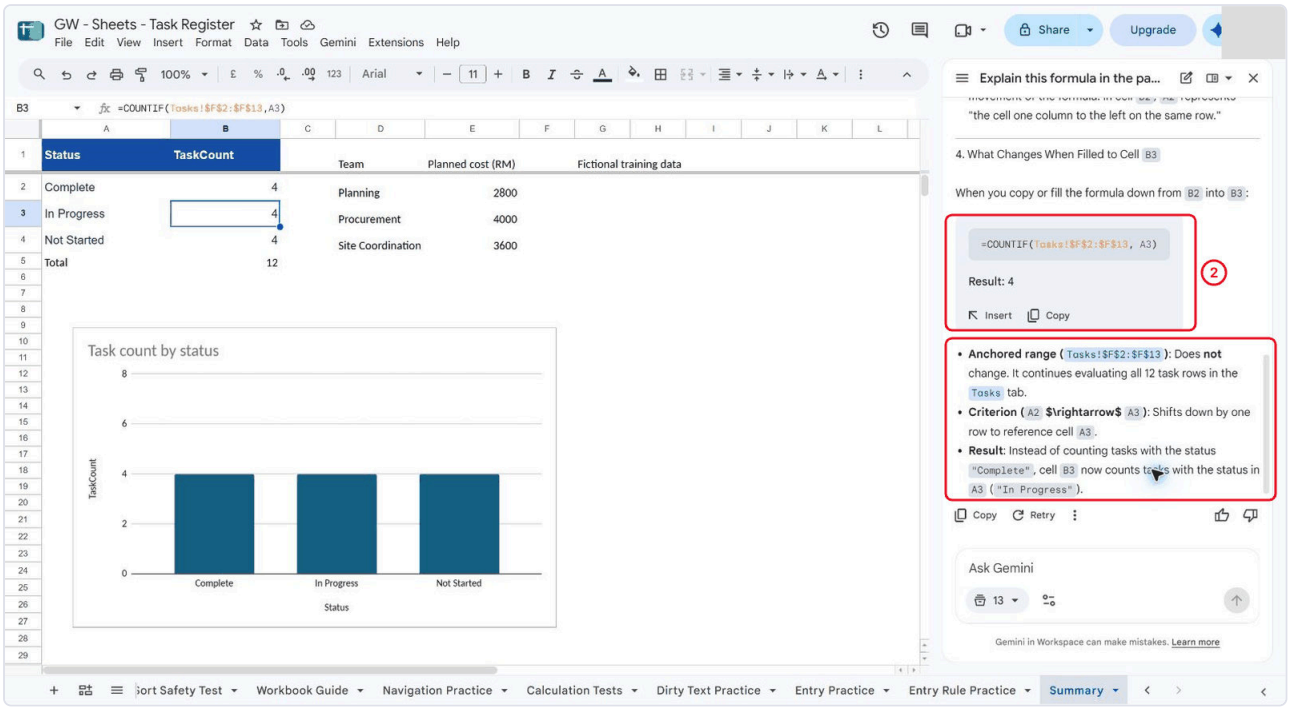
6.3 Explain and Check

1. If available, open **Ask Gemini** and request an explanation of the COUNTIF formula **in the panel only**. Ask about the anchored range, relative criterion and filling to B3. Do not apply Insert or other workbook changes. The exact prompt is in Part 3 of the prompts page. If Gemini is unavailable, explain the same references manually and record the fallback.



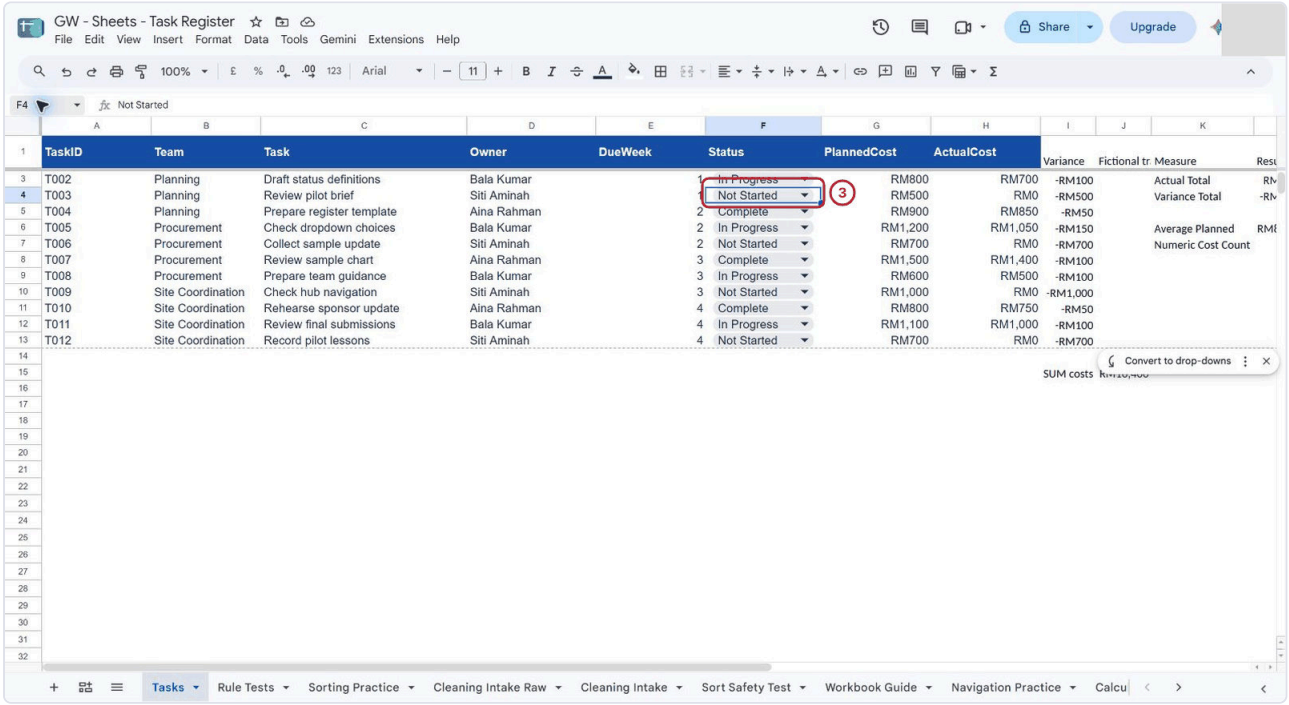
Ask for a panel-only formula explanation.

2. Compare the explanation with actual B3: **=COUNTIF(Tasks!$F$2:$F$13,A3)**. The twelve-row range stays fixed; the criterion shifts from A2 to A3. In this session Gemini showed the same B3 formula and result 4. Its prose explanation is supporting material, not proof that the workbook is correct. Close the panel.



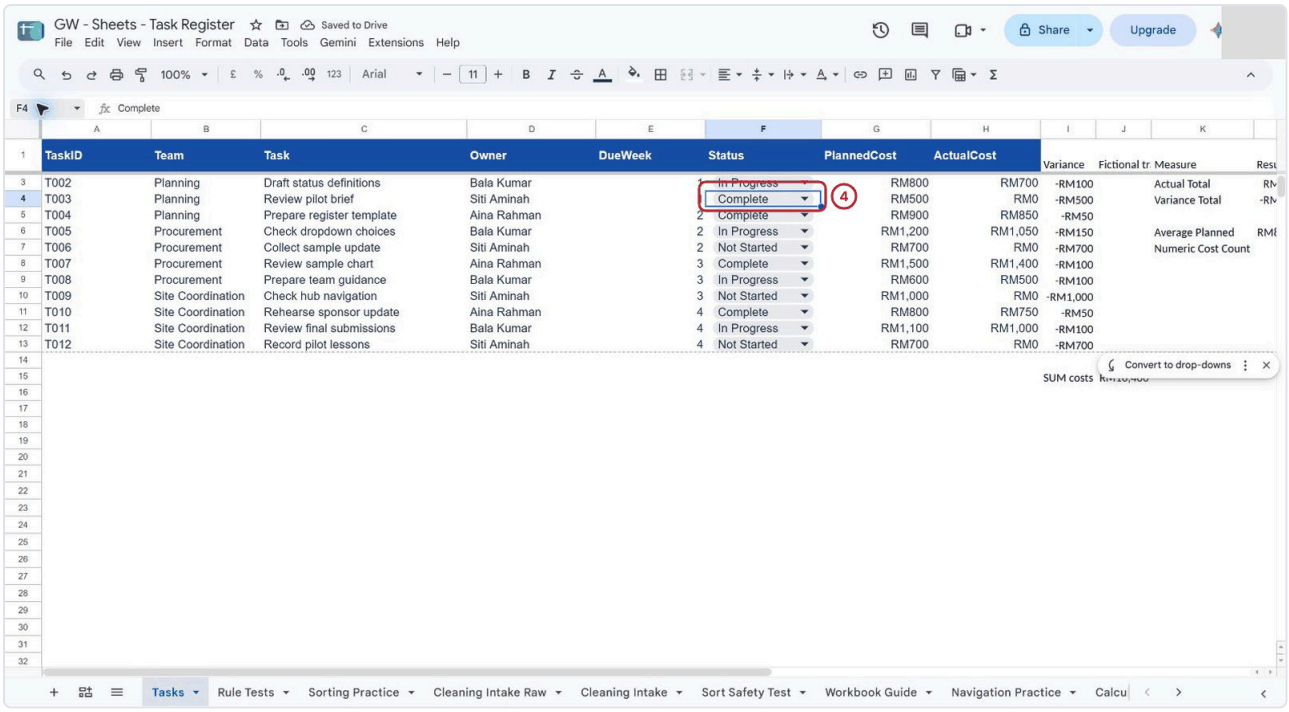
Compare Gemini explanation with the actual B3 formula.

3. On Tasks, identify **T003** in row 4 and record **F4 = Not Started** before editing. Do not change a different task or an entire status column.



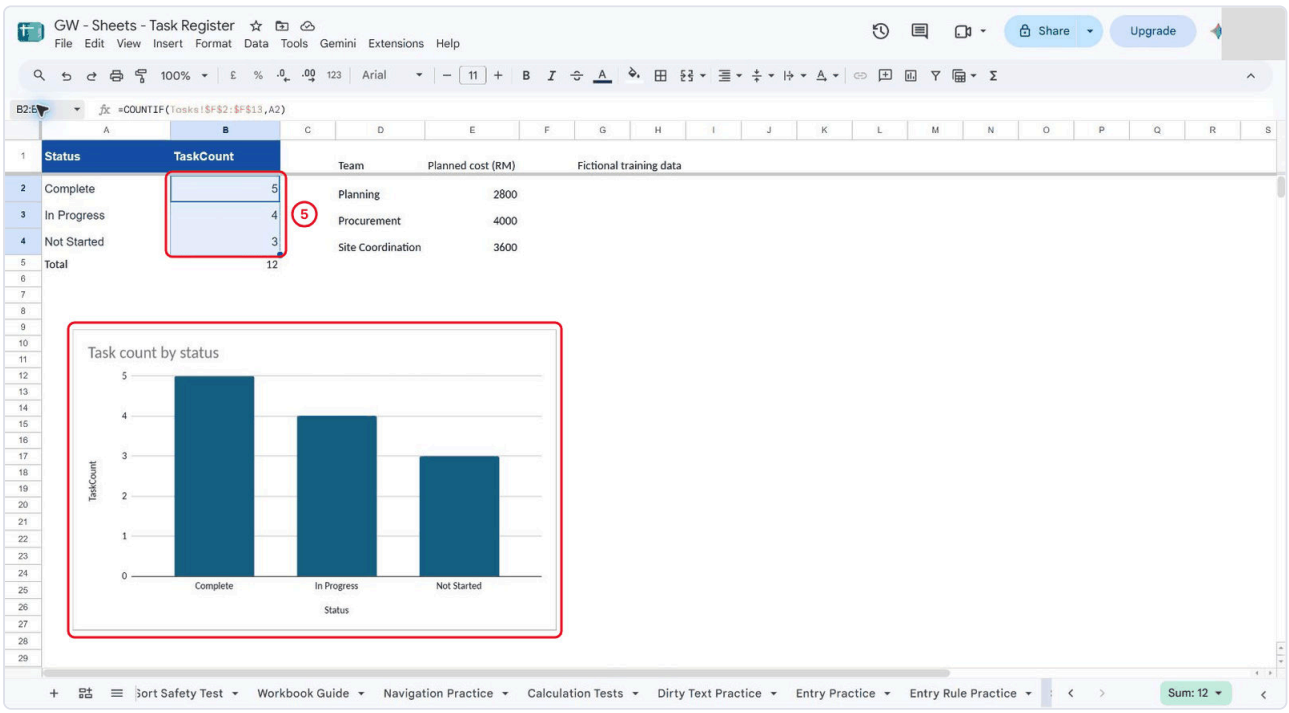
Record T003 original Not Started status.

4. Temporarily edit F4 to **Complete**, preserving its existing dropdown rule. Commit the cell edit.



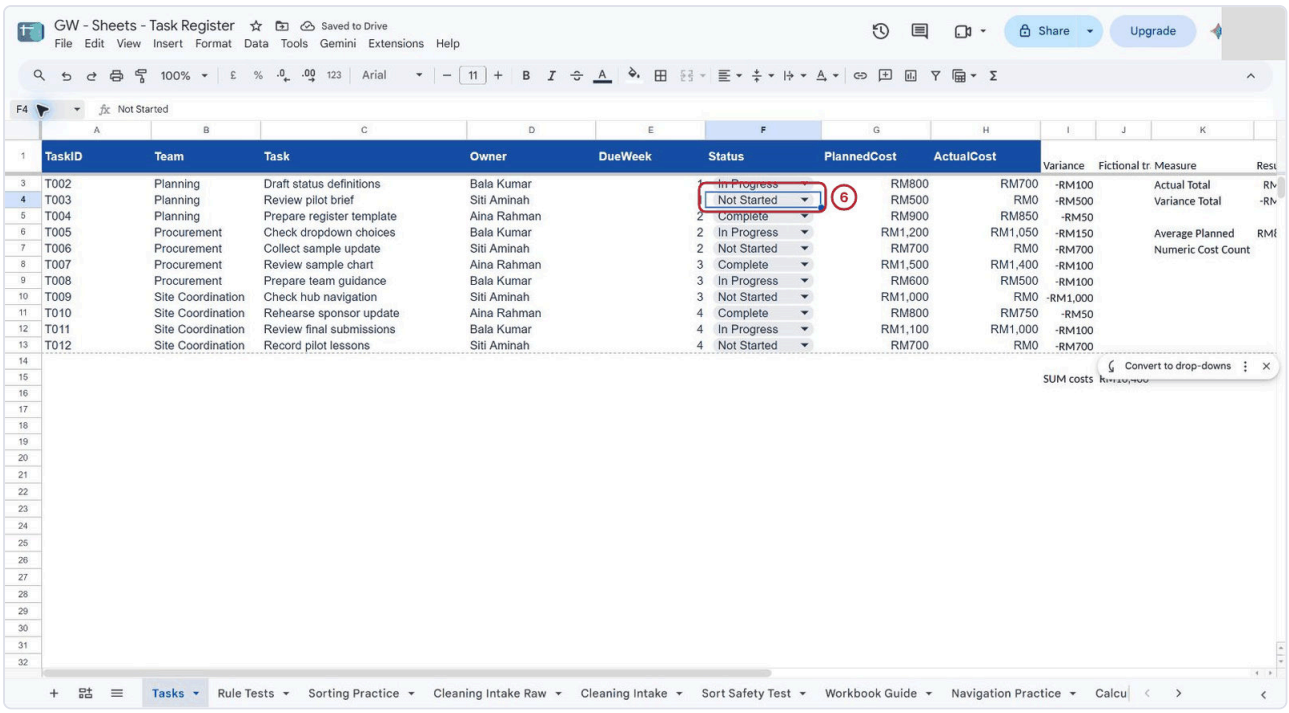
Temporarily set T003 Complete.

5. Return to Summary. Verify **Complete 5 / In Progress 4 / Not Started 3**, still total **12**, and the status chart changes with the source.



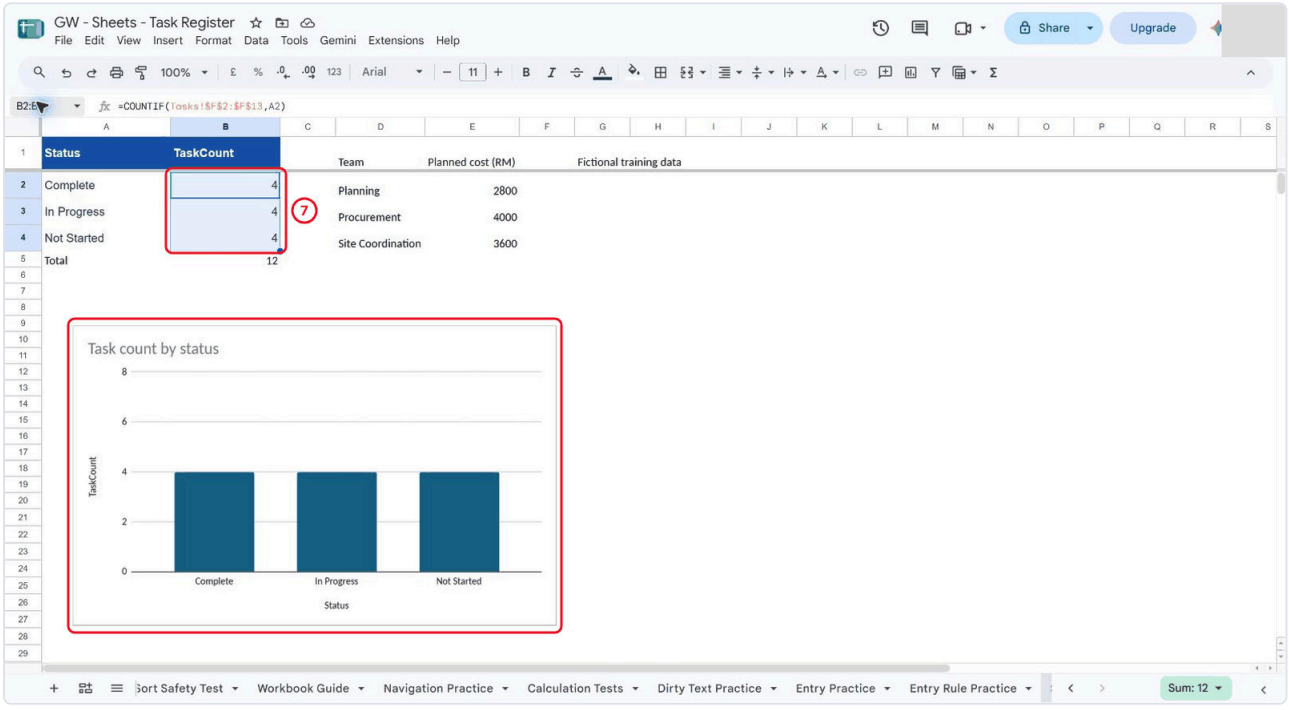
Verify counts and chart change to five four three.

6. Restore Tasks F4 to **Not Started** immediately.



Restore T003 Not Started.

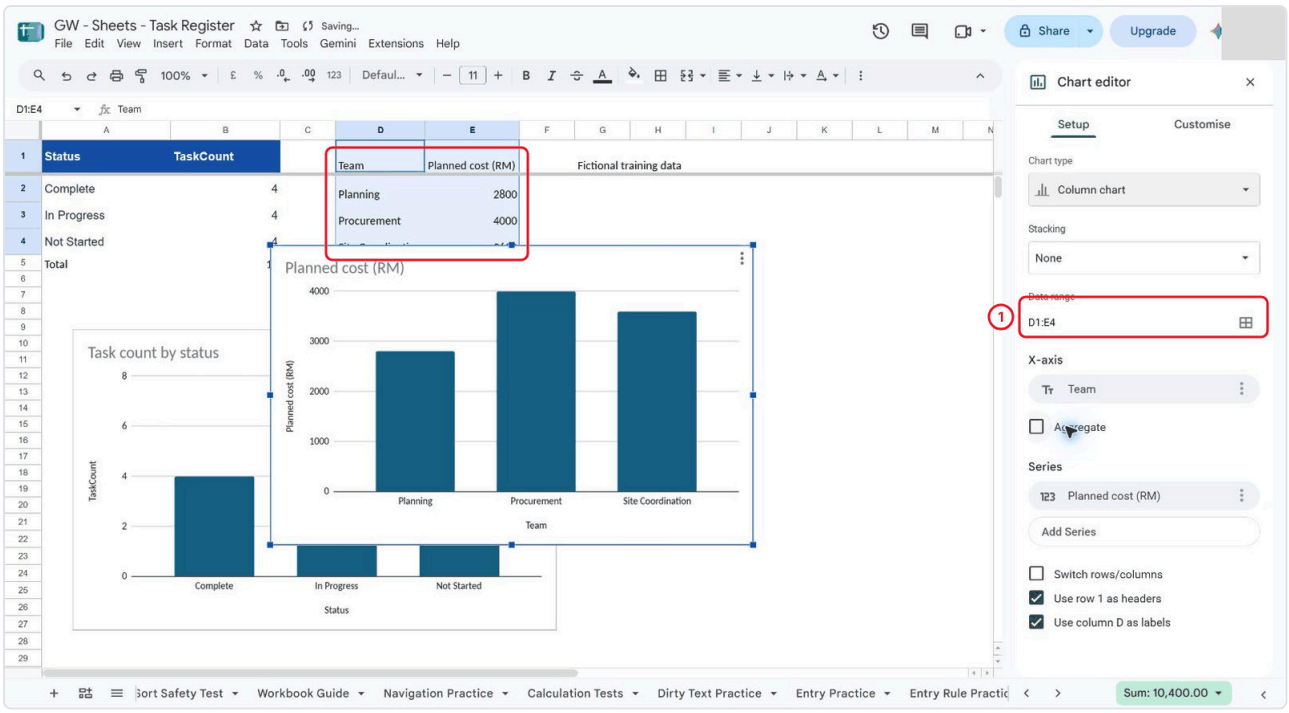
7. Return to Summary and verify **4 / 4 / 4** and the chart's three equal bars again. Both source and visual restoration matter.



Verify four four four baseline returns.

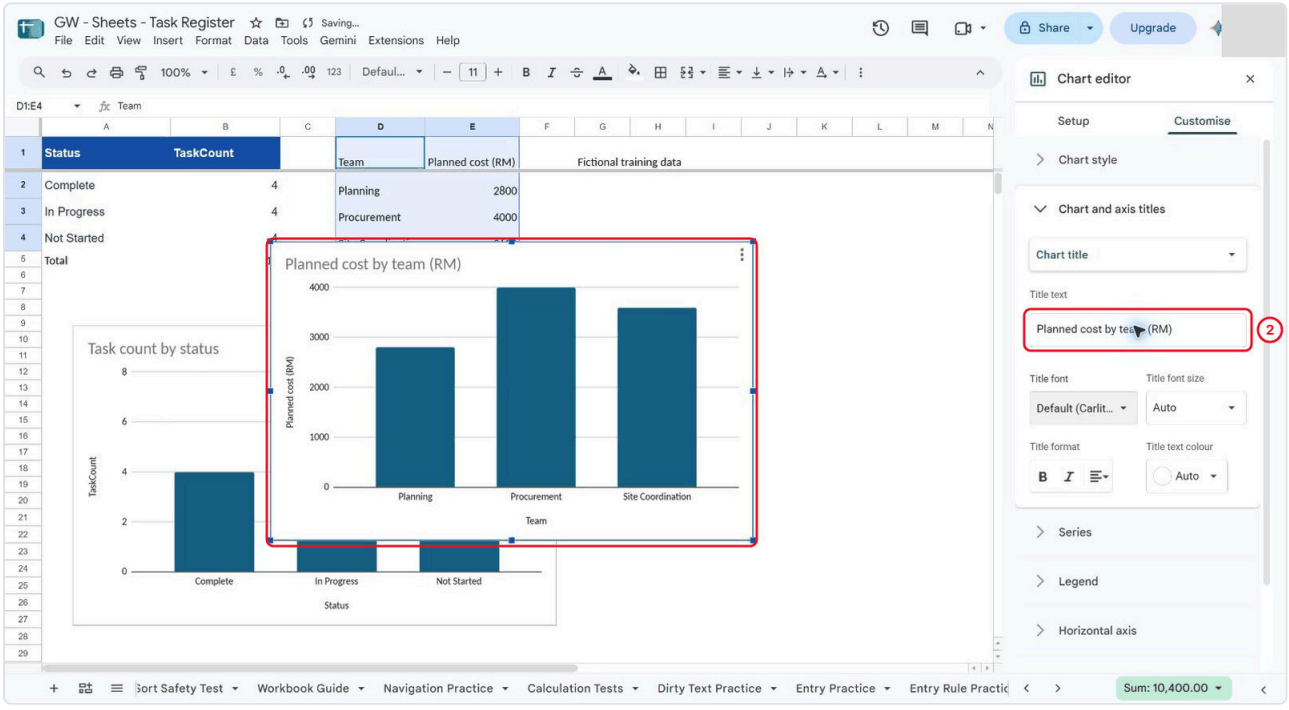
6.4 Build a Second Summary and Diagnose a Misleading Chart

1. Select **D1:E4** and insert another column chart. Confirm its range and category labels.



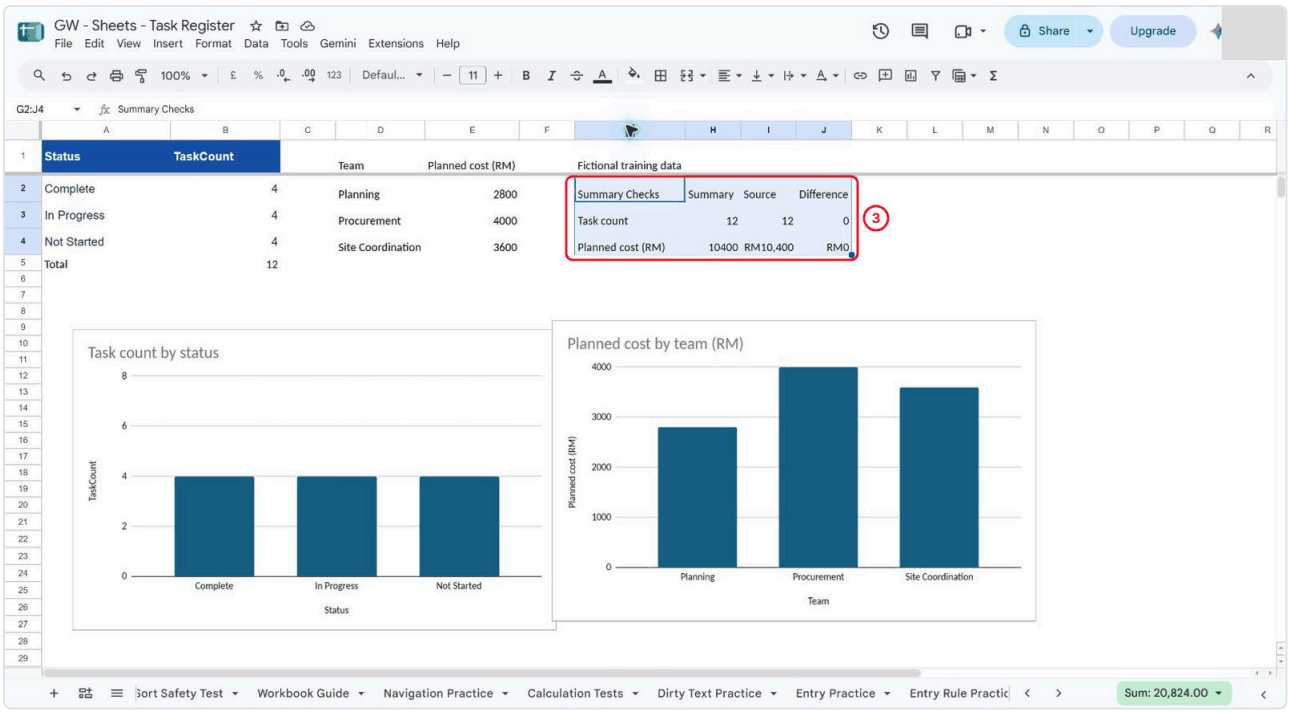
Build the team cost chart from D1:E4.

2. Title it **Planned cost by team (RM)**. The series/vertical axis reads **Planned cost (RM)**. Place it beside the status chart; keep currency amounts separate from task counts.



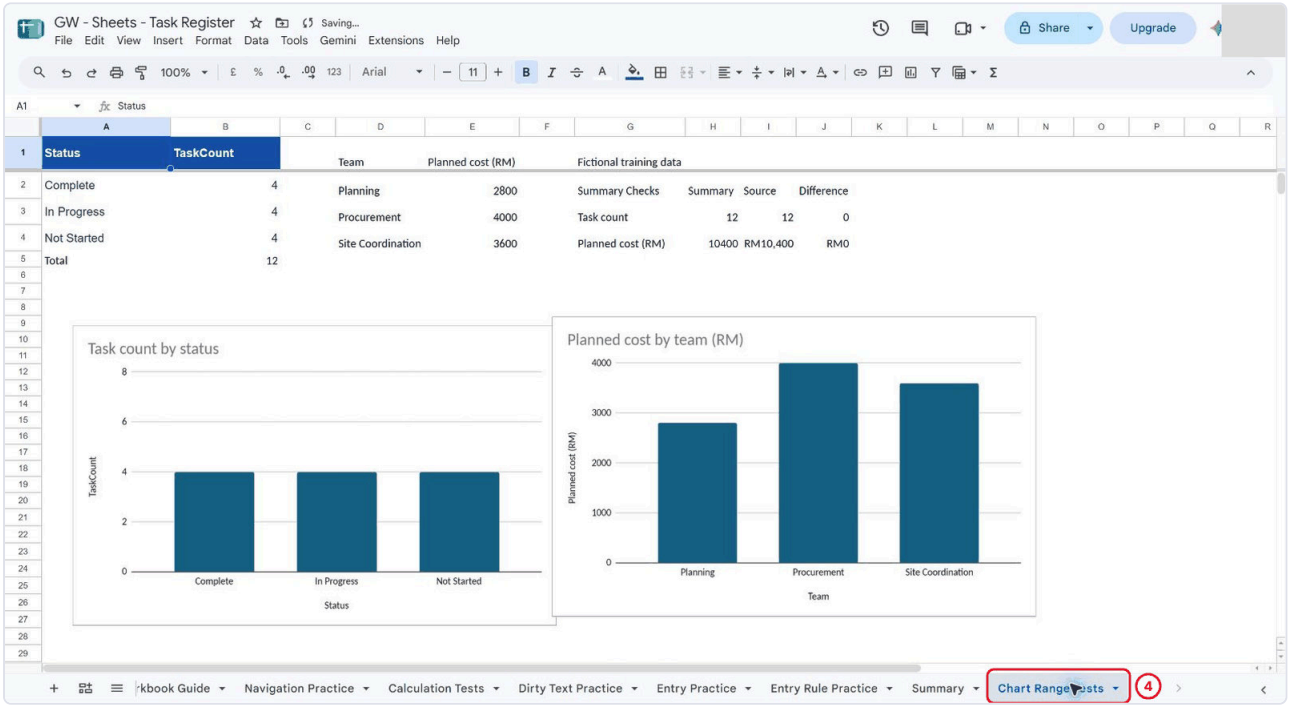
Label planned cost units as RM.

3. In G2:J4 create **Summary Checks / Summary / Source / Difference** (the block is in Part 4 of the prompts page). Row 3 is **Task count**, with H3 **=B5**, I3 **=COUNTA(Tasks!A2:A13)**, J3 **=H3-I3**. Row 4 is **Planned cost (RM)**, with H4 **=SUM(E2:E4)**, I4 **=SUM(Tasks!G2:G13)**, J4 **=H4-I4**. Expect **12 versus 12** and **10400 versus 10400**, both differences zero. Fit G so its labels are readable.



Reconcile summary and source totals.

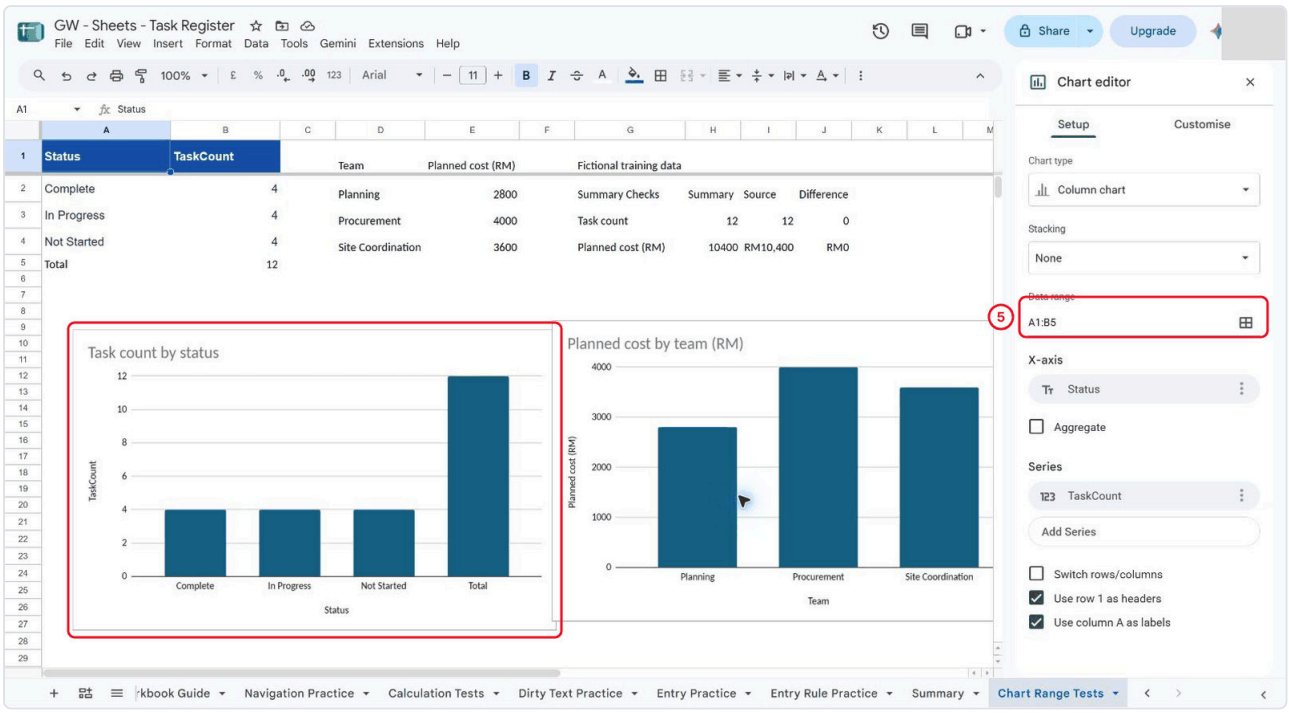
4. Duplicate Summary and name the copy **Chart Range Tests**. Keep baseline Summary intact for the fault tests. Its copied formulas still refer to Tasks.



Create a named chart fault-test copy.

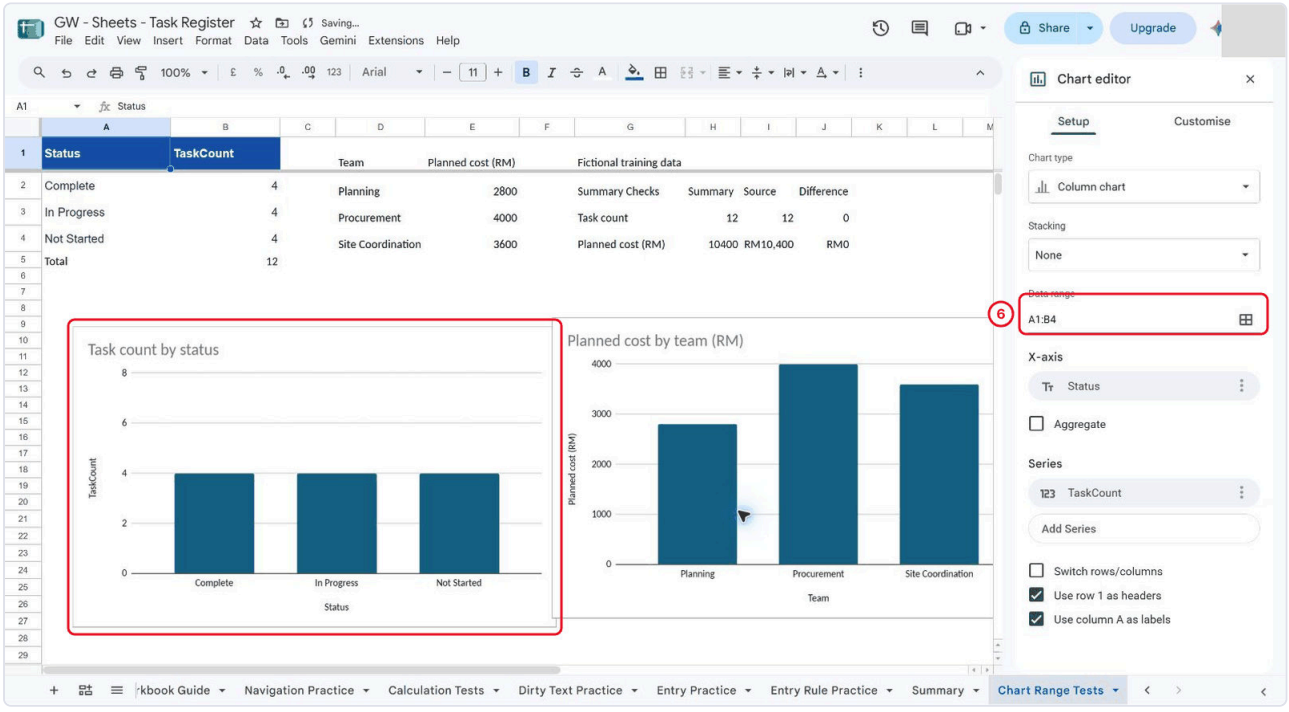
5. Edit the copied status chart in **Setup**. Use **Select data range**, set the single range **A1:B5**, and confirm **OK**. Verify the chart actually displays a fourth **Total = 12** category. If multiple ranges are listed, retain only the intended single range. The categories now represent 4+4+4+12=24 plotted counts even though only twelve tasks exist; Total duplicates the population already split into statuses.

> **Reference run:** The reference needed the range dialog: typing in the compact range field alone did not update the underlying category/series ranges.



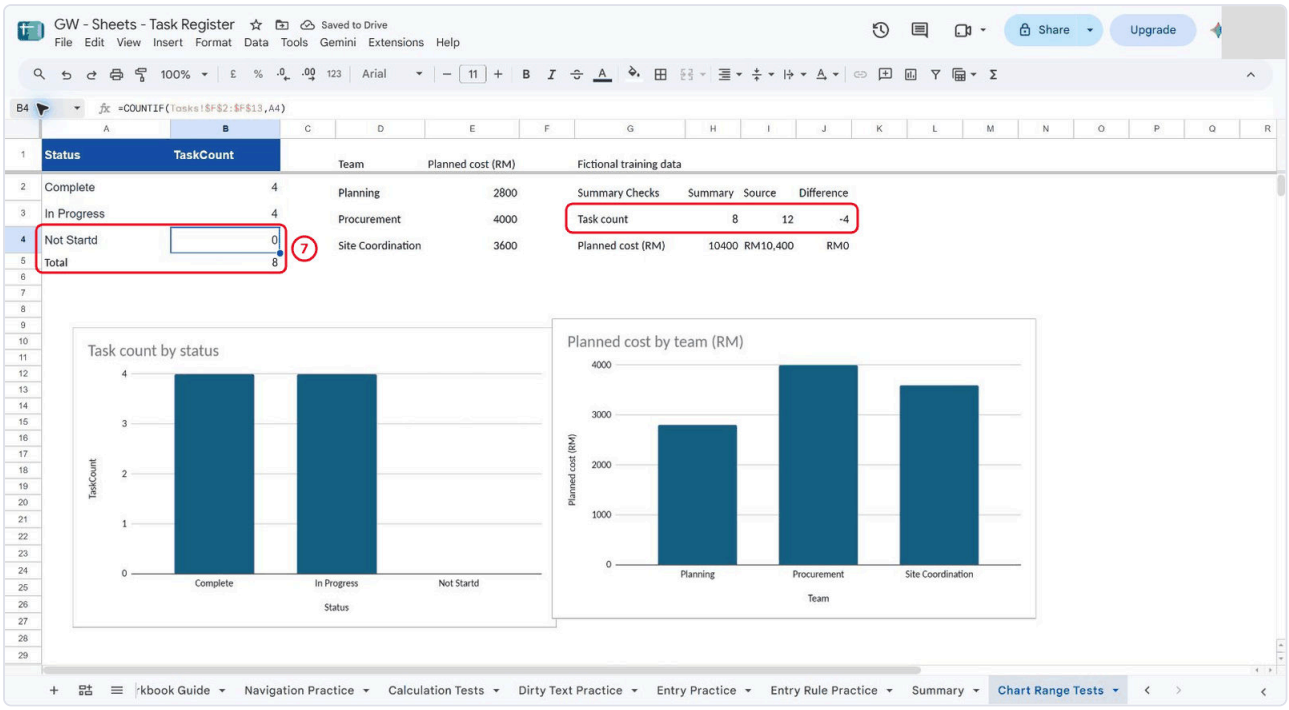
Observe Total incorrectly plotted as a fourth category.

6. Use the same range dialog to restore **A1:B4**. Verify three bars at four; changing the title or hiding a label would not repair the source selection.



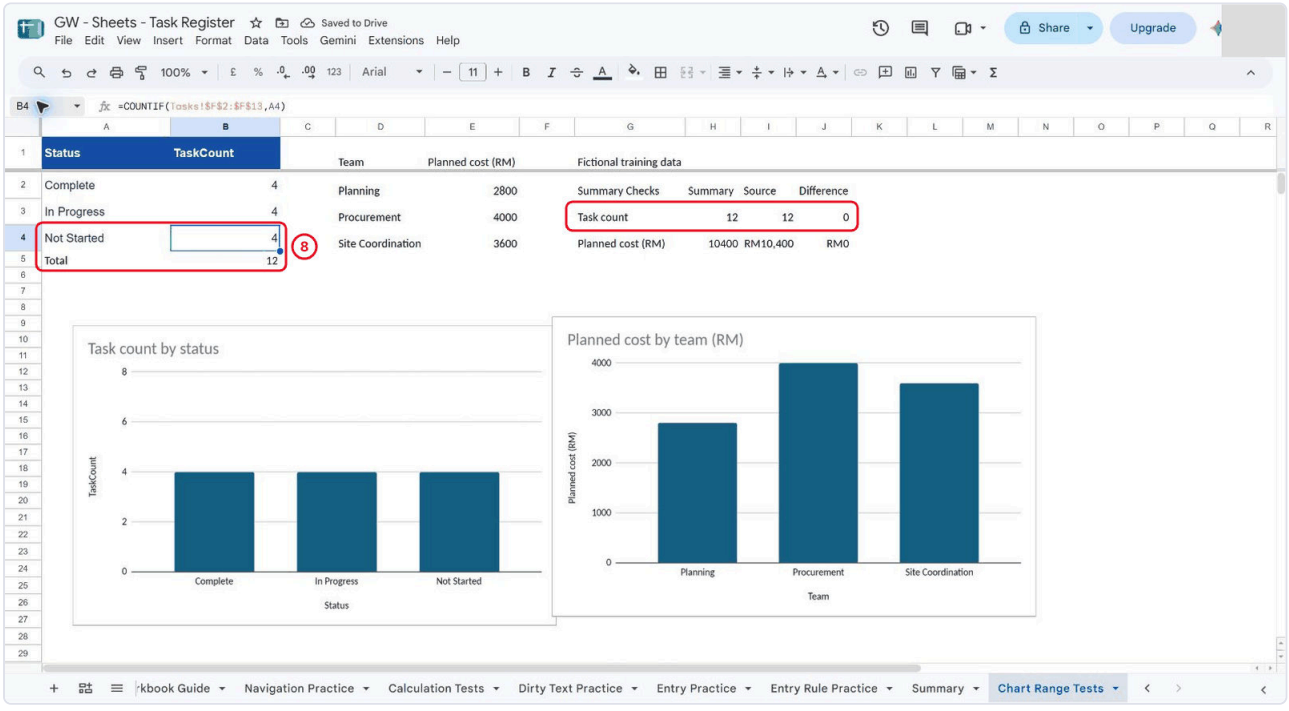
Repair chart range to exclude Total.

7. On **Chart Range Tests** only, temporarily misspell A4 as **Not Startd**. B4's COUNTIF becomes **0**, the category's bar has zero height, B5 falls to **8**, and J3 reports **-4** against twelve source TaskIDs. The source tasks are still present; the criterion is wrong.



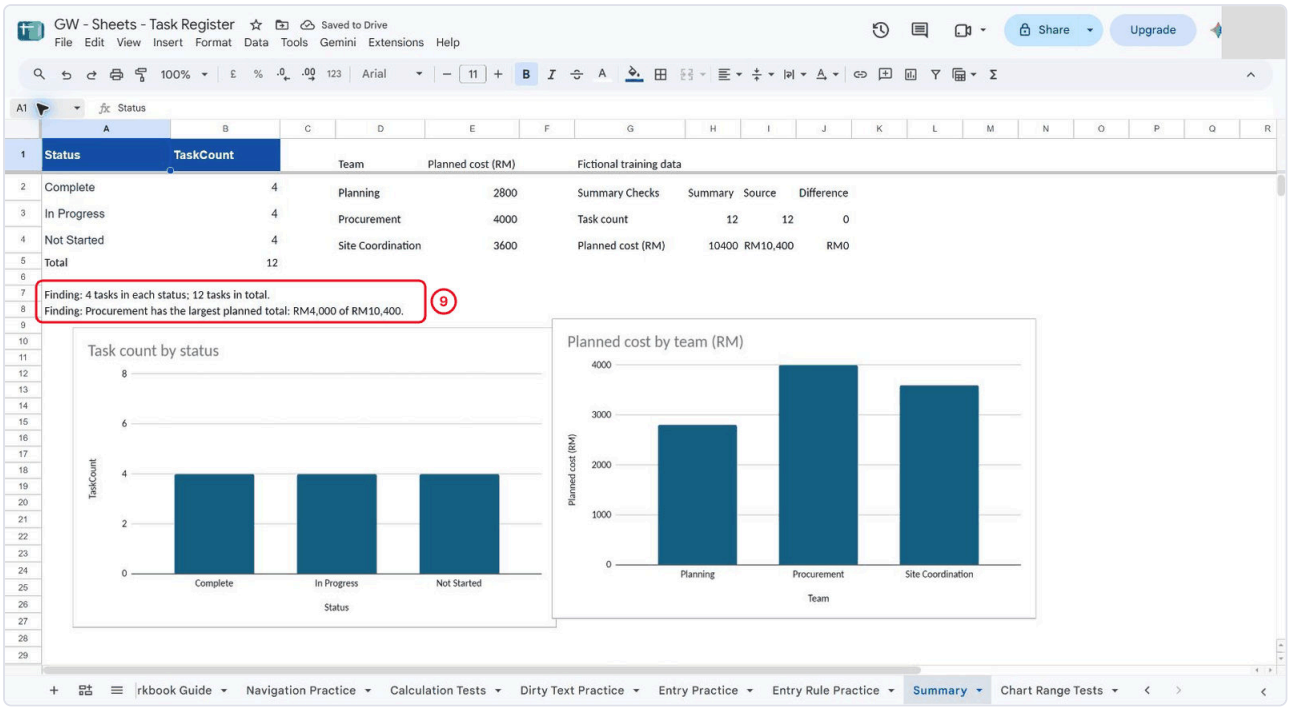
Observe misspelt status produce zero and a reconciliation difference.

8. Repair A4 to **Not Started**. B4 returns to **4**, total **12**, difference zero, and the chart returns to baseline. Retain the formula; never type a replacement count over it.



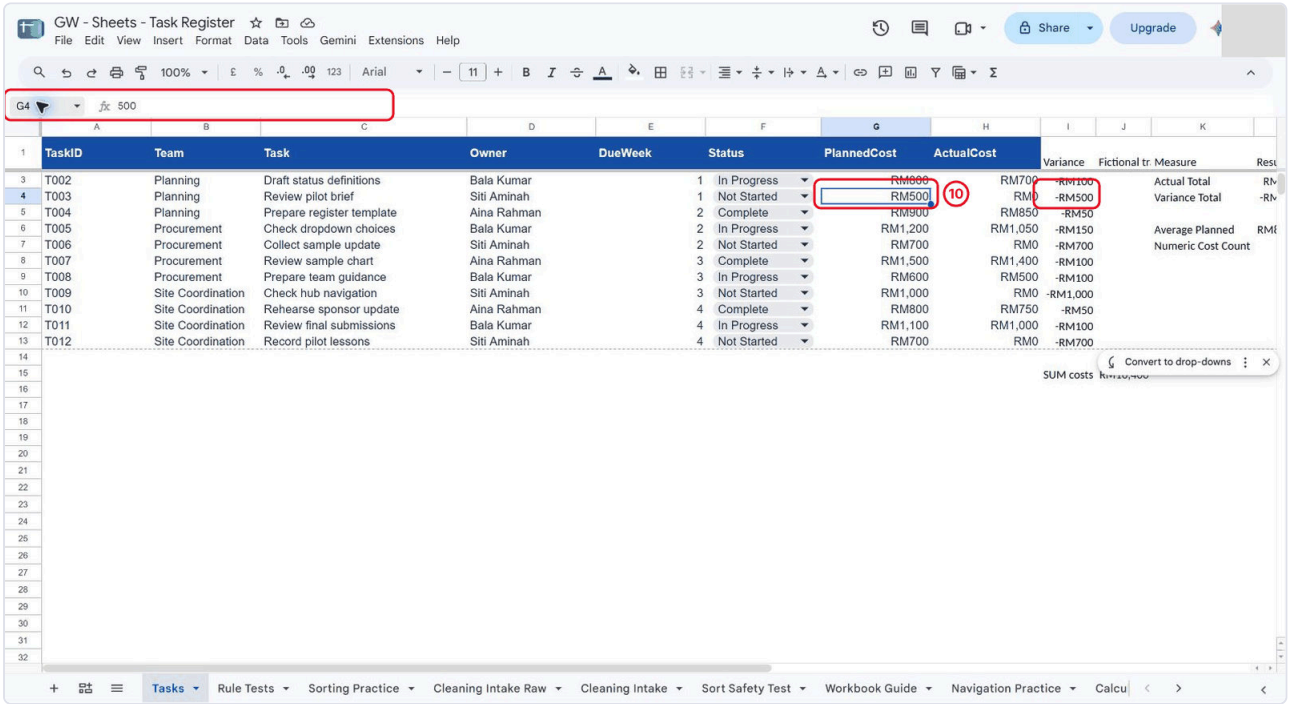
Repair criterion spelling and retain COUNTIF.

9. On baseline Summary, type two supported findings in A7 and A8: **4 tasks in each status; 12 in total**, and **Procurement has the largest planned total: RM4,000 of RM10,400**. Neither finding establishes on-time delivery or realised savings.



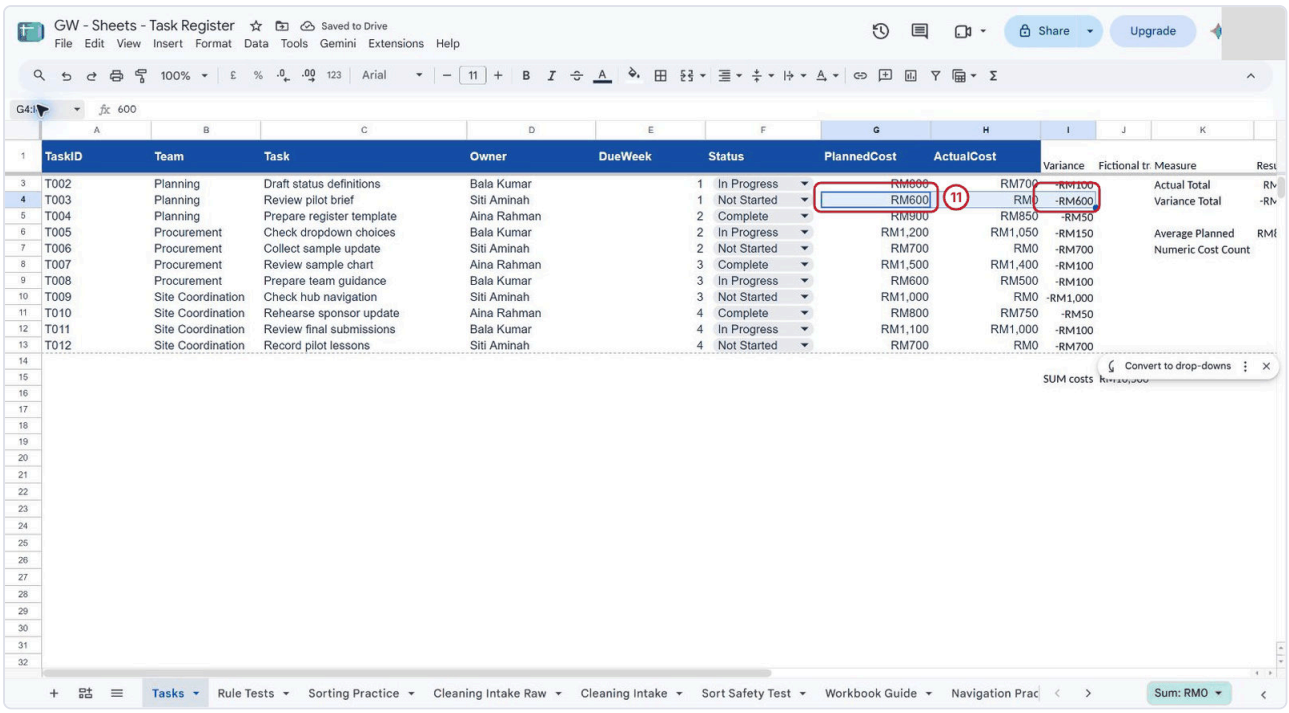
Write two evidence-supported findings.

10. For the cost propagation test, record original **Tasks G4 = 500** for T003. H4 is zero; I4's variance formula gives **-500**.



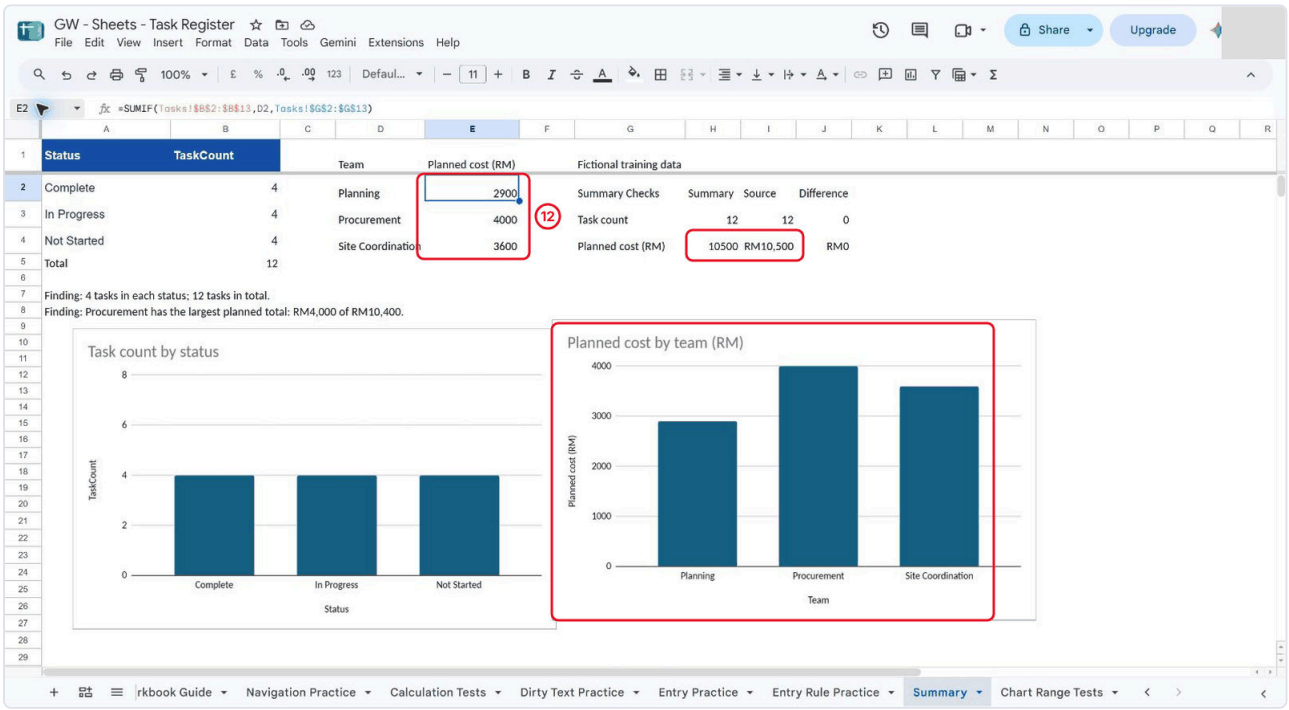
Record T003 original planned cost five hundred.

11. Temporarily edit G4 to **600**. Verify the row's I4 becomes **-600**; do not overwrite the variance formula.



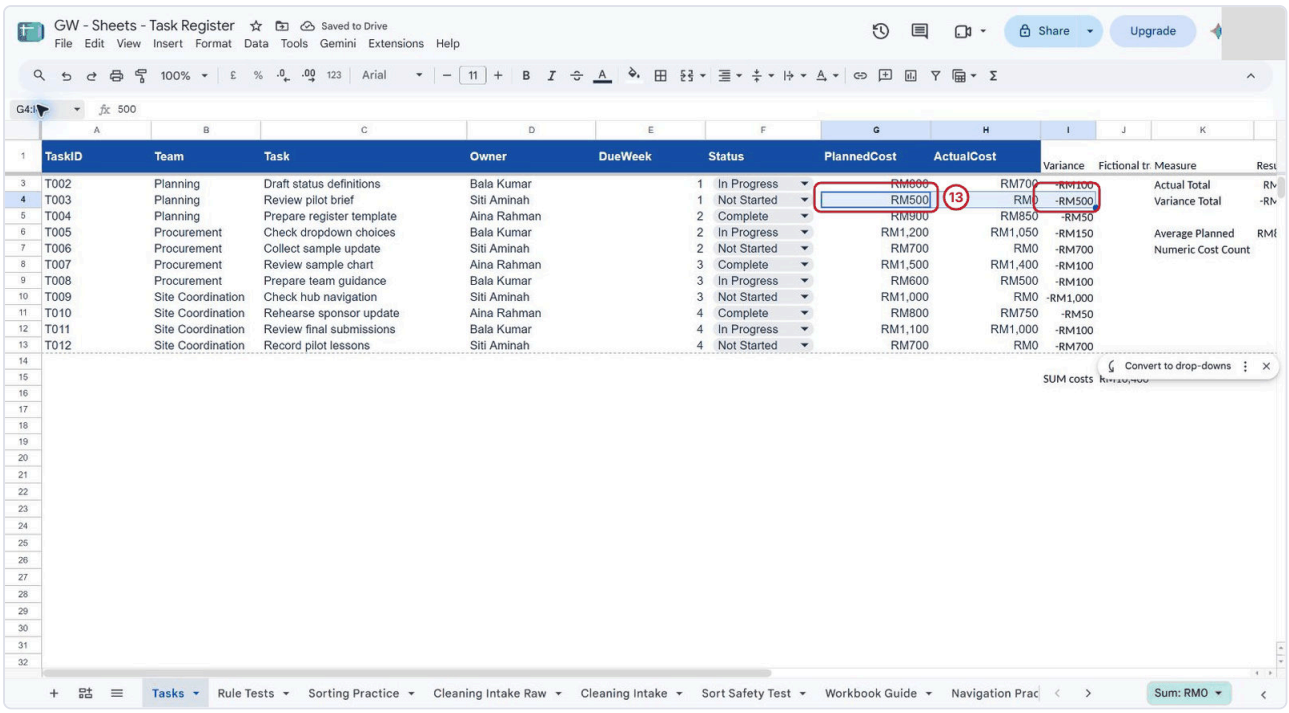
Temporarily increase planned cost by one hundred.

12. On Summary, Planning becomes **2900**, Procurement remains **4000**, Site Coordination remains **3600**, and both planned grand totals become **10500** with difference zero. Inspect the Planning chart bar. The baseline findings describe the restored baseline; do not treat them as automatically refreshed narrative.



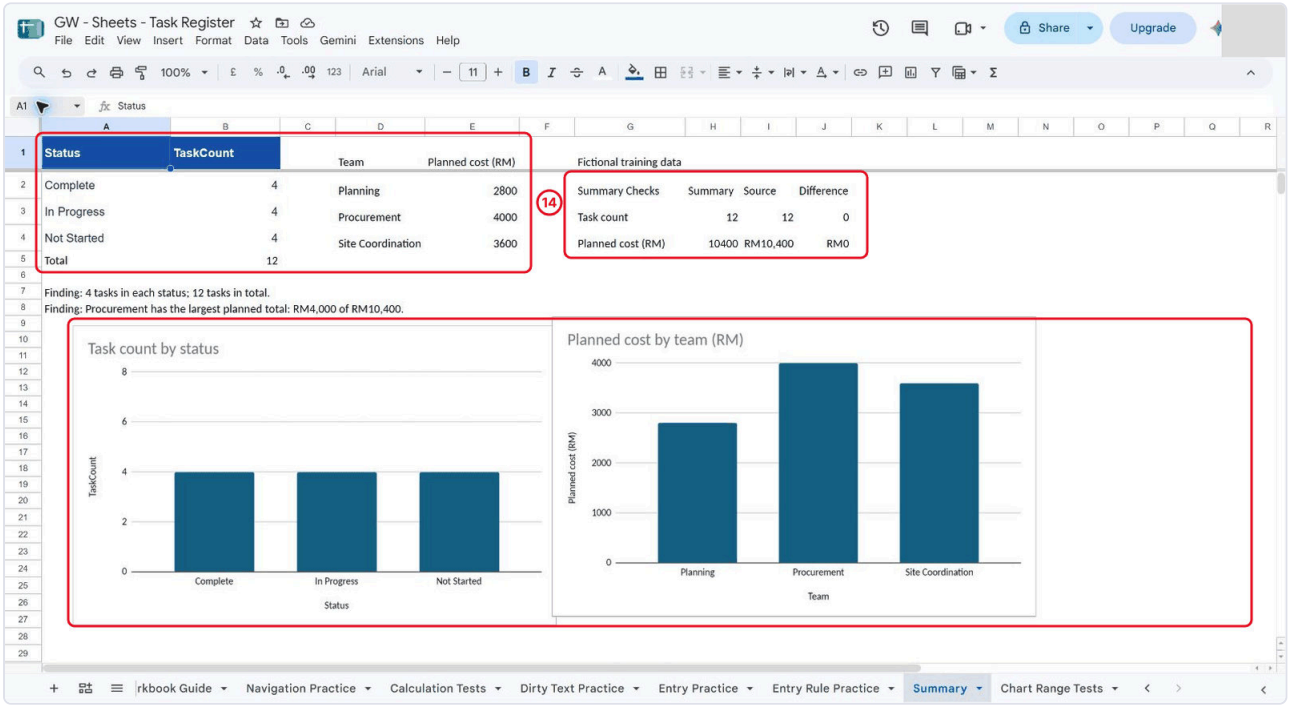
Verify Planning and grand totals and chart update.

13. Restore Tasks **G4 to 500**. Verify I4 returns to **-500** and the source planned total to **10400**.



Restore original planned cost.

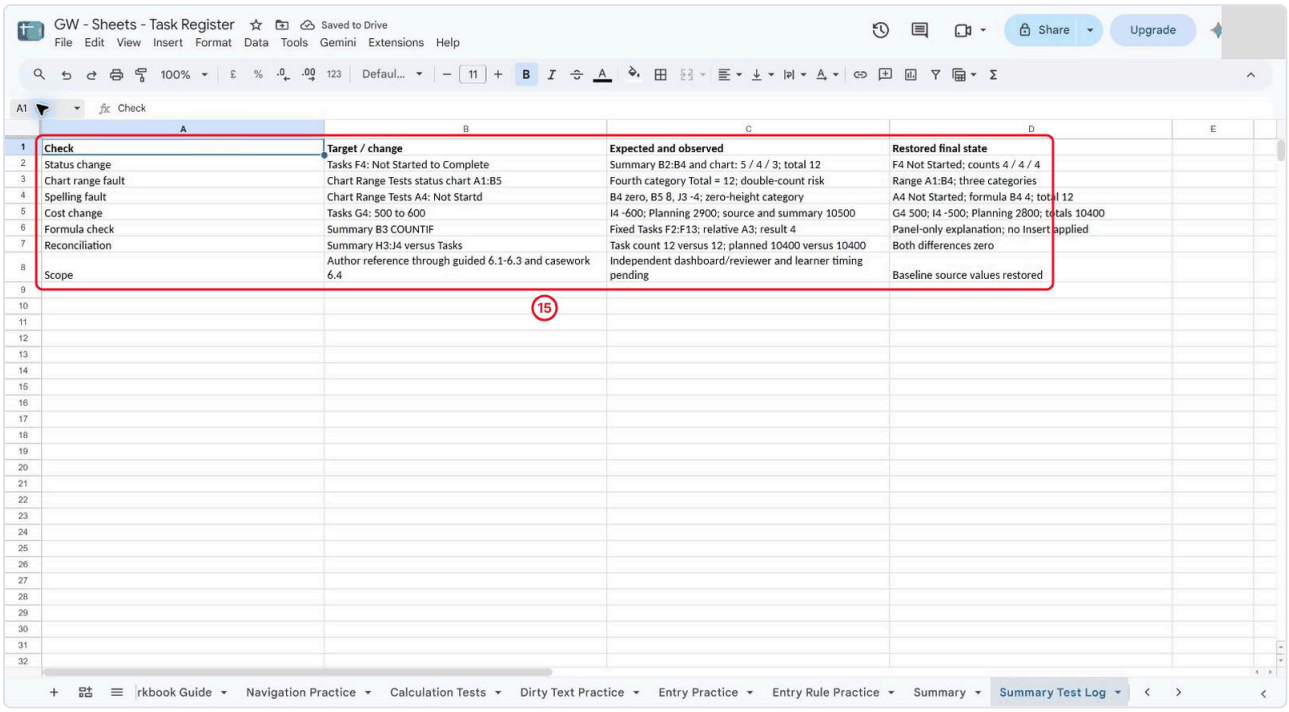
14. Verify Summary team totals **2800 / 4000 / 3600**, planned totals **10400**, status counts **4 / 4 / 4**, and zero reconciliation differences. Both charts and the baseline findings agree again.



Verify both baseline charts and zero differences.

15. Add a sheet named **Summary Test Log** with the headers **Check / Target / change / Expected and observed / Restored final state** and the row labels in Part 4 of the prompts page. Record the actual status, range, spelling, cost, formula and reconciliation checks. Wrap the log text and use readable

column widths. Record the independent challenge and learner timing as pending rather than claiming them from these guided author checks.



Record actual tests and restorations.

> **Validation boundary:** author UI reference through guided 6.1–6.3 and casework 6.4. All original Tasks A1:N15 values/formulas are restored; new summaries, charts, named copy and test log are retained. Independent dashboard, reviewer/fresh attempt and learner timing remain pending. Earlier CSV-import and values-only-format limitations remain separate.

Independent Practice

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Prepare an independent one-page dashboard on a new tab with two labelled charts, source links and two concise supported findings.
2. Diagnose the case input's (Part 2 of the prompts page) suspicious chart and summary symptoms using the source cells. Repair the cause, not just the visual appearance.
3. Use a reviewer or fresh read-through to check units, population and total reconciliation, then record one correction or evidence for a no-change decision.

Evidence to Hand Over

- Two reconciled summaries/charts with distinct units.
- Documented total-row and misspelt-category fault repairs.
- Independent dashboard, supported findings and restored source-change test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A missing count is hidden by typing a result	Repair the criterion or formula so later source changes still recalculate.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A checked summary and chart communicate the register accurately.

Check yourself: Why should the total row normally be excluded from this chart's categories?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=COUNTIF(Tasks!$F$2:$F$13,A2)
```

```
=SUM(B2:B4)
```

```
Task count by status
```

Part 2: Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Troubleshooting card: A status chart shows Complete, In Progress, Not Started and Total.
Another summary says 4, 4 and 0 even though 12 task rows remain. Investigate source range and
exact criterion spelling before altering the chart design.
```

Part 3: Captured Conditional Summary Reference

```
=SUMIF(Tasks!$B$2:$B$13,D2,Tasks!$G$2:$G$13)
```

Gemini prompt (panel explanation only):

```
Explain this formula in the panel only: =COUNTIF(Tasks!$F$2:$F$13,A2). Explain the anchored
range and the relative criterion cell, including what changes when filled to B3. Use only this
formula and this fictional workbook. Do not edit cells, create charts, or apply changes.
```

Baseline: status 4/4/4, total 12; team planned 2800/4000/3600, grand total 10400. Summary B2:B4 uses COUNTIF with fixed Tasks F2:F13 and relative A2/A3/A4; E2:E4 uses SUMIF with fixed Tasks B2:B13 and G2:G13 and relative D2/D3/D4.

Summary Checks G2:J4: Task count H3=B5, I3=COUNTA(Tasks!A2:A13), J3=H3-I3; Planned cost (RM) H4=SUM(E2:E4), I4=SUM(Tasks!G2:G13), J4=H4-I4. Both differences zero.

Chart titles: Task count by status (A1:B4), Planned cost by team (RM) (D1:E4). On Chart Range Tests, A1:B5 deliberately plots Total 12 as a fourth category; repair to A1:B4 using Select data range and OK. Misspell A4 Not Startd: COUNTIF 0, total 8, difference-4; repair label to Not Started, retaining the formula.

Temporary source tests: Tasks F4 Not Started→Complete gives 5/4/3, then restore Not Started. Tasks G4 500→600 gives I4-600, Planning 2900 and both planned totals 10500; restore 500 gives I4-500, Planning 2800 and 10400. Source values/formulas restored. Summary Test Log retains observations. Independent dashboard/reviewer and learner timing pending.

Part 4: Check Blocks

Columns are separated by tabs, so you can paste each block into its first cell, then add the formulas the chapter gives.

Summary Checks, G2:J4 (section 6.4, step 3). Put the formulas in H3:J4:

```
Summary Checks  Summary Source  Difference
Task count
Planned cost (RM)
```

Summary Test Log, A1:D8 (section 6.4, step 15). Record what you changed, expected, observed and restored in B:D:

```
Check   Target / change Expected and observed   Restored final state
Status change
Chart range fault
Spelling fault
Cost change
Formula check
Reconciliation
Scope
```

CHAPTER 7

Share and Prepare Output

In this chapter you will inspect access, print a selected range and check scaling and page breaks.

> **Prompts:** the prompts page has every value and prompt for this chapter, each with a Copy button.

Estimated time: 40 minutes, planned as 10 explanation and demonstration, 10 guided build and casework, 15 independent practice and 5 verification and repair, with 30 active minutes by design. Not yet timed with learners.

Your result: Checked access and readable register and summary print previews.

> **Status:** author reference with 29 staged captures through 7.4 (25 application views and four actual PDF pages). Restricted access/unpublished state inspected without changes; selected-range clipping repaired; three native PDF revisions checked, including a two-page register with repeated header; source heading restored and revision manifest recorded. Independent outputs/reviewer and learner timing remain pending.

What You Will Learn

- Inspect sharing without accidental exposure.
- Prepare a readable print range.
- Distinguish printing from public publishing.

Before You Begin

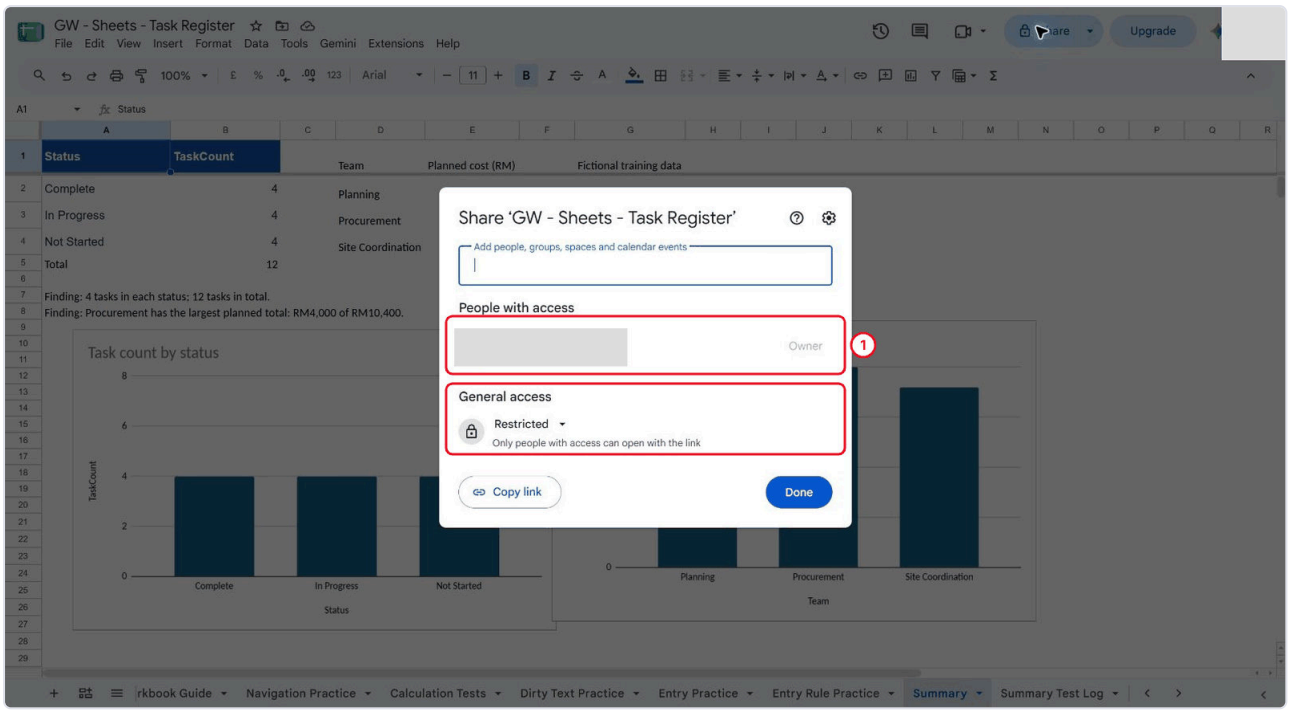
Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the checked register. No public publishing or email distribution is required.

7.1 Inspect Access

1. Open **Share** on the maintained workbook. Inspect **People with access** and **General access**. Record what is shown rather than assuming the same permissions in another account.

> **Reference run:** This reference shows **Restricted** and one listed owner; no additional people/groups or inherited group access is displayed in this dialog.



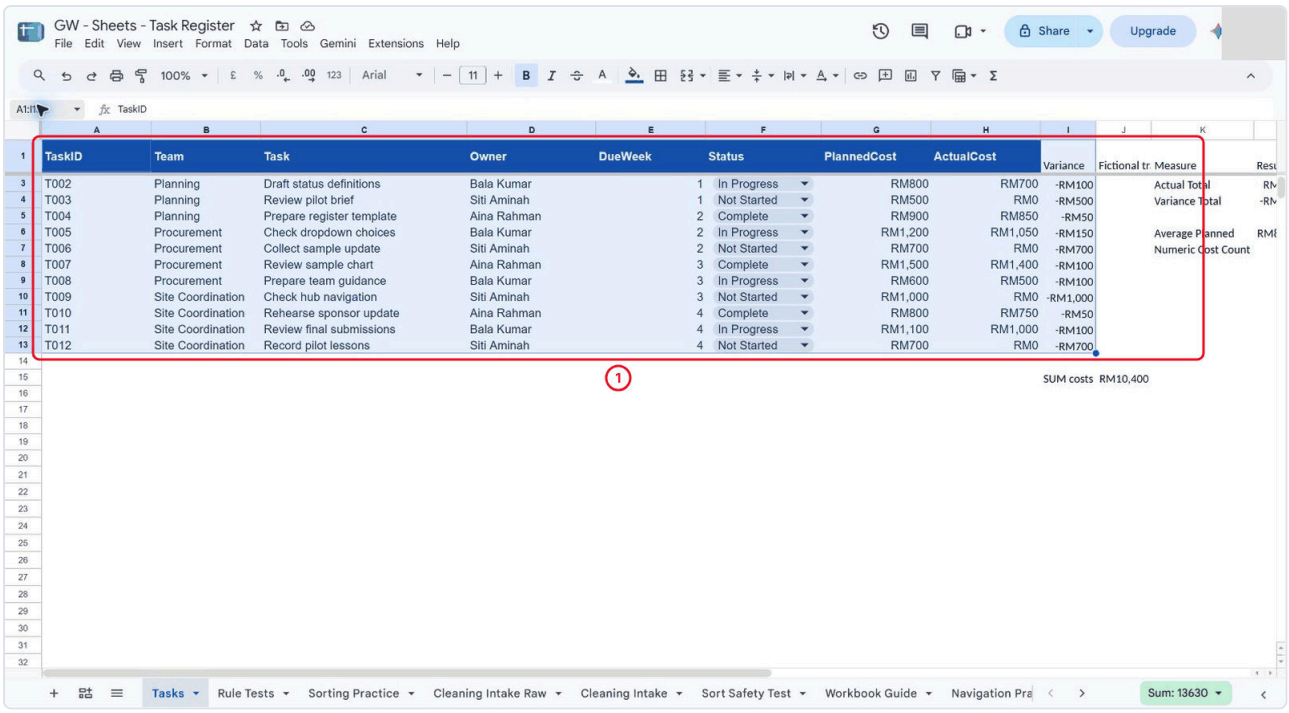
Inspect Restricted access and owner-only list without changes.

2. Explain the proposed roles: **Viewer** reads, **Commenter** reads and comments, **Editor** can edit. Keep these role explanations separate from actual recipient testing. Close with **Done**.

> **Reference run:** No named recipient was assigned in this reference, so no invitation or permission change was made.

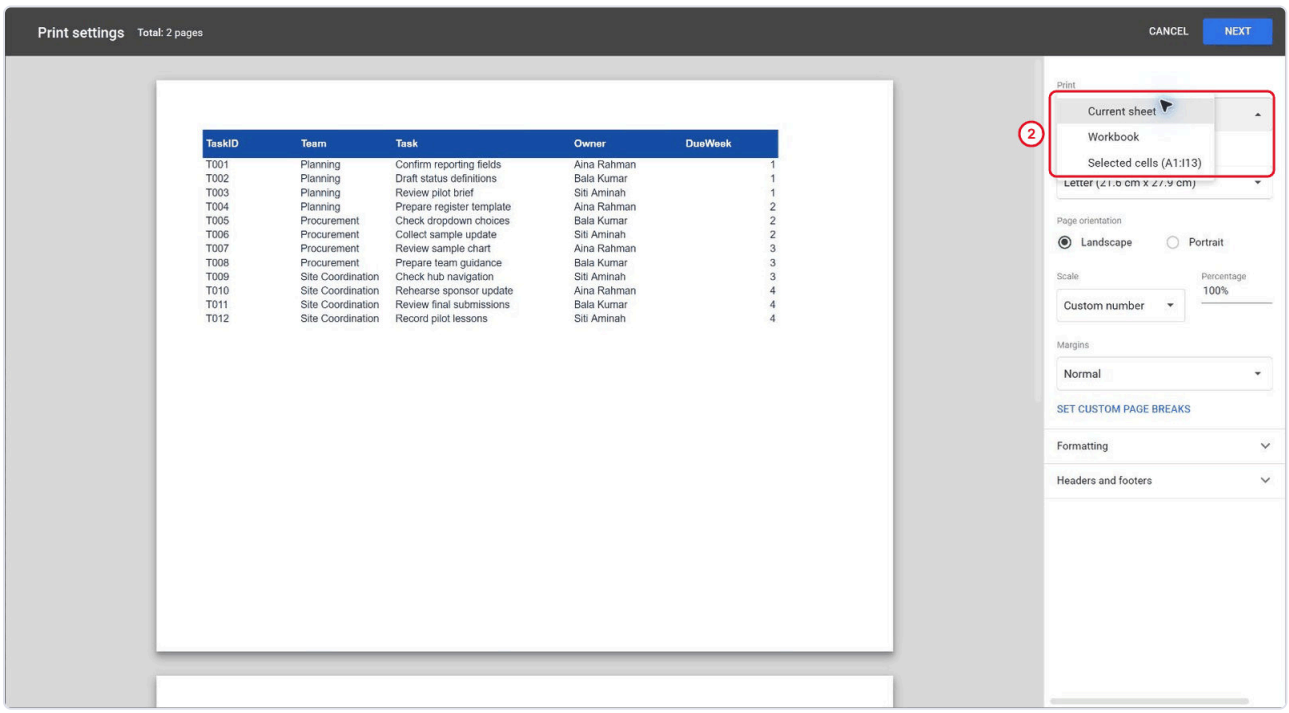
7.2 Prepare the Print Range

1. On Tasks, select **A1:I13**, including all twelve records and the variance column. This excludes K:L controls, the row 15 SUM, spare cells and unrelated tabs.



Select the bounded register A1:I13.

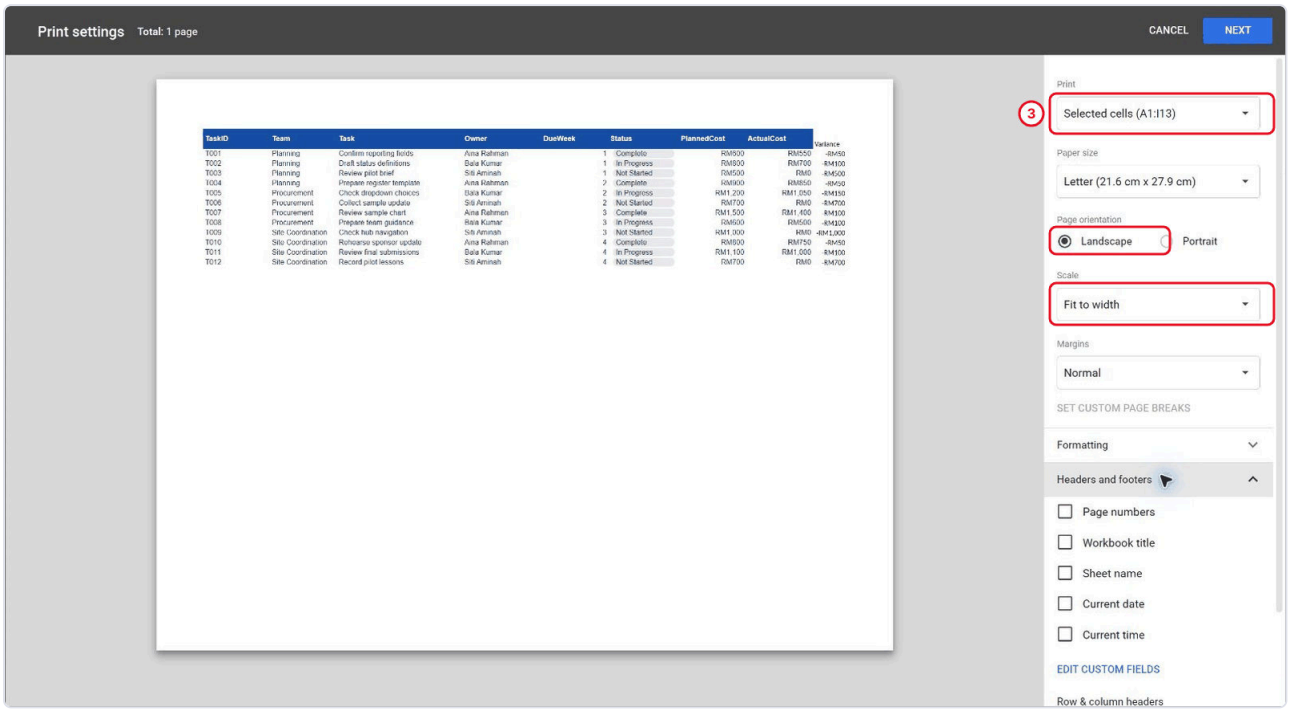
2. Open **File > Print**. Inspect the scope menu: it offers **Current sheet**, **Workbook**, and **Selected cells (A1:I13)**. A selection in the grid does not mean Selected cells is already chosen; this session initially used Current sheet.



Compare Current sheet Workbook and Selected cells scopes.

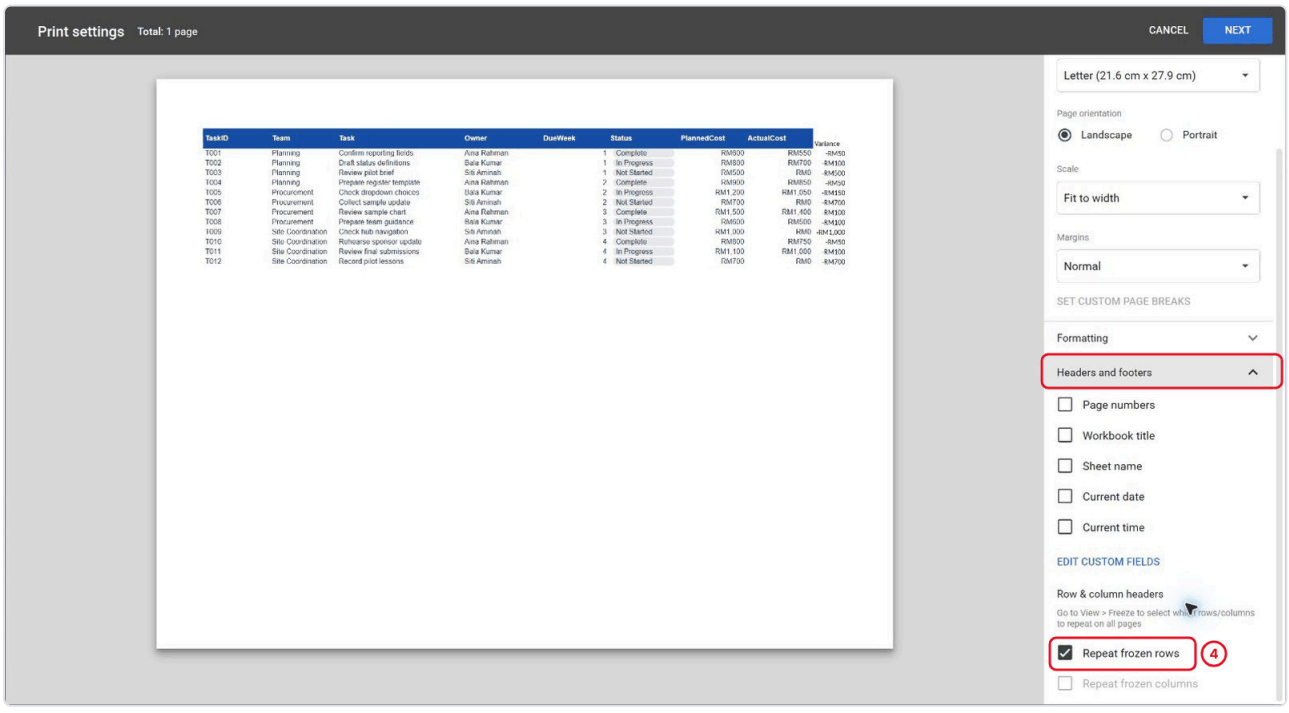
3. Explicitly choose **Selected cells (A1:I13)**, **Landscape**, and **Fit to width**. Inspect every preview page and all nine headers. The initial Letter preview fits one page but makes this wide register small; fitting

alone does not establish readable physical output. The later casework uses A3 at approximately 10.7 pt and a separate layout copy.



Inspect selected register at fit-to-width.

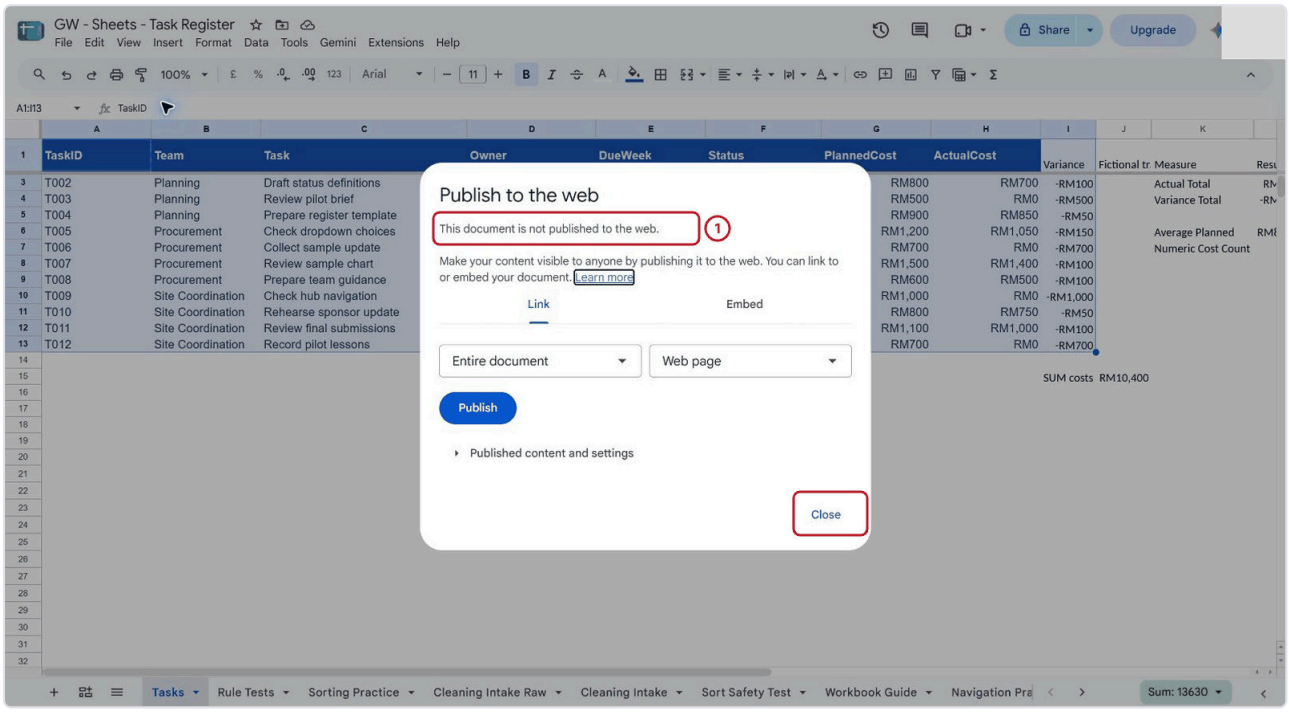
4. Expand **Headers and footers**. **Repeat frozen rows** is available because Tasks already has row 1 frozen; repeat frozen columns is disabled because none are frozen. This option must be tested on actual multi-page output. Cancel this print preview before continuing. If the browser's system print dialog opens after **Next**, cancel it; the PDF workflow below uses **Download as PDF** and **Export**.



Inspect repeat-frozen-rows option.

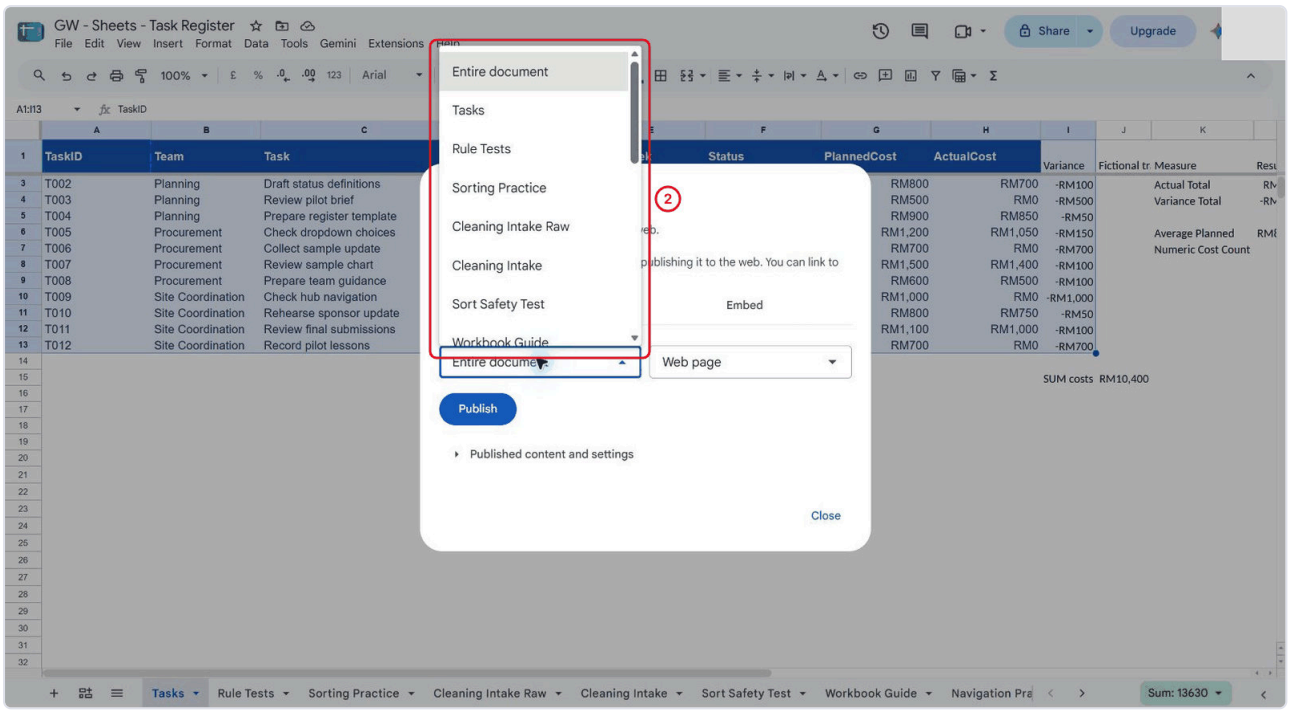
7.3 Compare Output Options

1. Inspect **File > Share > Publish to web**, or find **Publish to web** with menu search. The dialog explicitly says **This document is not published to the web**. Do not click Publish.



Verify the workbook is not published.

2. Inspect the sheet/chart scope selector. **Entire document** and individual sheets/charts are offered. Publication creates a web-facing view and can broaden the audience; it is separate from file access and from a fixed PDF snapshot. Close the menu and dialog without publishing. A fixed PDF is suitable when the recipient needs a checked revision rather than an editable live tracker; distribution still needs its own authorised recipient and scope.

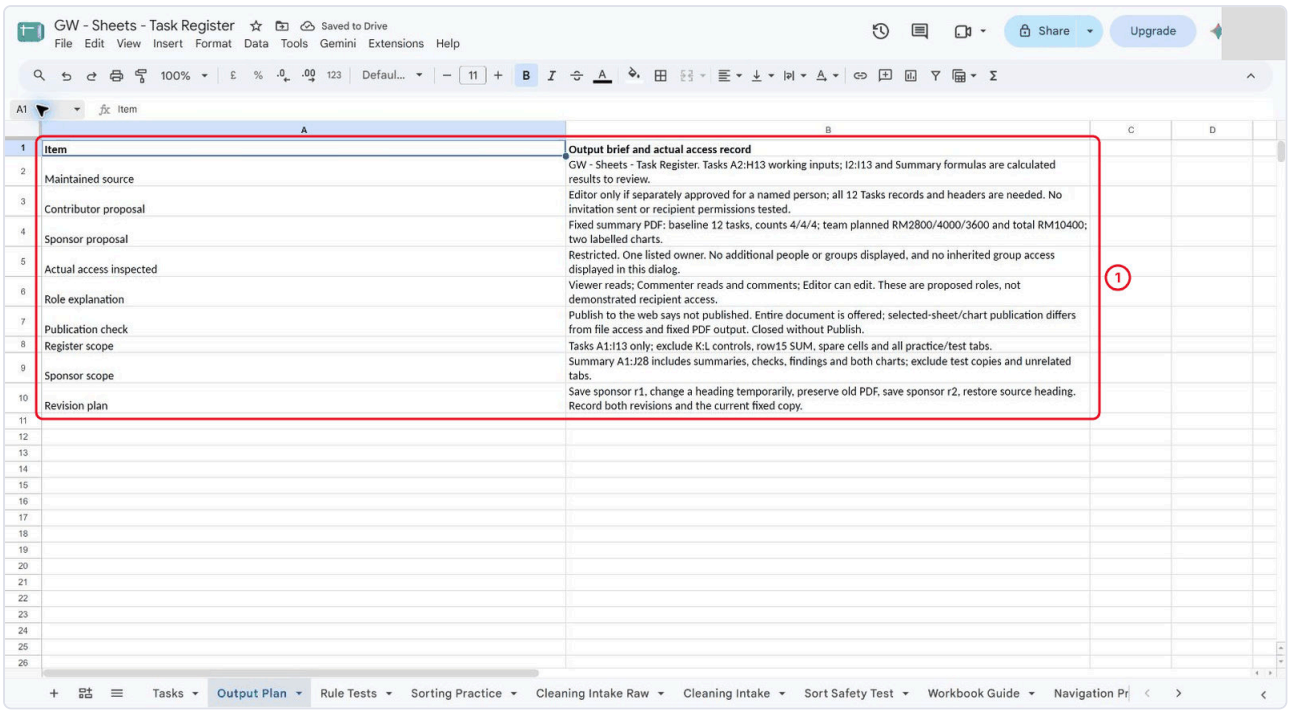


Inspect publication scope without publishing.

7.4 Prepare a Readable, Scoped and Reproducible Output

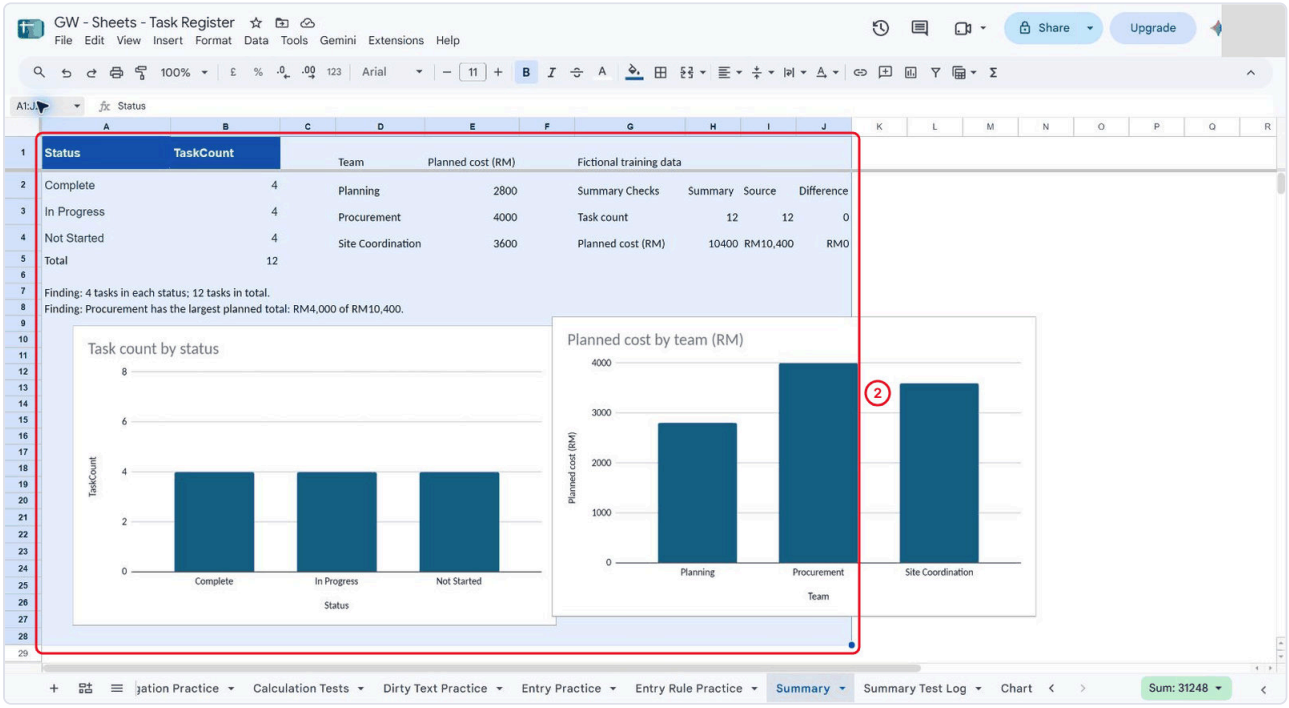
1. Add a sheet named **Output Plan**, with the row labels in Part 3 of the prompts page. Write the output brief (Part 1) and role proposal separately from the actual access record. A contributor needs all twelve Tasks rows and working input cells; a sponsor needs a fixed summary of **12 tasks**, counts **4/4/4**, team planned **RM2800/4000/3600**, total **RM10400**, and two labelled charts. Record Editor only as a proposal for a separately approved contributor; no recipient permission has been tested. Record the inspected Restricted/owner-only display and unpublished state.

> **Reference run:** In the reference, A:B are 650 pixels wide with wrapped text.



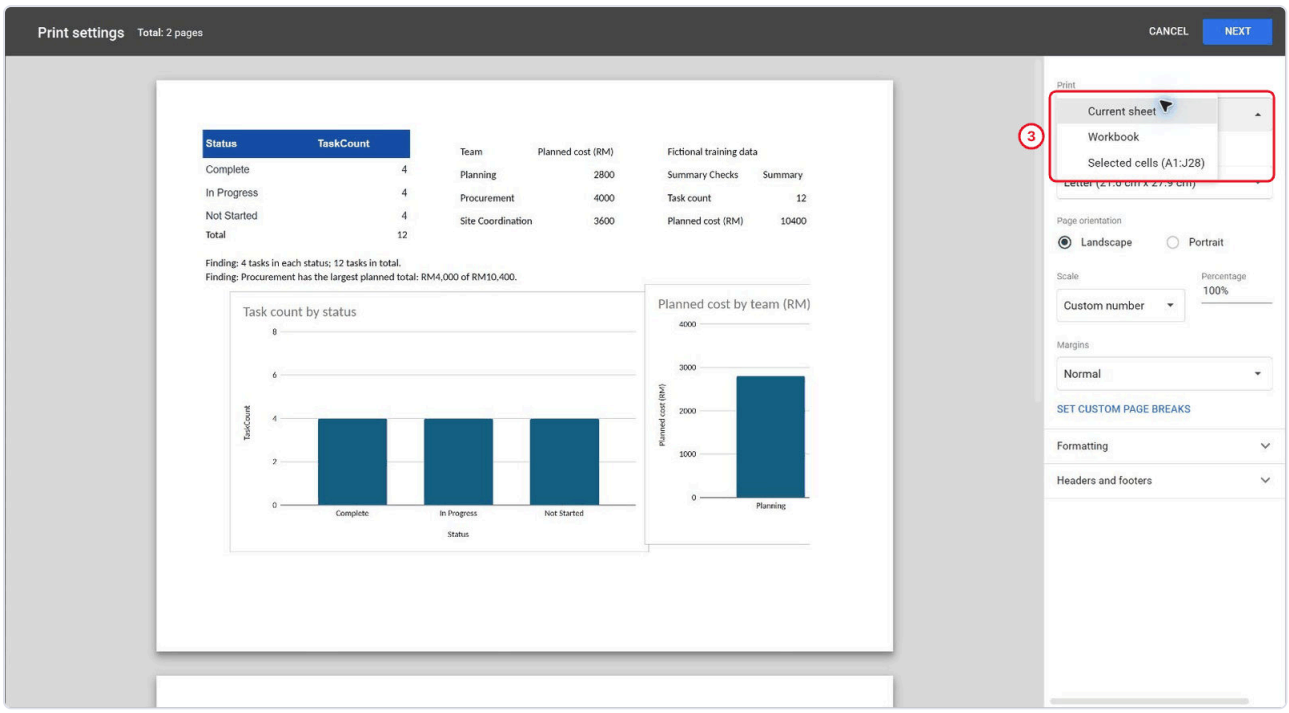
Write proposed roles and actual access separately.

2. On Summary, inspect the extent of the charts as well as the populated cells. The first attempted scope **A1:J28** covered the summary cells but did not extend far enough for the right-hand chart.



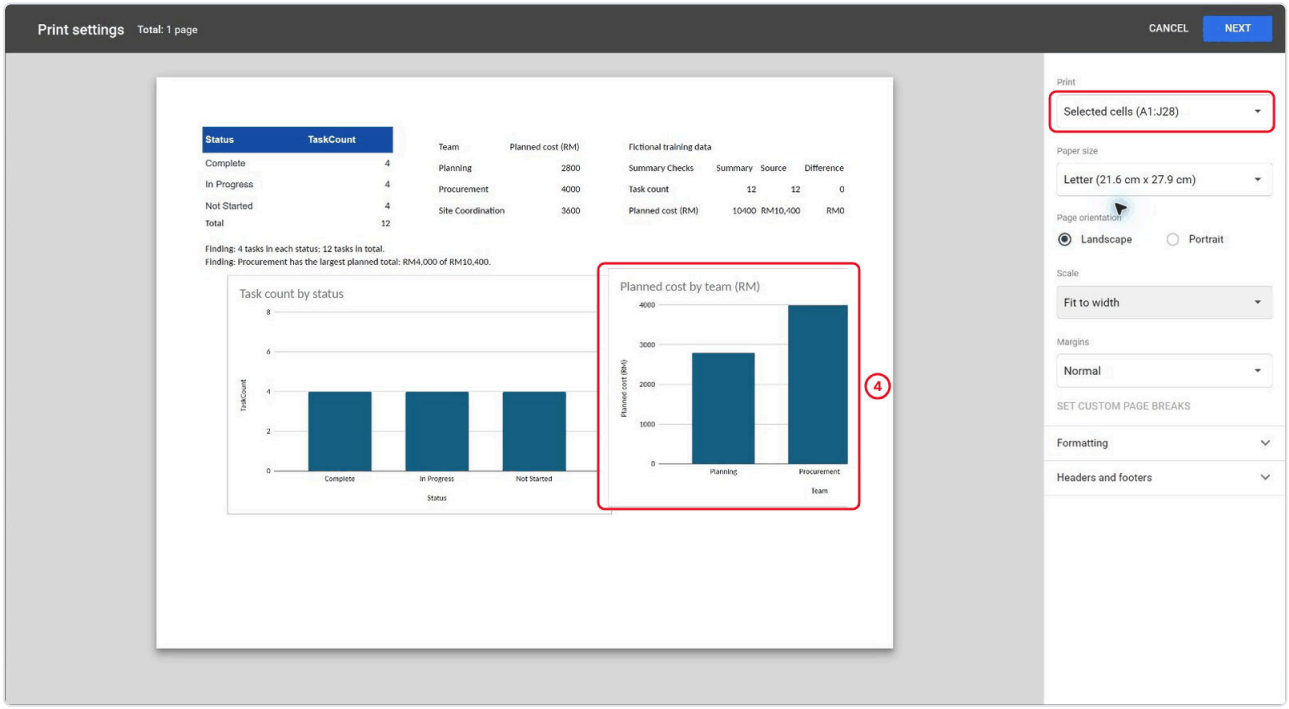
Select an initial summary scope.

3. Open Print and compare **Current sheet / Workbook / Selected cells**. Current sheet uses the used sheet area, Workbook risks including practice/test tabs, and Selected cells limits the exported region. Explicitly choose the intended scope each time.



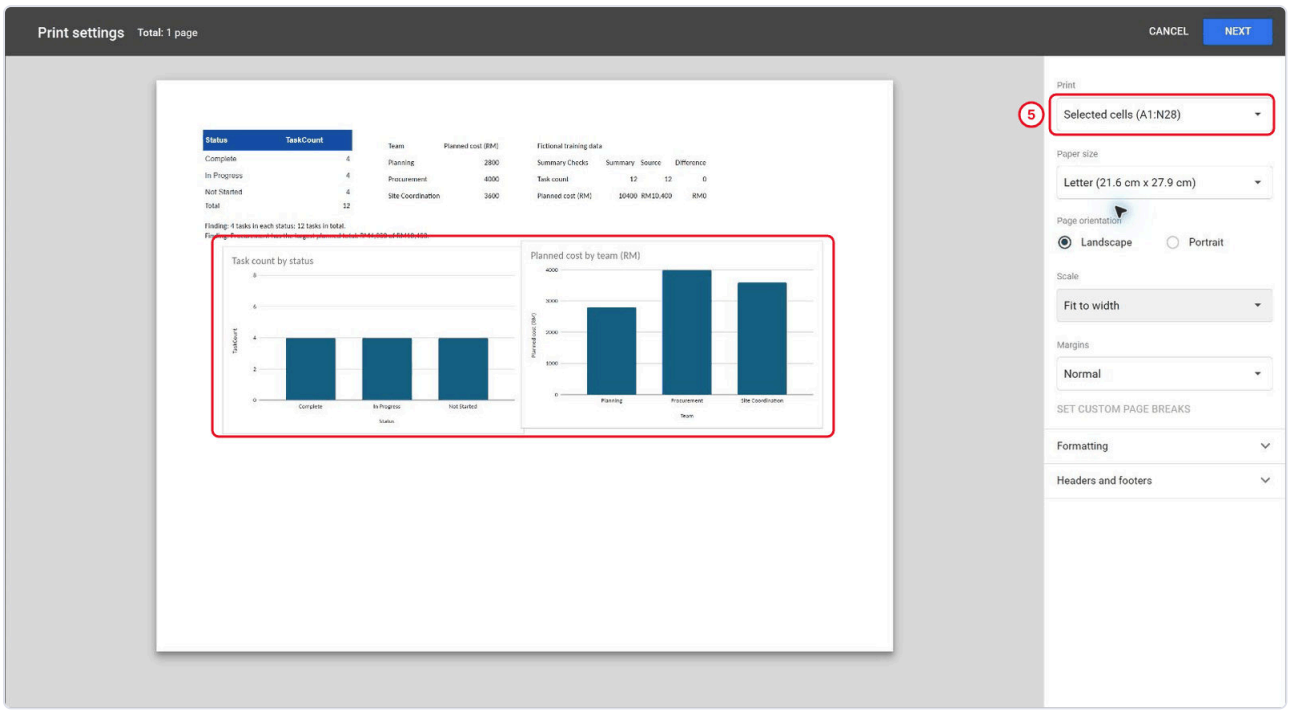
Compare summary print scopes.

4. The **A1:J28** selected-cells preview clips the cost chart: Site Coordination is missing from the printed right-hand side. This is a scope defect even though the source chart is complete. Cancel and repair the selection rather than accepting a partly visible chart.



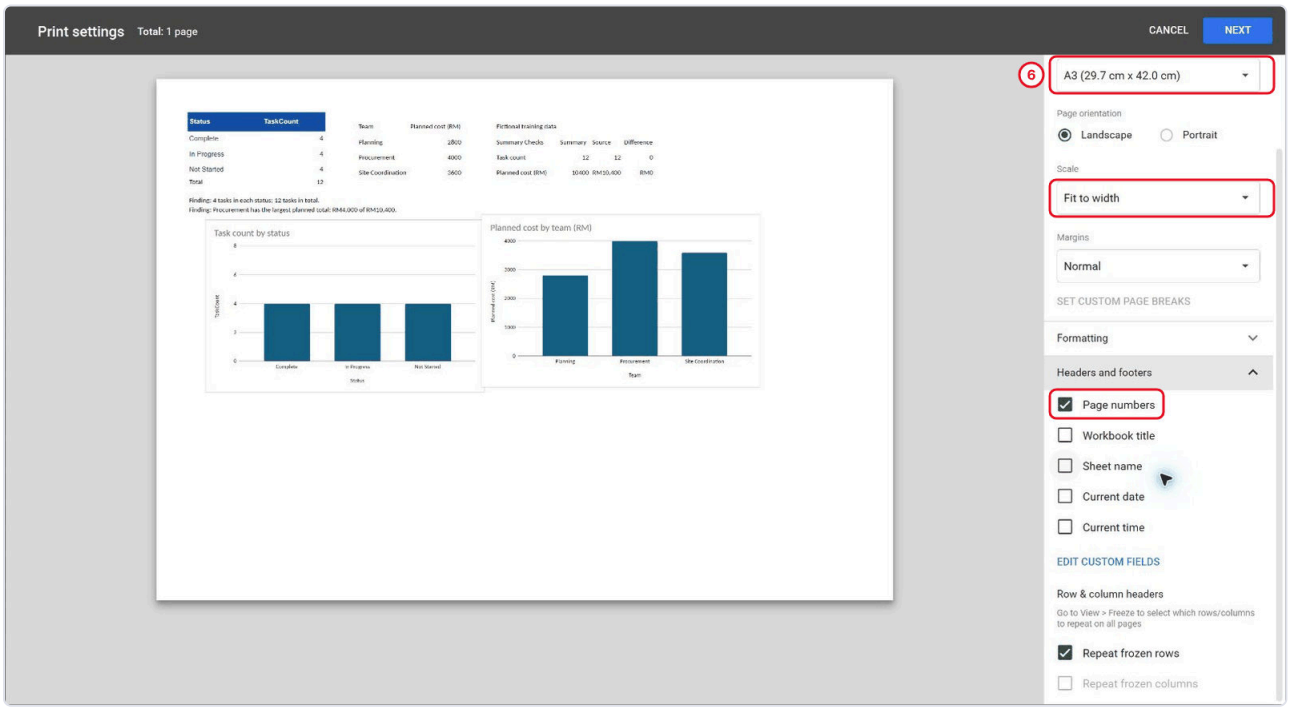
Observe a selected range clipping the cost chart.

5. Select **A1:N28** and reopen the preview with **Selected cells** and **Fit to width**. Both complete charts now fit inside the region. Inspect all three team labels, all three statuses, the totals and zero reconciliation differences. No test tabs belong in this sponsor handout.



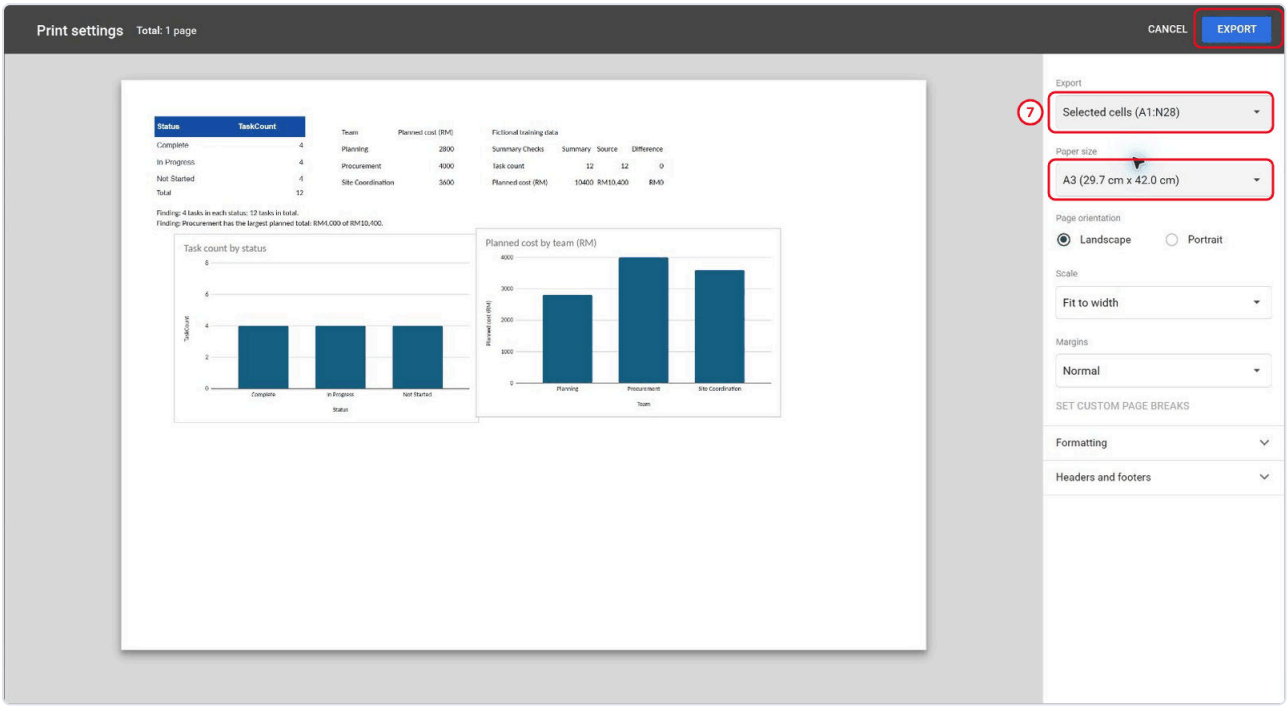
Repair scope to include both complete charts.

6. Choose **A3**, **Landscape**, and fit-to-width for this author reference. The saved text is approximately **10.7 pt** on A3; confirm the actual customer's output medium before adopting this layout. Do not claim that the same content remains readable when printed smaller. Page numbers can be selected under Headers and footers, but this session's saved PDFs did not contain visible footer numbers, so that option is not treated as verified output.



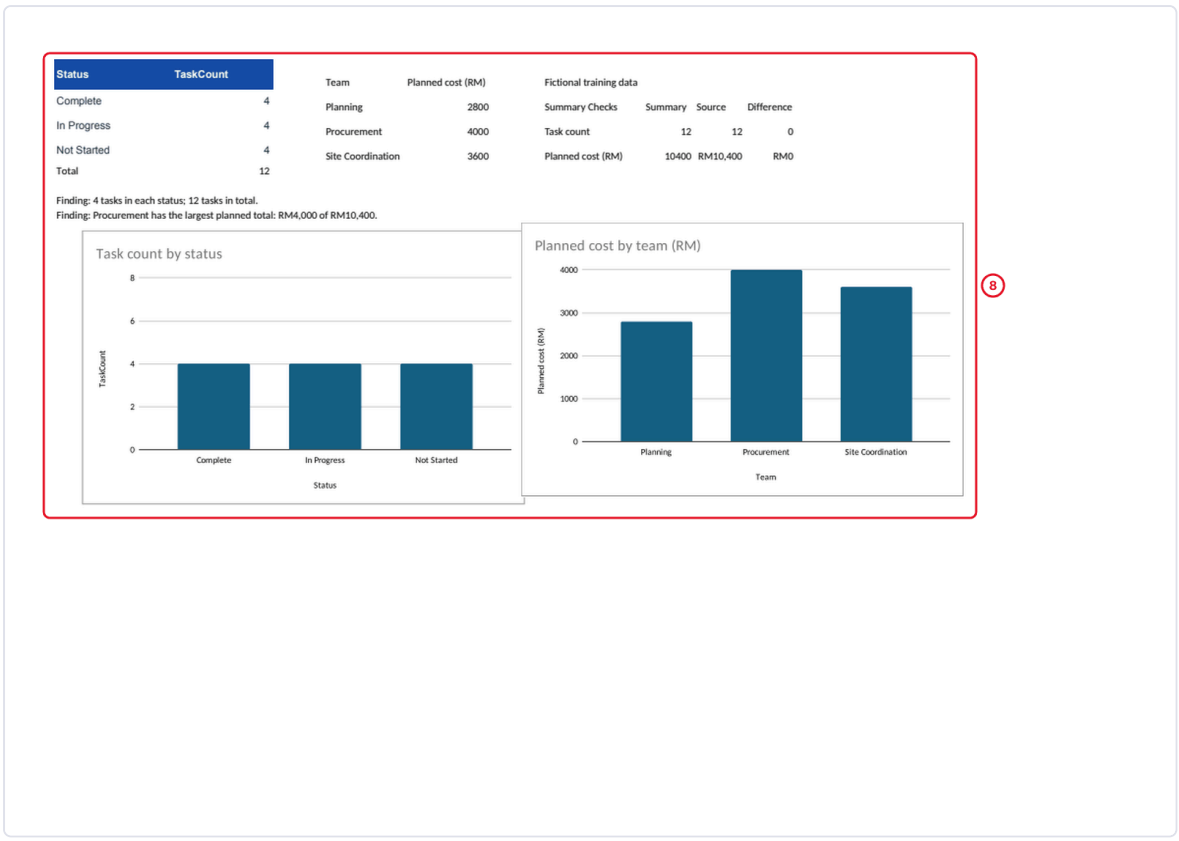
Choose A3 landscape for readable summary output.

7. Cancel the Print workflow and use **File > Download > PDF (.pdf)**, or menu search **Download as PDF**. This opens the same layout controls with **Export**, avoiding the browser system print dialog reached by Print's Next. Recheck **Selected cells (A1:N28)**, A3, Landscape, Fit to width, and the one-page preview; the export scope initially defaults to Current sheet again. Click **Export** and retain the downloaded file as **GW-Sponsor-Summary-r1.pdf**.



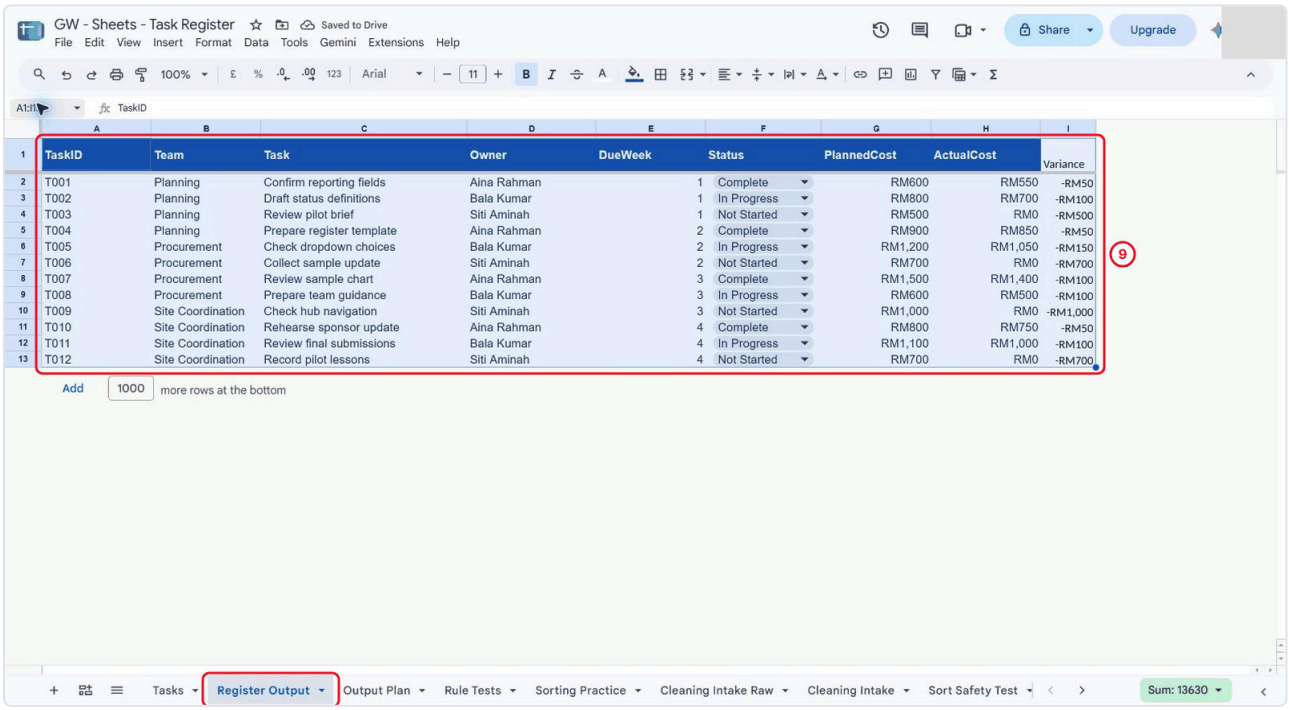
Use native Download as PDF and selected scope.

8. Open or render the actual saved PDF, not just its preview. Verify the counts, planned totals, findings and both complete charts. This full-page render is from the native downloaded PDF. Record its revision, page count, source range and file hash (Part 3 of the prompts page shows how, or use size and modified time) before changing the source.



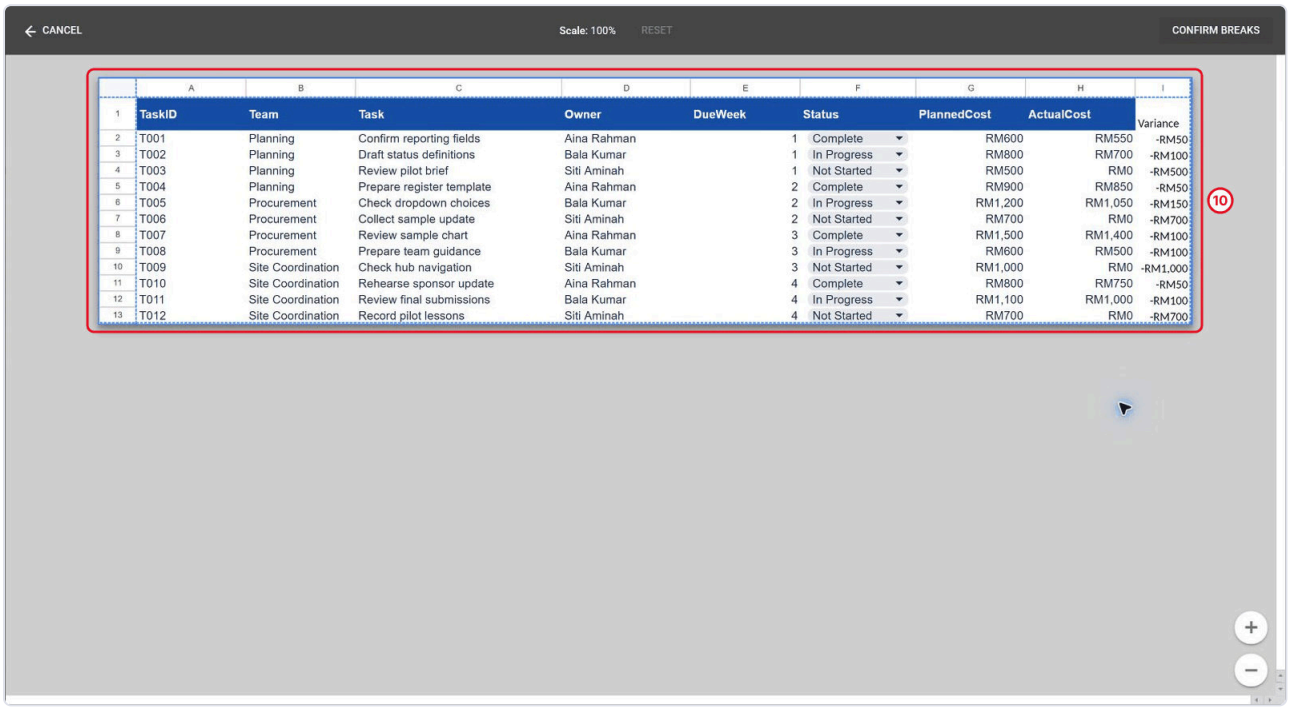
Inspect actual sponsor revision one PDF.

9. To test page breaks without changing the maintained register, duplicate Tasks as **Register Output**. On that copy only, remove columns **J:Z** and rows **14:1000**, leaving exactly **A1:I13** with the original twelve records and row 1 frozen. Preserve all nine fields and the variance formulas. This is a layout snapshot; contributors should continue editing the maintained Tasks sheet.



Create a bounded register layout copy.

10. Use **Download as PDF** on Register Output. Choose **Current sheet**, A3, Landscape and **Normal (100%)**, then **Set custom page breaks**. Selected-cells mode disables custom page breaks in the summary workflow; the bounded layout copy allows Current sheet without helper content.

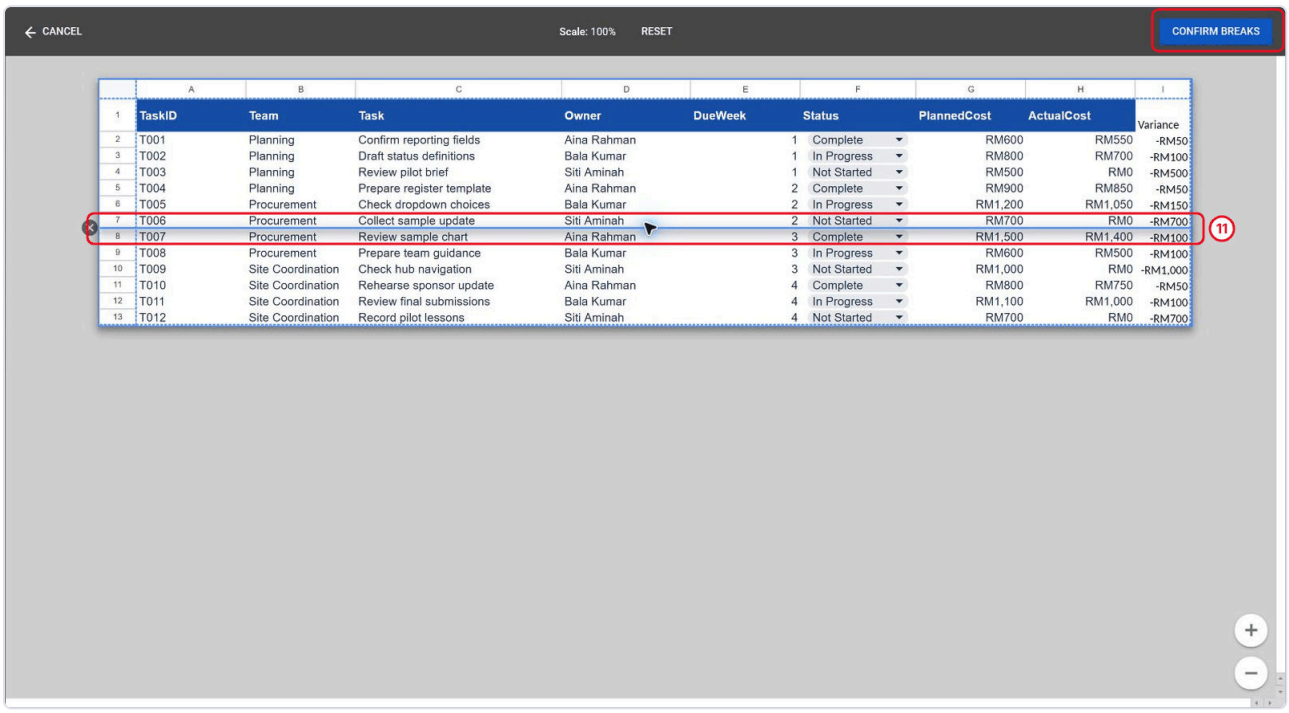


	A	B	C	D	E	F	G	H	I
1	TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost	Variance
2	T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550	-RM50
3	T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700	-RM100
4	T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0	-RM500
5	T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850	-RM50
6	T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050	-RM150
7	T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0	-RM700
8	T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400	-RM100
9	T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500	-RM100
10	T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0	-RM1,000
11	T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750	-RM50
12	T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000	-RM100
13	T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0	-RM700

Inspect the custom page-break editor.

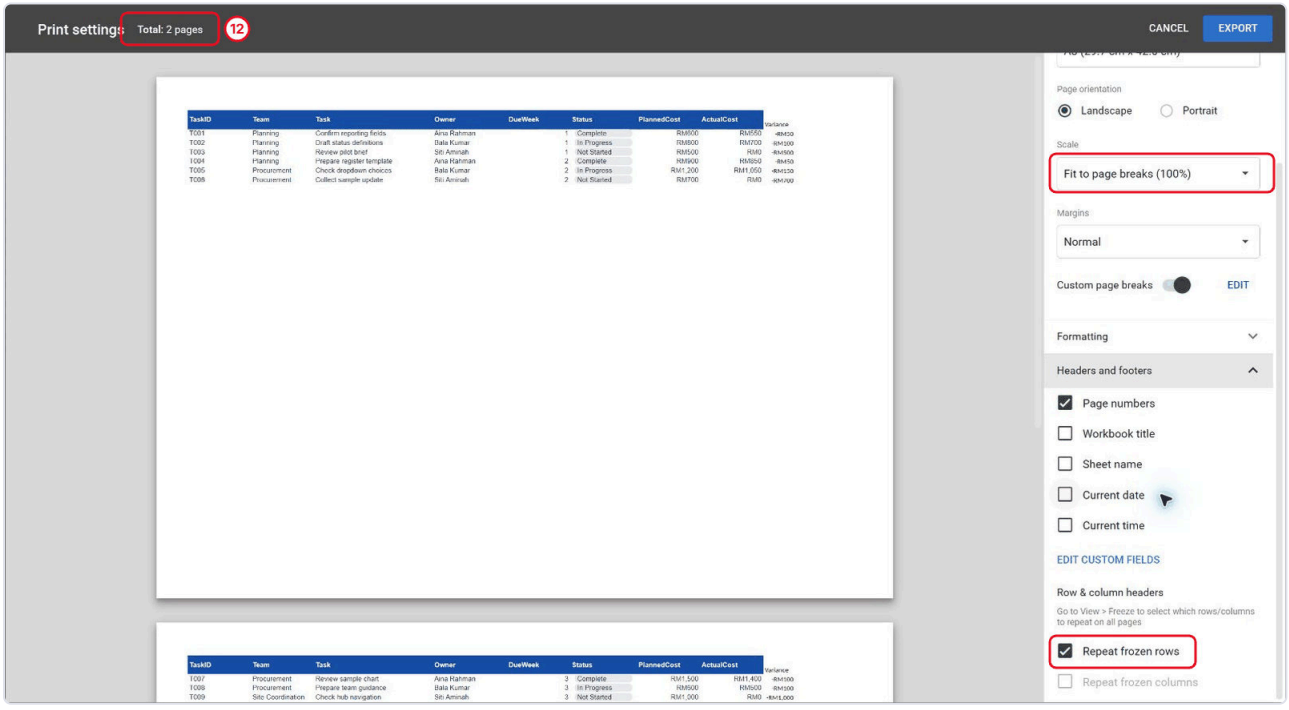
11. Drag the bottom row-page boundary to the line **after row 7**, between T006 and T007, then **Confirm breaks**. This deliberate split tests pagination and header repetition; it is not a requirement to waste a second page for every small register.

> **Reference run:** The reference now has two row-page regions.



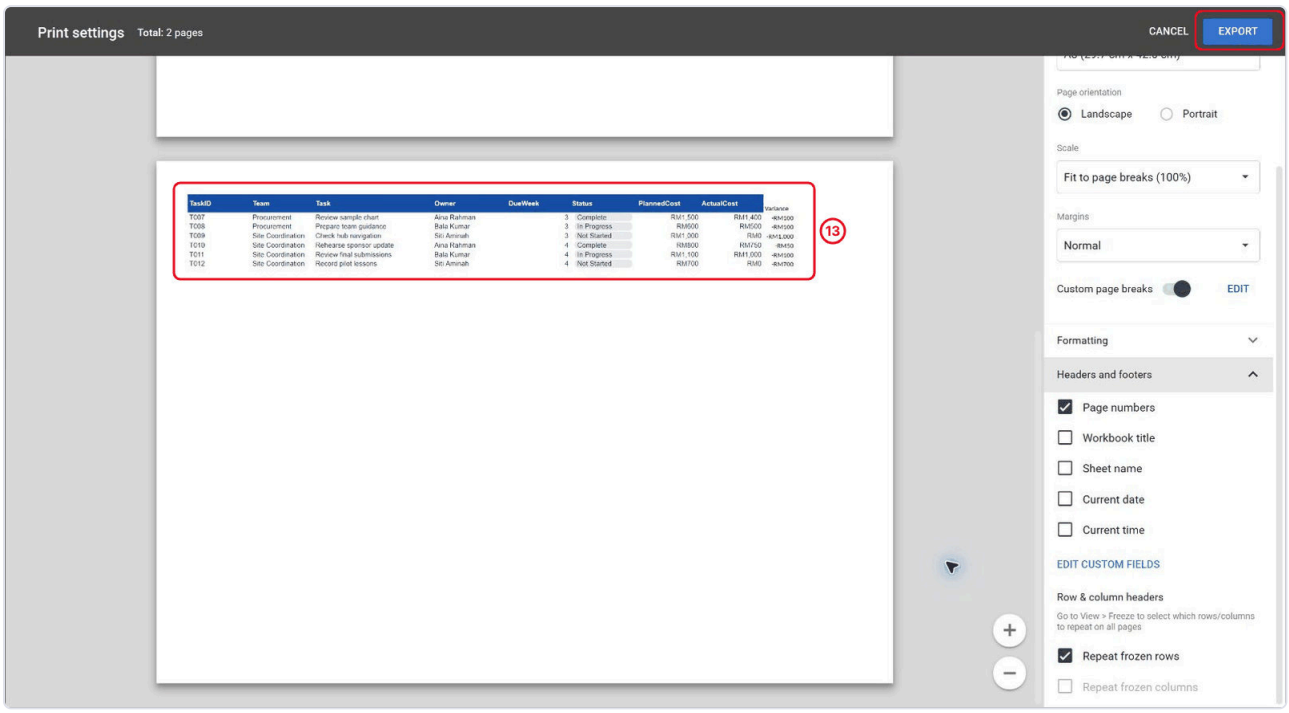
Split after six task records.

12. Under Headers and footers confirm **Repeat frozen rows**. The scale reads **Fit to page breaks (100%)**, and the preview reports **2 pages**. Inspect page 1 for T001–T006 and all column labels.



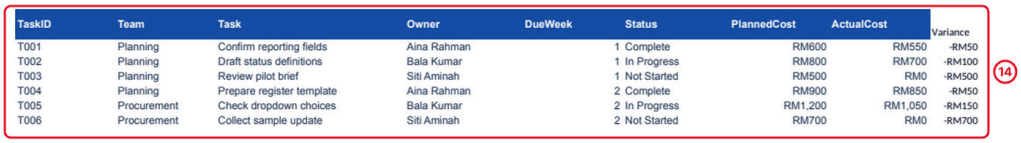
Enable repeated frozen header for two pages.

13. Scroll to the final preview page. It must repeat the header and show **T007–T012**, including the last task **Record pilot lessons** and its row values. No K:L controls, row 15 SUM or practice sheets should appear. Click **Export** and retain the file as **GW-Task-Listing-r1.pdf**.



Inspect the final preview page with T012.

14. Open or render **both actual PDF pages**. Page 1 contains six tasks and readable headers.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost	Variance
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550	-RM50
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700	-RM100
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0	-RM500
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850	-RM50
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050	-RM150
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0	-RM700

14

Inspect actual register PDF page one.

15. Page 2 contains the other six tasks, repeats the header and includes T012. Compare row labels and cost values with Tasks; all twelve records appear exactly once across the two pages. The PDF uses

A3 landscape and approximately 10.7 pt text.

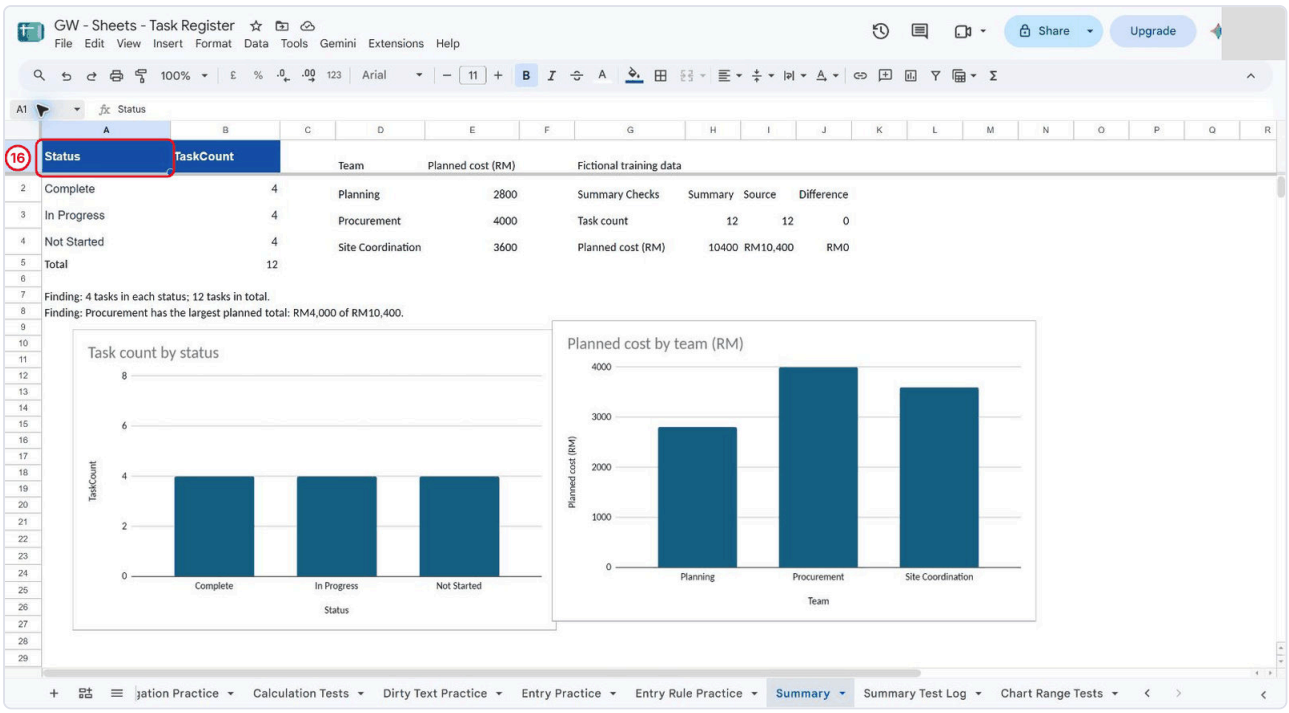
> **Reference run:** Footer numbers were not visible despite selecting the option; identify pages by the inspected content and PDF page count in this reference.

TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost	Variance
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400	-RM100
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500	-RM100
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0	-RM1,000
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750	-RM50
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000	-RM100
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0	-RM700

15

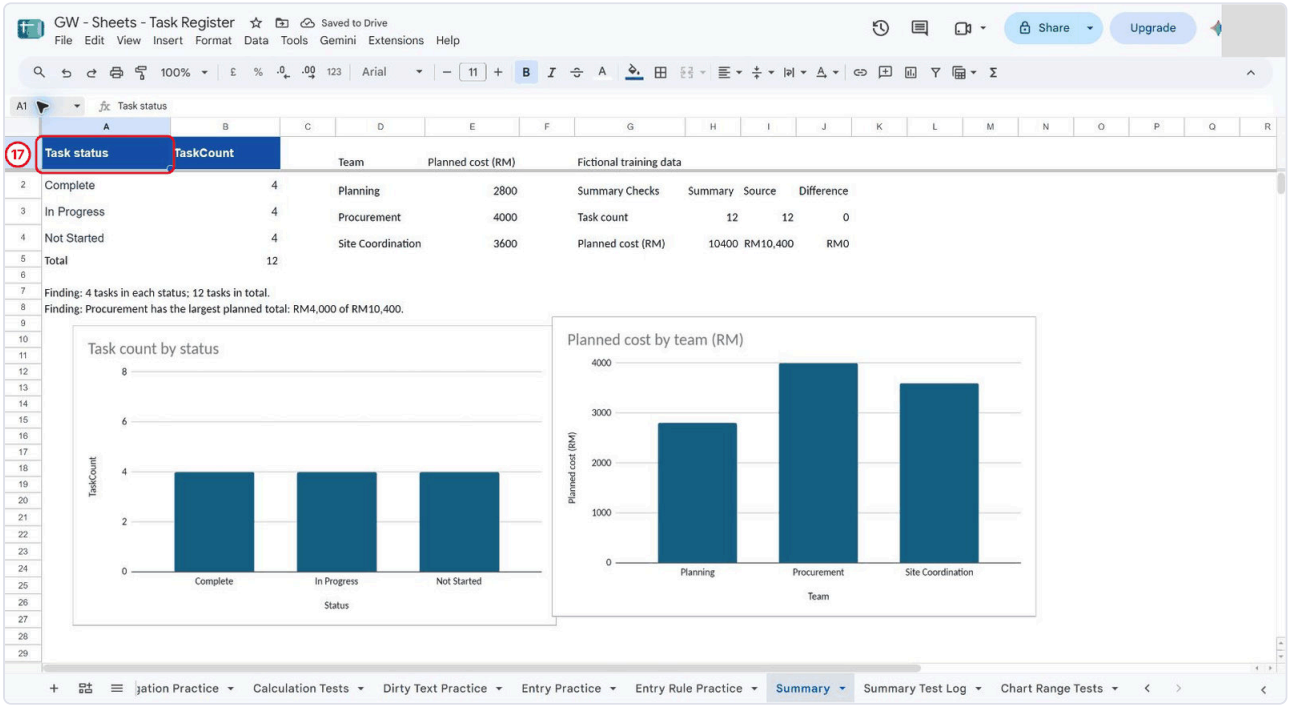
Inspect actual register PDF final page and repeated header.

16. On maintained Summary, record **A1 = Status**. Keep the saved revision 1 PDF untouched.



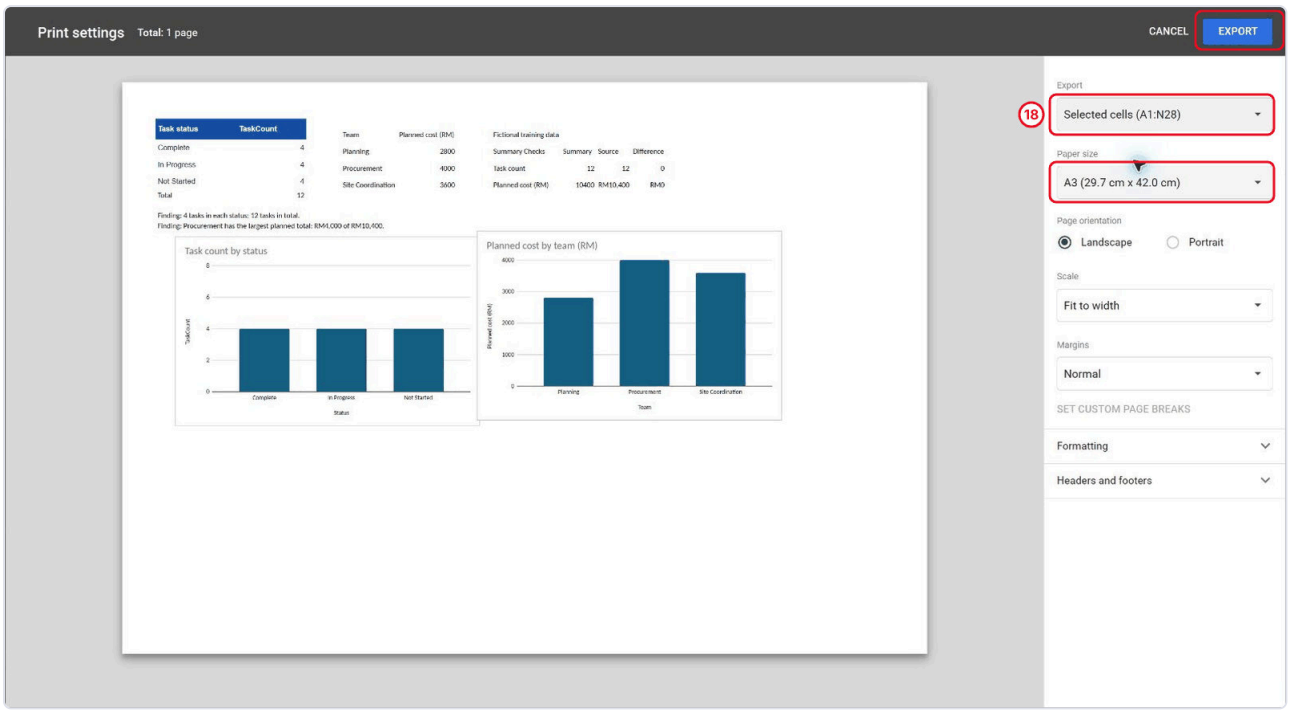
Record original summary heading.

17. Temporarily change that one heading to **Task status**. Counts, formulas, costs and chart data must remain unchanged. A saved PDF does not update when the working source changes.



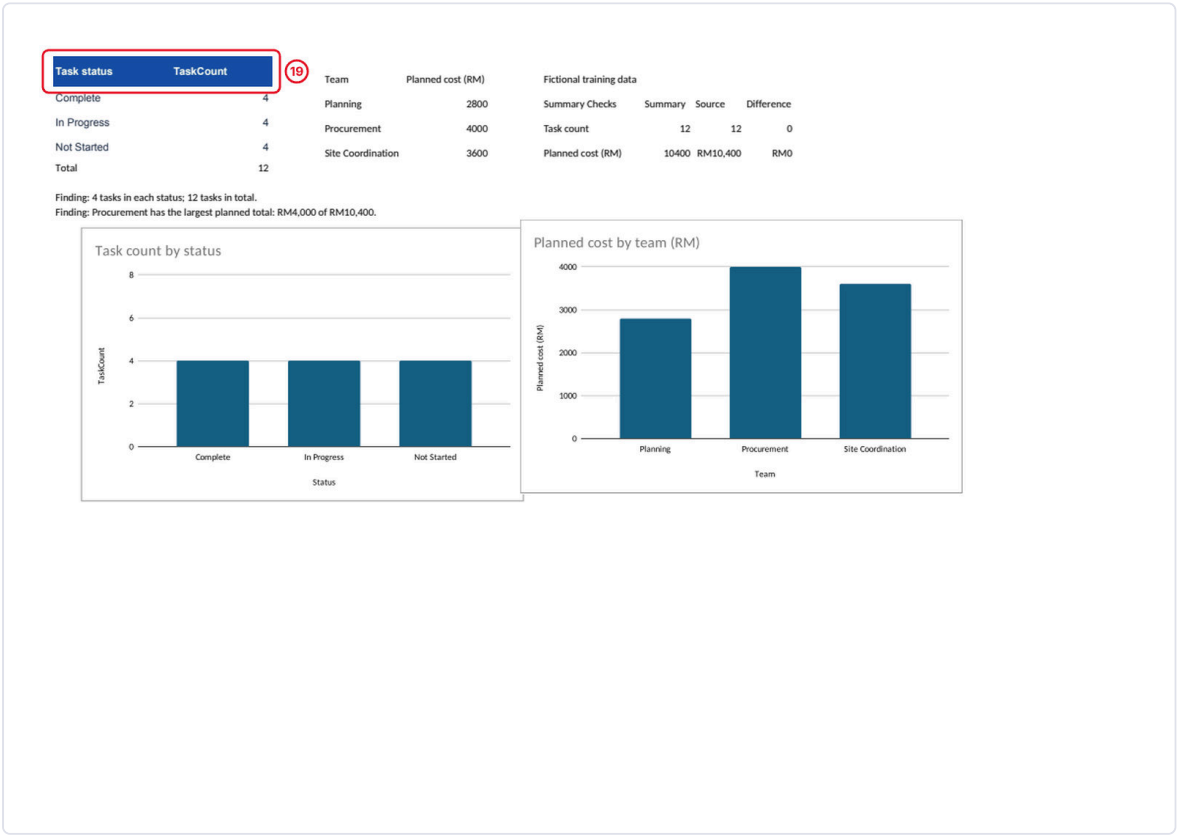
Change one heading for a revision test.

18. Export the same **A1:N28** range with the same A3/Landscape/Fit-to-width settings as **GW-Sponsor-Summary-r2.pdf**. Keep revision 1 as a separate file; do not overwrite it.



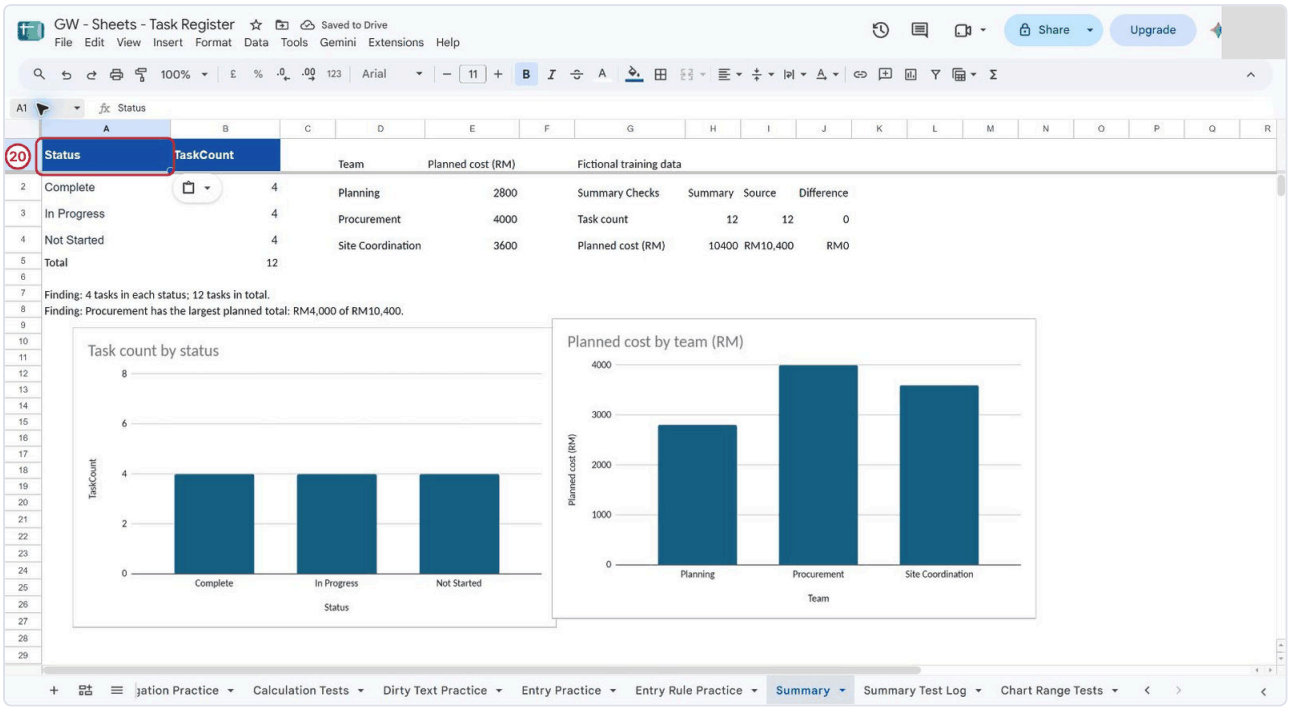
Export revision two with the same full scope.

19. Inspect revision 2: its heading is **Task status**, with the same 4/4/4, total 12, planned 10400 and complete charts. Reopen or compare revision 1: it still says **Status**, and its SHA256 (or size and modified time) remains unchanged. A newly generated revision and an old fixed copy can legitimately differ from one another.



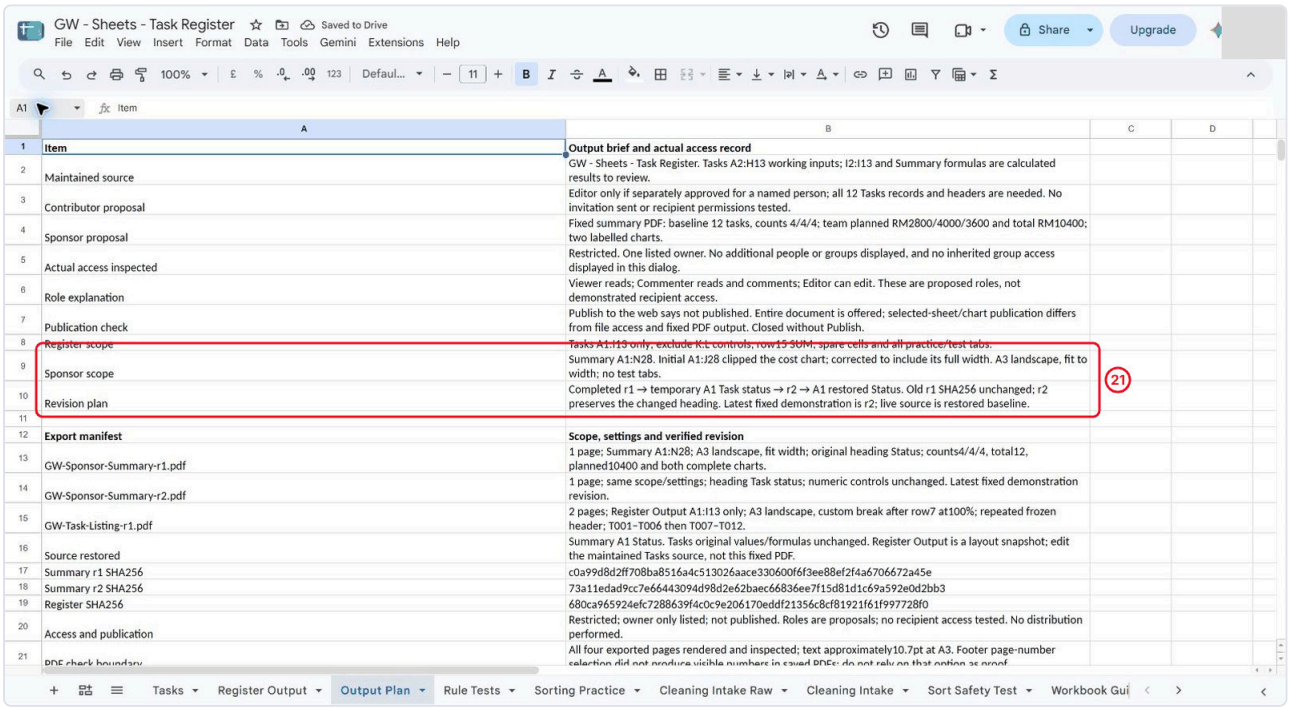
Inspect actual revision two PDF with changed heading.

20. Restore Summary A1 to **Status**. Verify the maintained source values/formulas match the pre-test baseline. Revision 2 remains the latest fixed demonstration with its changed heading; document that deliberate difference from the restored source rather than claiming the two are identical.



Restore original summary heading.

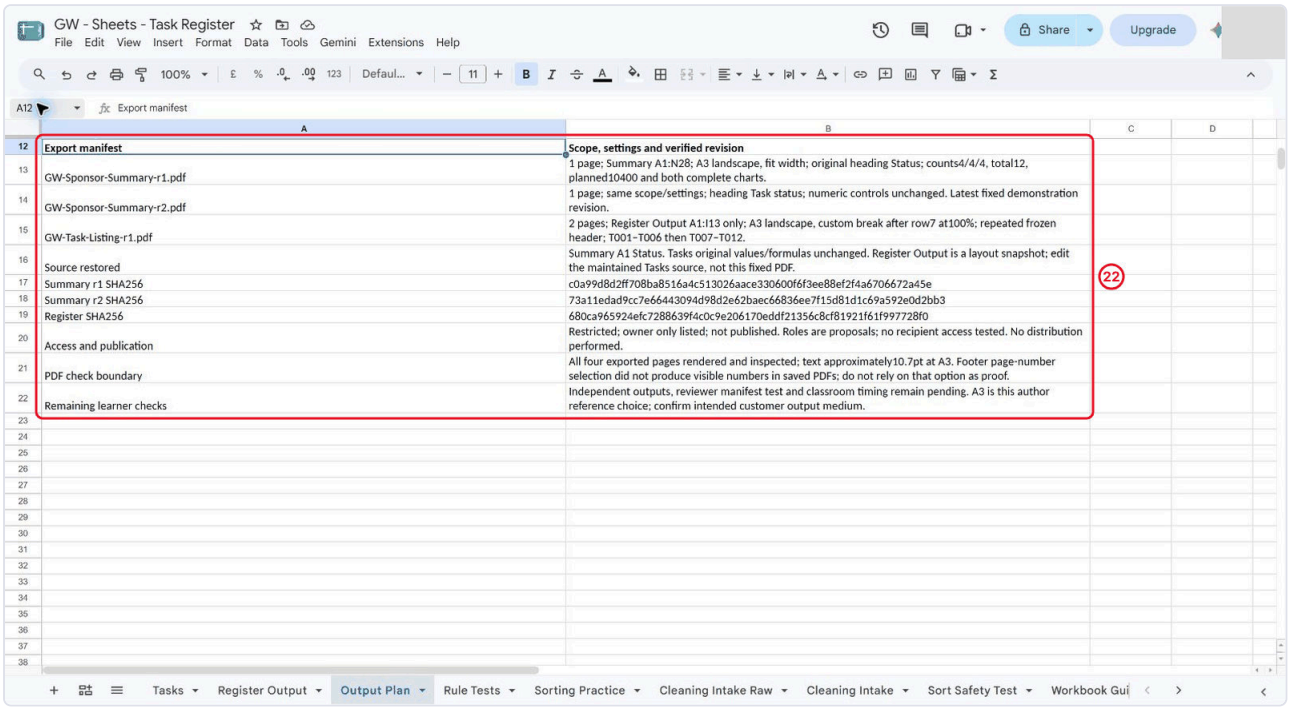
21. Update Output Plan with the repaired **A1:N28** scope and completed revision sequence. The maintained workbook is the live source; Register Output is the bounded layout snapshot; the PDFs are fixed outputs.



Update output brief with checked scopes and revisions.

22. Below the plan on Output Plan, add an **Export manifest** (labels in Part 3 of the prompts page) identifying each filename, source range, paper/orientation/scale, page count, inspected final-page content, hash, and revision status. Record source restoration, actual access versus proposed roles, the footer-number limitation and pending independent reviewer/timing checks.

> **Reference run:** The downloadable examples below are the native author reference exports.



Record the export revision manifest and validation limits.

- Sponsor summary r1 — original heading, one A3 page
- Sponsor summary r2 — changed heading, one A3 page
- Task listing r1 — repeated header, two A3 pages
- Export manifest with hashes and scope

> **Validation boundary:** guided 7.1–7.3 and casework 7.4 are author reference checks. The three PDFs contain four inspected pages. No sharing, publication, email distribution or recipient-role testing was performed. The original Tasks values/formulas and Summary heading are restored. Independent outputs, reviewer manifest test, customer output medium and classroom timing remain pending; earlier import/paste limitations remain separate.

Independent Practice

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Prepare two independent outputs: a concise sponsor summary and a complete task listing with readable headings. Do not force both into the same one-page layout.
2. Have a reviewer use the manifest to identify the current source and current fixed copy. Repair ambiguous filenames or scope labels.

3. Explain which ranges should be edited, which calculated cells should be reviewed and why a fixed PDF is not a live tracker.

Evidence to Hand Over

- Inspected access settings and proposed-role plan.
- Two scope-checked readable PDFs with the correct final-page content.
- Source-to-export comparison and a revision manifest that identifies the current copy.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The PDF contains practice/helper content	Inspect selected-cells versus current-sheet/workbook scope and regenerate the intended output.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Checked access and readable output complete the register handover.

Check yourself: Which output would automatically reflect later source changes, and which is a snapshot?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Part 1: Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Output brief: Sponsor wants a readable summary of the 12-task baseline and planned costs.
Contributor needs all task rows and labels. Exclude Entry Practice, temporary rule tests and
unrelated helper notes from the sponsor handout.
```

Part 2: Captured Output Reference

Actual access display: Restricted, one listed owner; no additional people/groups or inherited group access displayed. Roles are proposed only; no invitation or recipient access test. Publish to the web says not published and was closed without Publish.

Register logical scope: Tasks A1:I13, twelve records/nine fields. Register Output is a separate layout copy bounded by removing J:Z and rows 14:1000. Current sheet PDF, A3 landscape, custom break after row 7 at 100%, repeated frozen row 1: page 1 T001–T006, page 2 T007–T012. GW-Task-Listing-r1.pdf has 2 pages.

Sponsor logical scope: Summary A1:N28. Initial A1:J28 clips the cost chart. A3 landscape / Fit to width; all counts 4/4/4, total 12, team planned 2800/4000/3600, total 10400, zero differences, two complete charts. Native Download as PDF opens Export; Print's Next opens a browser dialog. Recheck Selected cells after each opening.

Revision test: GW-Sponsor-Summary-r1.pdf, 1 page, heading Status; temporarily change Summary A1 to Task status; export GW-Sponsor-Summary-r2.pdf, 1 page, changed heading/same numeric controls. Original r1 SHA256 unchanged. Restore source A1 Status. Latest fixed demonstration r2 retains its changed heading; restored live source and fixed copy are documented separately in Output Plan.

Native exports and JSON manifest are linked in the lesson. All four pages were rendered and checked at approximately 10.7 pt on A3. Page-number option did not produce visible footers in the downloaded PDFs; repeated header was verified. Independent outputs/reviewer and learner timing remain pending.

Part 3: Output Plan Labels

Section 7.4. Put the plan in A1:B10 and the export manifest from A12 down. Write your own entry for each row in column B.

Output Plan, A1:A10 (step 1), with `Output brief and actual access record` in B1:

```
Item
Maintained source
Contributor proposal
Sponsor proposal
Actual access inspected
Role explanation
Publication check
Register scope
Sponsor scope
Revision plan
```

Export manifest, A12:A22 (step 22), with `Scope, settings and verified revision` in B12:

```
Export manifest
GW-Sponsor-Summary-r1.pdf
GW-Sponsor-Summary-r2.pdf
GW-Task-Listing-r1.pdf
Source restored
Summary r1 SHA256
Summary r2 SHA256
Register SHA256
Access and publication
PDF check boundary
Remaining learner checks
```

The SHA256 rows are optional on Windows: `certutil -hashfile GW-Sponsor-Summary-r1.pdf SHA256` in Command Prompt prints the hash. If you can't run it, record each file's size and modified time instead.

CHAPTER 8

Complete the Register Challenge

In this chapter you will extend a separate register, repair a formula and reconcile its totals before an independent changed-row challenge.

> **Prompts:** the prompts page has every value and prompt for this chapter, each with a Copy button.

Estimated time: 45 minutes, planned as 10 explanation and demonstration, 15 guided build and casework, 15 independent practice and 5 verification and repair, with 35 active minutes by design. Not yet timed with learners.

Your result: A separate 13-record challenge copy with reconciled totals and chart.

> **Status:** author reference with 32 staged captures through 8.4 (31 application views and one native PDF page). Separate 13-record workbook, extended ranges, repaired reference, actual validation rejection, two-method controls, cost correction, focused views, current PDF and operating guide checked. Original baseline preserved. Partner explanation/handover, independent T014/range omission and learner timing remain pending.

What You Will Learn

- Extend a register deliberately.
- Repair a formula and reconcile totals.
- Explain a checked chart.

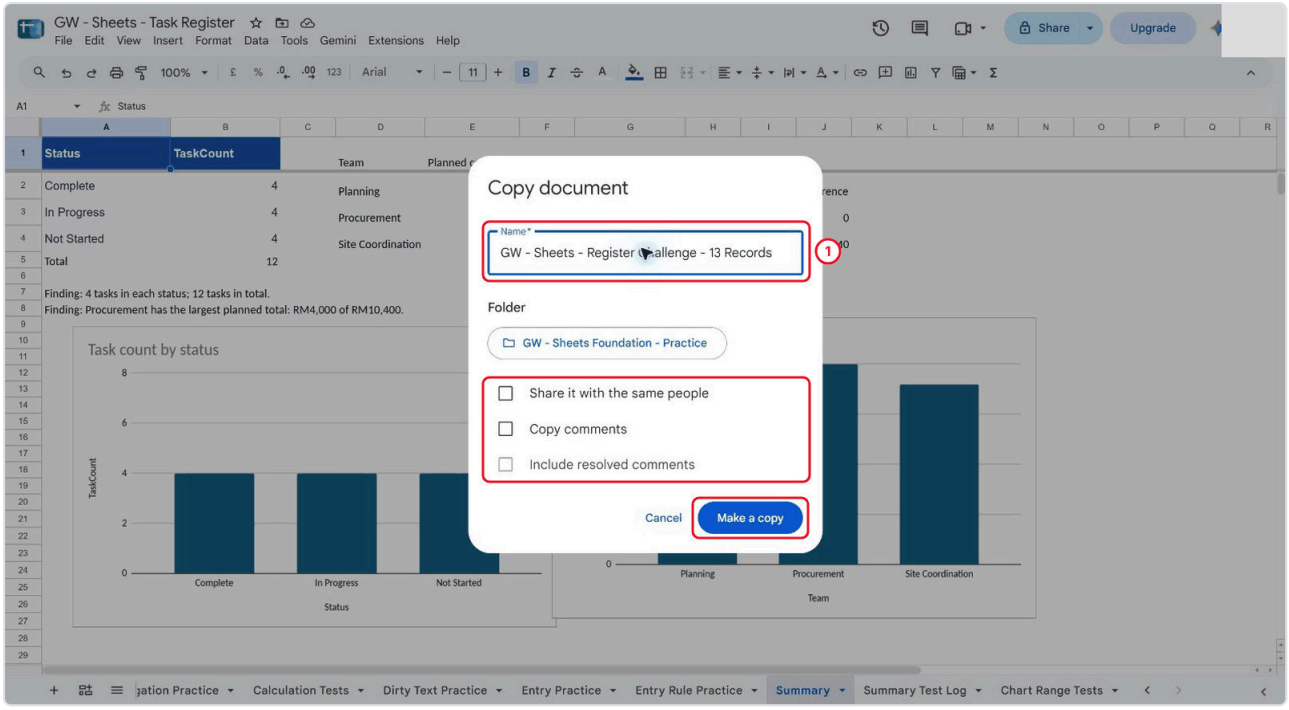
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Make a separate challenge copy of your workbook. The original 12-record baseline must stay intact.

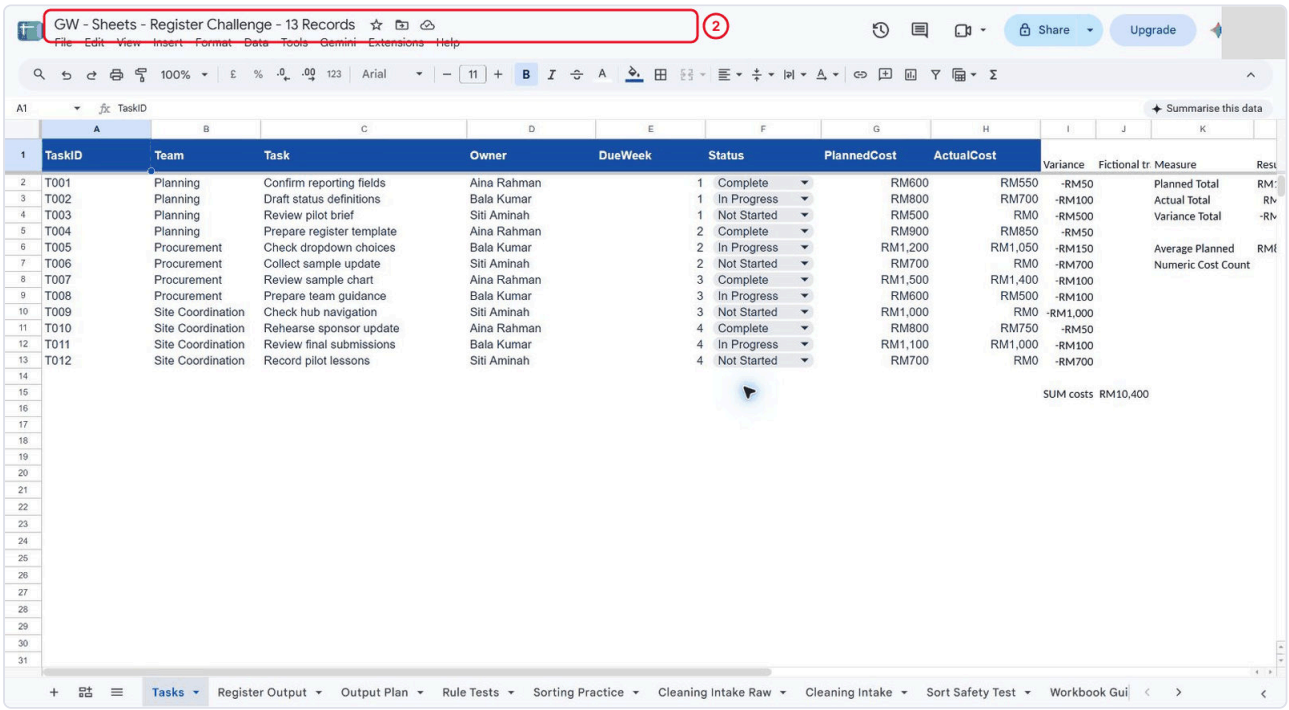
8.1 Add a New Record

1. On the maintained workbook use **File > Make a copy**. Name the new file **GW - Sheets - Register Challenge - 13 Records**. Keep **Share it with the same people** and **Copy comments** unchecked. Verify the new title and distinct workbook identity before editing. This is a whole-workbook scenario copy, not a replacement for the original baseline.



Name a separate private workbook copy.

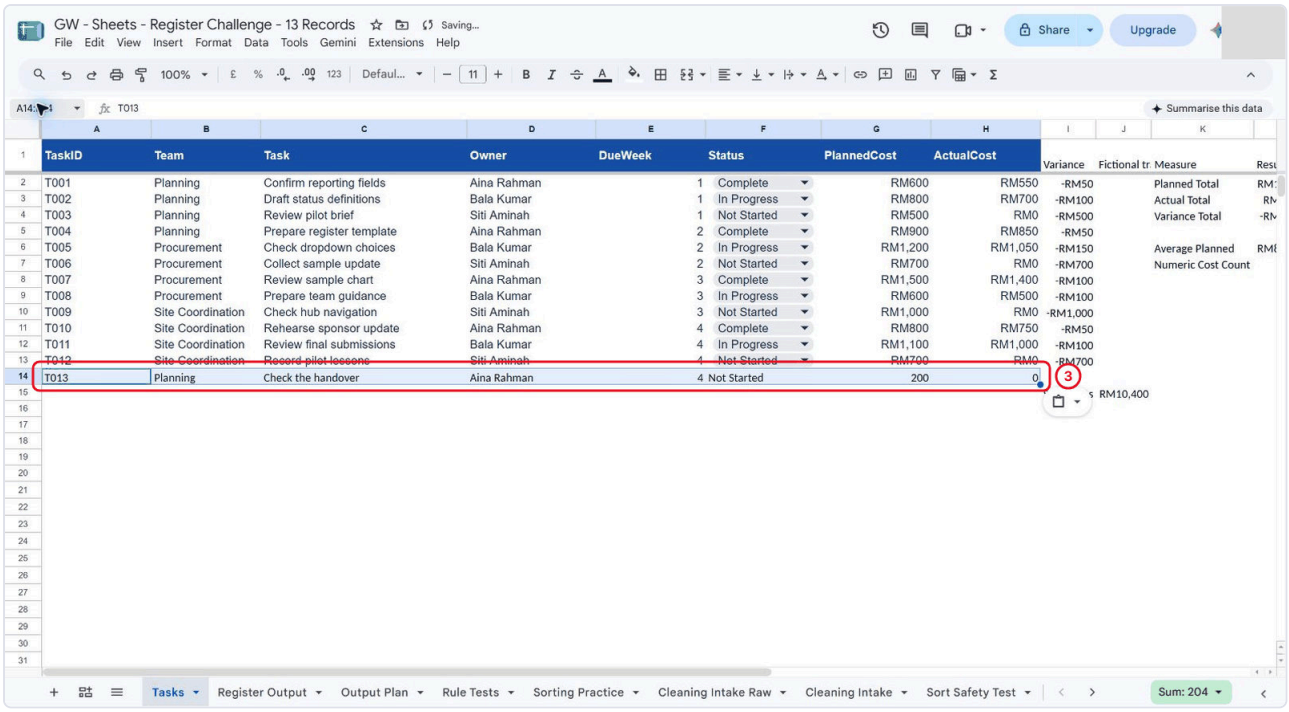
2. Inspect Tasks and its original twelve records. The copied workbook also contains earlier practice tabs and output plans; these are training history. The current challenge handover will use **Tasks / Summary / Change Log** only. Do not present an inherited twelve-record PDF or practice control as the corrected challenge output.



Inspect the named twelve-record challenge starting state.

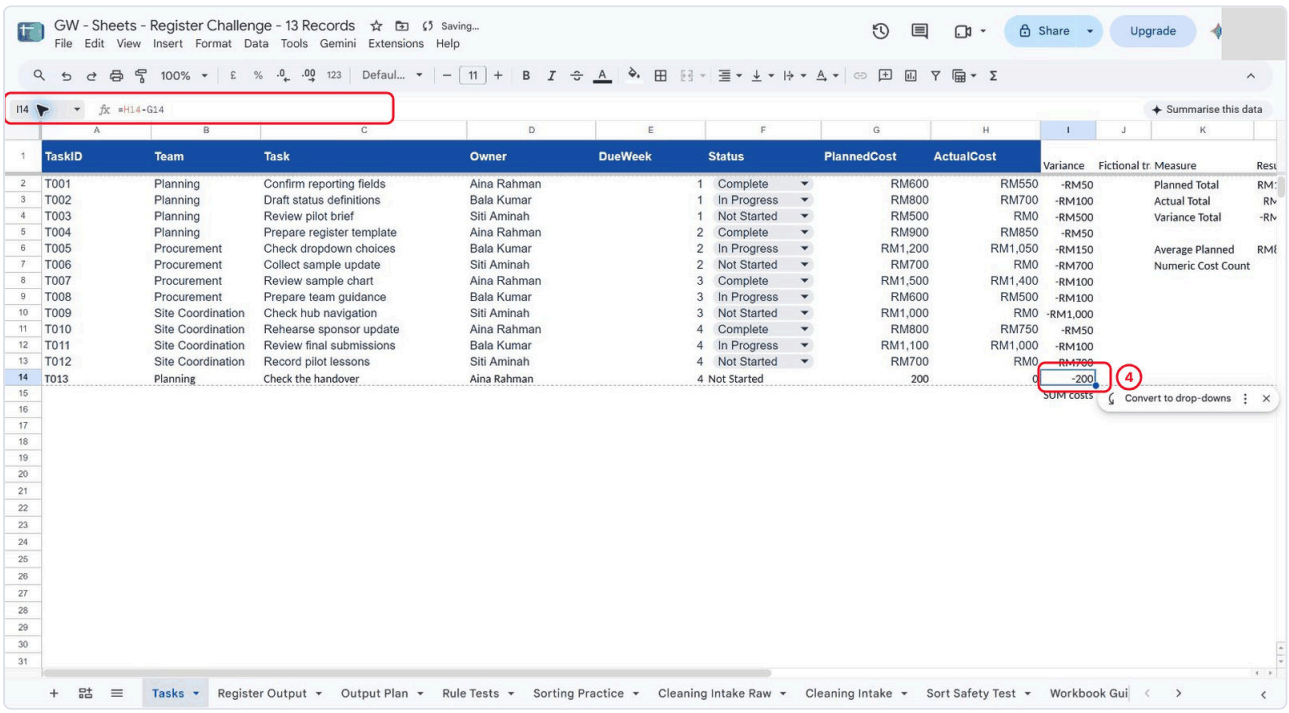
3. In Tasks A14:H14 enter **T013 / Planning / Check the handover / Aina Rahman / 4 / Not Started / 200 / 0** (paste the tab-separated line from Part 3 of the prompts page into A14). The new costs are numeric;

the existing bounded controls still omit row 14 until extended.



Add T013 before extending dependent ranges.

4. Select I13:I14 and **Fill down**. Inspect **I14 =H14-G14**, result **-200**. It must calculate from its own record rather than a prior task.

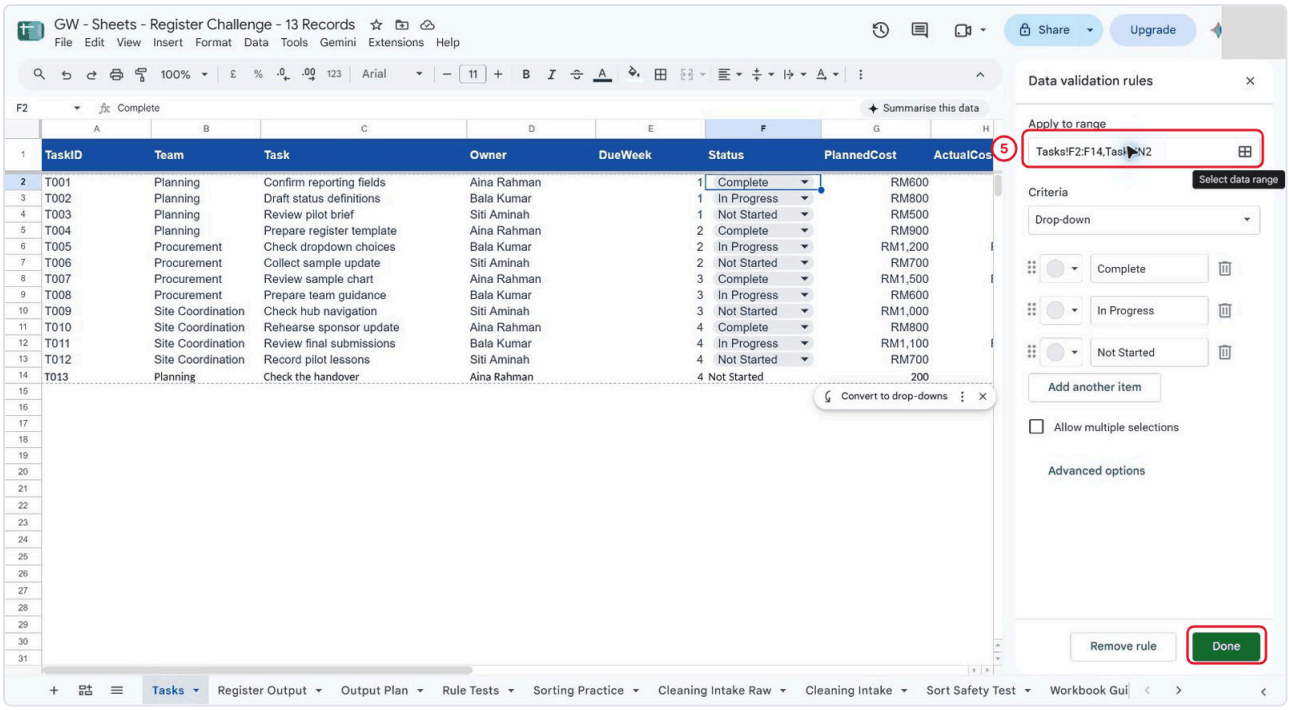


Copy the new row variance formula.

5. Inspect the status rule through **Data > Data validation**. Its intended range is **Tasks!F2:F14**, retaining the separate N2 spare rule. Use the **Select data range** picker, extend the F range and confirm **OK /**

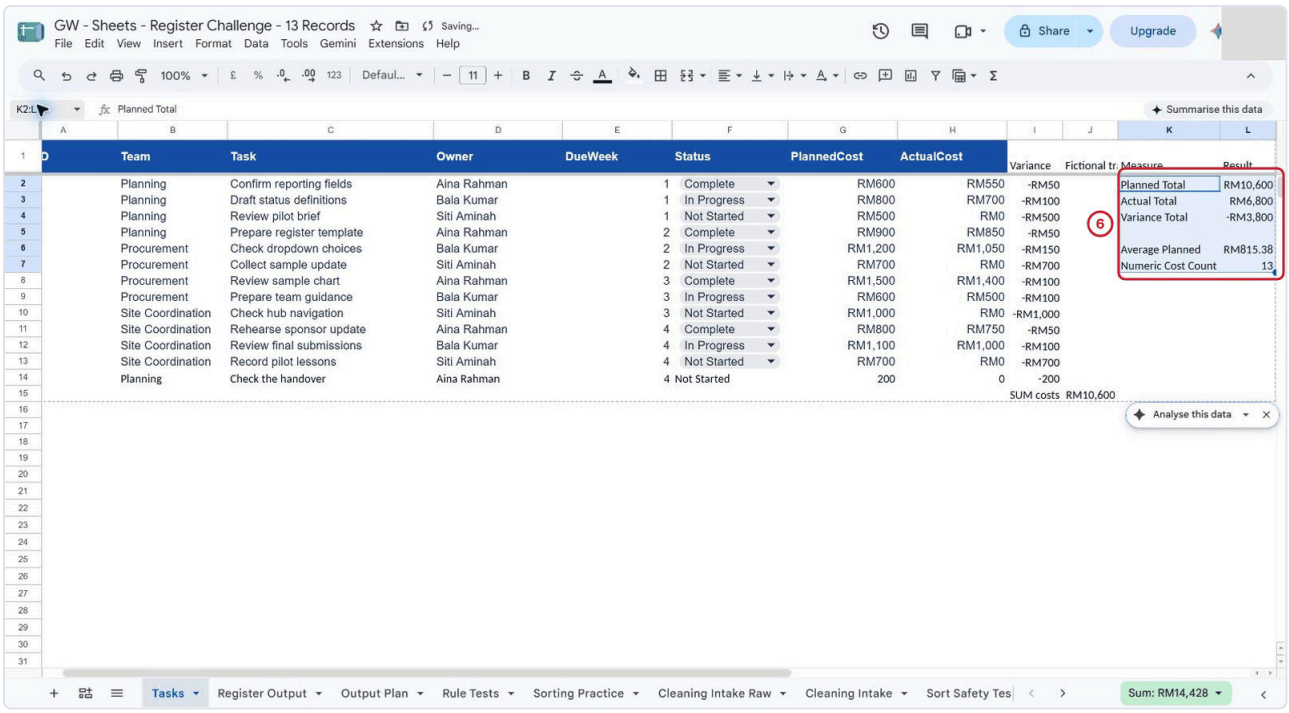
Done; preserve the exact canonical choices and Reject input. The actual new-row verification and committed repair are shown in 8.3. Do not accept a typed range field alone as proof of coverage.

> **Reference run:** The captured field edit below shows the intended extension, but this author's first text-field attempt did not persist.



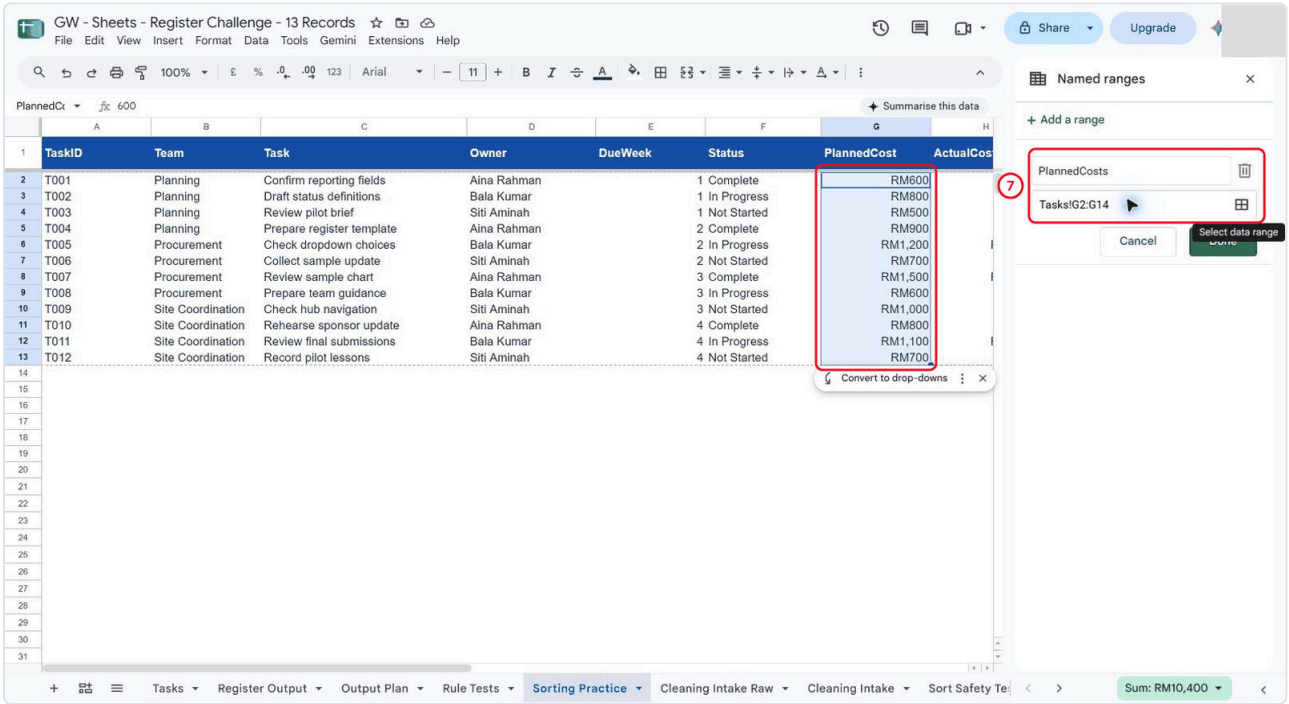
Inspect the intended status rule extension.

6. Extend the live Tasks controls: L2 **=SUM(G2:G14)**, L3 **=SUM(H2:H14)**, L4 **=SUM(I2:I14)**, L6 **=AVERAGE(G2:G14)**, L7 **=COUNT(G2:G14)**, and J15 **=SUM(G2:G14)**. Expected values before correction are **10600 / 6800 / -3800**, average planned **815.384615...**, numeric count **13**. The original twelve task values remain unchanged.



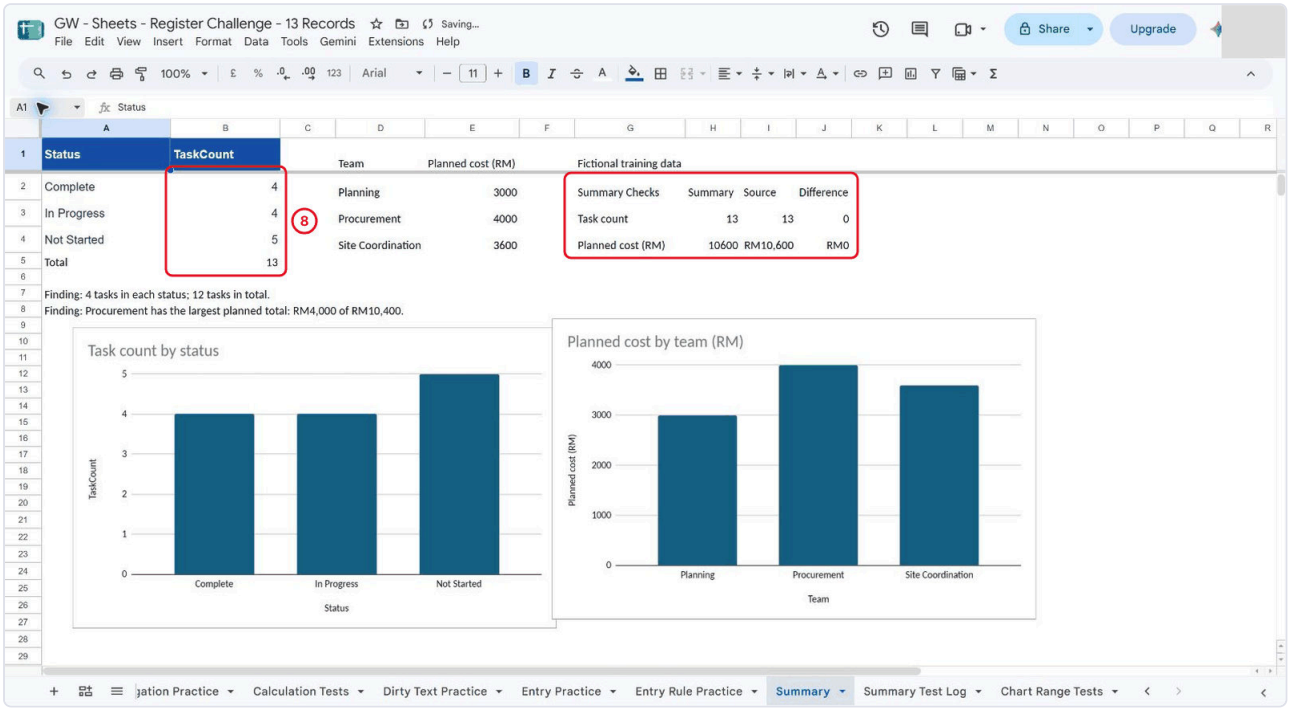
Extend Tasks totals average and numeric count.

7. Inspect **Named ranges**. In the inherited training workbook, PlannedCosts referred to the earlier **Sorting Practice!G2:G13**. In this scenario copy, edit it to the authoritative **Tasks!G2:G14** and save. Navigating to PlannedCosts selects Tasks and reports **RM10600**. Record both the sheet change and new last row; simply extending a historical practice table would leave the named control tied to the wrong source.



Retarget the challenge planned-cost named range.

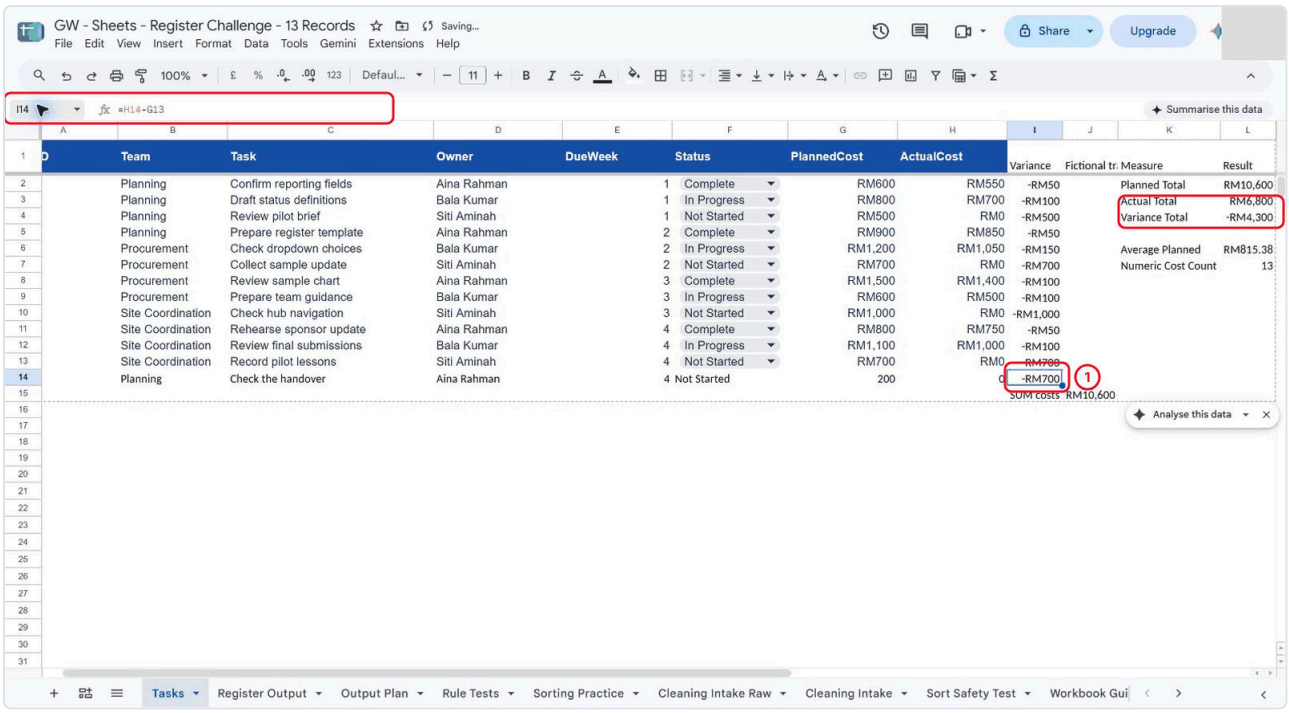
8. On Summary, enter **=COUNTIF(Tasks!$F$2:$F$14,A2)** in B2 and fill to B4. Enter **=SUMIF(Tasks!$B$2:$B$14,D2,Tasks!$G$2:$G$14)** in E2 and fill to E4. Extend I3 to **=COUNTA(Tasks!A2:A14)** and I4 to **=SUM(Tasks!G2:G14)**. The existing B5/H3/H4/difference formulas can remain because their summary-category population has not gained a row. Expect **4 / 4 / 5**, total **13**, team planned **3000 / 4000 / 3600**, total **10600**, zero differences. The inherited written findings are updated in the next section; text does not automatically refresh with a chart.



Extend conditional summary and source checks.

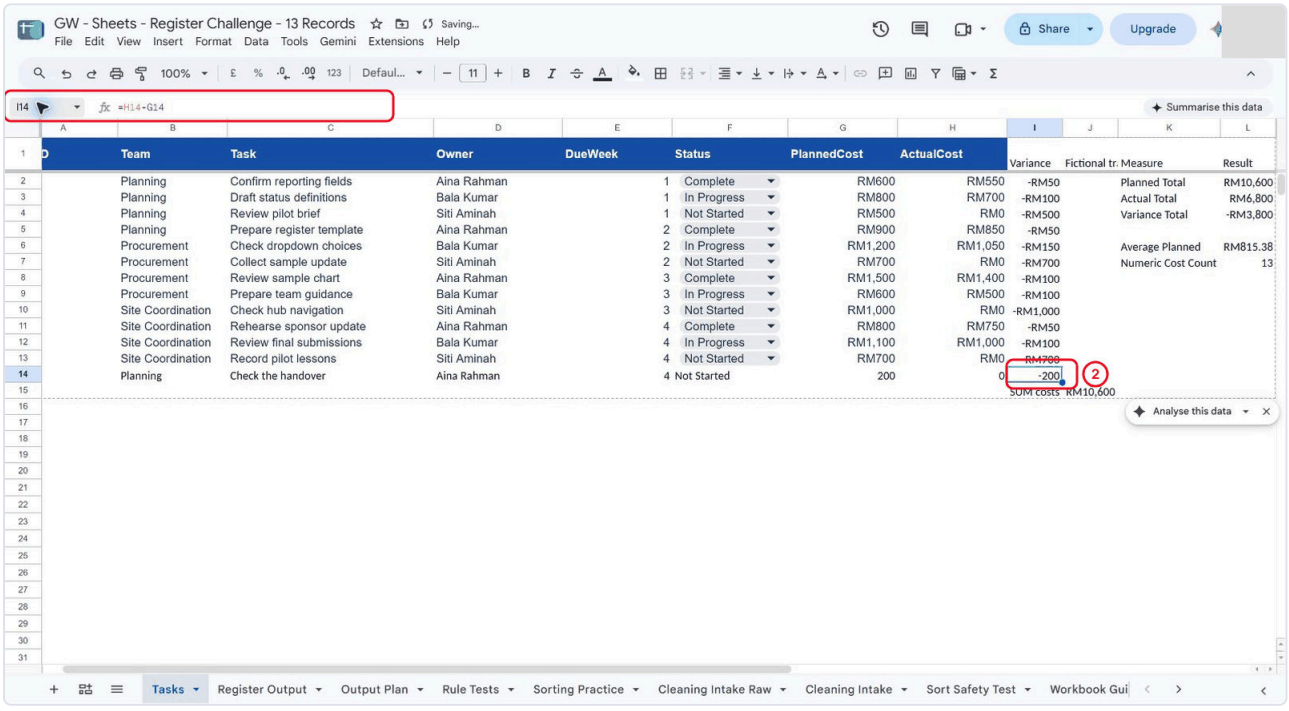
8.2 Repair and Reconcile

1. Predict a deliberate error before applying it. Change challenge I14 to **=H14-G13**. G13 is T012's planned **700**, so the predicted row result is **-700**, summed variance **-4300**, and discrepancy **-500** from the correct actual-minus-planned **-3800**. Inspect the actual row and total; the captured grid shows that predicted fault.



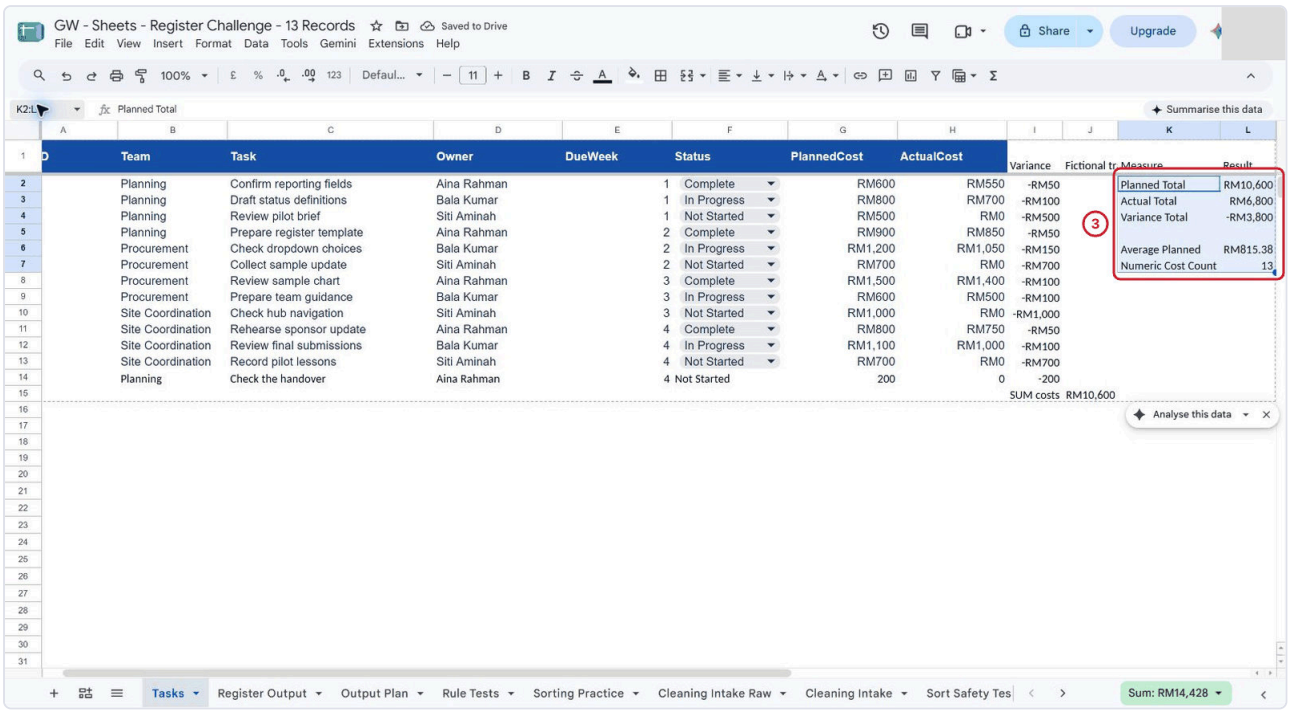
Observe the deliberately wrong row reference and total.

2. Repair I14 to **=H14-G14**. The row returns to **-200**; do not replace the formula with a typed result.



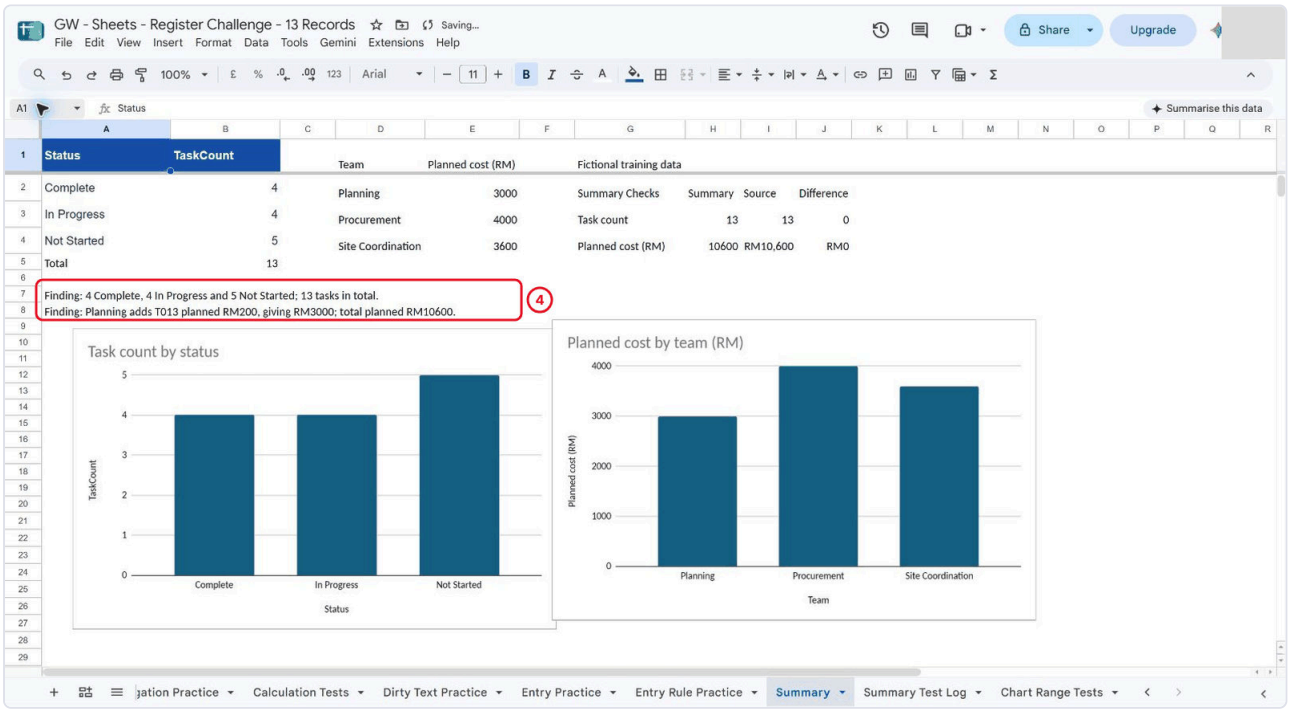
Repair the variance reference to its own row.

3. Reconcile planned **10600**, actual **6800**, and variance **-3800**. Compare the sum of row variances with actual total minus planned total. Numeric planned-cost count is thirteen.



Reconcile thirteen-record cost totals.

4. Update the written findings to **4 Complete / 4 In Progress / 5 Not Started; 13 tasks** and explain that T013 adds planned **RM200**, increasing Planning to **RM3000** and planned grand total to **RM10600**. Inspect both charts. Their category source ranges stay **A1:B4 / D1:E4**; the conditional formulas underneath them now cover the thirteenth source record.

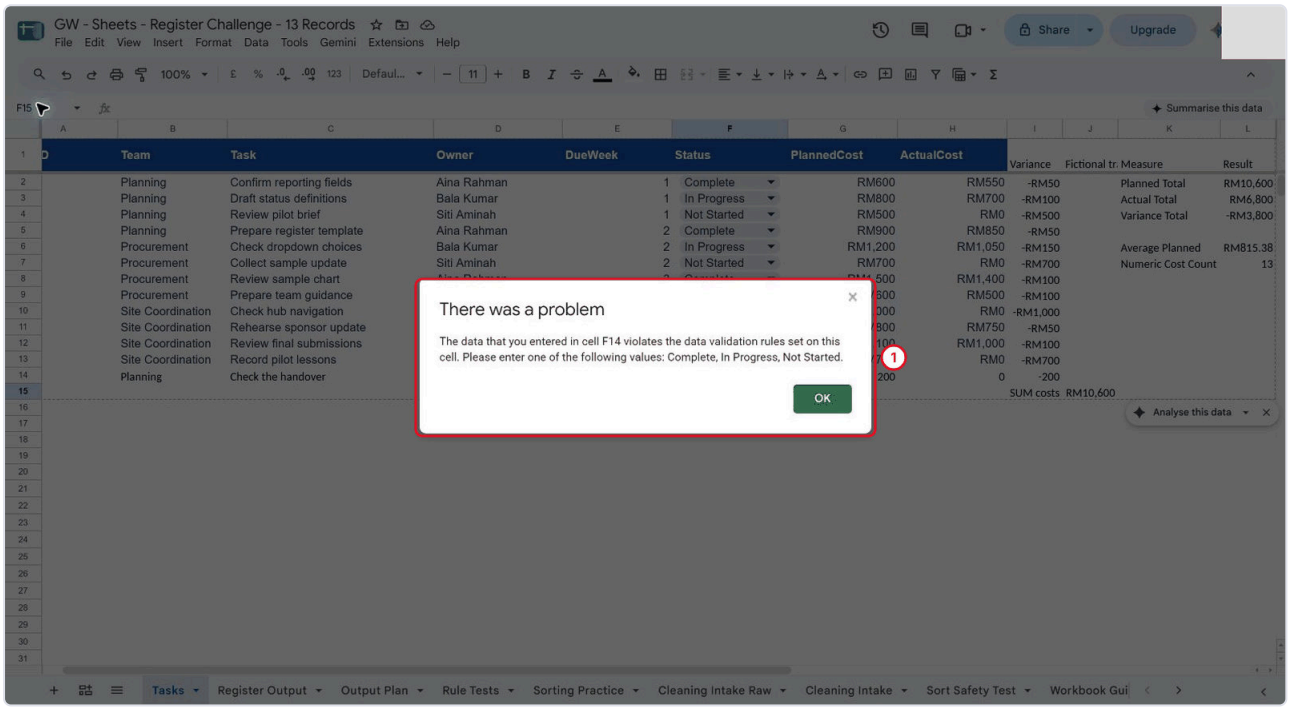


Update findings for the extended population.

8.3 Explain the Result

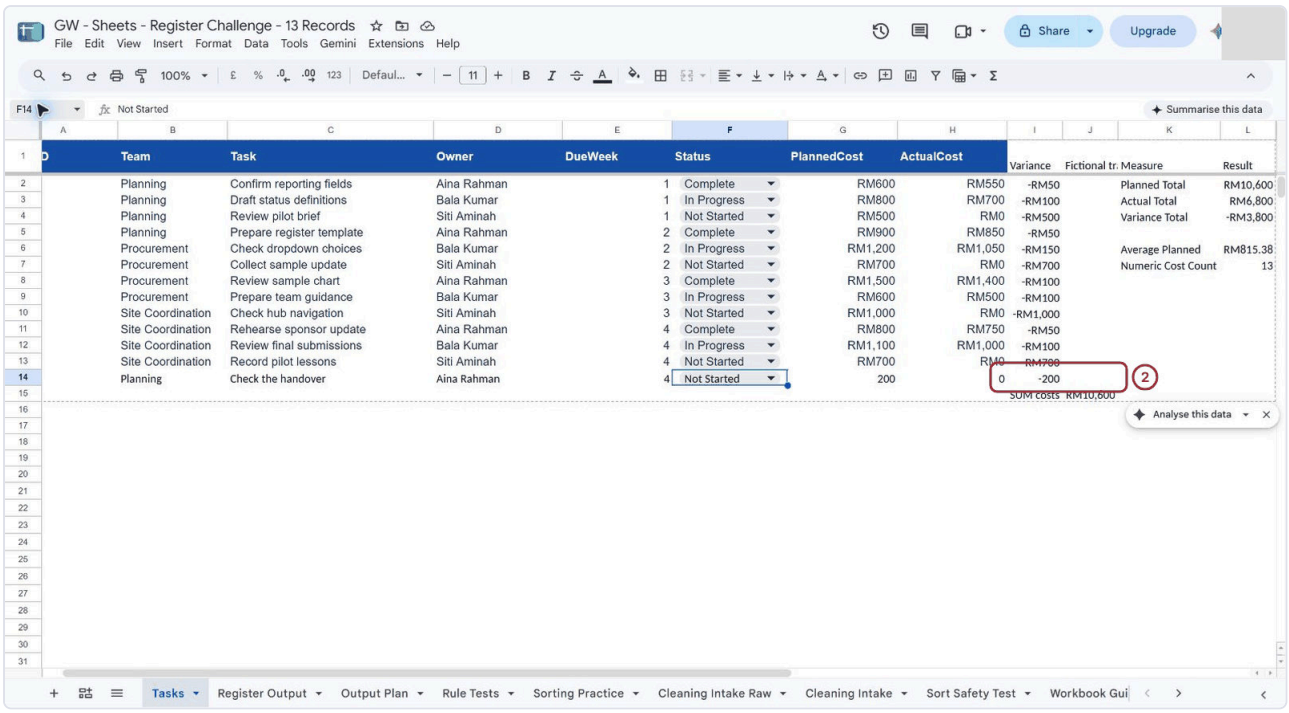
1. Verify the new row's validation before handover. Entering **Done** in F14 through cell-edit mode is now actually rejected with the three allowed statuses.

> **Reference run:** In this run the first compact range-text edit had not saved F14. The author reopened the rule, used **Select data range**, changed the first range to **Tasks!F2:F14**, preserved **Tasks!N2**, then confirmed **OK / Done**.



Verify actual invalid-status rejection on T013.

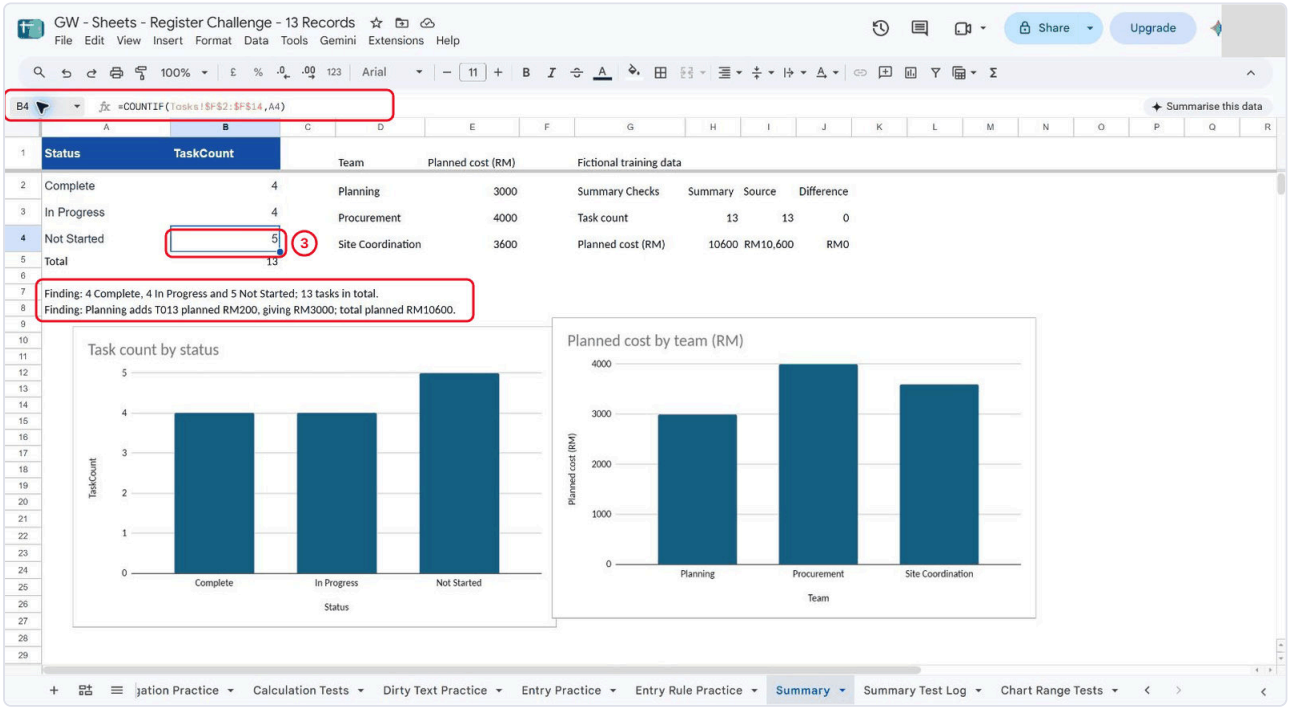
2. Dismiss the message and exit editing. F14 retains **Not Started**. This is evidence of the tested rule on the new row; a dropdown elsewhere in the workbook would not prove that coverage. The repaired range is also confirmed in native cell metadata.



Retain the canonical status after rejection.

3. Explain using the cells: I14's H14/G14 references follow its own row; Summary B4 holds **=COUNTIF(Tasks!$F$2:$F$14,A4)**, with a fixed source range and relative criterion; F14 uses the canonical strict status rule. Identify the new record's RM200 planned contribution and its effect on population and variance.

> **Reference run:** This reference provides the cell-supported author explanation; the requested partner explanation remains a learner/human check and was not performed.

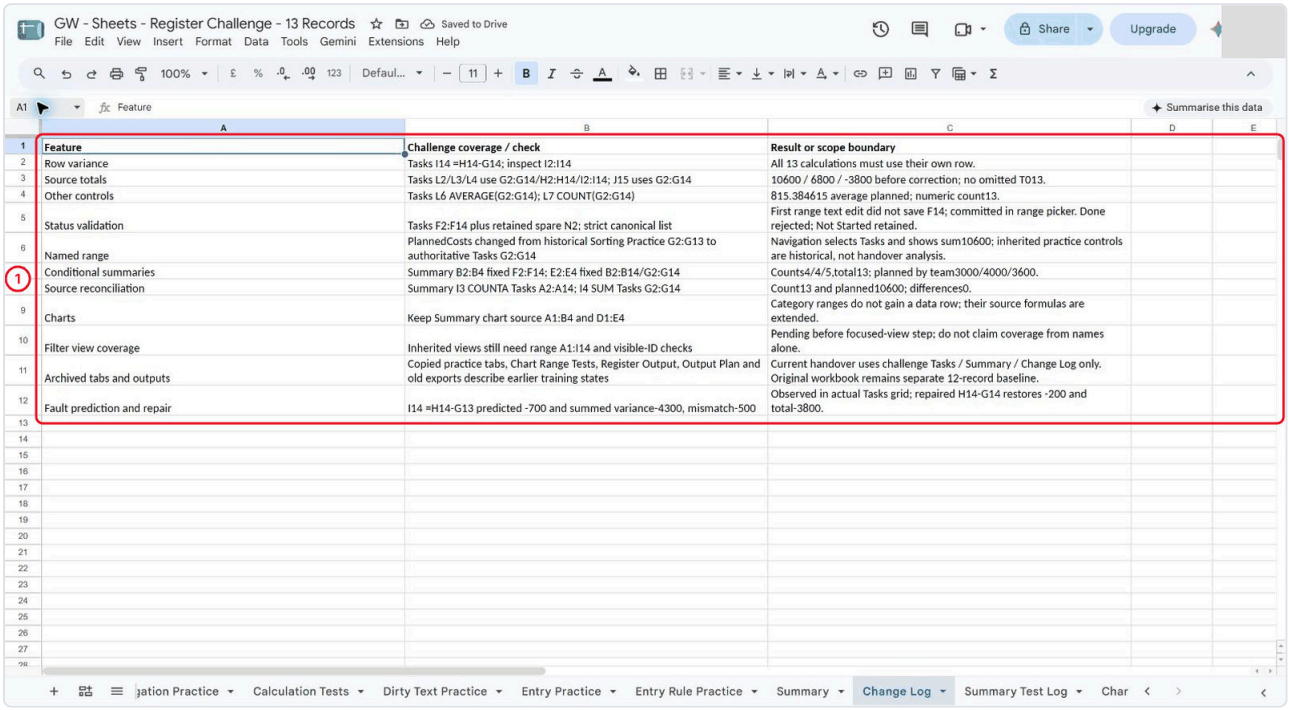


Explain an anchored range and relative criterion with actual cells.

8.4 Stress-Test the Extended Register and Complete the Handover

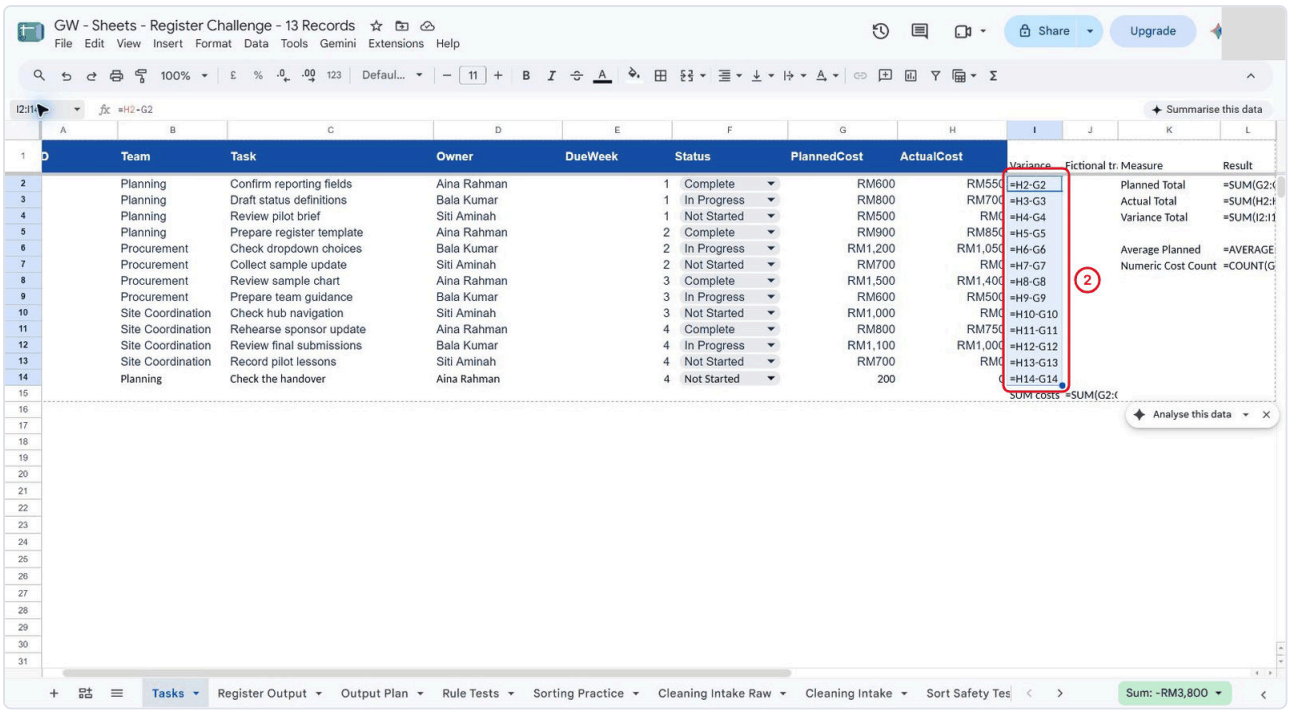
1. Add a sheet named **Change Log** with the headers and feature labels in Part 4 of the prompts page, then record every active range changed for T013: row variance, totals, average/count, status validation, PlannedCosts, conditional summaries, source checks, chart dependency and filter-view coverage. Distinguish inherited training tabs and baseline exports from the current scenario. Leave filter-view verification pending until reopened below.

> **Reference run:** The reference uses wrapped 450-pixel A:C columns so descriptions remain readable.



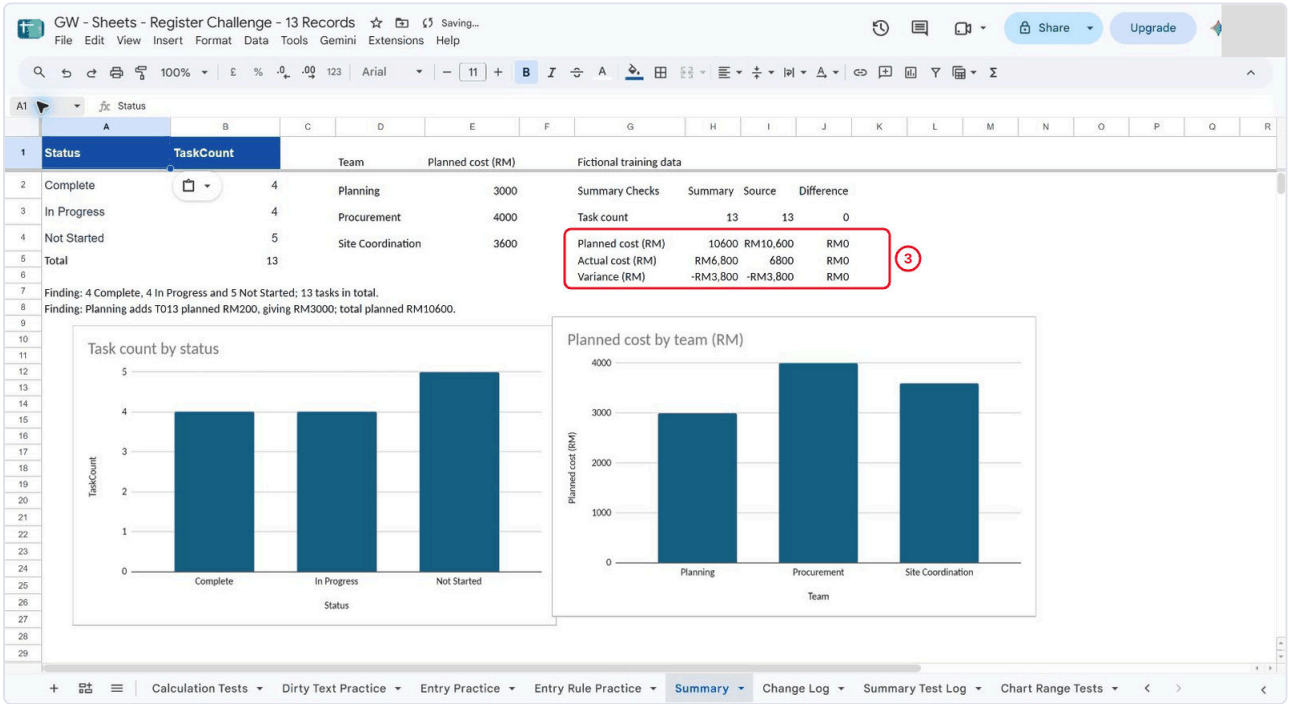
Record full range coverage and historical-tab boundaries.

2. On Tasks use **View > Show > Formulae**, or menu search **Show all formulae**. Inspect every cell I2:I14: each is **=Hrow-Grow**, including repaired I14. Turn formula display off afterward. If a sort is required, sort whole records and repeat the row-reference check; this reference did not perform an additional sort in Lab 8.



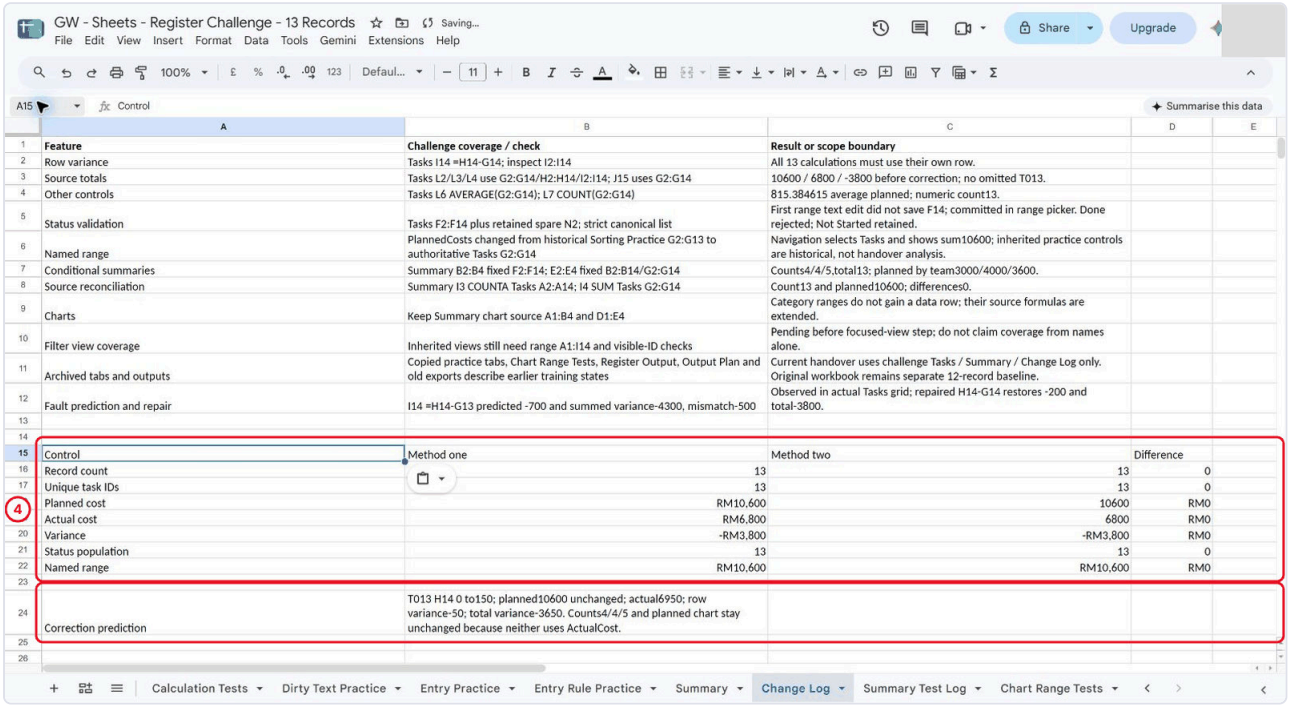
Inspect every variance formula in the live grid.

3. Extend Summary's control block with actual and variance checks in G5:J6 (type **Actual cost (RM)** in G5 and **Variance (RM)** in G6). H5 is **=SUM(Tasks!H2:H14)**. I5 independently adds SUMIF actual totals for Planning, Procurement and Site Coordination over B2:B14/H2:H14; J5 is H5-I5. H6 is **=SUM(Tasks!I2:I14)**; I6 is H5-H4, and J6 is H6-I6. Before the correction these reconcile to **6800 / -3800** and zero differences.



Add actual and variance reconciliation before correction.

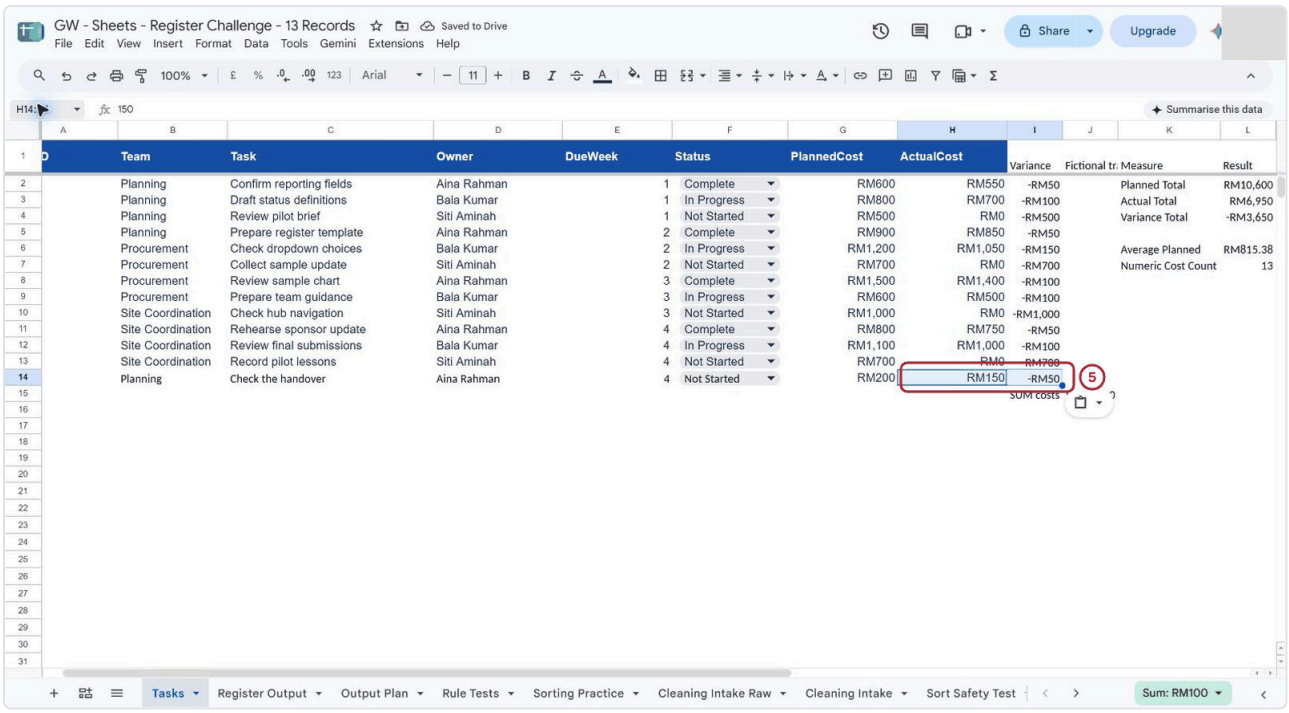
4. In Change Log A15:D22 make a control table **Control / Method one / Method two / Difference**, with the row labels in Part 4 of the prompts page. Compare COUNTA task IDs with COUNT numeric planned costs; COUNTUNIQUE with COUNTA; SUM planned with the team SUMIF total; SUM actual with the three-team actual total; summed variance with actual-minus-planned; status-count sum with record count; and SUM(PlannedCosts) with the direct source sum. Each difference should be zero. Record the next prediction in A24:B24 before changing the source: **T013 ActualCost 0→150** should give row variance **-50**, actual grand total **6950**, variance grand total **-3650**, with planned **10600** unchanged. Status and planned-cost charts should remain unchanged because neither uses ActualCost.



	A	B	C	D	E
1	**Feature**	**Challenge coverage / check**	**Result or scope boundary**		
2	Row variance	Tasks I14 =H14-G14; inspect I2:I14	All 13 calculations must use their own row.		
3	Source totals	Tasks L2/L3/L4 use G2:G14/H2:H14/I2:I14; J15 uses G2:G14	10600 / 6800 / -3800 before correction; no omitted T013.		
4	Other controls	Tasks L6 AVERAGE(G2:G14); L7 COUNT(G2:G14)	815.384615 average planned; numeric count13.		
5	Status validation	Tasks F2:F14 plus retained spare N2; strict canonical list	First range text edit did not save F14; committed in range picker. Done rejected; Not Started retained.		
6	Named range	PlannedCosts changed from historical Sorting Practice G2:G13 to authoritative Tasks G2:G14	Navigation selects Tasks and shows sum10600; inherited practice controls are historical, not handover analysis.		
7	Conditional summaries	Summary B2:B4 fixed F2:F14; E2:E4 fixed B2:B14/G2:G14	Counts4/4/5,total13; planned by team3000/4000/3600.		
8	Source reconciliation	Summary I3 COUNTA Tasks A2:A14; I4 SUM Tasks G2:G14	Count13 and planned10600; differences0.		
9	Charts	Keep Summary chart source A1:B4 and D1:E4	Category ranges do not gain a data row; their source formulas are extended.		
10	Filter view coverage	Inherited views still need range A1:I14 and visible-ID checks	Pending before focused-view step; do not claim coverage from names alone.		
11	Archived tabs and outputs	Copied practice tabs, Chart Range Tests, Register Output, Output Plan and old exports describe earlier training states	Current handover uses challenge Tasks / Summary / Change Log only. Original workbook remains separate 12-record baseline.		
12	Fault prediction and repair	I14 =H14-G13 predicted -700 and summed variance-4300, mismatch-500	Observed in actual Tasks grid; repaired H14-G14 restores -200 and total-3800.		
13					
14					
15	Control	Method one	Method two	Difference	
16	Record count	13	13	0	
17	Unique task IDs	13	13	0	
4	Planned cost	RM10,600	10600	RM0	
19	Actual cost	RM6,800	6800	RM0	
20	Variance	-RM3,800	-RM3,800	RM0	
21	Status population	13	13	0	
22	Named range	RM10,600	RM10,600	RM0	
23					
24	Correction prediction	T013 H14 0 to150; planned10600 unchanged; actual6950; row variance-50; total variance-3650. Counts4/4/5 and planned chart stay unchanged because neither uses ActualCost.			
25					
26					

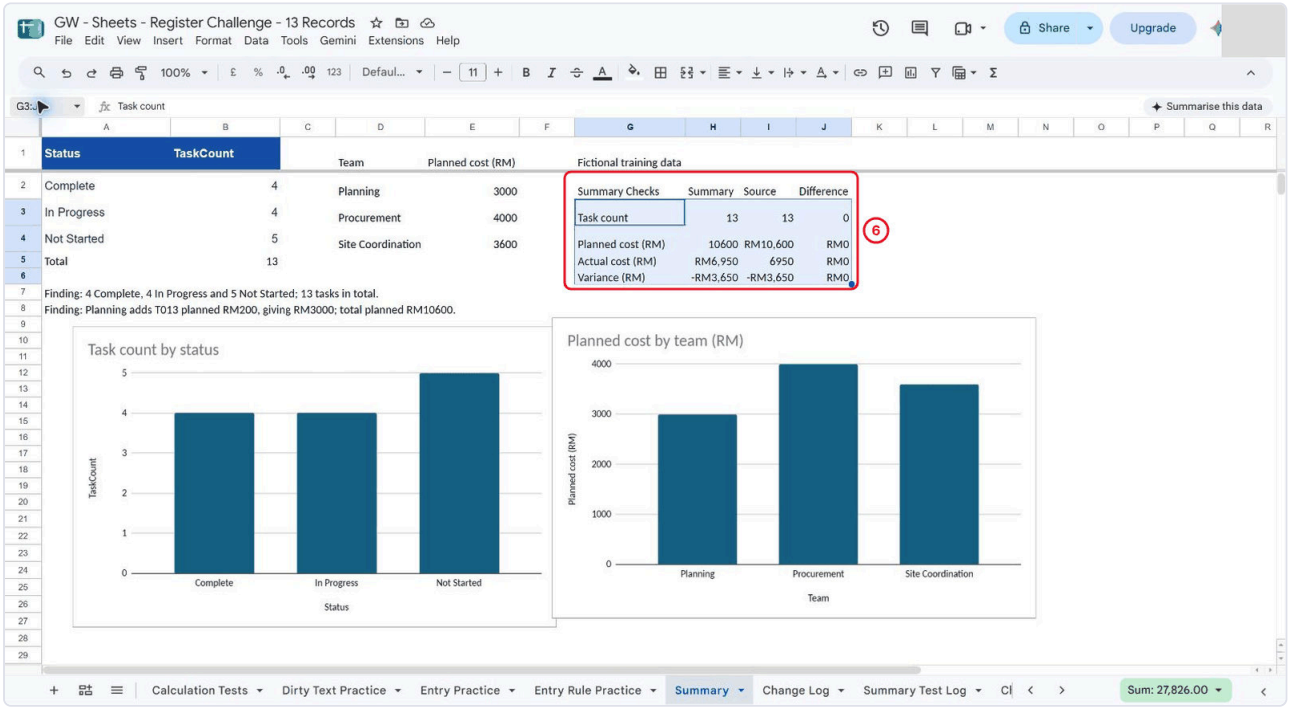
Record two-method controls and correction prediction.

5. Apply **150** to challenge Tasks H14 only. Verify **I14 =H14-G14 = -50**. For consistent display, copy G13:H13 and paste **Format only** into G14:H14 without overwriting numeric 200/150. Preserve the clipboard while selecting the destination; pasting an address would replace the copied cells.



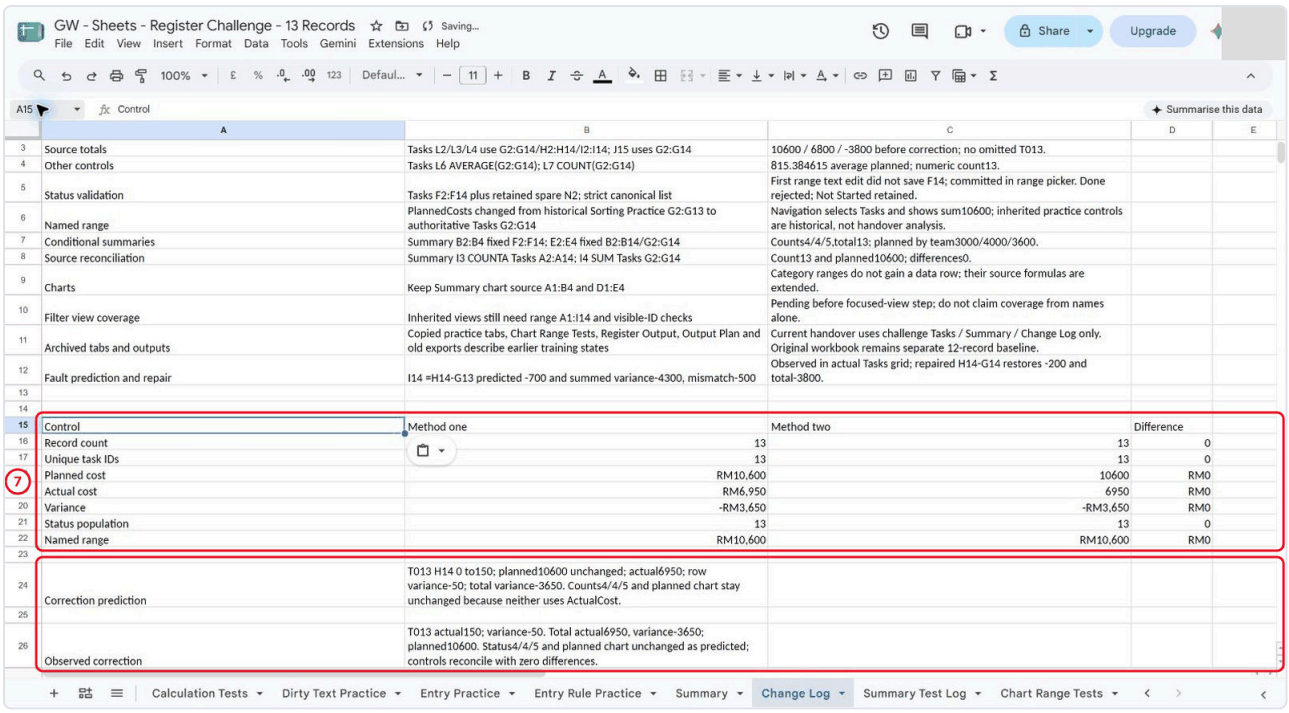
Correct T013 ActualCost to one hundred fifty.

6. Inspect Summary: actual **6950**, variance **-3650**, planned **10600**, counts **4/4/5**, zero reconciliation differences. The unchanged charts are correct for their selected measures; the actual/variance controls communicate the correction.



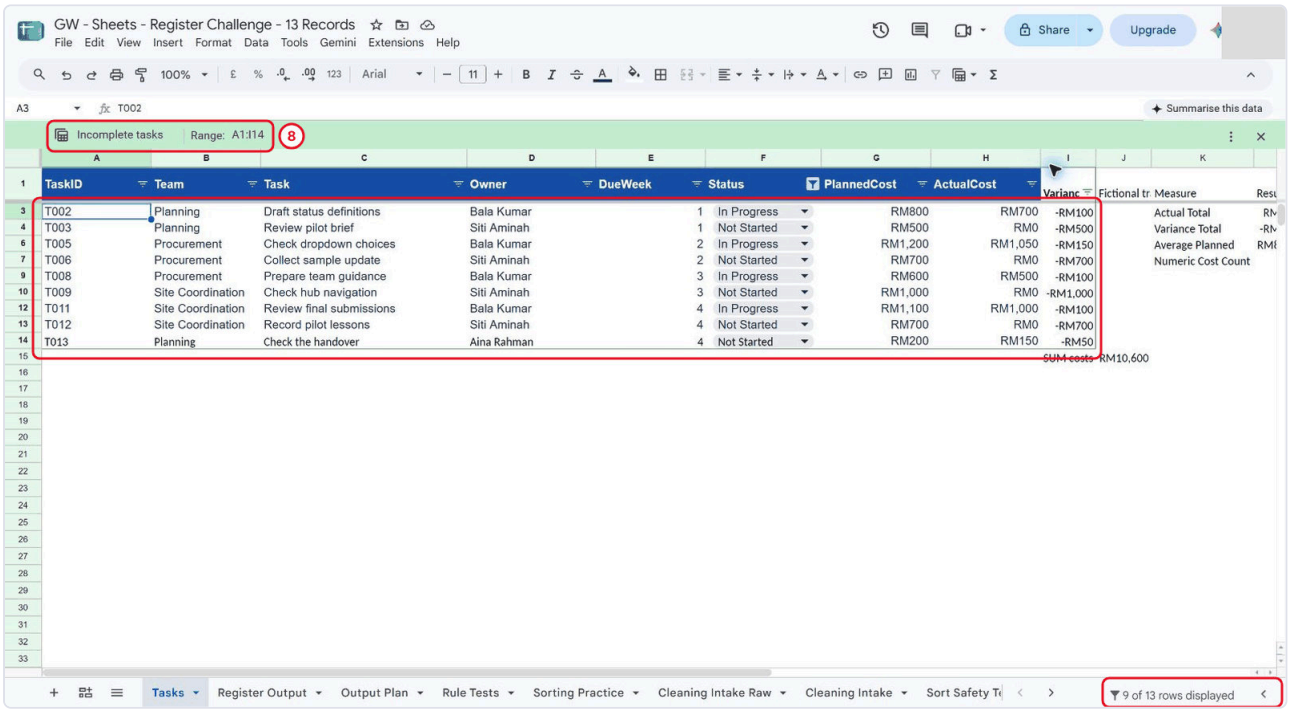
Verify refreshed actual and variance controls.

7. Recheck the two-method table and record observed versus predicted effects in A26:B26. The corrected state is retained in this scenario copy; the original baseline is not changed to 13 records.



Reconcile the corrected state by two methods.

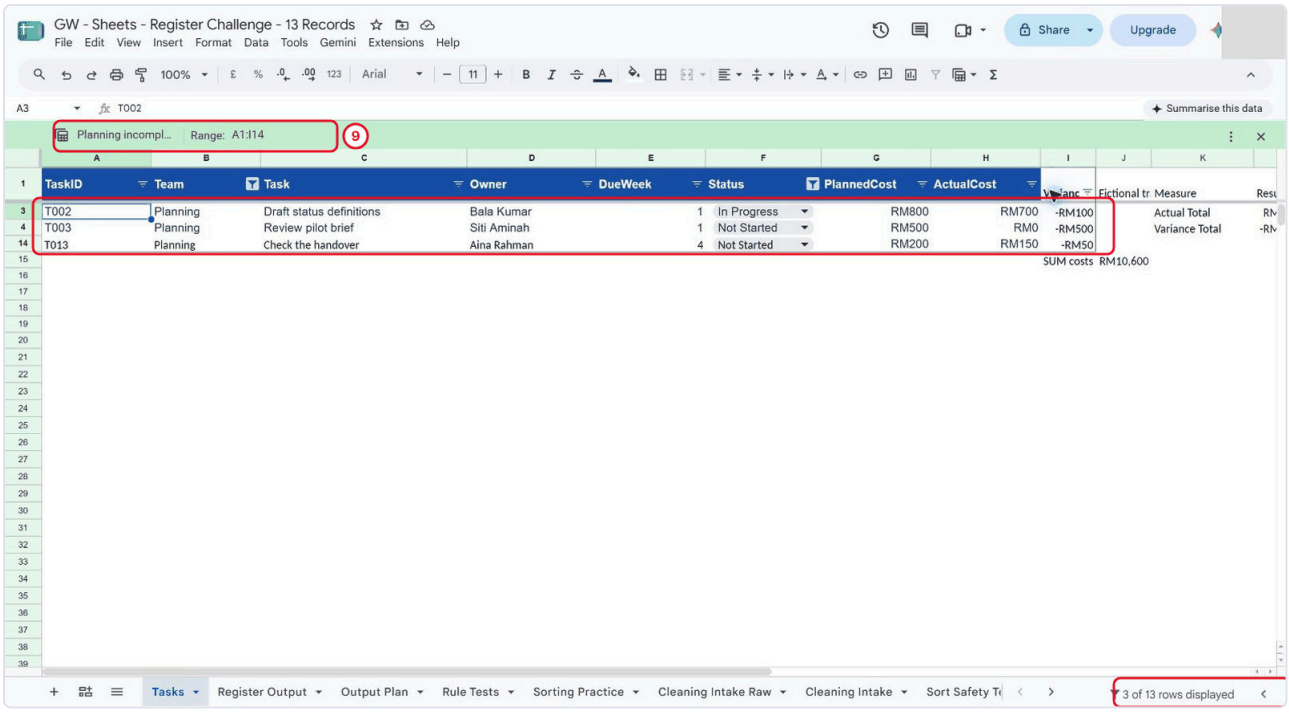
8. Reopen **Incomplete tasks** and inspect its range and record IDs. In this session the inherited view already reopened over **A1:I14**, so no manual range edit was needed. Verify nine visible IDs **T002, T003, T005, T006, T008, T009, T011, T012, T013**. Check bounds explicitly; do not assume the application will expand a bounded view in every future run.



Verify nine incomplete records in the extended view.

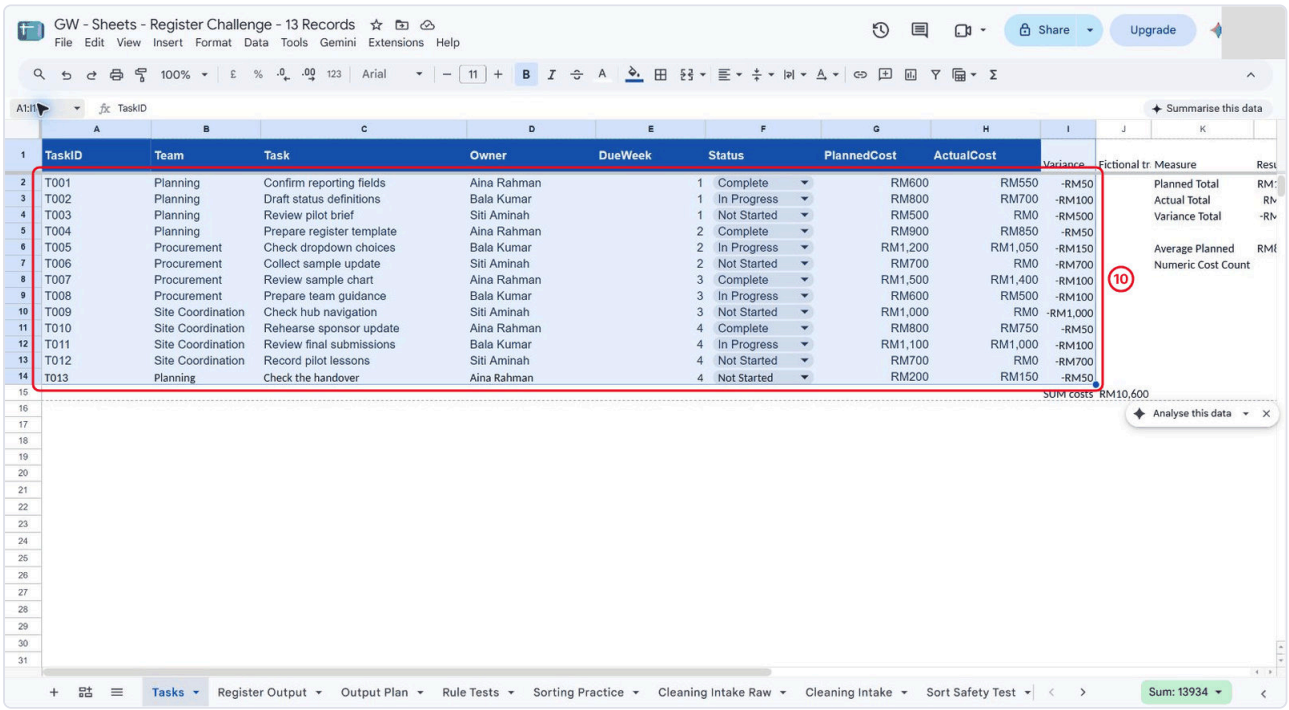
9. Close that view and reopen **Planning incomplete tasks**. It also shows **A1:I14**, with **3 of 13 rows displayed**: **T002, T003, T013**. The new task must be included. A saved view name alone is not a

population check.



Verify the three incomplete Planning IDs include T013.

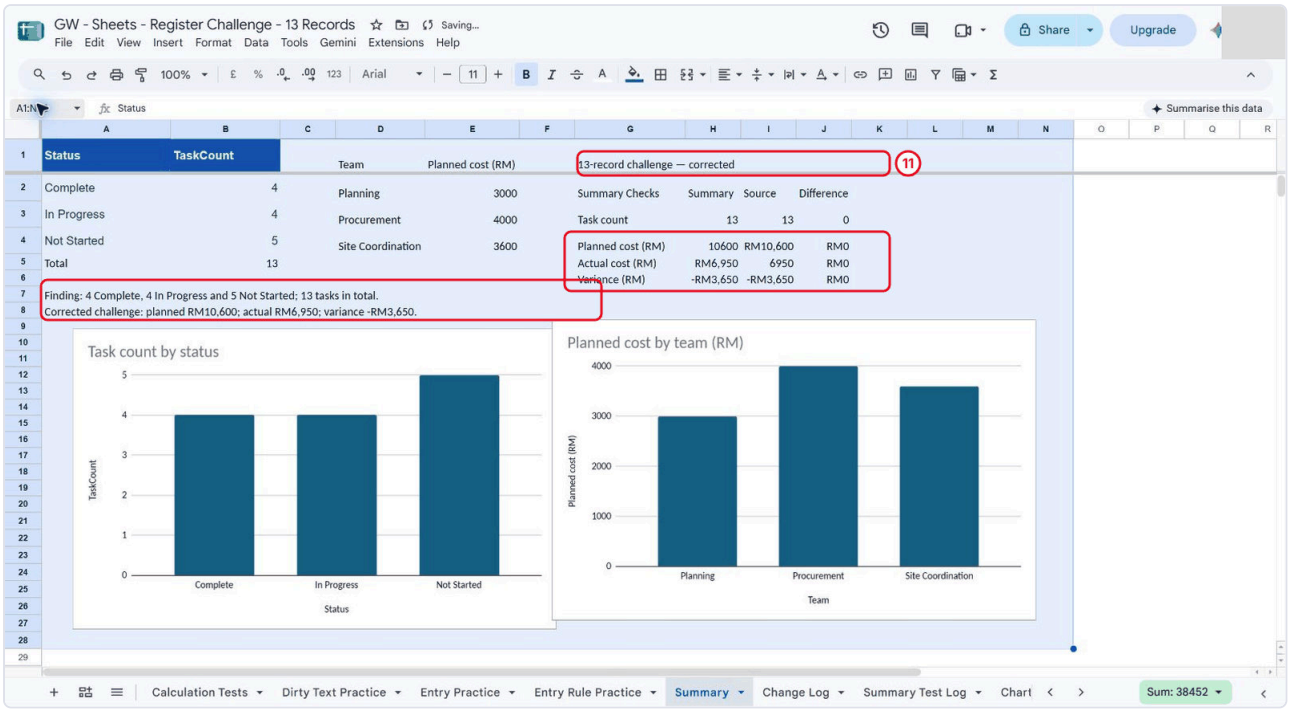
10. Exit the view and leave all thirteen records visible. Update Change Log with the observed ranges and IDs, noting that both views were closed. Ordinary SUM still includes the full source population while a view hides rows.



Close views and restore all thirteen records.

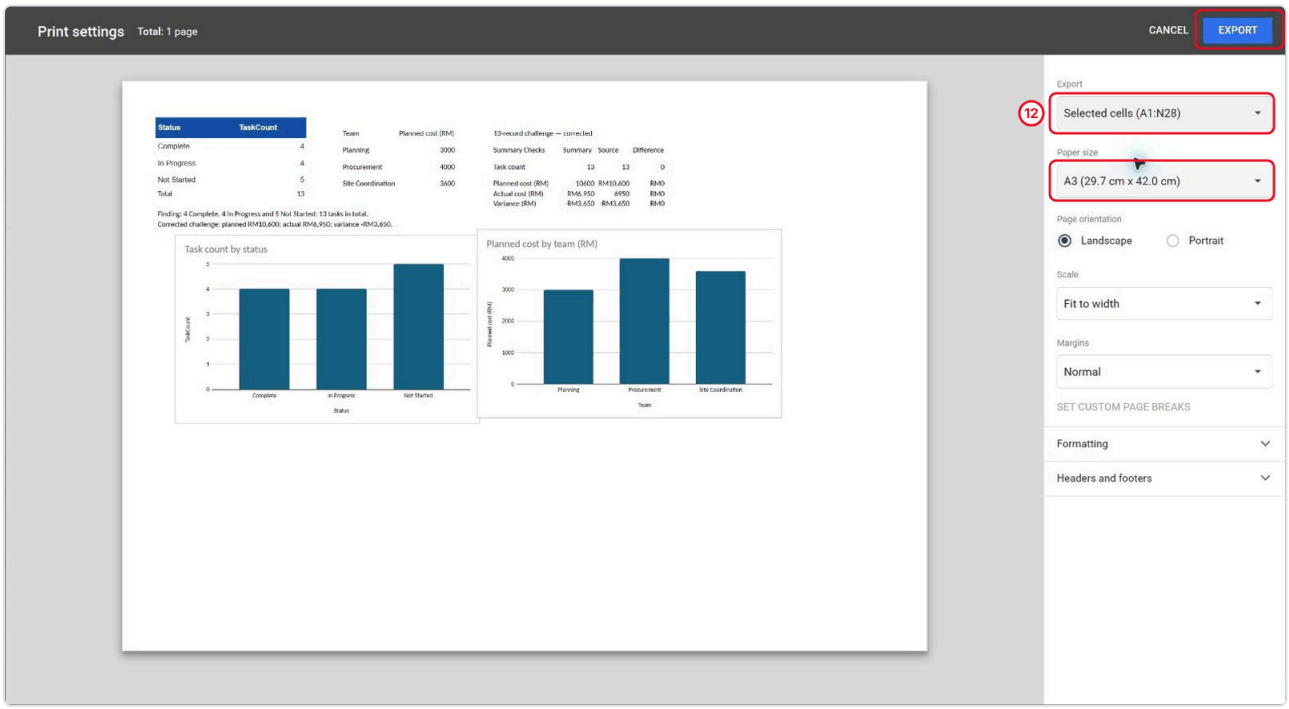
11. Label Summary **13-record challenge - corrected** and write a current finding with **planned RM10600 / actual RM6950 / variance -RM3650**. Keep the task-count finding at 4/4/5, thirteen total. Include the

new actual/variance controls in the sponsor output so its corrected state can be distinguished from the old baseline PDF.



Prepare the labelled corrected challenge summary.

12. Select **Summary A1:N28**, use **Download as PDF**, and explicitly choose **Selected cells**, A3, Landscape, Fit to width. Inspect the complete labels, charts and current control values. Export as **GW-Register-Challenge-Corrected-r1.pdf**; do not overwrite a baseline file or export the whole workbook's historical practice tabs.

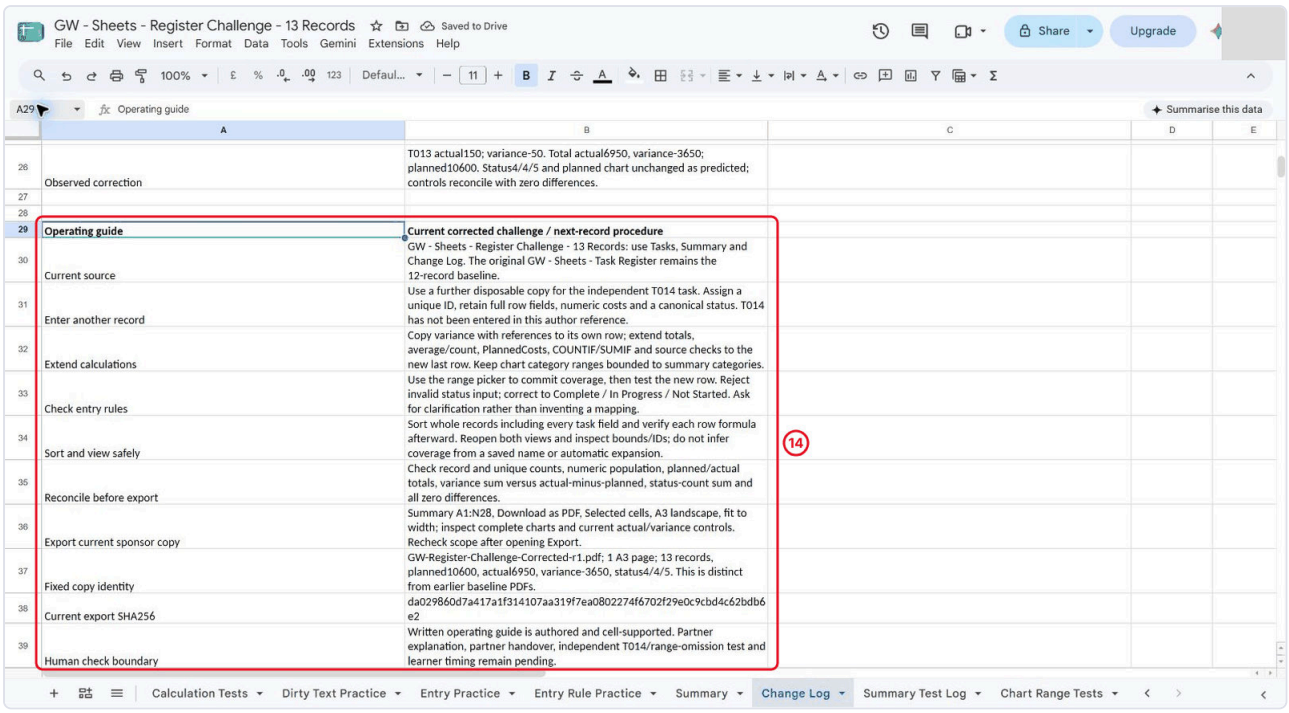


Choose only the corrected summary for PDF export.

13. Inspect the actual saved page. It is a one-page A3 native PDF with approximately 10.7 pt text, thirteen tasks, status **4/4/5**, Planning **3000**, planned **10600**, actual **6950**, variance **-3650**, zero differences and both complete charts. It is distinct from the untouched baseline exports.

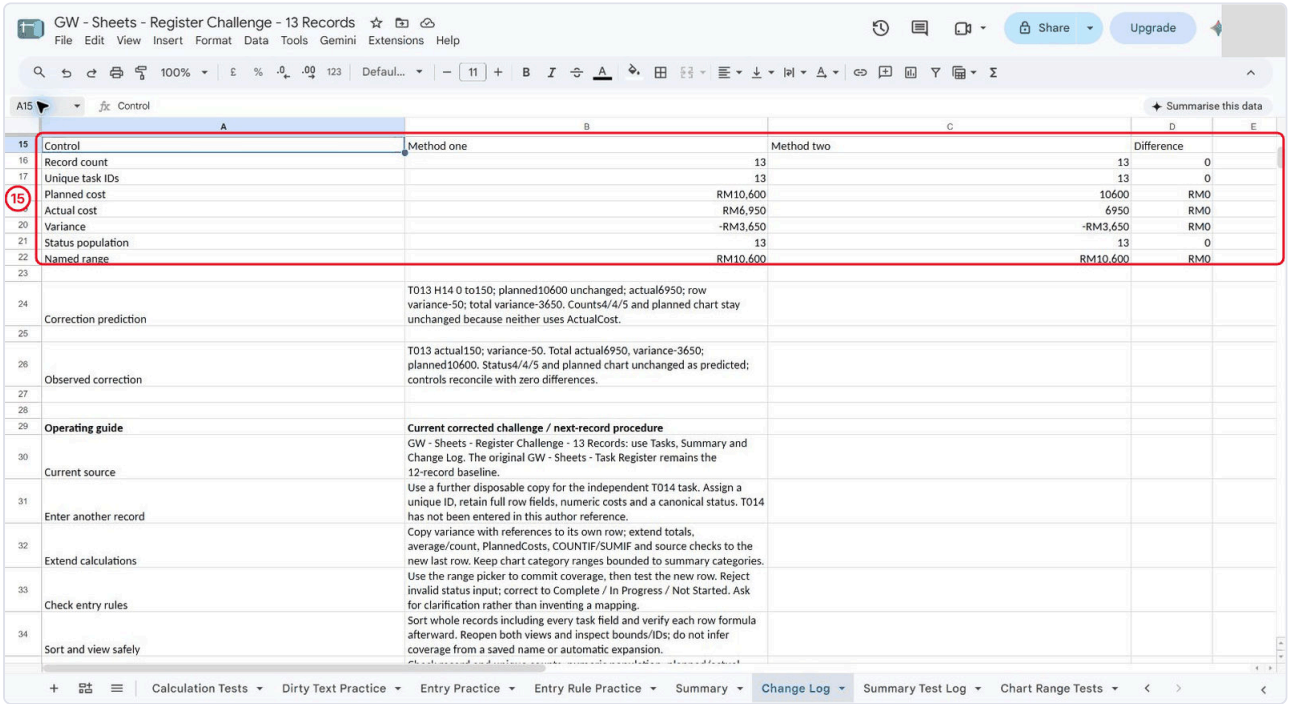
Inspect the actual corrected challenge PDF.

14. Write an operating guide in Change Log from A29 down (row labels in Part 4 of the prompts page): identify the maintained challenge source; enter the next record on a further disposable copy; retain unique IDs/numeric costs/canonical statuses; copy a same-row formula; extend every active bounded range; commit and test rule coverage; inspect whole-record sorts and view IDs; reconcile by two methods; then export a newly named current summary. Record the PDF's scope, state, page count and hash. Partner handover and independent T014 remain pending rather than being inferred from this written guide.



Write the operating guide and fixed-copy identity.

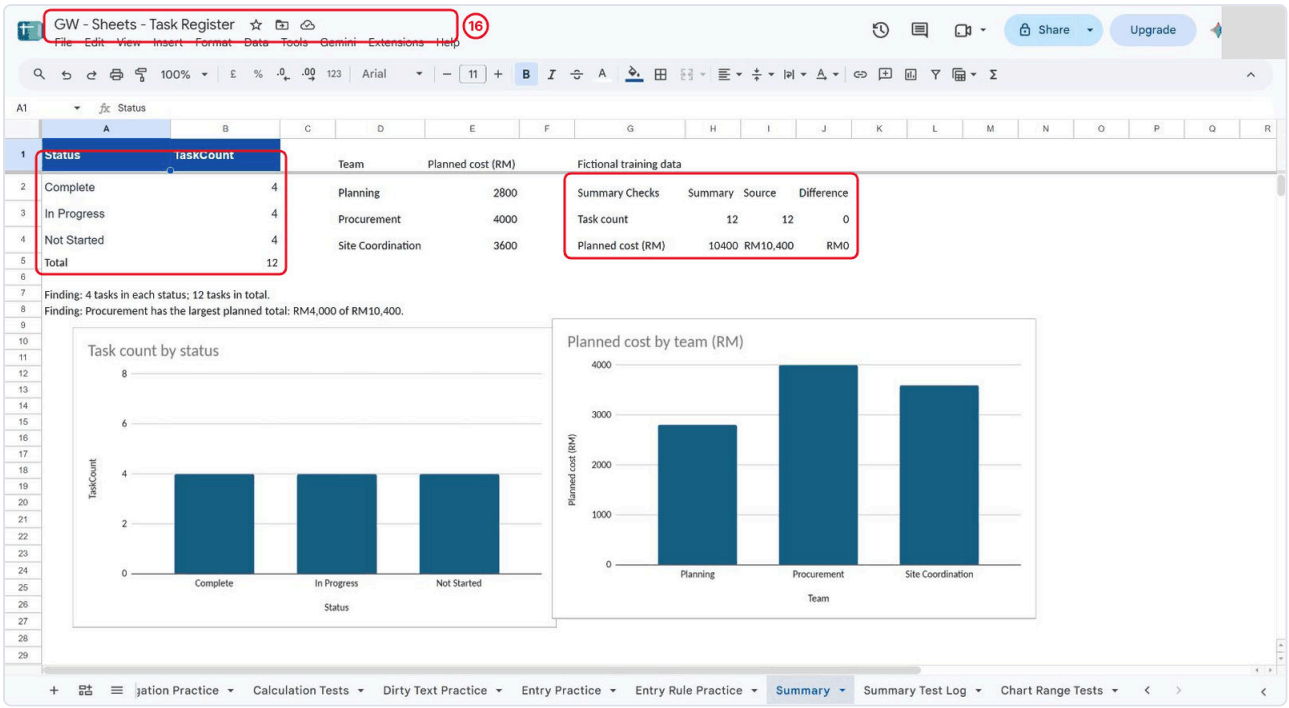
15. Inspect the completed coverage log and control table. Current handover sources are **Tasks / Summary / Change Log**; copied practice tabs, prior Register Output, Output Plan and old PDFs describe earlier training states. The PlannedCosts definition now belongs to the authoritative challenge Tasks source, so historical practice controls must not be presented as the current handover analysis.



Inspect the final range and control table.

16. Reopen the original **GW - Sheets - Task Register** and compare its twelve-record state: **4/4/4**, planned **10400**, actual **6800**, variance **-3600**.

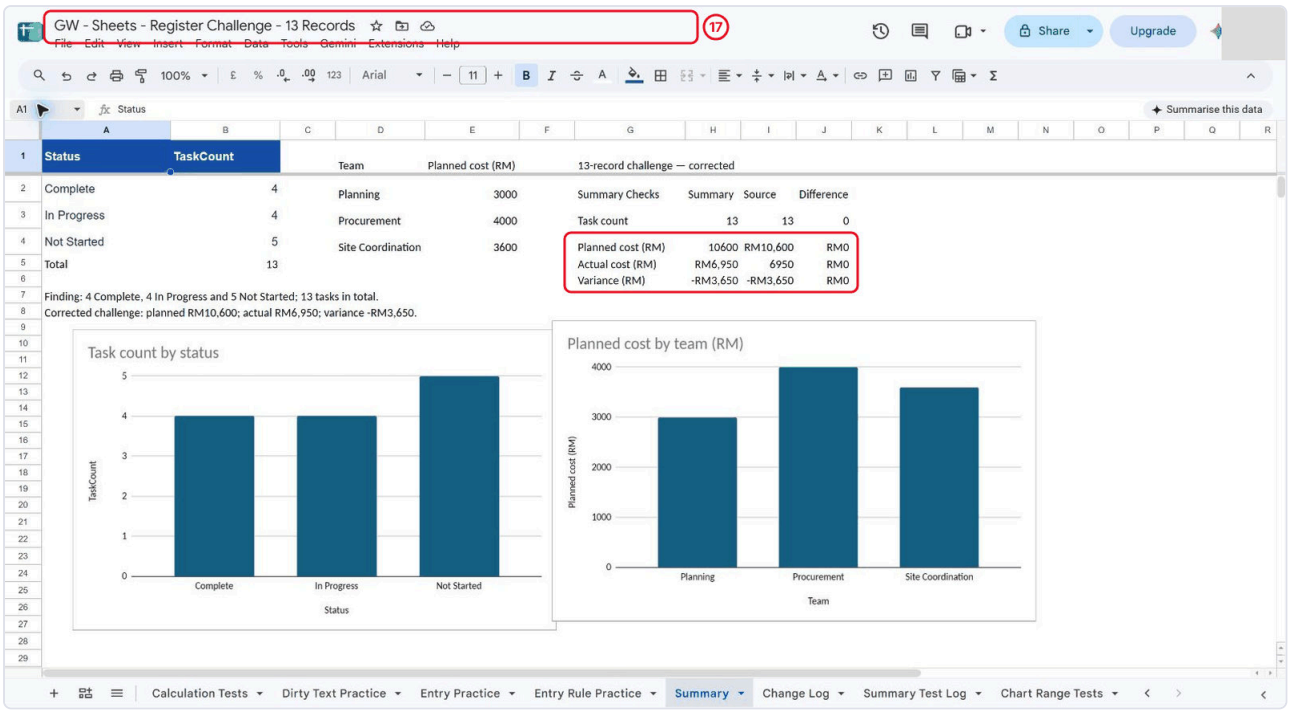
> **Reference run:** Native readback confirms its Tasks and Summary values/formulas remain unchanged from Lab 7.



Verify the original twelve-record workbook remains unchanged.

17. Leave **GW - Sheets - Register Challenge - 13 Records** clearly named and saved in the corrected state.

> **Reference run:** Its 13-record source, controls, export and written guide are retained; there is no T014 author-reference copy yet.



Leave the corrected challenge ready for handover.

- Corrected challenge sponsor PDF
- Current export manifest

> **Validation boundary:** guided 8.1–8.2, cell-supported author explanation and casework 8.4 are checked. Partner explanation/handover, independent T014 and shortened-range diagnosis, and learner timing remain pending. All eight Foundation labs now have author-reference captures; this does not establish a classroom-ready release or resolve earlier CSV/import and values-only-format limitations.

Independent Practice

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Close the earlier click instructions. Add T014 on a further disposable copy using the fields in the case input (Part 2 of the prompts page), then extend and verify all affected features independently.
2. Use your control table to catch at least one deliberately shortened range. Correct it and explain the cause with cell evidence.
3. Give a partner the operating guide and ask them to find the current source and identify the checks for the next row. Record a handover improvement and leave all scenario copies clearly named.

Evidence to Hand Over

- 13-record cost-corrected challenge and a full coverage/change log.
- Separate 14-record independent test with reconciled totals and repaired range omission.
- Readable current export plus a tested operating guide and preserved baseline.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The challenge contaminates the next baseline lesson	Keep named scenario copies and compare their population and totals before proceeding.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You have built, checked and extended a usable register.

Check yourself: Which ranges needed extending after T013 was added?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Check the handover
```

Part 2: Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Corrections and extension: T013 ActualCost becomes 150. Further-copy T014: Procurement |
Verify submission checklist | Bala Kumar | week 4 | In Progress | planned 300 | actual 100.
Keep all three states separate: original baseline, 13-record corrected challenge, 14-record
extension.
```

Part 3: Captured Challenge Reference

T013 TSV for Tasks A14:H14:

```
T013    Planning        Check the handover      Aina Rahman     4       Not Started     200
0
```

I14 =H14-G14. Live source totals/average/count and COUNTIF/SUMIF bounds include row 14. PlannedCosts in this challenge is Tasks!G2:G14, retargeted from historical Sorting Practice!G2:G13. Status range Tasks F2:F14 (plus retained spare N2), strict canonical choices Complete/In Progress/Not Started. Commit through range picker and actually test the new cell; first compact text edit did not persist in the author run.

Before correction: 13 unique records, planned 10600, actual 6800, variance -3800, status 4/4/5; Planning planned 3000. Deliberate I14=H14-G13 gives -700 and total -4300 (mismatch -500), repaired to own-row H14-G14. After H14=150: row variance -50, actual 6950, variance -3650, planned 10600 and status 4/4/5 unchanged. No extra sort was performed; all 13 variance formulas were inspected in the grid and metadata.

Summary actual check:

```
=SUMIF(Tasks!B2:B14,"Planning",Tasks!H2:H14)+SUMIF(Tasks!B2:B14,"Procurement",Tasks!H2:H14)+SU
MIF(Tasks!B2:B14,"Site Coordination",Tasks!H2:H14)
```

Change Log controls compare count/unique IDs, planned and actual team totals, variance sum versus actual-minus-planned, status population and named range. All differences 0. Both saved views reopened with A1:I14: Incomplete tasks shows 9 IDs T002,T003,T005,T006,T008,T009,T011,T012,T013; Planning incomplete tasks shows 3 IDs T002,T003,T013. No manual view-range edit needed in this session; both closed with 13 source rows visible.

Current PDF: GW-Register-Challenge-Corrected-r1.pdf, Summary A1:N28, one A3 landscape page, 13-record corrected state. Current native workbook's Tasks/Summary/Change Log are authoritative; inherited practice tabs and old outputs are history. Original 12-record baseline unchanged at 10400/6800/-3600/status 4/4/4. Independent T014 and partner/timing checks pending.

Part 4: Change Log Layout

Columns are separated by tabs, so you can paste each block into its first cell, then write your own entries.

Coverage log, A1:C12 (section 8.4, step 1):

```
Feature Challenge coverage / check      Result or scope boundary
Row variance
Source totals
Other controls
Status validation
Named range
Conditional summaries
Source reconciliation
Charts
Filter view coverage
Archived tabs and outputs
Fault prediction and repair
```

Control table, A15:D22 (step 4). Put each method's formula in B and C, and B minus C in D:

```
Control Method one      Method two      Difference
Record count
Unique task IDs
Planned cost
Actual cost
Variance
Status population
Named range
```

Prediction and observation labels, A24 and A26 (steps 4 and 7):

```
Correction prediction
```

```
Observed correction
```

Operating guide, A29:A39 (step 14), with `Current corrected challenge / next-record procedure` in B29:

```
Operating guide
Current source
Enter another record
Extend calculations
Check entry rules
Sort and view safely
Reconcile before export
Export current sponsor copy
Fixed copy identity
Current export SHA256
Human check boundary
```

Google Sheets Fundamental and Intermediate

A one-day course review draft with fictional inputs, substantial casework and expected-result checks. Lab 1 includes captured native conversion, a source map, separate practice tabs, record reconstruction and Drive reopening. Browser upload required instructor handoff. Labs 2–8 and independent learner validation remain pending.

Asraf Jaafar Sidik · Microsoft Certified Trainer ·